A Course Material on

Linear Integrated Circuits

By

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QUALITY CERTIFICATE

This is to certify that the e-course material

Subject Code **: EC6404**

Subject **: Linear Integrated Circuits**

Class **: II Year ECE**

Being prepared by me and it meets the knowledge requirement of the university curriculum.

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EC6404 LINEAR INTEGRATED CIRCUITS **L T P C 3 0 0 3**

OBJECTIVES:

- To introduce the basic building blocks of linear integrated circuits.
- To learn the linear and non-linear applications of operational amplifiers.
- To introduce the theory and applications of analog multipliers and PLL.
- To learn the theory of ADC and DAC.
- To introduce the concepts of waveform generation and introduce some special function ICs.

UNIT I BASICS OF OPERATIONAL AMPLIFIERS 9
Current mirror and current sources, Current sources as active loads, Voltage sources, Voltage References, BJT Differential amplifier with active loads, Basic information about op-amps – Ideal Operational Amplifier - General operational amplifier stages -and internal circuit diagrams of IC 741, DC and AC performance characteristics, slew rate, Open and closed loop configurations.

UNIT II APPLICATIONS OF OPERATIONAL AMPLIFIERS 9
Sign Changer, Scale Changer, Phase Shift Circuits, Voltage Follower, V-to-I and I-to-V converters, adder, subtractor, Instrumentation amplifier, Integrator, Differentiator, Logarithmic amplifier, Antilogarithmic amplifier, Comparators, Schmitt trigger, Precision rectifier, peak detector, clipper and clamper, Low-pass, high-pass and band-pass Butterworth filters.

UNIT III ANALOG MULTIPLIER AND PLL 9
Analog Multiplier using Emitter Coupled Transistor Pair - Gilbert Multiplier cell – Variable transconductance technique, analog multiplier ICs and their applications, Operation of the basic PLL, Closed loop analysis, Voltage controlled oscillator, Monolithic PLL IC 565, application of PLL for AM detection, FM detection, FSK modulation and demodulation and Frequency synthesizing.

UNIT IV ANALOG TO DIGITAL AND DIGITAL TO ANALOG CONVERTERS 9 Analog and Digital Data Conversions, D/A converter – specifications - weighted resistor type, R-2R Ladder type, Voltage Mode and Current-Mode *R* □ *2R* Ladder types - switches for D/A converters, high speed sample-and-hold circuits, A/D Converters – specifications - Flash type - Successive Approximation type - Single Slope type – Dual Slope type - A/D Converter using Voltage-to-Time Conversion - Over-sampling A/D Converters.

UNIT V WAVEFORM GENERATORS AND SPECIAL FUNCTION ICS 9 Sine-wave generators, Multivibrators and Triangular wave generator, Saw-tooth wave generator, ICL8038 function generator, Timer IC 555, IC Voltage regulators – Three terminal fixed and adjustable voltage regulators - IC 723 general purpose regulator - Monolithic switching regulator, Switched capacitor filter IC MF10, Frequency to Voltage and Voltage to Frequency converters, Audio Power amplifier, Video Amplifier, Isolation Amplifier, Opto-couplers and fibre optic IC.

TOTAL: 45 PERIODS

OUTCOMES: Upon Completion of the course, the students will be able to:

- Design linear and non linear applications of op – amps.
- Design applications using analog multiplier and PLL.
- Design ADC and DAC using op – amps.
- Generate waveforms using op – amp circuits.
- Analyze special function ICs.

TEXT BOOKS:

1. D.Roy Choudhry, Shail Jain, “Linear Integrated Circuits”, New Age International Pvt. Ltd., 2000.
2. Sergio Franco, “Design with Operational Amplifiers and Analog Integrated Circuits”, 3rd Edition, Tata Mc Graw-Hill, 2007.

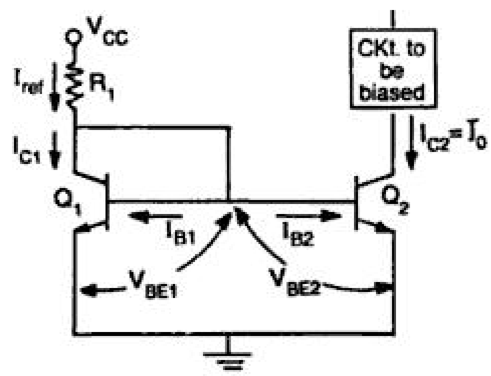
REFERENCES:

1. Ramakant A. Gayakwad, “OP-AMP and Linear ICs”, 4th Edition, Prentice Hall / Pearson Education, 2001.
2. Robert F.Coughlin, Frederick F.Driscoll, “Operational Amplifiers and Linear Integrated Circuits”, Sixth Edition, PHI, 2001.
3. B.S.Sonde, “System design using Integrated Circuits” , 2nd Edition, New Age Pub, 2001
4. Gray and Meyer, “Analysis and Design of Analog Integrated Circuits”, Wiley International, 2005.
5. Michael Jacob, “Applications and Design with Analog Integrated Circuits”, Prentice Hall of India, 1996.
6. William D.Stanley, “Operational Amplifiers with Linear Integrated Circuits”, Pearson Education, 2004.
7. S.Salivahanan & V.S. Kanchana Bhaskaran, “Linear Integrated Circuits”, TMH, 2008.

UNIT-1
BASICS OF OPERATIONAL AMPLIFERS

1.1 Constant current source (Current Mirror):

A constant current source makes use of the fact that for a transistor in the active mode of operation, the collector current is relatively independent of the collector voltage. In the basic circuit shown in fig 1 and collector characteristics of a CE Transistor as in fig.2



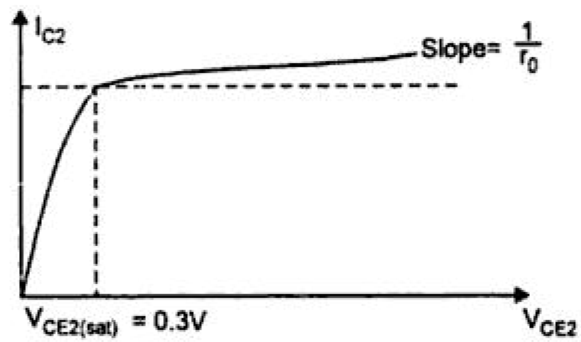


Fig. 1.1 Current mirror circuit Fig 1.2 Current source output current characteristics

Transistors Q_1&Q_2 are matched as the circuit is fabricated using IC technology. Base and emitter of Q_1& Q_2 are tied together and thus have the same V_{BE}. In addition, transistor Q1 is connected as a diode by shorting its collector to base. The input current I_{ref} flows through the diode connected transistor Q1 and thus establishes a voltage across Q_1. This voltage in turn appears between the base and emitter of Q_2 .Since Q_2 is identical to Q_1, the emitter current of Q_2 will be equal to emitter current of Q_1 which is approximately equal to I_{ref}. As long as Q2 is maintained in the active region ,its collector current $I_{C2}=I_O$ will be approximately equal to I_{ref} . Since the output current Io is a reflection or mirror of the reference current I_{ref}, the circuit is often referred to as a current mirror.

Analysis:

The collector current I_{C1} and I_{C2} for the transistor Q1 and Q2 can be approximately expressed as

$$I_{C1}(t) = \alpha\, I_{ES} e^{V_{BE1}/V_T} \text{ ----------------------------- (1)}$$

$$I_{C2}(t) = \alpha\, I_{ES} e^{V_{BE2}/V_T} \text{ ------------------------------ (2)}$$

Where I_{ES} is reverse saturation current in emitter junction and V_T is temperature equivalent of voltage.

From equation (1) & (2)

Since $V_{BE1}=V_{BE2}$ we obtain $I_{C2}=I_{C1}=I\backslash_C=I_O$

Also since both the transistors are identical, $I_{C1} = I_{C2}$
KCL at the collector of Q1 gives

$$I_{ref} = I_{C1}+I_{B1}+I_{B2}$$
$$= I_{C1} + \frac{I_{C1}}{\beta} + \frac{I_{C2}}{\beta} = I_{C1} + 2\frac{I_{C1}}{\beta} \quad \text{When } I_{C1} = I_{C2} = I_C = I_o \quad \text{---------------- (4)}$$

Solving Eq (4)

$$I_{ref} = (V_{CC} - V_{BE(ON)})/R$$
$$I_{ref} = I_{C1} + 2\frac{I_{C1}}{\beta} = I_{C1}(1 + \frac{2}{\beta})$$
$$I_C = I_{C1} = I_{C2} = \frac{I_{ref}}{1+\frac{2}{\beta}} = \frac{\beta}{\beta+2}(V_{CC} - V_{BE(ON)})/R \quad \text{----------------- (5)}$$
$$I_0 = I_{ref}$$

From Eq.5 for $\beta>>1$, $\frac{\beta}{\beta+2}$ is almost unity and the output current I0 is equal to the reference current, ref which for a given R1 is constant. Typically I_O varies by about 3% for $50 \le \beta \le 200$.
The circuit however operates as a constant current source as long as Q2 remains in the active region.

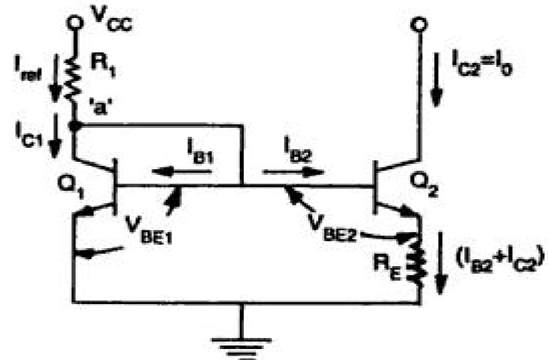


Fig.1.3 Simple current source

1.1.1 Widlar current source:

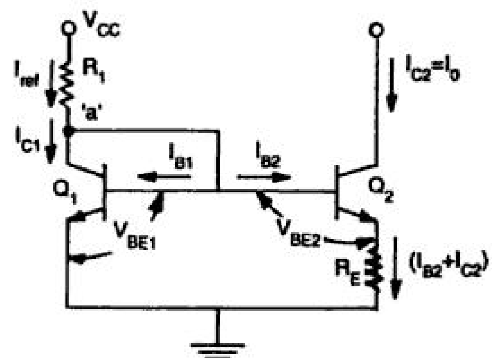


Fig.1.4 Widlar current source

Widlar current source which is particularly suitable for low value of currents. The circuit differs from the basic current mirror only in the resistance R_E that is included in the emitter lead of Q_2. It can be seen that due to R_E the base-emitter voltage V_{BE2} is less than V_{BE1} and consequently current Io is smaller than I_{C1}

The ratio of collector currents I_{C1}&I_{C2} using

$$\frac{I_{C1}}{I_{C2}} = e^{\frac{V_{BE1}-V_{BE2}}{V_T}} \quad \textbf{(1)}$$

Taking natural logarithm of both sides, we get

$$V_{BE1} - V_{BE2} = V_T \ln \frac{I_{C1}}{I_{C2}} \quad \textbf{----- (2)}$$

Writing KVL for the emitter base loop

$$V_{BE1}=V_{BE2}+(I_{B2}+I_{C2})R_E \text{ ----------------(3)}$$

Or $$V_{BE1}-V_{BE2}=(1/\beta+1)I_{C2}R_E \text{ -----------(4)}$$

From eqn. (2) & (4) we obtain

$$V_T \ln \frac{I_{C1}}{I_{C2}} = (1/\beta+1)\, I_{C2}R_E \text{ -------------------------------------(5)}$$

A relation between I_{C1} and the reference current I_{ref} is obtained by writing KCL at the collector point of Q_1

$$I_{ref}= I_{C1}+I_{B1}+I_{B2}$$

$$I_{ref} = I_{C1} + I_{C1}/\beta + I_{C2}/\beta$$

Neglecting I_{C2}/β,

$$I_{ref} = I_{C1}(1+\frac{1}{\beta})$$

$$I_{ref} = \frac{V_{cc}-V_{BE}}{R_1}$$

When $\beta >> 1$, $I_{C1} = I_{ref}$

1.1.2 Wilson current source:

The Wilson current source shown in figure

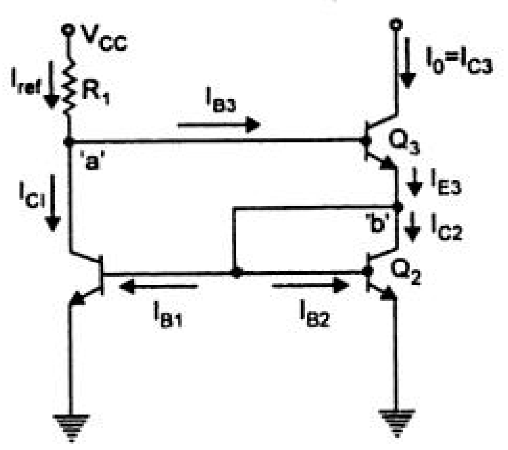


Fig.1.5 Wilson current source

It provides an output current I_0 which is very nearly equal to V_{ref} and also exhibits a very high output resistance.

Analysis:

Since $V_{BE1}=V_{BE2}$

$I_{C1}=I_{C2}$ and $I_{B1}=I_{B2}=I_B$

At node 'b'

$I_{E3}=2I_B+I_{C2}=(\frac{2}{\beta}+1)I_{C2}$------------------ (1)

I_{E3} is equal to

$I_{E3}=I_{C3}+I_{B3}=I_{C3}(1+1/\beta)$ ------------------ (2)

From (1) and (2)

$I_{C3}(1+1/\beta)=I_{C2}(1+2/\beta)$

From Eqn. (1) & (2) we obtain

$I_{C3}=I_0=I_{C2}(\beta \square 2)/(\beta+1)=I_{C1}(\beta \square 2)/(\beta+1)$ Since $I_{C1}=I_{C2}$

At node 'a' $I_{ref}=I_{C1}+I_{B3}=\frac{\beta+1}{\beta+2}I_0+\frac{I_0}{\beta}=$

$Iref=\frac{\beta^2+2\beta+2}{\beta^2+2\beta}I_0$ and $Iref=\frac{V_{cc}-2V_{BE}}{R_1}$; $I_0-Iref=\frac{2}{\beta^2+2\beta+2}Iref$ is very small for modest β.

But output resistance is greater than Widlar source.

1.2 Current sources as Active loads

The current source can be used as an active load in both analog and digital IC's. The active load realized using current source in place of the passive load (i.e. a resistor) in the collector arm of differential amplifier makes it possible to achieve high voltage gain without requiring large power supply voltage. The active load so achieved is basically R_0 of a PNP transistor.

1.3 Voltage Sources

A voltage source is a circuit that produces an output voltage V_0, which is independent of the load driven by the voltage source, or the output current supplied to the load. The voltage source is the circuit dual of the constant current source.

A number of IC applications require a voltage reference point with very low ac impedance and a stable dc voltage that is not affected by power supply and temperature variations. There are two methods which can be used to produce a voltage source, namely,

1. Using the impedance transforming properties of the transistor, which in turn determines the current gain of the transistor and
2. Using an amplifier with negative feedback.

1.3.1 Voltage source circuit using Impedance transformation:

The voltage source circuit using the impedance transforming property of the transistor is shown in figure. The source voltage V_S drives the base of the transistor through a series resistance R_S and the output is taken across the emitter. From the circuit, the output ac resistance looking Into emitter is given by

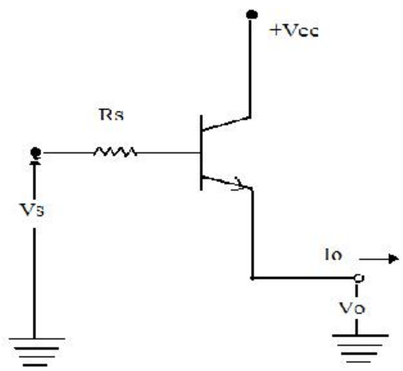


Fig.1.6 Voltage source circuit using Impedance transformation:

$$\frac{d_o}{dI_o} = R_0 = \frac{R_S}{\beta+1} + r_{eb} \;;$$

With β >> 100, $R_0 = \frac{R_S}{\beta+1}$

It is to be noted that, equation is applicable only for small changes in the output current. The load regulation parameter indicates the changes in V_0 resulting from large changes in output current I_0 , Reduction in V_0 occurs as I_0 goes from no-load current to full-load current and this factor determines the output impedance of the voltage sources.

1.3.2 Emitter– follower or Common Collector Type Voltage source:

The figure shows an emitter follower or common collector type voltage source.

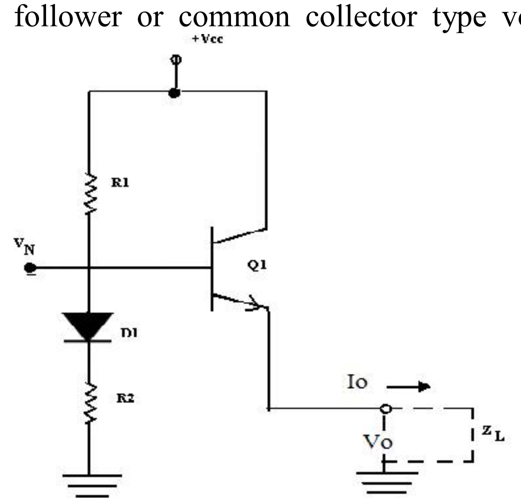


Fig.1.7 Emitter– follower or Common Collector Type Voltage source

This voltage source is suitable for the differential gain stage used in op-amps. This circuit has the advantages of

1. Producing low ac impedance and
2. Resulting in effective decoupling of adjacent gain stages.

The low output impedance of the common-collector stage simulates a low impedance voltage source with an output voltage level of V_0 represented by

$$V_0 = V_{CC} \frac{R_2}{R_1 + R_2}$$

The diode D_1 is used for offsetting the effect of dc value V_{BE} , across the E-B junction of the transistor, and for compensating the temperature dependence of V_{BE} drop of Q_1. The load Z_L shown in dotted line represents the circuit biased by the current through Q_1.

The impedance R_0 looking into the emitter of Q_1 derived from the hybrid π model is given by

$$R_o = \frac{V_T}{I_1} + \frac{R_1 R_2}{\beta(R_1 + R_2)}$$

1.3.3 Voltage Source Using Temperature compensated Avalanche Diode

The voltage source using common collector stage has the limitations of its vulnerability for changes in bias voltage V_N and the output voltage V_0 with respect to changes in supply voltage Vcc. This is overcome in the voltage source circuit using the breakdown voltage of the base- emitter junction shown below.

The emitter – follower stage of common – collector is eliminated in this circuit, since the impedance seen looking into the bias terminal N is very low. The current source I_1 is normally simulated by a resistor connected between Vcc and node n. Then, the output voltage level V_0 at node N is given by $V_0 = V_B + V_{BE}$ Where V_B is the breakdown voltage of diode D_B and V_{BE} is the diode drop across D_1.

The breakdown diode D_B is normally realized using the base-emitter junction of the transistor. The diode D_1 provides partial compensation for the positive temperature coefficient effect of V_B. In a monolithic IC structure, D_B and D_1 can be conveniently realized as a single transistor with two individual emitters as shown in figure.

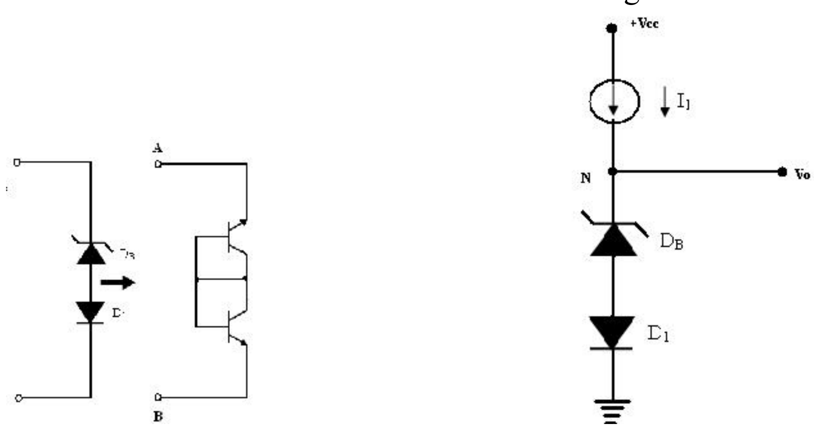


Fig.1.8 Temperature Compensated avalanche diode

1.3.4 Voltage source using breakdown voltage of the base- emitter junction

The structure consists of composite connection of two transistors which are diode-connected back-to back. Since the transistors have their base to collector terminals common, they can be designed as a single transistor with two emitters.

The output resistance R_0 looking into the output terminal in figure is given by $R_o=R_B+V_T/I_1$ where R_B and V_T/I_1 are the ac resistances of the base–emitter resistance of diode D_B and D_1 respectively. Typically R_B is in the range of 40Ω to 100Ω, and V_0 in the range of 6.5V to 9V.

1.3.5 Voltage Source using V_{BE} as a reference:

The output stage of op-amp requires stabilized bias voltage source, which can be obtained using a forward-biased diode connected transistor. The forward voltage drop for such a connection is approximately 0.7V, and it changes slightly with current.

When a voltage level greater than 0.7V, is needed, several diodes can be connected in series, which can offer integral multiples of 0.7V. Alternatively, the figure shows a multiplier circuit, which can offer voltage levels that need not be integral multiplied of 0.7V. The drop across R_2 equals V_{BE} drop of Q_1. Considering negligible base current for Q_1, current through R_2 is the same as that flowing through R_1.

Therefore, the output voltage V_0 can be expressed as

$$V_0 = I_2(R_1 + R_2) = \frac{V_{BE}}{R_2}(R_1 + R_2) = V_{BE}(\frac{R_1}{R_2} + 1)$$

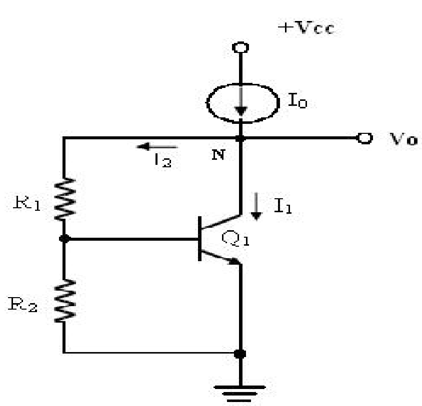


Fig.1.9 V_{BE} multiplier Circuit

Hence, the voltage V_0 can be any multiple of V_{BE} by properly selecting the resistors R_1 and R_2. Due to the shunt feedback provided by R_1, the transistor current I_1 automatically adjusts itself, towards maintaining I_2 and V_0 relatively independent of the changes in supply voltage.

The ac output resistance of the circuit R_0 is given by,

$$R_O = \frac{dV_0}{dI_o} = \frac{R_1+R_2}{1+g_m R_2} = \frac{(R_1+R_2)}{R_2 g_m} \quad \text{When } g_m R_2 >> 1$$

$$R_O = \frac{V_0}{V_{BE}}\frac{1}{g_m} = \frac{V_O}{V_{BE}}\frac{V_1}{I_C} \quad \text{as } \frac{V_O}{V_{BE}} = \frac{(R_1+R_2)}{R_2}$$

1.4 Voltage References

The circuit that is primarily designed for providing a constant voltage independent of changes in temperature is called a voltage reference. The most important characteristic of a voltage reference is the temperature coefficient of the output reference voltage T_{CR} , and it is expressed as

$$T_{CR} = \frac{dV_R}{dT}$$

The desirable properties of a voltage reference are:

1. Reference voltage must be independent of any temperature change.
2. Reference voltage must have good power supply rejection which is as independent of the supply voltage as possible and
3. Output voltage must be as independent of the loading of output current as possible, or in other words, the circuit should have low output impedance.

The voltage reference circuit is used to bias the voltage source circuit, and the combination can be called as the voltage regulator. The basic design strategy is producing a zero TC_R at a given temperature, and thereby achieving good thermal ability. Temperature stability of the order of 100ppm/^{0} C is typically expected.

1.4.1 Voltage Reference circuit using temperature compensation scheme

The voltage reference circuit using basic temperature compensation scheme is shown below. This design utilizes the close thermal coupling achievable among the monolithic components and this technique compensates the known thermal drifts by introducing an opposing and compensating drift source of equal magnitude.

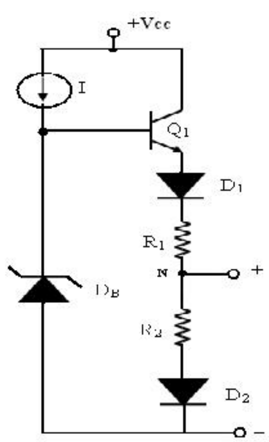


Fig.1.10 Voltage reference circuit using temperature compensation scheme

A constant current I is supplied to the avalanche diode D_B and it provides a bias voltage of V_B to the base of Q_1. The temperature dependence of the V_{BE} drop across Q_1 and those across D_1 and D_2 results in respective temperature coefficients. Hence, with the use of resistors R_1 and R_2 with tapping across them at point N compensates for the temperature drifts in the base-emitter loop of Q_1. This results in generating a voltage reference V_R with normally zero temperature coefficient.

1.4.2 Voltage Reference circuit using Avalanche Diode Reference:

A voltage reference can be implemented using the breakdown phenomenon condition of a heavily doped PN junction. The Zener breakdown is the main mechanism for junctions, which breakdown at a voltage of 5V or less. For integrated transistors, the base-emitter breakdown voltage falls in the range of 6 to 8V. Therefore, the breakdown in the junctions of the integrated transistor is primarily due to avalanche multiplication. The avalanche breakdown voltage VB of a transistor incurs a positive temperature coefficient, typically in the range of 2mV/0 C to 5mV/0 C.

Figure depicts a current reference circuit using avalanche diode reference. The base bias for transistor Q1 is provided through register R1 and it also provides the dc current needed to bias DB, D1 and D2 .The voltage at the base of Q_1 is equal to the Zener voltage V_B added with two diode drops due to D_1 and D_2. The voltage across R_2 is equal to the voltage at the base of Q_1 less the sum of the base – emitter voltages of Q_1 and Q_2.

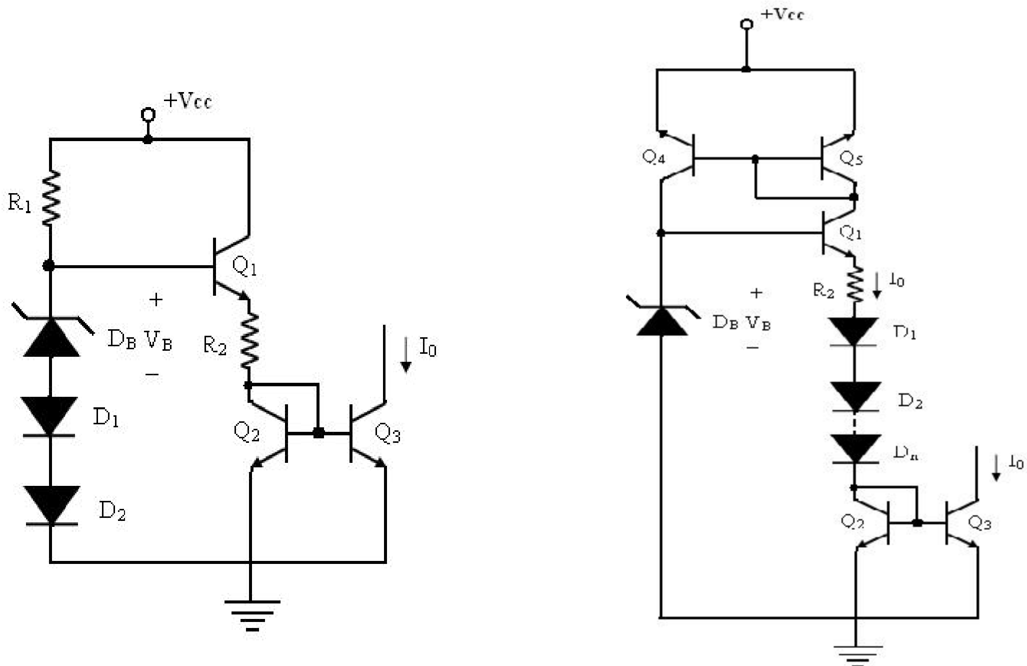


Fig. 1.11 Voltage reference using avalanche diodes and temperature compensated

Hence, the voltage across R_2 is approximately equal to that across $D_B = V_B$. Since Q_2 and Q_3 act as a current mirror circuit, current I_0 equals the current through R_2.

$$I_O = \frac{V_B}{R_2}$$

It shows that, the output current I_0 has low temperature coefficient, if the temperature coefficient of R_2 is low, such as that produced by a diffused resistor in IC fabrication.

The zero temperature coefficients for output current can be achieved, if diodes are added in series with R_2, so that they can compensate for the temperature variation of R_2 and V_B. The temperature compensated avalanche diode reference source circuit is shown in figure. The transistor Q_4 and Q_5 form an active load current mirror circuit. The base voltage of Q_1 is the voltage V_B across Zener D_B.

Then, $V_B = (V_{BE} * n) + V_{BE}$ across Q_1 + V_{BE} across Q_2 + drop across R_2. Here, n is the number of diodes.

It can be expressed as $V_B = (n+2)\ V_{BE} + I_0 * R_2$

Differentiating for V_B, I_0, R_2 and V_{BE} partially, with respect to temperature T, we get

$$\frac{\partial V_B}{\partial T} = n + 2\frac{\partial V_{BE}}{\partial T} + R_2\frac{\partial I_0}{\partial T} + I_0\frac{\partial R_2}{\partial T}$$

Dividing throughout by I_0 R_2, we get

$$\frac{I}{I_O}\frac{\partial I_O}{\partial T} = 0 = \frac{1}{R_2 I_0}[\frac{\partial V_B}{\partial T} - (n+2)\frac{\partial V_{BE}}{\partial T} - \frac{1}{R_2}\frac{\partial R_2}{\partial T}$$

Therefore, zero temperature coefficient of I_0 can be obtained, if the above condition is satisfied.

1.5 Differential amplifier

The function of a differential amplifier is to amplify the difference between two signals. The need for differential amplifier arises in many physical measurements where response from DC to many MHz of frequency is required. This forms the basic input stage of an integrated amplifier. The basic differential amplifier has the following important properties of

Excellent stability
High versatility and
High immunity to interference signals

The differential amplifier as a building block of the op-amp has the advantages of

Lower cost
Easier fabrication as IC component and
closely matched components.

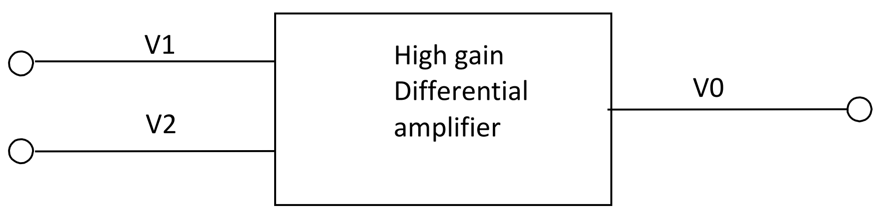


Fig. 1.12 Block diagram of Differential amplifier

The above figure shows the basic block diagram of a differential amplifier, with two input terminals and one output terminal. The output signal of the differential amplifier is proportional to the difference between the two input signals.

$$V_0 = A_{dm}(V_1 - V_2)$$

If $V_1 = V_2$, then the output voltage is zero. A non-zero output voltage V_0 is obtained when V_1 and V_2 are not equal. The difference mode input voltage is defined as $V_m = V_1 - V_2$ and the common mode input voltage is defined as

$$V_{CM=}\frac{V1+V2}{2}$$

These equation show that if $V_1 = V_2$, then the differential mode input signal is zero and common mode input signal is $V_{cm} = V_1 = V_2$.

1.5.1 Differential Amplifier with Active load:

Differential amplifier is designed with active loads to increase the differential mode voltage gain. The open circuit voltage gain of an op-amp is needed to be as large as possible. This is got by cascading the gain stages which increase the phase shift and the amplifier also becomes vulnerable to oscillations. The gain can be increased by using large values of collector resistance. For such a circuit, the voltage gain is given by

$$A_{dm} = g_m R_C$$

To increase the gain the $I_C R_C$ product must be made very large. However, there are limitations in IC fabrication such as,

1. A large value of resistance needs a large chip area.
2. For large R_C, the quiescent drop across the resistor increase and a large power supply will be required to maintain a given operating current.
3. Large monolithic resistor introduces large parasitic capacitances which limits the frequency response of the amplifier.
4. for linear operation of the differential pair, the devices should not be allowed to enter into saturation. This limits the max input voltage that can be applied to the bases of transistors Q_1 and Q_2 the base-collector junction must be allowed to become forward-biased by more than 0.5V. The large value of load resistance produces a large dc voltage drop $(I_{EE} / 2) R_C$, so that the collector voltage will be $V_C = V_{cc} - (I_{EE}/2) R_C$ and it will be substantially less than the supply voltage Vcc. This will reduce the input voltage range of the differential amplifier. Due to the reasons cited above, an active load is preferred in the differential amplifier configurations.

- ✓ **BJT Differential Amplifier using active loads:**

A simple active load circuit for a differential amplifier is the current mirror active load as shown in figure. The active load comprises of transistors Q3 and Q4 with the transistor Q3 connected as a Diode with its base and collector shorted. The circuit is shown to drive a load R_L. When an ac input voltage is applied to the differential amplifier, the various currents of the circuit are given by

$I_{C4} = I_{C3} = I_{C1} = g_m V_{id}/2$ where $I_{C4} = I_{C3}$ due to current mirror action.

$I_{C2} = - g_m V_{id}/2$.

We know that the load current I_L entering the next stage is

$I_L = I_{C2} - I_{C4} = - g_m V_{id}/2 - g_m V_{id}/2 = - g_m V_{id}$

Then, the output voltage from the differential amplifier is given by $V_0 = - I_L R_L = g_m R_L V_{id}$. The ac voltage gain of the circuit is given by $A_V = \frac{V_0}{V_{id}} = g_{mR_L}$. The amplifier can amplify the differential input signals and it provides single-ended output with a ground reference since the load R_L is connected to only one output terminal. This is made possible by the use of the current mirror active load. The output resistance R_0 of the circuit is that offered by the parallel combination of transistors Q_2 (NPN) and Q_4 (PNP). It is given by $R_r = r_{02} \parallel r_{04}$.

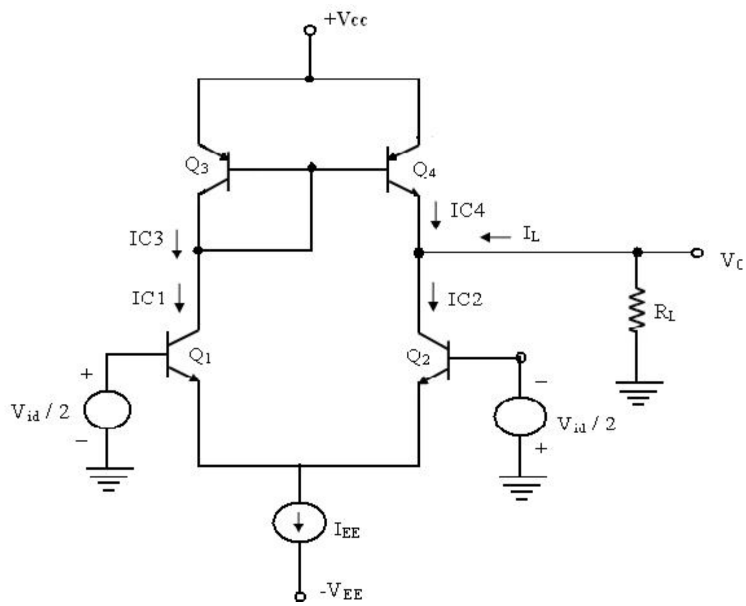


Fig. 1.13 BJT differential amplifier with current mirror active load

Analysis of BJT differential amplifier with active load:

The collector currents of all the transistors are equal.

$$I_{C1} = I_{C2} = I_{C3} = I_{C4} = I_{EE}/2 .$$

The Collector -emitter voltages of Q_1 and Q_2 are given by

$$V_{CE1}-V_{CE2} = V_C-V_E = V_{CC} - V_{EB}-(-V_{EB}) = V_{CC}$$

Eqn. shows that, the offset is higher than that of a resistive loaded differential amplifier A. This can be reduced by the use of emitter resistors for Q_3 and Q_4, and a transistor Q_5 in the current mirror load.

- ✓ **CMRR of the differential amplifier using active load:**

The differential amplifier using active load provides high voltage gain to the differential input signal and a single – ended output that is referenced to the ground is obtained. The differential amplifier which provides conversion for a differential signal to a single ended signal is necessary in differential input signal ended output amplifiers. The op-amp is one such circuit. The changes in the common-mode signal of the bias current source. This induces a change in I_{C2} and an identical change in I_{C1}. The change in I_{C1} will then produce a change in the PNP load devices, and thereby a change in I_{C4}, which is the collector current Q_4, The current I_{C4} is in such a direction as to cancel the change in I_{C2}. As a result of this, any common mode input does not cause a change in output.

The voltage gain of the differential amplifier is independent of the quiescent current I_{EE}. This makes it possible to use very small value of I_{EE} as low as 20μA, while still maintaining a large voltage gain. Small value of I_{EE} is preferred, since it results in a small value of bias current and a large value for the input resistance. A limitation in choosing a small I_{EE} is, however, the fact that, it will result in a poor frequency response of the amplifier.

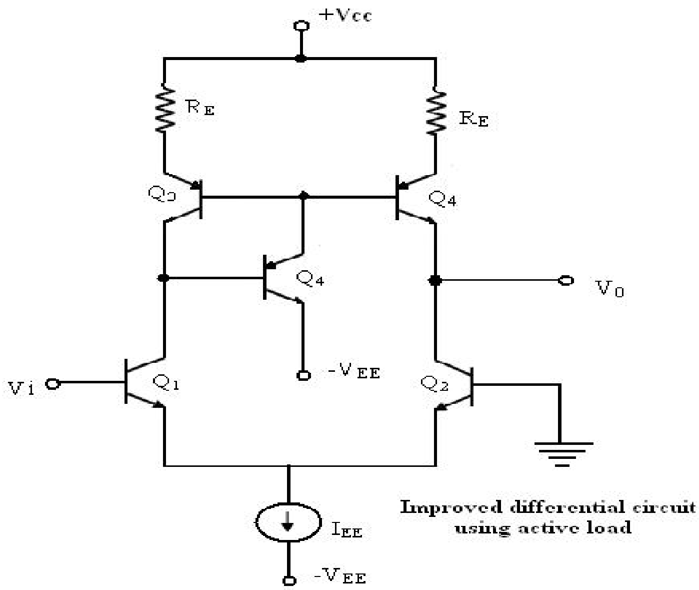


Fig.1.14. Improved differential circuit using active load

When a small value of bias current is required, the best approach is to use a JFET or MOSFET differential amplifier that is operated at comparatively higher values of I_{EE}.

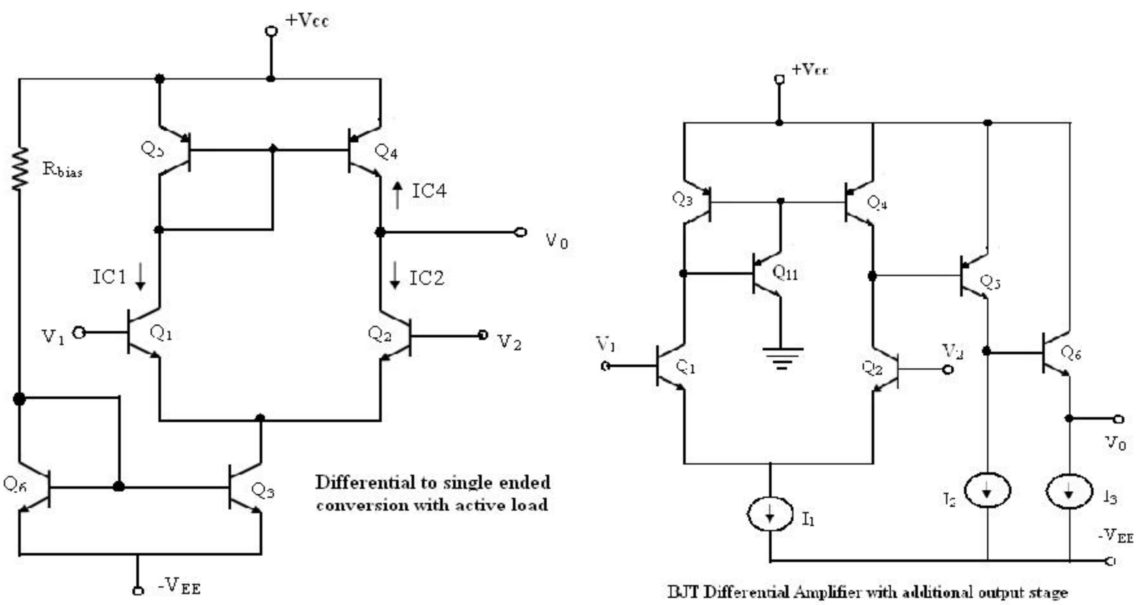


Fig.1.15. Differential to single ended conversion and output stage

Differential Mode signal analysis:

The ac analysis of the differential amplifier can be made using the circuit model as shown below. The differential input transistor pair produces equal and opposite currents whose amplitude us given by gm2 V_{id} /2 at the collector of Q_1 and Q_2. The collector current I_{c1} is fed by the transistor Q3 and it is mirrored at the output of Q_4. Therefore, the total current i0 flowing through the load resistor R_L is given by $i_0 = \frac{2g_{m2}V_{id}}{2} = g_{m2}V_{id}$.

Then the output voltage is $v_0 = i_0 R_L = g_{m2} R_L v_{id}$ and the differential mode gain Ad of the differential amplifier is

$$A_{dm} = \frac{v_0}{v_{dm}} = g_{m2} R_L$$

This current mirror provides a single ended output which has a voltage equal to the maximum gain of the common emitter amplifier.

The power of the current mirror can be increased by including additional common collector stages at the o/p of the differential input stage. A bipolar differential amplifier structure with additional stages is shown in figure. The resistance at the output of the differential stage is now given by the parallel combination of transistors Q2 and Q4 and the input resistance is offered by Q5. Then, the equivalent resistance is expressed by $R_{eq} = r_{o2} \| r_{04} \| r_{i5} = r_{i5}$.

The gain of the differential stage then becomes $A_{dm} = g_m 2 R_{eq} = g_{m2} r_{i5} = \beta I_{C2}/I_{C5}$.

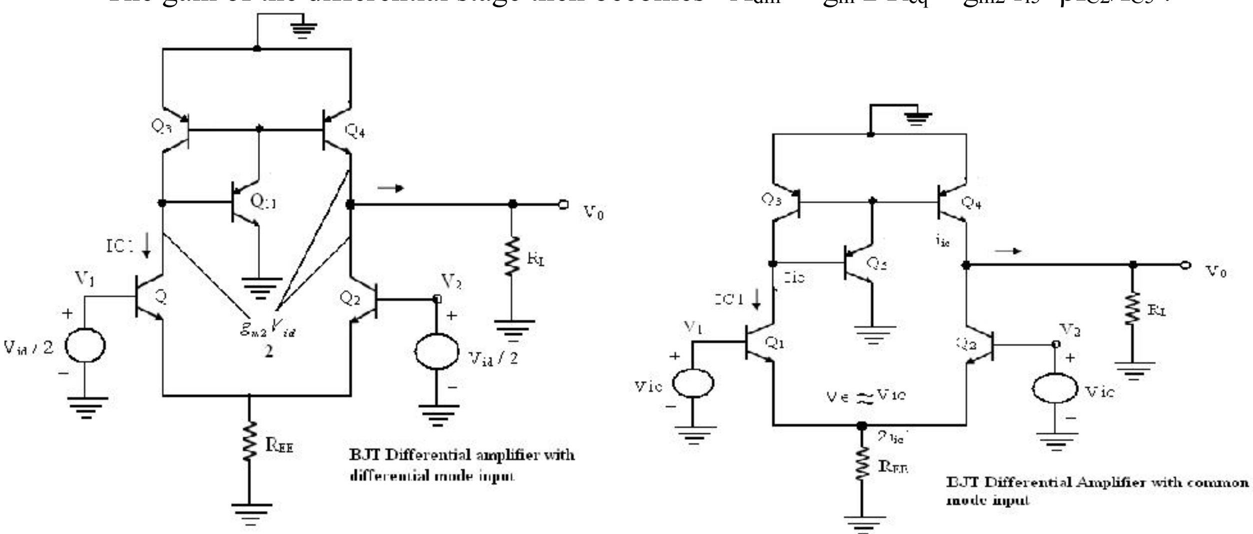


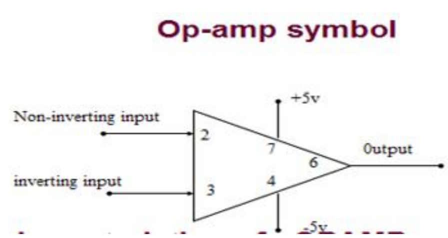
Fig. 1.16 Differentail amplifier with differential mode input and common mode input

1.6 Basic information about operational amplifiers

An operational amplifier is a direct coupled high gain amplifier consisting of one or more differential amplifiers, followed by a level translator and an output stage.

It is a versatile device that can be used to amplify ac as well as dc input signals & designed for computing mathematical functions such as addition, subtraction ,multiplication, integration & differentiation

1.7 Ideal operational Amplifiers



Ideal op-amp characteristics:

- ✓ Infinite voltage gain A.
- ✓ Infinite input resistance R_i, so that almost any signal source can drive it and there is no loading of the proceeding stage.
- ✓ Zero output resistance R_O, so that the output can drive an infinite number of other devices.
- ✓ Zero output voltage, when input voltage is zero.
- ✓ Infinite bandwidth, so that any frequency signals from o to ∞ HZ can be amplified with out attenuation.
- ✓ Infinite common mode rejection ratio, so that the output common mode noise voltage is zero.
- ✓ Infinite slew rate, so that output voltage changes occur simultaneously with input voltage changes.

1.8 General Operational Amplifier stages and internal circuit diagrams of IC 741

An operational amplifier generally consists of three stages, namely

1. A differential amplifier
2. Additional amplifier stages to provide the required voltage gain and dc level shifting.
3. An emitter-follower or source follower output stage to provide current gain and low output resistance.

A low-frequency or dc gain of approximately 104 is desired for a general purpose op-amp and hence, the use of active load is preferred in the internal circuitry of op-amp.

The output voltage is required to be at ground, when the differential input voltages are zero, and this necessitates the use of dual polarity supply voltage. Since the output resistance of op-amp is required to be low, a complementary push-pull emitter – follower or source follower output stage is employed. Moreover, as the input bias currents are to be very small of the order of pico amperes, an FET input stage is normally preferred.

Input stage:

The input differential amplifier stage uses p-channel JFETs M_1 and M_2. It employs a three-transistor active load formed by Q_3, Q_4, and Q_5. The bias current for the stage is provided by a two-transistor current source using PNP transistors Q_6 and Q_7. Resistor R_1 increases the o utput resistance seen looking into the collector of Q_4 as indicated by R_{04}. This is necessary to provide bias current stability against the transistor parameter variations. Resistor R_2 establishes a definite bias current through Q_5. A single ended output is taken out at the collector of Q_4.
MOSFET‘s are used in place of JFETs with additional devices in the circuit to prevent any damage for the gate oxide due to electrostatic discharges.

Gain stage:

The second stage or the gain stage uses Darlington transistor pair formed by Q_8 and Q_9 as shown in figure. The transistor Q_8 is connected as an emitter follower, providing large input resistance.
Therefore, it minimizes the loading effect on the input differential amplifier stage. The transistor Q_9 provides an additional gain and Q_{10} acts as an active load for this stage. The current mirror

formed by Q_7 and Q_{10} establishes the bias current for Q_9. The V_{BE} drop across Q_9 and drop across R_5 constitute the voltage drop across R_4, and this voltage sets the current through Q_8. It can be set to a small value, such that the base current of Q_8 also is very less.

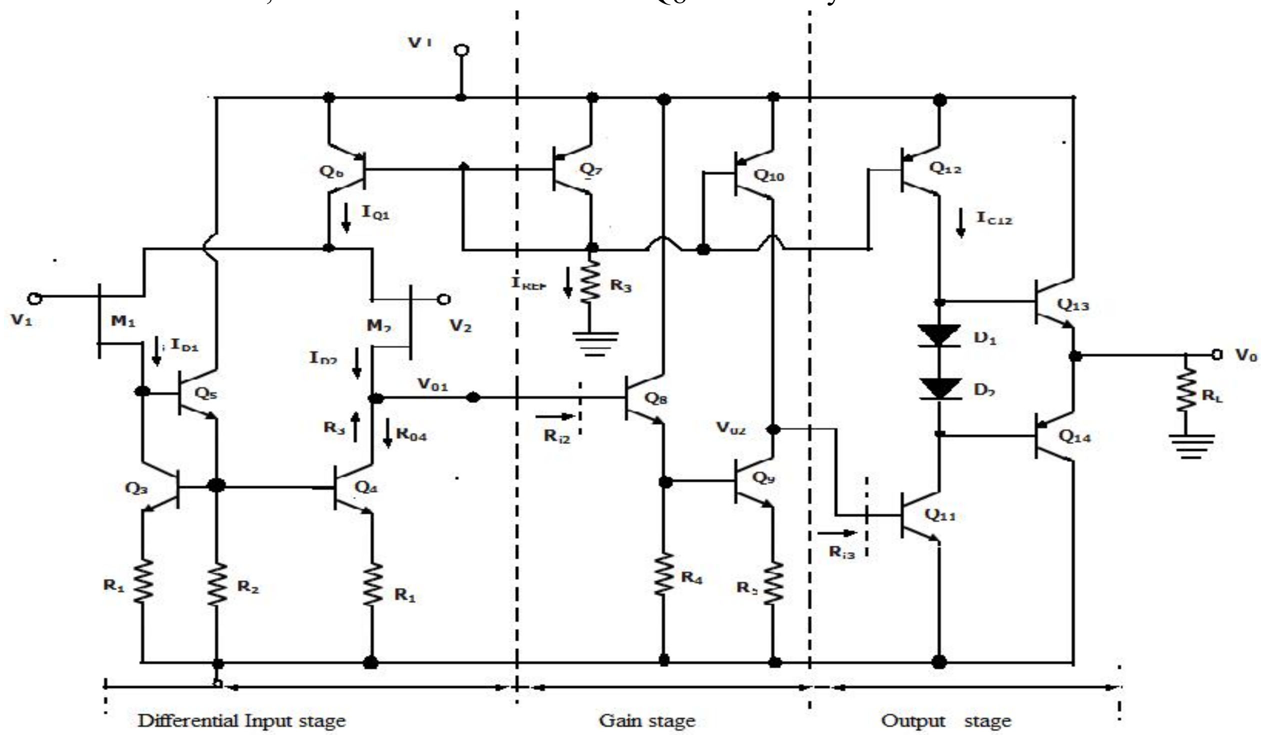


Fig. 1.17 Internal stages of Op-amp

Output stage:

The final stage of the op-amp is a class AB complementary push-pull output stage. Q_{11} is an emitter follower, providing a large input resistance for minimizing the loading effects on the gain stage. Bias current for Q_{11} is provided by the current mirror formed by Q_7 and Q_{12}, through Q_{13} and Q_{14} for minimizing the cross over distortion. Transistors can also be used in place of the two diodes.

The overall voltage gain A_V of the op-amp is the product of voltage gain of each stage as given by

$A_V=|A_d|\,|A_2||A_3|$

Where A_d is the gain of the differential amplifier stage, A_2 is the gain of the second gain stage and A_3 is the gain of the output stage.

IC 741 Bipolar operational amplifier:

The IC 741 produced since 1966 by several manufactures is a widely used general purpose operational amplifier. Figure shows that equivalent circuit of the 741 op-amp, divided into various individual stages. The op-amp circuit consists of three stages.

1. The input differential amplifier
2. The gain stage
3. the output stage.

A bias circuit is used to establish the bias current for whole of the circuit in the IC. The op-amp is supplied with positive and negative supply voltages of value ± 15V and the supply voltages as low as ±5V can also be used.

Bias Circuit:

The reference bias current I_{REF} for the 741 circuit is established by the bias circuit consisting of two diodes-connected transistors Q_{11} and Q_{12} and resistor R_5. The Widlar current source formed by Q_{11}, Q_{10} and R_4 provide bias current for the differential amplifier stage at the collector of Q_{10}. Transistors Q_8 and Q_9 form another current mirror providing bias current for the differential amplifier. The reference bias current I_{REF} also provides mirrored and proportional current at the collector of the double –collector lateral PNP transistor Q_{13}. The transistor Q_{13} and Q_{12} thus form a two-output current mirror with Q_{13A} providing bias current for output stage and Q_{13B} providing bias current for Q_{17}. The transistor Q_{18} and Q_{19} provide dc bias for the output stage. Formed by Q_{14} and Q_{20} and they establish two V_{BE} drops of potential difference between the bases of Q_{14} and Q_{18}.

Input stage:

The input differential amplifier stage consists of transistors Q_1 through Q_7 with biasing provided by Q_8 through Q_{12}. The transistor Q_1 and Q_2 form emitter – followers contributing to high differential input resistance, and whose output currents are inputs to the common base amplifier using Q_3 and Q_4 which offers a large voltage gain. The transistors Q5, Q6 and Q7 along with resistors R_1, R_2 and R_3 from the active load for input stage. The single-ended output is available at the collector of Q_6. The two null terminals in the input stage facilitate the null adjustment. The lateral PNP transistors Q_3 and Q_4 provide additional protection against voltage breakdown conditions. The emitter-base junction Q_3 and Q_4 have higher emitter-base breakdown voltages of about 50V. Therefore, placing PNP transistors in series with NPN transistors provide protection against accidental shorting of supply to the input terminals.

Gain Stage:

The Second or the gain stage consists of transistors Q_{16} and Q_{17}, with Q_{16} acting as an emitter – follower for achieving high input resistance. The transistor Q_{17} operates in common emitter configuration with its collector voltage applied as input to the output stage. Level shifting is done for this signal at this stage.

Internal compensation through Miller compensation technique is achieved using the feedback capacitor C_1 connected between the output and input terminals of the gain stage.

Output stage:

The output stage is a class AB circuit consisting of complementary emitter follower transistor pair Q_{14} and Q_{20}. Hence, they provide an effective loss output resistance and current gain. The output of the gain stage is connected at the base of Q_{22}, which is connected as an emitter follower providing a very high input resistance, and it offers no appreciable loading effect on the gain stage. It is biased by transistor Q_{13A} which also drives Q_{18} and Q_{19}, that are used for establishing a quiescent bias current in the output transistors Q_{14} and Q_{20}.

1.9 AC Characteristics:

For small signal sinusoidal (AC) application one has to know the ac characteristics such as frequency response and slew-rate.

1.9.1 Frequency Response:

The variation in operating frequency will cause variations in gain magnitude and its phase angle. The manner in which the gain of the op-amp responds to different frequencies is called the frequency response. Op-amp should have an infinite bandwidth BW =∞ (i.e.) if its open loop gain in 90dB with dc signal its gain should remain the same 90 dB through audio and onto high radio frequency. The op-amp gain decreases (roll-off) at higher frequency what reasons to decrease gain after a certain frequency reached. There must be a capacitive component in the equivalent circuit of the op-amp. For an op-amp with only one break (corner) frequency all the capacitors effects can be represented by a single capacitor C. Below fig is a modified variation of the low frequency model with capacitor C at the output.

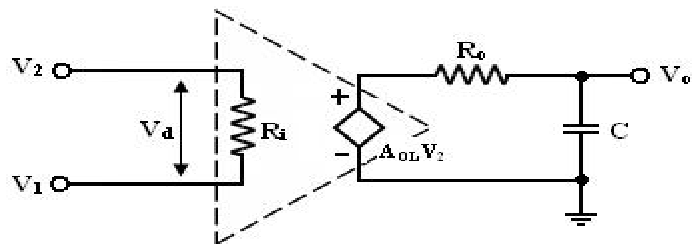


Fig 1.18 Equivalent circuit of practical circuit

There is one pole due to R_0 C and one -20dB/decade. The open loop voltage gain of an op-amp with only one corner frequency is obtained from above fig.
f1 is the corner frequency or the upper 3 dB frequency of the op-amp. The magnitude and phase angle of the open loop volt gain are f1 of frequency can be written as,
The magnitude and phase angle characteristics:

1. For frequency f<< f_1 the magnitude of the gain is 20 log A_{OL} in db.
2. At frequency f = f_1 the gain in 3 dB down from the dc value of A_{OL} in db. This frequency f_1 is called corner frequency.
3. For f>> f_1 the fain roll-off at the rate off -20dB/decade or -6dB/decade.

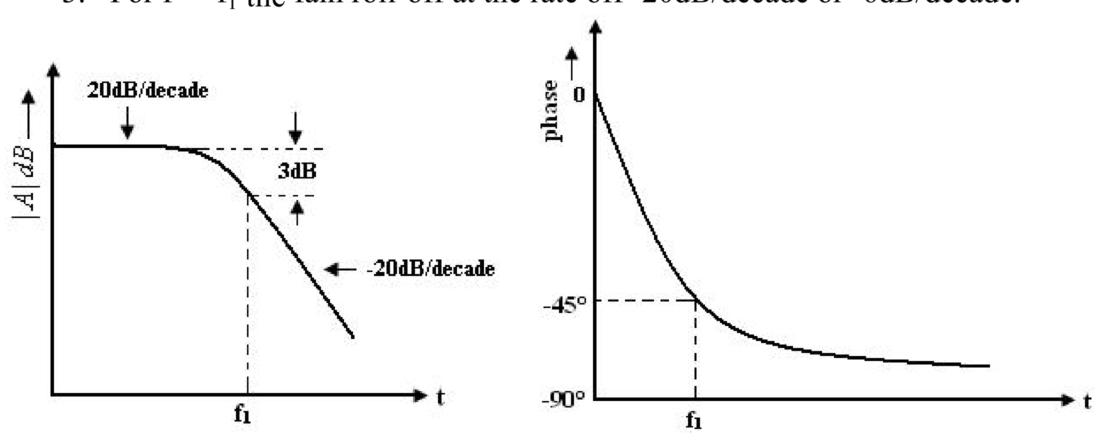


Fig 1.19 Frequency response of op amp

From the phase characteristics that the phase angle is zero at frequency f = 0. At the corner frequency f1 the phase angle is -45° (lagging and an infinite frequency the phase angle is -90° . It shows that a maximum of 90° phase change can occur in an op-amp with a single capacitor C. Zero frequency is taken as the decade below the corner frequency and infinite frequency is one decade above the corner frequency.

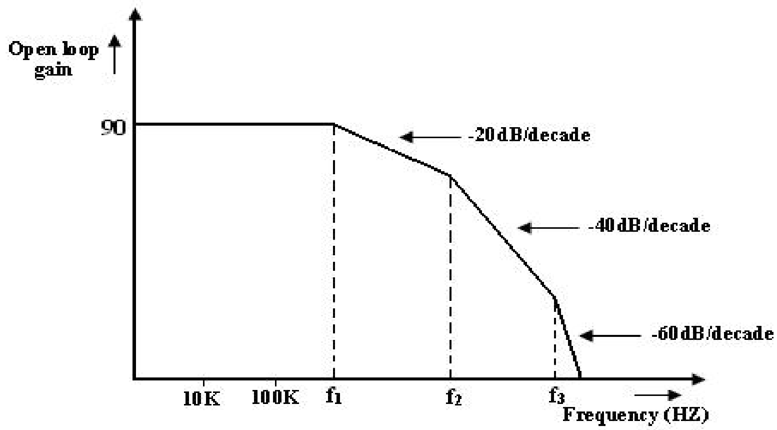


Fig. 1.20 Roll off rate of op amp gain

1.9.2 Circuit Stability:

A circuit or a group of circuit connected together as a system is said to be stable, if its o/p reaches a fixed value in a finite time. A system is said to be unstable, if its o/p increases with time instead of achieving a fixed value. In fact the o/p of an unstable sys keeps on increasing until the system break down. The unstable system is impractical and need be made stable. The criterion gn for stability is used when the system is to be tested practically. In theoretically, always used to test system for stability, ex: Bode plots.
Bode plots are compared of magnitude Vs Frequency and phase angle Vs frequency. Any system whose stability is to be determined can represented by the block diagram.

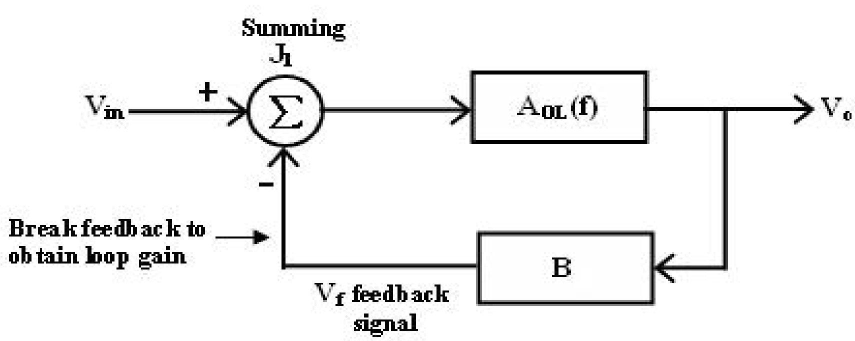


Fig. 1.21 Feedback loop system

The block between the output and input is referred to as forward block and the block between the output signal and f/b signal is referred to as feedback block. The content of each block is referred as transfer frequency. From fig. we represented it by A_{OL} (f) which is given by

A_{OL} (f) = V_0 /Vin if V_f = 0 ----- (1)

where A_{OL} (f) = open loop volt gain.

The closed loop gain A_f is given by A_F = V_0 /Vin

= A_{OL} / (1+(A_{OL}) (B) ----(2)

B = gain of feedback circuit.

B is a constant if the feedback circuit uses only resistive components.

Once the magnitude Vs frequency and phase angle Vs frequency plots are drawn, system stability may be determined as follows

1. Method 1:

Determine the phase angle when the magnitude of (AOL) (B) is 0dB (or) 1.

If phase angle is >-180°, the system is stable. However, the some systems the magnitude may never be 0, in that cases method 2, must be used.

2. Method 2:

Determine the phase angle when the magnitude of (AOL) (B) is 0dB (or) 1.

If phase angle is > - 180°, If the magnitude is –ve decibels then the system is stable. However, the some systems the phase angle of a system may reach -1800, under such conditions method 1 must be used to determine the system stability.

1.9.3 DC Characteristics of op-amp:

Current is taken from the source into the op-amp inputs respond differently to current and voltage due to mismatch in transistor.

DC output voltages are,

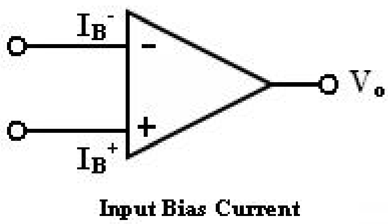
- ✓ Input bias current
- ✓ Input offset current
- ✓ Input offset voltage
- ✓ Thermal drift

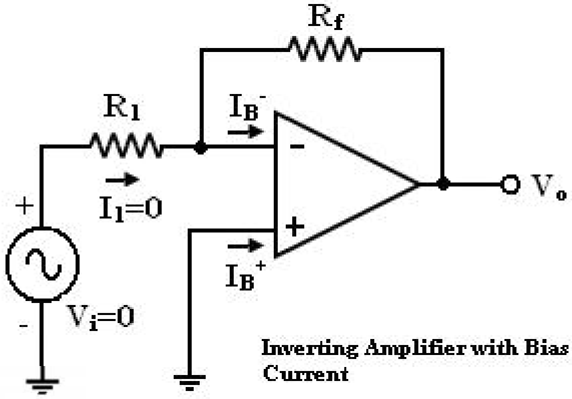
Input bias current:

The op-amp‘s input is differential amplifier, which may be made of BJT or FET.

In an ideal op-amp, we assumed that no current is drawn from the input terminals the base currents entering into the inverting and non-inverting terminals (IB- & IB+ respectively).

Even though both the transistors are identical, IB- and IB+ are not exactly equal due to internal imbalance between the two inputs. Manufacturers specify the input bias current I_B





$$I_B = \frac{I_B^+ + I_B^-}{2}$$

If input voltage Vi = 0V. The output Voltage Vo should also be (Vo = 0) but for IB = 500nA
We find that the output voltage is offset by Op-amp with a 1M feedback resistor
Vo = 500nA X 1M = 500mV
The output is driven to 500mV with zero input, because of the bias currents.
In application where the signal levels are measured in mV, this is totally unacceptable. This can be compensated by a compensation resistor Rcomp has been added between the non-inverting input terminal and ground as shown in the figure below.

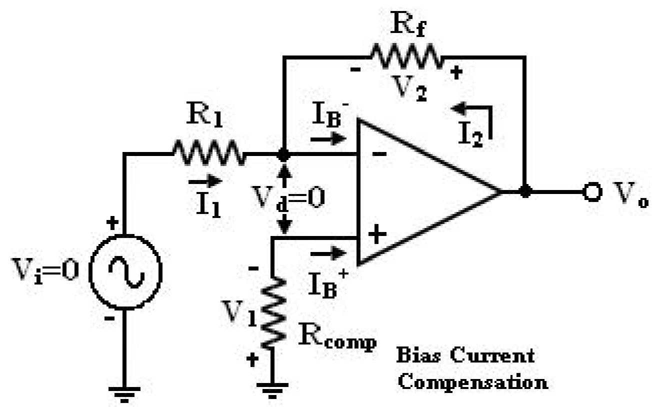


Fig. 1.22 Bias compensated circuit

Current I_B^+ flowing through the compensating resistor R_{comp}, then by KVL we get,

$-V_1+0+V_2-V_o = 0$ (or)

$V_o = V_2 - V_1$ --------- (1)

By selecting proper value of R_{comp}, V_2 can be cancelled with V_1 and the $V_o = 0$. The value of R_{comp} is derived as

$V_1 = I_B^+ R_{comp}$ (or)

$I_B^+ = V_1/R_{comp}$ ---------------------- (2)

The node ‘a’ is at voltage ($-V_1$). Because the voltage at the non-inverting input terminal is ($-V_1$). So with $V_i = 0$ we get,

$I_1 = V_1/R_1$ ------------------------ (3)

$I_2 = V_2/R_f$ ------------------------ (4)

For compensation, V_o should equal to zero ($V_o = 0$, $V_i = 0$). i.e. from equation (3) $V_2 = V_1$. So that, $I_2 = V_1/R_f$ ——> (5)

KCL at node 'a' gives,

$I_B^- = I_2 + I_1 = (V_1/R_f) + (V_1/R_1) = V_1(R_1+R_f)/R_1R_f$ ------------------------ (5)

Assume $I_B^- = I_B^+$ and using equation (2) & (5) we get

$V_1 (R_1+R_f)/R_1R_f = V_1/R_{comp}$

$R_{comp} = R_1 \parallel R_f$ ------------------------ (6)

i.e. to compensate for bias current, the compensating resistor, R_{comp} should be equal to the parallel combination of resistor R_1 and R_f.

Input offset current:

- ✓ Bias current compensation will work if both bias currents I_B^+ and I_B^- are equal.
- ✓ Since the input transistor cannot be made identical. There will always be some small difference between I_B^+ and I_B^-. This difference is called the offset current

$|I_{os}| = I_B^+ - I_B^-$------------------------ (7)

Offset current I_{os} for BJT op-amp is 200nA and for FET op-amp is 10pA. Even with bias current compensation, offset current will produce an output voltage when $V_i = 0$.

$V_1 = I_B^+ R_{comp}$ ------------------------ (11)

And $I_1 = V_1/R_1$------------------------ (12)

KCL at node a gives,

$$I_2 = (I_B^- - I_1) = I_B^- - (I_B^+ \frac{R_{comp}}{R_1})$$

Again $V_0 = I_2 R_f - V_1$

$V_o = I_2 R_f - I_B^+ R_{comp}$

$V_o = 1M\,\Omega$ X 200nA

$V_o = 200mV$ with $V_i = 0$

Equation (16) the offset current can be minimized by keeping feedback resistance small.

- ✓ Unfortunately to obtain high input impedance, R_1 must be kept large.
- ✓ R_1 large, the feedback resistor R_f must also be high. So as to obtain reasonable gain.

The T-feedback network is a good solution. This will allow large feedback resistance, while keeping the resistance to ground low (in dotted line).

- ✓ The T-network provides a feedback signal as if the network were a single feedback resistor.

By T to Π conversion,

$$R_f = \frac{R_t^2 + 2R_tR_s}{R_s}$$

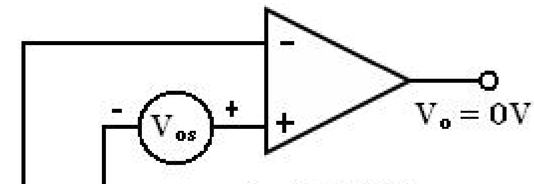
To design T- network first pick $R_t \ll R_f/2$ and calculate

$$R_s = \frac{R_t^2}{R_f - 2R_t}$$

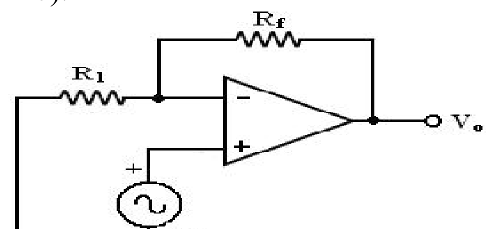
Input offset voltage:

In spite of the use of the above compensating techniques, it is found that the output voltage may still not be zero with zero input voltage [$V_O \neq 0$ with $V_i = 0$]. This is due to unavoidable imbalances inside the op-amp and one may have to apply a small voltage at the input terminal to make output (V_O) = 0.

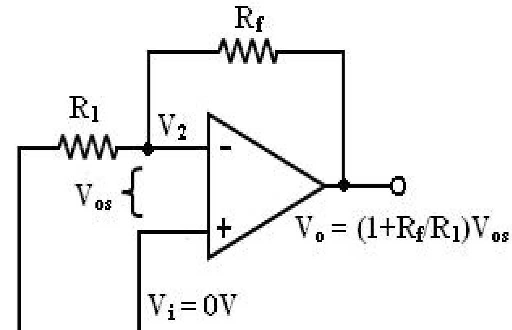
This voltage is called input offset voltage V_{OS}. This is the voltage required to be applied at the input for making output voltage to zero ($V_O = 0$).



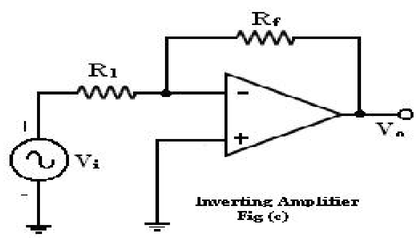
Input Offset Voltage
Fig (a)



Non-Inverting Amplifier
Fig (b)



Equivalent circuit for Vi = 0V
Fig (d)



Inverting Amplifier
Fig (c)

Let us determine the V_{OS} on the output of inverting and non-inverting amplifier. If $V_i = 0$ (Fig (b) and (c)) become the same as in figure (d).

Total output offset voltage:

The total output offset voltage V_{OT} could be either more or less than the offset voltage produced at the output due to input bias current (I_B) or input offset voltage alone(V_{OS}). This is because I_B and V_{OS} could be either positive or negative with respect to ground. Therefore the maximum offset voltage at the output of an inverting and non-inverting amplifier (figure b, c) without any compensation technique used is given by many op amps provide offset compensation pins to nullify the offset voltage. A 10K potentiometer is placed across offset null pins 1&5. The wipes connected to the negative supply at pin 4. The position of the wipes is adjusted to nullify the offset voltage.

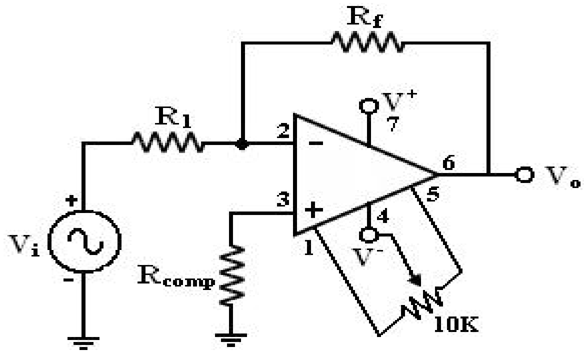


Fig.1.23 Compensation circuit for offset voltage

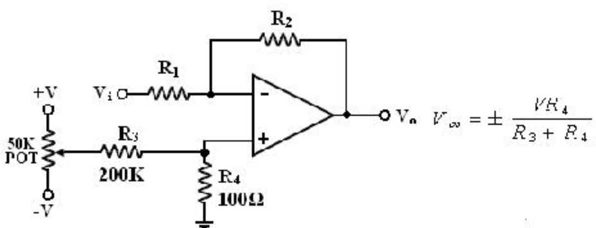
When the given (below) op-amps does not have these offset null pins, external balancing techniques are used.

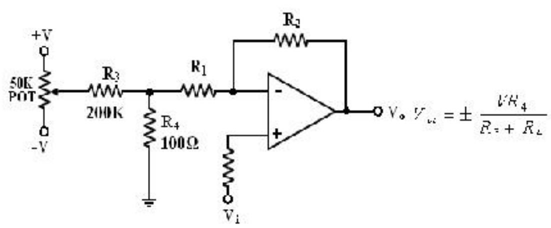
$$V_{OT} = \left(1 + \frac{R_f}{R_1}\right) V_{OS} + R_f I_B$$

With R_{comp}, the total output offset voltage

$$V_{OT} = \left(1 + \frac{R_f}{R_1}\right) V_{OS} + R_f I_{OS}$$

Balancing circuit: Inverting amplifier: **Non-inverting amplifier:**





Thermal drift:

Bias current, offset current, and offset voltage change with temperature. A circuit carefully nulled at 25ºC may not remain. So when the temperature rises to 35ºC. This is called drift. Offset current drift is expressed in nA/ºC. These indicate the change in offset for each degree Celsius change in temperature.

<u>1.10 Slew Rate</u>

Slew rate is the maximum rate of change of output voltage with respect to time. Specified in V/µs.

Reason for Slew rate:

There is usually a capacitor within 0, outside an op-amp oscillation. It is this capacitor which prevents the o/p voltage from fast changing input. The rate at which the volt across the capacitor increases is given by

dVc/dt = I/C --------(1)

I -> Maximum amount furnished by the op-amp to capacitor C.

Op-amp should have the either a higher current or small compensating capacitors.

For 741 IC, the maximum internal capacitor charging current is limited to about 15µA. So the

slew
rate of 741 IC is

$$SR = dV_c/dt\ |_{max} = I_{max}/C$$

For a sine wave input, the effect of slew rate can be calculated as consider volt follower. The input is large amp, high frequency sine wave.
If $V_s = V_m \sin wt$ then output $V_0 = V_m \sin wt$.
The rate of change of output is given by $dV_0/dt = V_m\, w \cos wt$.

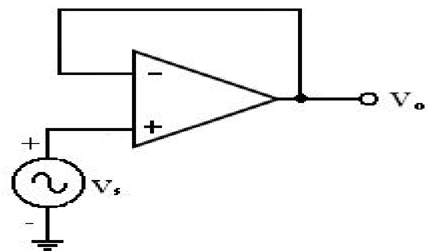


Fig. 1.22 Voltage Follower Circuit

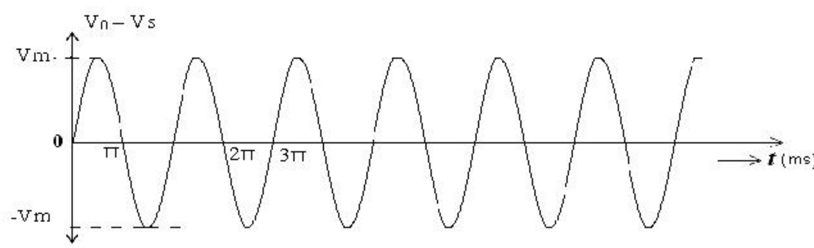


Fig. 1.23 Input and output waveforms of a voltage follower

The max rate of change of output across when coswt =1

(i.e) $SR = dV_0/dt\ |_{max} = wV_m$.

$SR = 2\pi f V_m$ V/s $= 2\pi f V_m$ v/ms.

Thus the maximum frequency fmax at which undistorted output volt of peak value Vm is given by fmax (Hz) = Slew rate/6.28 * Vm called the full power response. It is maximum frequency of a large amplitude sine wave with which op-amp can have without distortion.

1.11. Open – loop op-amp Configuration:

The term open-loop indicates that no feedback in any form is fed to the input from the output. When connected in open – loop the op-amp functions as a very high gain amplifier. There are three open – loop configurations of op-amp namely,

1. Differential amplifier
2. Inverting amplifier
3. Non-inverting amplifier

The above classification is made based on the number of inputs used and the terminal to which the input is applied. The op-amp amplifies both ac and dc input signals. Thus, the input signals can be either ac or dc voltage.

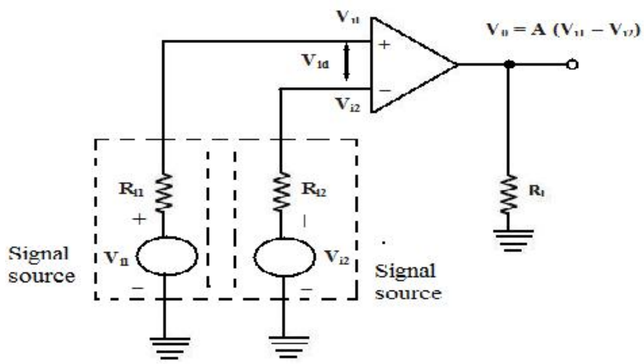
1.11.1 Loop Differential Amplifier:

In this configuration, the inputs are applied to both the inverting and the non-inverting input terminals of the op-amp and it amplifies the difference between the two input voltages. Figure shows the open-loop differential amplifier configuration.

The input voltages are represented by V_{i1} and V_{i2}. The source resistance R_{i1} and R_{i2} are negligibly small in comparison with the very high input resistance offered by the op-amp, and thus the voltage drop across these source resistances is assumed to be zero. The output voltage V_0 is given by

$$V_0 = A\,(V_{i1} - V_{i2})$$

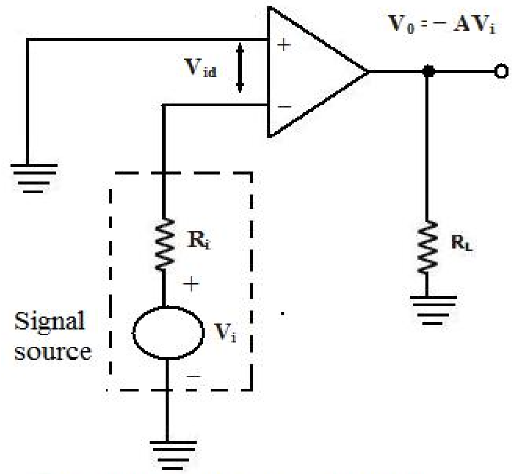
where A is the large signal voltage gain. Thus the output voltage is equal to the voltage gain A times the difference between the two input voltages. This is the reason why this configuration is called a differential amplifier. In open – loop configurations, the large signal voltage gain A is also called open-loop gain A.



Open - loop Differential Amplifier

Inverting amplifier:

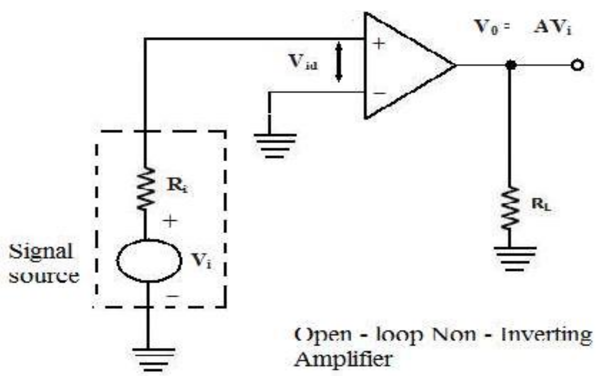
In this configuration the input signal is applied to the inverting input terminal of the op- amp and the non-inverting input terminal is connected to the ground. Figure shows the circuit of an open – loop inverting amplifier. The output voltage is 180° out of phase with respect to the input and hence, the output voltage V0 is given by, V0 = -AVi. Thus, in an inverting amplifier, the input signal is amplified by the open-loop gain A and in phase shifted by 180°



Open - loop Inverting Amplifier

Non-inverting Amplifier:

Figure shows the open – loop non- inverting amplifier. The input signal is applied to the non-inverting input terminal of the op-amp and the inverting input terminal is connected to the ground. The input signal is amplified by the open – loop gain A and the output is in-phase with input signal. $V_0 = AV_i$



Open - loop Non - Inverting Amplifier

In all the above open-loop configurations, only very small values of input voltages can be applied. Even for voltages levels slightly greater than zero, the output is driven into saturation, which is observed from the ideal transfer characteristics of op-amp shown in figure. Thus, when operated in the open-loop configuration, the output of the op-amp is either in negative or positive saturation, or switches between positive and negative saturation levels. This prevents the use of open – loop configuration of op-amps in linear applications.

Limitations of Open – loop Op – amp configuration:

Firstly, in the open – loop configurations, clipping of the output waveform can occur when the output voltage exceeds the saturation level of op-amp. This is due to the very high open – loop gain of the op-amp. This feature actually makes it possible to amplify very low frequency signal of the order of microvolt or even less, and the amplification can be achieved accurately without any distortion. However, signals of such magnitudes are susceptible to noise and the amplification for that application is almost impossible to obtain in the laboratory.

Secondly, the open – loop gain of the op – amp is not a constant and it varies with changing temperature and variations in power supply. Also, the bandwidth of most of the open- loop op amps is negligibly small. This makes the open – loop configuration of op-amp unsuitable for ac applications. The open – loop bandwidth of the widely used 741 IC is approximately 5Hz. But in almost all ac applications, the bandwidth requirement is much larger than this.

For the reason stated, the open – loop op-amp is generally not used in linear applications. However, the open – loop op amp configurations find use in certain non – linear applications such as comparators, square wave generators and astable multivibrators.

1.11.2 Closed – loop op-amp configuration:

The op-amp can be effectively utilized in linear applications by providing a feedback from the output to the input, either directly or through another network. If the signal feedback is out- of-phase by 1800 with respect to the input, then the feedback is referred to as negative feedback or degenerative feedback. Conversely, if the feedback signal is in phase with that at the input, then the feedback is referred to as positive feedback or regenerative feedback.

An op – amp that uses feedback is called a closed – loop amplifier. The most commonly used closed – loop amplifier configurations are 1. Inverting amplifier (Voltage shunt amplifier) 2. Non- Inverting amplifier (Voltage – series Amplifier)

Inverting Amplifier:

The inverting amplifier is shown in figure and its alternate circuit arrangement is shown in figure, with the circuit redrawn in a different way to illustrate how the voltage shunt feedback is achieved. The input signal drives the inverting input of the op – amp through resistor R1.

The op – amp has an open – loop gain of A, so that the output signal is much larger than the error voltage. Because of the phase inversion, the output signal is 1800 out – of – phase with the input signal. This means that the feedback signal opposes the input signal and the feedback is negative or degenerative.

Practical Inverting amplifier:

The practical inverting amplifier has finite value of input resistance and input current, its open voltage gain A_0 is less than infinity and its output resistance R_0 is not zero, as against the ideal inverting amplifier with finite input resistance, infinite open – loop voltage gain and zero output resistance respectively.

Figure shows the low frequency equivalent circuit model of a practical inverting amplifier. This circuit can be simplified using the Thevenin‘s equivalent circuit shown in figure. The signal source Vi and the resistors R_1 and R_i are replaced by their Thevenin‘s equivalent values. The closed – loop gain A_V and the input impedance R_{if} are calculated as follows.

The input impedance of the op- amp is normally much larger than the input resistance R_1.

Therefore, we can assume $V_{eq} \approx V_i$ and $R_{eq} \approx R_1$. From the figure

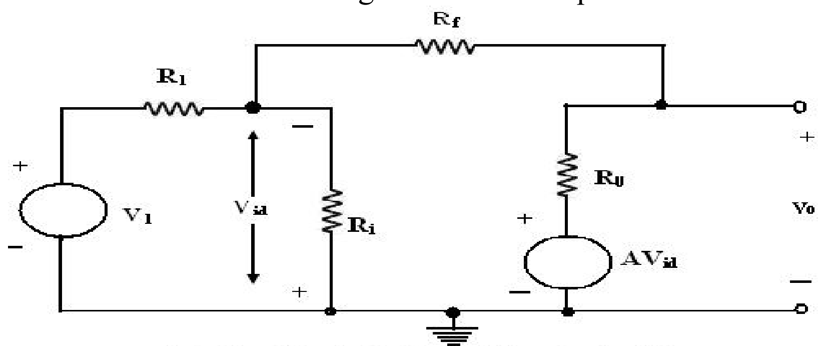
$$V_O = IR_O = AV_{id} \quad \text{and} \quad V_{id} = IR_f = AV_{id}$$

$$V_0 = IR_0 = AV_{id}$$

Substituting the value of I derived from above eqn. and obtaining the closed loop gain. It can be observed from above eqn. that when A>> 1, R_0 is negligibly small and the product $AR_1 >> R_0 + R_f$, the closed loop gain is given by

$$Av = -\frac{R_f}{R_1}$$

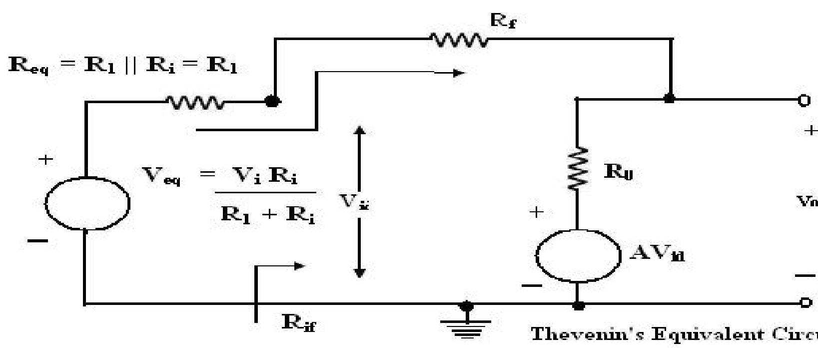
Which as the same form as given in above eqn for an ideal inverter.



Equivalent Circuit of a Practical Inerting Amplifier

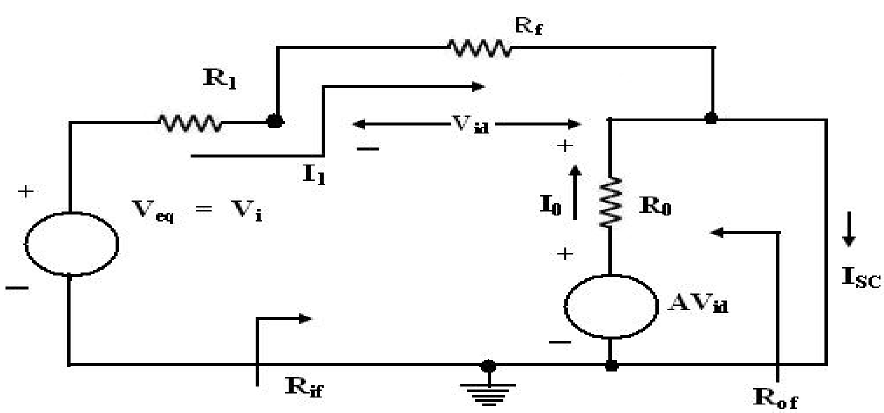
Input Resistance:

$R_{if} = V_{id}/ I_1 = (R_f+R_0)/(1+A)$



Thevenin's Equivalent Circuit

Output Resistance:



Equivalent circuit to determine Rof

Figure shows the equivalent circuit to determine R_{of}. The output impedance R_{of} without the load resistance factor R_L is calculated from the open circuit output voltage V_{oc} and the short circuit output current I_{SC}.

$$R_{of} = \frac{\dfrac{R_0(R_1 + R_f)}{R_0 + R_1 + R_F}}{1 + \dfrac{R_1 A}{R_0 + R_1 + R_f}}$$

Non –Inverting Amplifier:

The non – inverting Amplifier with negative feedback is shown in figure. The input signal drives the non – inverting input of op-amp. The op-amp provides an internal gain A. The external resistors R_1 and R_f form the feedback voltage divider circuit with an attenuation factor of β. Since the feedback voltage is at the inverting input, it opposes the input voltage at the non – inverting input terminals, and hence the feedback is negative or degenerative.
The differential voltage V_{id} at the input of the op-amp is zero, because node A is at the same voltage as that of the non- inverting input terminal. As shown in figure, R_f and R_1 form a potential divider. Therefore,

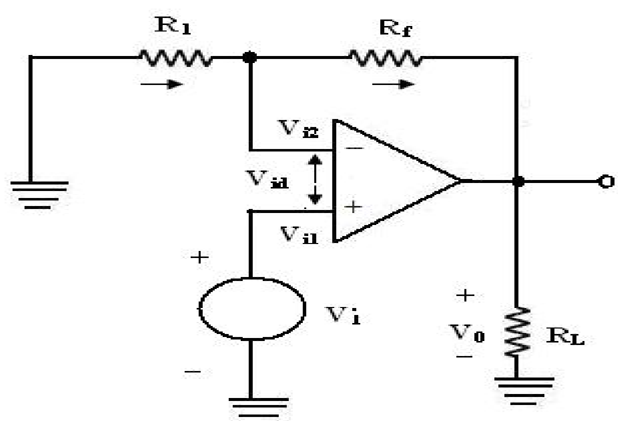


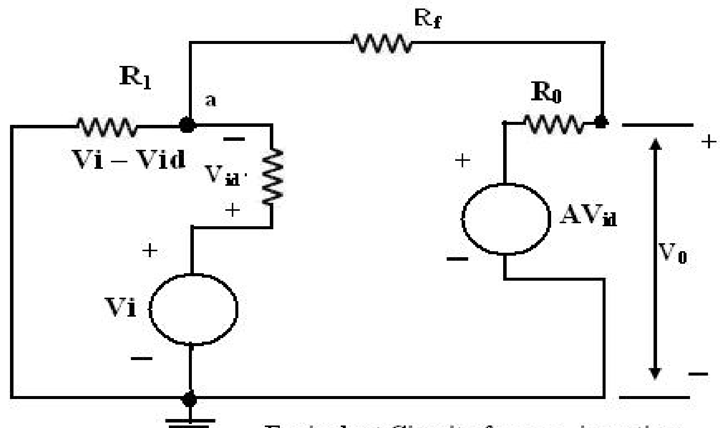
Fig. 1.24 Non –Inverting Amplifier:

Closed Loop Non – Inverting Amplifier

The input resistance of the op – amp is extremely large (approximately infinity,) since the op – amp draws negligible current from the input signal.

Practical Non –inverting amplifier:

The equivalent circuit of a non- inverting amplifier using the low frequency model is shown below in figure. Using Kirchhoff's current law at node a,



Equivalent Circuit of a non - inverting Amplifier using low frequency

$$Av = 1 + \frac{R_f}{R_1}$$

The difference volt is equal to the input volt minus the f/b volt. (or) The feedback volt always opposes the input volt (or out of phase by 180^0 with respect to the input voltage) hence the feedback is said to be negative.

It will be performed by computing

1. Closed loop volt gain
2. Input and output resistance
3. Bandwidth

1. Closed loop volt gain:

The closed loop volt gain is $A_F = V_0 / V_{in}$

$$V_0 = Avid = A(V_1 - V_2)$$

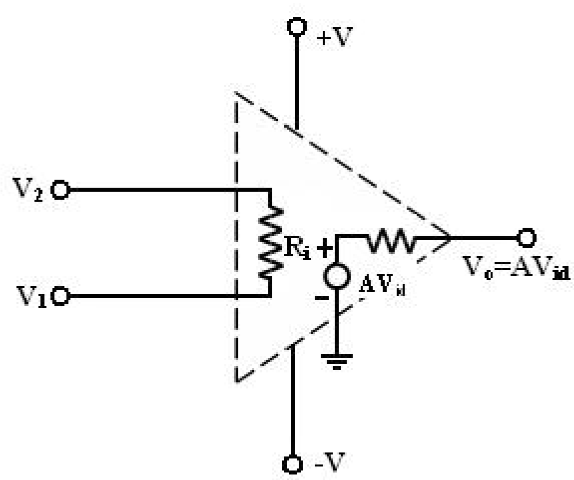


Fig.1.25 equivalent circuit of practical op amp

A = large signal voltage gain.

From the above eqn. $V_0 = A(V_1 - V_2)$

Refer fig, we see that, $V_1 = V_{in}$

$$V_2 = V_f = \frac{R_1}{R_1+R_f} V_0 \quad \text{Since } R_i >> R_1$$

$$V_0 = AV_{in} - \frac{R_1 V_0}{R_1 + R_f}$$

$$V_0 + \frac{R_1}{R_1+R_f} V_0 = AV_{in}$$

UNIT – II
APPLICATIONS OF OPERATIONAL AMPLIFIER

2.1 Sign Changer (Phase Inverter)

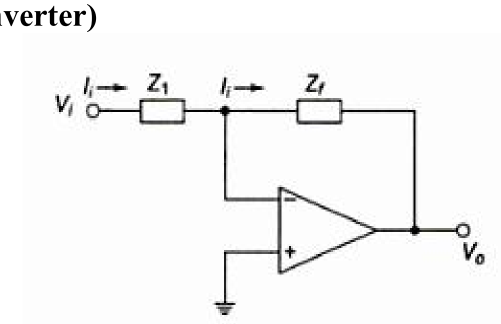


Fig 2.1 Basic inverting configuration

The basic inverting amplifier configuration using an op-amp with input impedance Z1 and feedback impedance Z_f . If the impedance Z1 and Z f are equal in magnitude and phase, then the closed loop voltage gain is -1, and the input signal will undergo a 180° phase shift at the output. Hence, such circuit is also called phase inverter. If two such amplifiers are connected in cascade, then the output from the second stage is the same as the input signal without any change of sign. Hence, the outputs from the two stages are equal in magnitude but opposite in phase and such a system is an excellent paraphase amplifier.

2.2 Scale Changer:

Referring the above diagram, if the ratio $Z_f / Z_1 = k$, a real constant, then the closed loop gain is –k, and the input voltage is multiplied by a factor –k and the scaled output is available at the output. Usually, in such applications, Z_f and Z_1 are selected as precision resistors for obtaining precise and scaled value of input voltage.

2.3 Phase Shift Circuits

The phase shift circuits produce phase shifts that depend on the frequency and maintain a constant gain. These circuits are also called constant-delay filters or all-pass filters. That constant delay refers to the fact the time difference between input and output remains constant when frequency is changed over a range of operating frequencies.

This is called all-pass because normally a constant gain is maintained for all the frequencies within the operating range. The two types of circuits, for lagging phase angles and leading phase angles.

Phase-lag circuit:

Phase log circuit is constructed using an op-amp, connected in both inverting and non inverting modes. To analyze the circuit operation, it is assumed that the input voltage v1 drives a simple inverting amplifier with inverting input applied at(-)terminal of op-amp and a non inverting amplifier with a low-pass filter.

It is also assumed that inverting gain is -1 and non-inverting gain after the low-pass circuit is $1+\frac{R_f}{R_1}=1+1=2$ Since $R_f = R_1$.

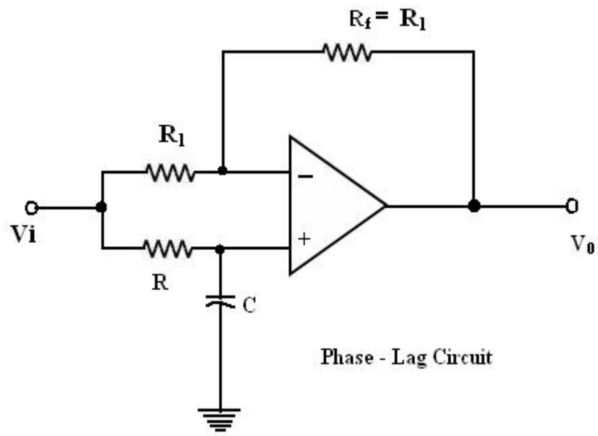


Fig. 2.2 Phase lag circuit

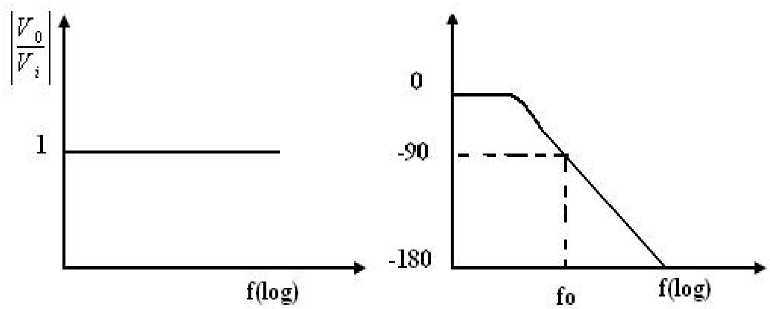


Fig 2.3 Bode plot of phase lag circuit

For the circuit fig 2.2, it can be written as

$$V_O(jw) = -V_i(jw)\left(-1+\frac{2}{1+jwRC}\right)$$

and the relationship between output and input can be expressed by

$$\frac{V_O(jw)}{V_i(jw)} = \frac{(1-jwRC)}{(1+jwRC)}$$

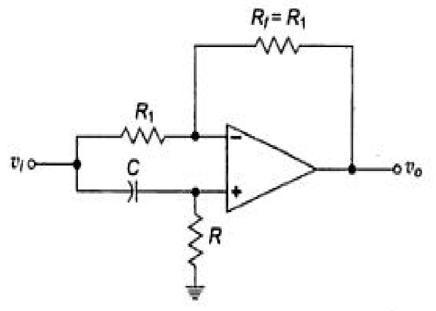
The relationship is complex as defined above equation and it shows that it has both magnitude and phase. Since the numerator and denominator are complex conjugates, their magnitudes are identical and the overall phase angle equals the angle of numerator less the angle of the denominator.

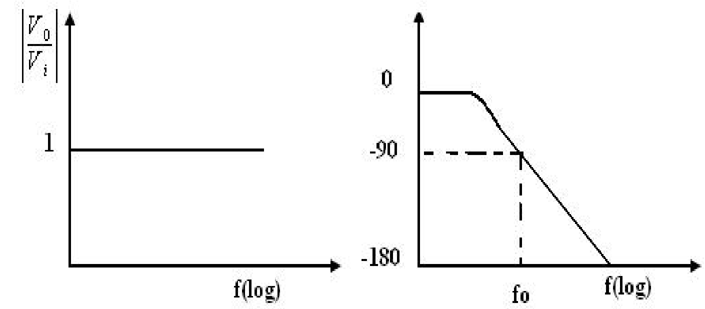
$$\theta = -2\ \tan^{-1} RC\omega$$

Phases-lead circuit:

$$\frac{V_O(jw)}{V_i(jw)} = -\frac{(1-jwRC)}{(1+jwRC)}$$

$$\theta = 180° - 2\ \tan^{-1} RC\omega$$





Figs 2.4 Phase lead circuit **Fig 2.5 Bode plot of Phase lead circuit**

2.4 Voltage follower:

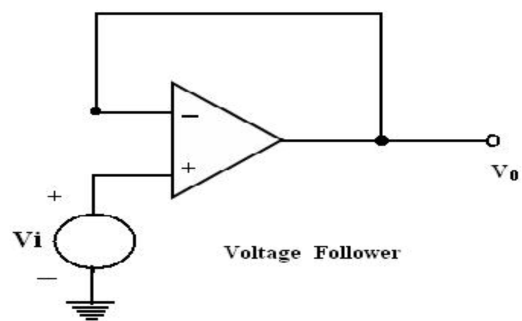


Fig 2.6 Voltage follower

If $R_1=\infty$ and $R_f=0$ in the non inverting amplifier configuration. The amplifier act as a unity-gain amplifier or voltage follower.
The circuit consists of an op-amp and a wire connecting the output voltage to the input, i.e. the output voltage is equal to the input voltage, both in magnitude and phase. $V_0=V_i$.Since the output voltage of the circuit follows the input voltage, the circuit is called voltage follower. It offers very high input impedance of the order of MΩ and very low output impedance.

Therefore, this circuit draws negligible current from the source. Thus, the voltage follower can be used as a buffer between a high impedance source and a low impedance load for impedance matching applications.

2.5 Voltage to Current Converter with floating loads (V/I):

Voltage to current converter in which load resistor R_L is floating (not connected to ground). V_{in} is applied to the non- inverting input terminal, and the feedback voltage across R_1 devices the inverting input terminal. This circuit is also called as a current – series negative feedback amplifier. Because the feedback voltage across R_1 (applied Non-inverting terminal) depends on the output current i_0 and is in series with the input difference voltage Vid.

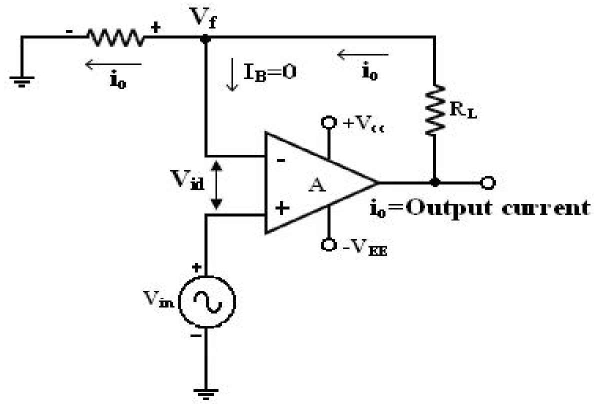


Fig. 2.7 Voltage to Current Converter with floating loads (V/I):

Writing KVL for the input loop,

Voltage $V_{id} = V_f$ and $I_B = 0$, $V_i = R_L i_o$ where $i_o = \frac{V_i}{R_L}$

From the fig input voltage Vin is converted into output current of Vin/R_L [Vin -> i_0]. In other words, input volt appears across R_1. If R_L is a precision resistor, the output current (i_0 = Vin/R_1) will be precisely fixed.

Applications:

1. Low voltage ac and dc voltmeters
2. Diode match finders
3. LED and Zener diode testers.

Voltage – to current converter with Grounded load:

This is the other type V – I converter, in which one terminal of the load is connected to ground.

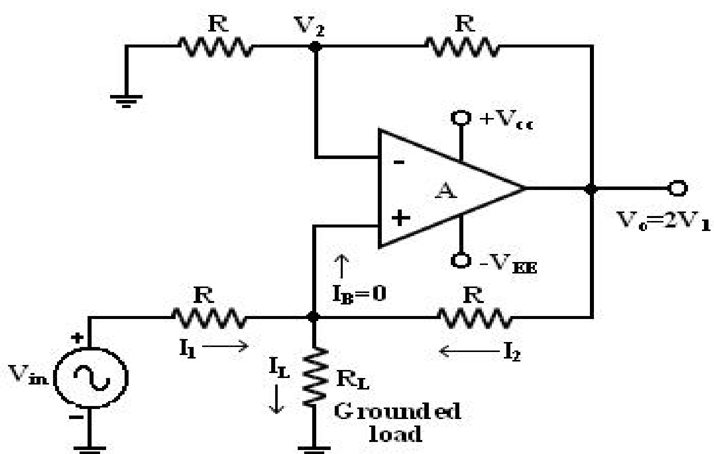


Fig 2.8 V – I converter with grounded load

Analysis of the circuit:
The analysis of the circuit can be done by following 2 steps.

1. To determine the voltage V_1 at the non-inverting (+) terminals and
2. To establish relationship between V_1 and the load current I_L. Applying KCL at node a,

$$R = R_f$$
$$I_1 + I_2 = I_L$$
$$(V_i+V_a)/R + (V_o - V_a)/R = I_L$$
$$V_o = (V_i + V_o - I_L R)/2 \text{ and gain } = 1 + R/R = 2.$$
$$\therefore V_i = I_L R \;; I_L = V_i/R$$

Current to Voltage Converter (I –V):

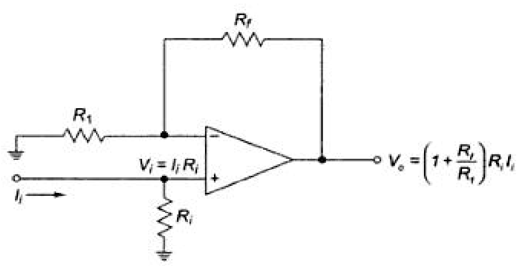


Fig. 2.9 Non inverting current to voltage convertor

Open – loop gain A of the op-amp is very large. Input impedance of the op amp is very high.

Sensitivity of the I – V converter:

1. The output voltage $V_0 = -R_F I_{in}$.
2. Hence the gain of this converter is equal to $-R_F$. The magnitude of the gain (i.e.) is called as sensitivity of I to V converter.
3. The amount of change in output volt ΔV_0 for a given change in the input current ΔI_{in} is decide by the sensitivity of I-V converter.
4. By keeping R_F variable, it is possible to vary the sensitivity as per the requirements.

Applications of V-I converter with Floating Load:
1. Diode Match finder:

In some applications, it is necessary to have matched diodes with equal voltage drops at a particular value of diode current. The circuit can be used in finding matched diodes and is obtained from fig (V-I converter with floating load) by replacing R_L with a diode.

When the switch is in position 1: (Diode Match Finder) Rectifier diode (IN 4001) is placed in the f/b loop, the current through this loop is set by input voltage Vin and Resistor R_1. For Vin = 1V and $R_1 = 100\Omega$, the current through this $I_0 = Vin/R_1 = 1/100 = 10mA$. As long as V_0 and R_1 constant, I_0 will be constant. The Voltage drop across the diode can be found either by measuring the volt across it or o/p voltage.

The output voltage is equal to $(Vin + V_D)$ $V_0 = Vin + V_D$.

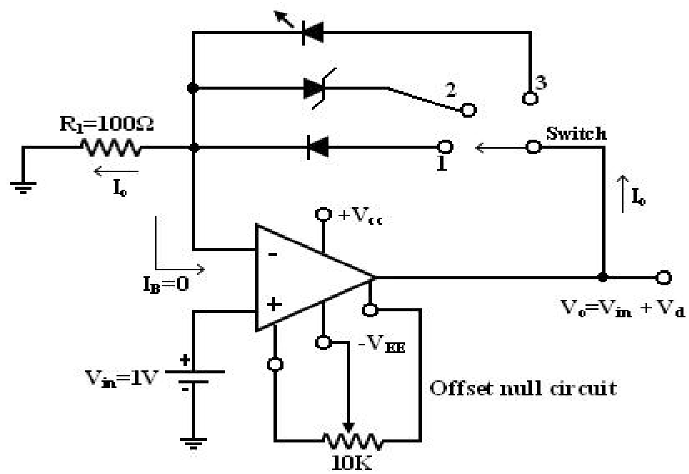


Fig. 2.10 Diode Match finder:

To avoid an error in output voltage the op-amp should be initially nulled. Thus the matched diodes can be found by connecting diodes one after another in the feedback path and measuring voltage across them.

2. Zener diode Tester:

(When the switch position 2) when the switch is in position 2, the circuit becomes a Zener diode tester. The circuit can be used to find the breakdown voltage of Zener diodes. The Zener current is set at a constant value by Vin and R_1. If this current is larger than the knee current (I_{ZK}) of the Zener, the Zener blocks (V_Z) volts. For Ex: I_{ZK} = 1mA , V_Z = 6.2V, Vin = 1V, R_1 = 100Ω Since the current through the Zener is , I_0 = Vin/R_1 = 1/100 =10mA > I_{ZK} the voltage across the Zener will be approximately equal to 6.2V.

3. When the switch is in position 3: (LED)

The circuit becomes a LED when the switch is in position 3. LED current is set at a constant value by Vin and R_1. LEDs can be tested for brightness one after another at this current. Matched LEDs with equal brightness at a specific value of current are useful as indicates and display devices in digital applications.

Applications of I – V Converter:

One of the most common uses of the current to voltage converter is

1. Digital to analog Converter (DAC)
2. Sensing current through Photo detector. Such as photo cell, photo diodes and photovoltaic cells.

Photoconductive devices produce a current that is proportional to an incident energy or light (i.e). It can be used to detect the light.

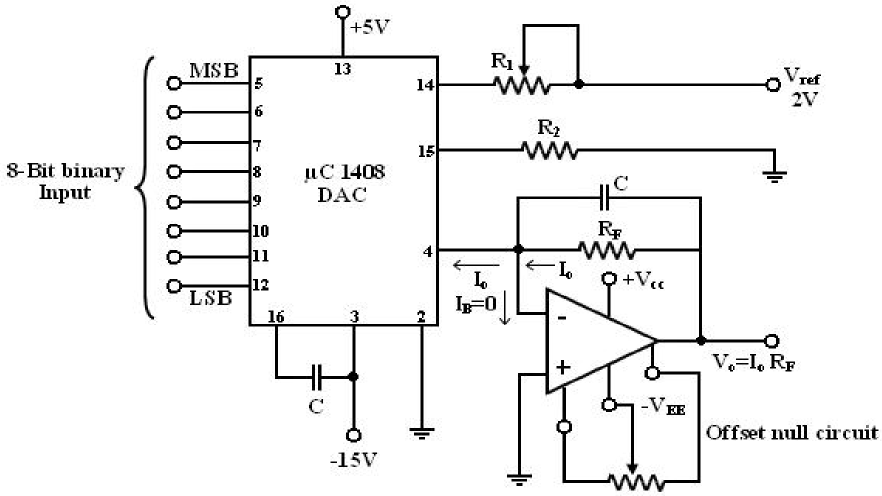


Fig. 2.11 I – V Converter DAC

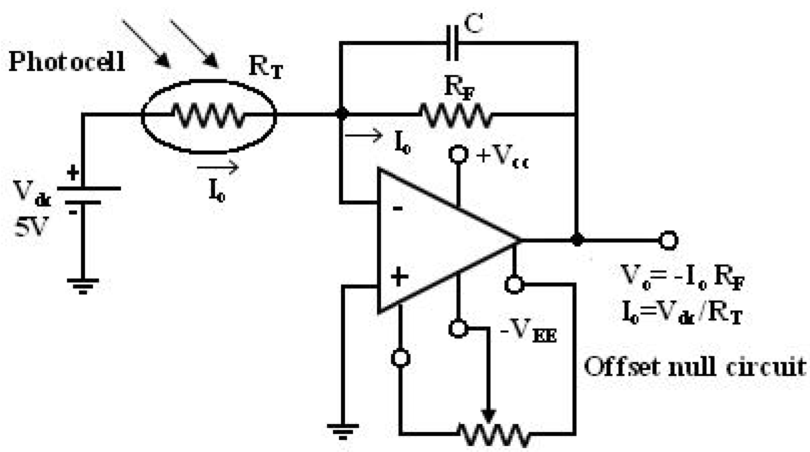


Fig. 2.12 Photo cell detector

Photocells, photodiodes, photovoltaic cells give an output current that depends on the intensity of light and independent of the load. The current through these devices can be converted to voltage by I – V converter and it can be used as a measure of the amount of light. In this fig photocell is connected to the I – V Converter. Photocell is a passive transducer it requires an external dc voltage (Vdc). The dc voltage can be eliminated if a photovoltaic cell is used instead of a photocell. The Photovoltaic Cell is a semiconductor device that converts the radiant energy to electrical power. It is a self-generating circuit because it does not require dc voltage externally. Ex of Photovoltaic Cell: used in space applications and watches.

2.6 Adder:

Op-amp may be used to design a circuit whose output is the sum of several input signals. Such a circuit is called a summing amplifier or a summer or adder.

An inverting summer or a non-inverting summer may be discussed now.

Inverting Summing Amplifier:

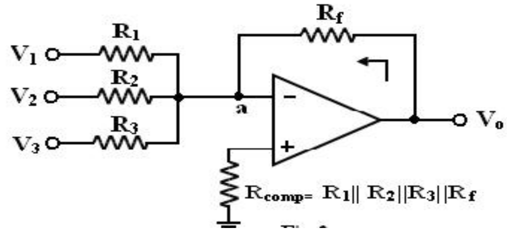


Fig. 2.13 inverting summer

A typical summing amplifier with three input voltages V_1, V_2 and V_3 three input resistors R1, R_2, R_3 and a feedback resistor R_f is shown in figure 2.
The following analysis is carried out assuming that the op-amp is an ideal one, $A_{OL} = \infty$.
Since the input bias current is assumed to be zero, there is no voltage drop across the resistor R_{comp} and hence the non-inverting input terminal is at ground potential.

$$I = V_1/R_1 + V_2/R_2 \ldots.. + V_n/R_n;$$

$$V_o = - R_f\, I = R_f/R(V_1 + V_2 + \ldots . V_n).$$

To find R_{comp}, make all inputs $V_1 = V_2 = V_3 = 0$.
So the effective input resistance $R_i = R_1 \parallel R_2 \parallel R_3$.
Therefore, $R_{comp} = R_i \parallel R_f = R_1 \parallel R_2 \parallel R_3 \parallel R_{,f}$.

Non-Inverting Summing Amplifier:

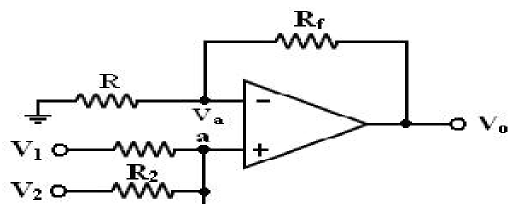


Fig.2.14 Non inverting summer

A summer that gives a non-inverted sum is the non-inverting summing amplifier of figure Let the voltage at the (-) input terminal be V_a. which is a non-inverting weighted sum of inputs.
Let $R_1 = R_2 = R_3 = R = R_f/2$, then $V_o = V_1 + V_2 + V_3$

<u>2.7 Subtractor:</u>

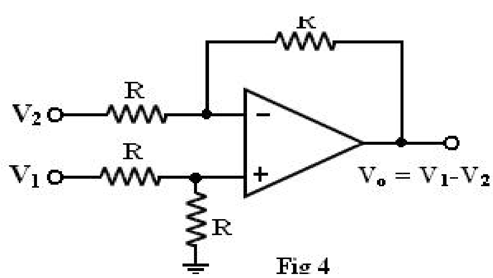


Fig. 2.15 Subtractor

A basic differential amplifier can be used as a subtractor as shown in the above figure. If all resistors are equal in value, then the output voltage can be derived by using superposition principle.

To find the output V_{01} due to V_1 alone, make $V_2 = 0$.

Then the circuit of figure as shown in the above becomes a non-inverting amplifier having input voltage $V_1/2$ at the non-inverting input terminal and the output becomes

$V_{01} = V_1/2(1+R/R) = V1$ when all resistances are R in the circuit.

Similarly the output V_{02} due to V_2 alone (with V_1 grounded) can be written simply for an inverting amplifier as

$$V_{02} = -V_2$$

Thus the output voltage V_o due to both the inputs can be written as

$$V_0 = V_{01} - V_{02} = V_1 - V_2$$

Adder/Subtractor:

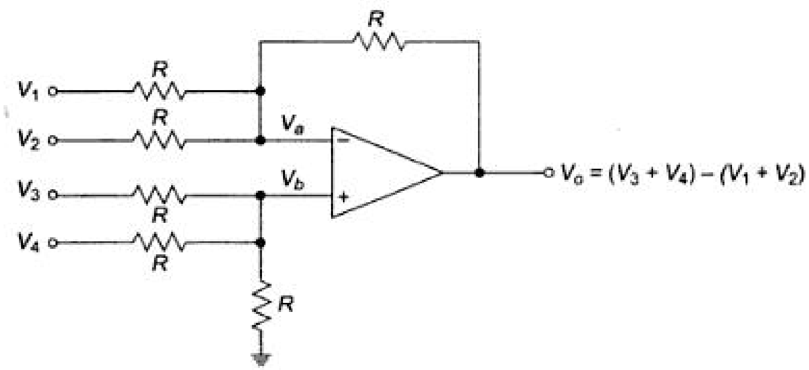


Fig. 2.16 Adder-Subtractor

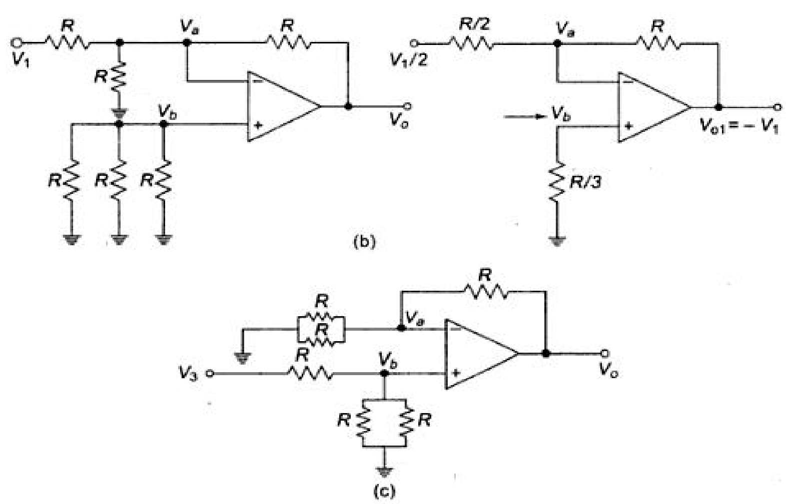


Fig. 2.17 (b) equivalent circuit for V_2=V_3=V_4=0 and (c) for V_1=V_2=V_4=0

It is possible to perform addition and subtraction simultaneously with a single op-amp using the circuit shown in figure 2.16.

The output voltage V_O can be obtained by using superposition theorem. To find output voltage V_{01} due to V_1 alone, make all other input voltages V_2, V_3 and V_4 equal to zero.
The simplified circuit is shown in figure 2.17. This is the circuit of an inverting amplifier and its output voltage is, $V_{01}= -R/(R/2) * V_1/2= - V_1$ by Thevenin's equivalent circuit at inverting input terminal).

Similarly, the output voltage V_{02} due to V_2 alone is,

$V_{02}= - V_2$

Now, the output voltage V_{03} due to the input voltage signal V_3 alone applied at the (+) input terminal can be found by setting V_1, V_2 and V_4 equal to zero.

$V_{03}=V_3$

The circuit now becomes a non-inverting amplifier as shown in fig.(c).

So, the output voltage V_{03} due to V_3 alone is

$$V_{03} = V3$$

Similarly, it can be shown that the output voltage V_{04} due to V_4 alone is

$$V_{04} = V_4$$

Thus, the output voltage V_O due to all four input voltages is given by

$$V_o = V_{01} = V_{02} = V_{03} = V_{04}$$

$$V_o = -V_1 - V_2 + V_{3} + V_4$$

$$V_o = (V_3 + V_4) - (V_1 + V_2)$$

So, the circuit is an adder-subtractor.

2.8 Instrumentation Amplifier:

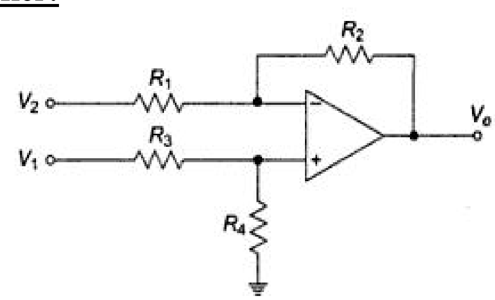


Fig. 2.18 Basic Differential Amplifier

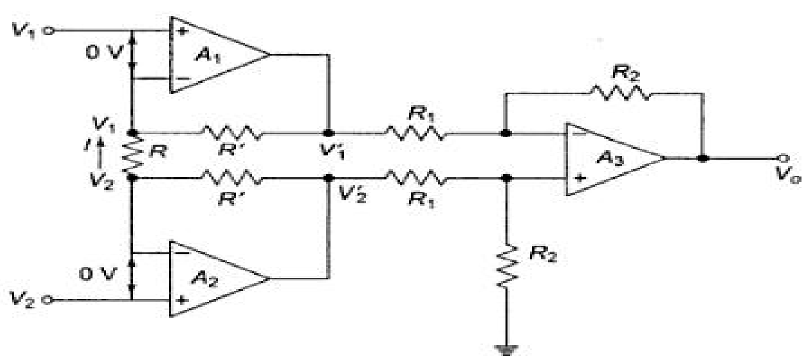


Fig. 2.19 Instrumentation Amplifier

Current flowing in resistor R is $I= (V_1-V_2)/R$ and it flow through R' in the direction shown, Voltage at non-inverting terminal op-amp A3 is $R_2V_1'/(R_1+R_2)$. By superposition theorem,

$$V_o= (R_2/R_1)V_1+(1+R_2/R_1)(R_2V_2/(R_1+R_2)=R_2/R_1(V_1'-V_2');$$

$$V_1'= R'I+V_1=R'/R(V_1-V_2)+V_1$$

$$V_2'= R'I+V_1=R'/R(V_1-V_2)+V_2;$$

$$V0= (R_2/R_1)[(2R'/R(V_2-V1)+ (V_2-V1)]= (R_2/R_1)[(1+2R'/R)(V_2-V1)$$

In a number of industrial and consumer applications, one is required to measure and control physical quantities.

Some typical examples are measurement and control of temperature, humidity, light intensity, water flow etc. these physical quantities are usually measured with help of transducers.

The output of transducer has to be amplified so that it can drive the indicator or display system. This function is performed by an instrumentation amplifier. The important features of an instrumentation amplifier are

1. High gain accuracy
2. High CMRR
3. High gain stability with low temperature coefficient
4. Low output impedance

There are specially designed op-amps such as μA725 to meet the above stated requirements of a good instrumentation amplifier. Monolithic (single chip) instrumentation amplifier are also available commercially such as AD521, AD524, AD620, AD624 by Analog Devices, LM363.XX (XX -->10,100,500) by National Semiconductor and INA101, 104, 3626, 3629 by Burr Brown.

In the circuit of figure 6(a), source V_1 sees an input impedance = R_3+R_4 (=101K) and the impedance seen by source V_2 is only R_1 (1K). This low impedance may load the signal source heavily.

Therefore, high resistance buffer is used preceding each input to avoid this loading effect as shown in figure

The op-amp A_1 and A_2 have differential input voltage as zero. For $V_1=V_2$, that is, under common mode condition, the voltage across R will be zero. As no current flows through R and R‘ the non-inverting amplifier.

A_1 acts as voltage follower, so its output $V_2'=V_2$. Similarly op-amp A_2 acts as voltage follower having output $V_1'=V_1$. However, if $V_1 \neq V_2$, current flows in R and R‘, and $(V_2'-V_1')>(V_2- V_1)$. Therefore, this circuit has differential gain and CMRR more compared to the single op-amp circuit of figure 2.10.

The difference gain of this instrumentation amplifier R, however should never be made zero, as this will make the gain infinity. To avoid such a situation, in a practical circuit, a fixed resistance in series with a potentiometer is used in place of R.

Figure (c) shows a differential instrumentation amplifier using Transducer Bridge. The circuit uses a resistive transducer whose resistance changes as a function of the physical quantity to be measured.

The bridge is initially balanced by a dc supply voltage V_{dc} so that $V_1=V_2$. As the physical quantity changes, the resistance R_T of the transducer also changes, causing an unbalance in the bridge $(V_1 \neq V_2)$. This differential voltage now gets amplified by the three op-amp differential instrumentation amplifier.

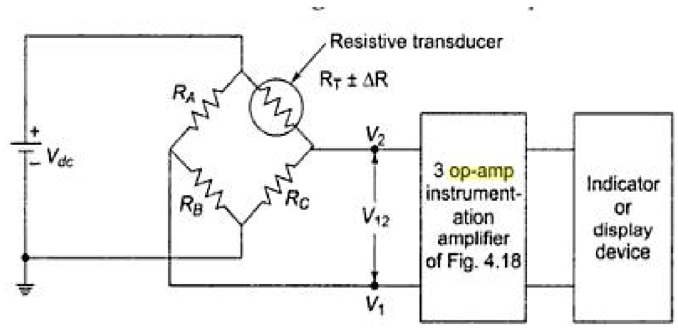


Fig.2.20 Instrumentation bridge using transducer Bridge

$R_B(V_{dc})/(R_B+R_A) = R_C V_{dc}/(R_C+R_T)$

Applications of instrumentation amplifier with the transducer bridge,

- temperature indicator,
- temperature controller and
- light intensity meter .
-

2.9 Integrator:

A circuit in which the output voltage waveform is the integral of the input voltage waveform is the integrator or Integration Amplifier. Such a circuit is obtained by using a basic inverting amplifier configuration if the feedback resistor R_F is replaced by a capacitor C_F.

The expression for the output voltage V_0 can be obtained by KVL eqn. at node V_2.

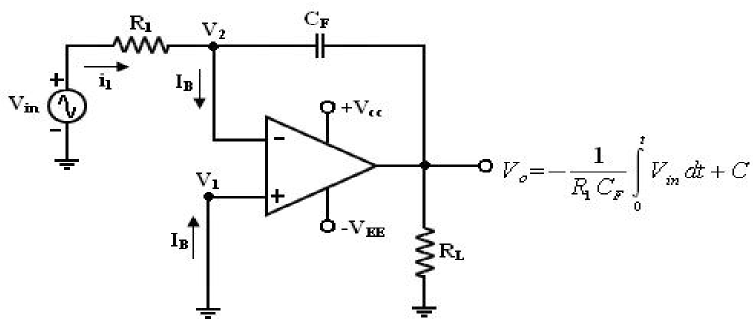


Fig 2.21 Integrator Circuit

$$i_1 = I_B + i_f$$

Since I_B is negligible small, $i_1 = i_F$

Relation between current through and voltage across the capacitor is

$i_{C}(t) = C dv_c(t)/dt$

$V_1 = 0$ because A is very large,

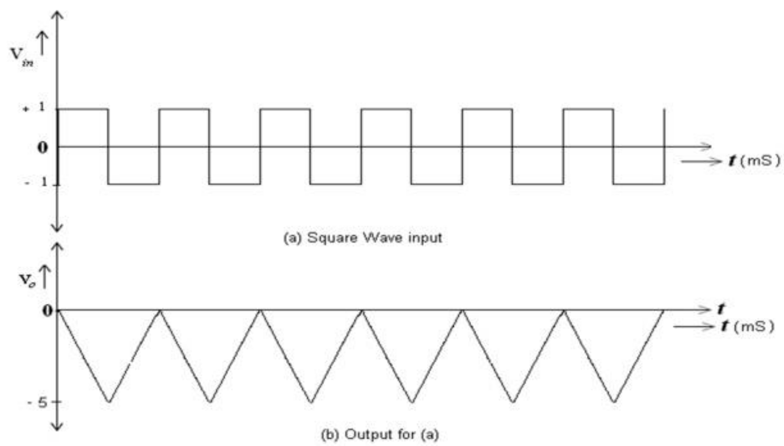
The output voltage can be obtained by integrating both sides with respect to time

$$V_0(jw) = \frac{1}{jwR_1C_f} V_i\ (jw)$$

Indicates that the output is directly proportional to the negative integral of the input volts and inversely proportional to the time constant $R_1 C_F$.

Ex: If the input is sine wave -> output is cosine wave.

If the input is square wave -> output is triangular wave.



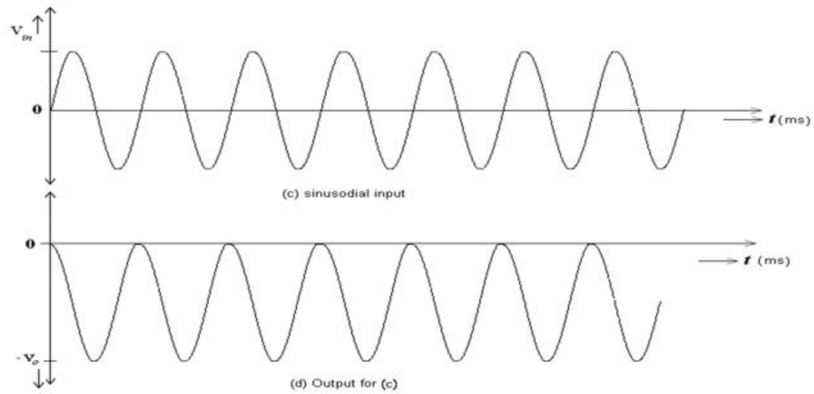


Fig.2.22 Waveforms from Integrator

These waveform with assumption of R_1 C_f = 1, Vout =0V (i.e) C =0.
When Vin = 0 the integrator works as an open loop amplifier because the capacitor C_F acts an open circuit to the input offset voltage Vio.
The Input offset voltage V_{io} and the part of the input is charging capacitor C_F produce the error voltage at the output of the integrator.

Practical Integrator:

Practical Integrator to reduce the error voltage at the output, a resistor R_F is connected across the feedback capacitor C_F.
Thus R_F limits the low frequency gain and hence minimizes the variations in the output voltages. The frequency response of the basic integrator, shown from this fb is the frequency at which the gain is dB and is given by

$$f_b = \frac{1}{2\pi R_1 C_F}$$

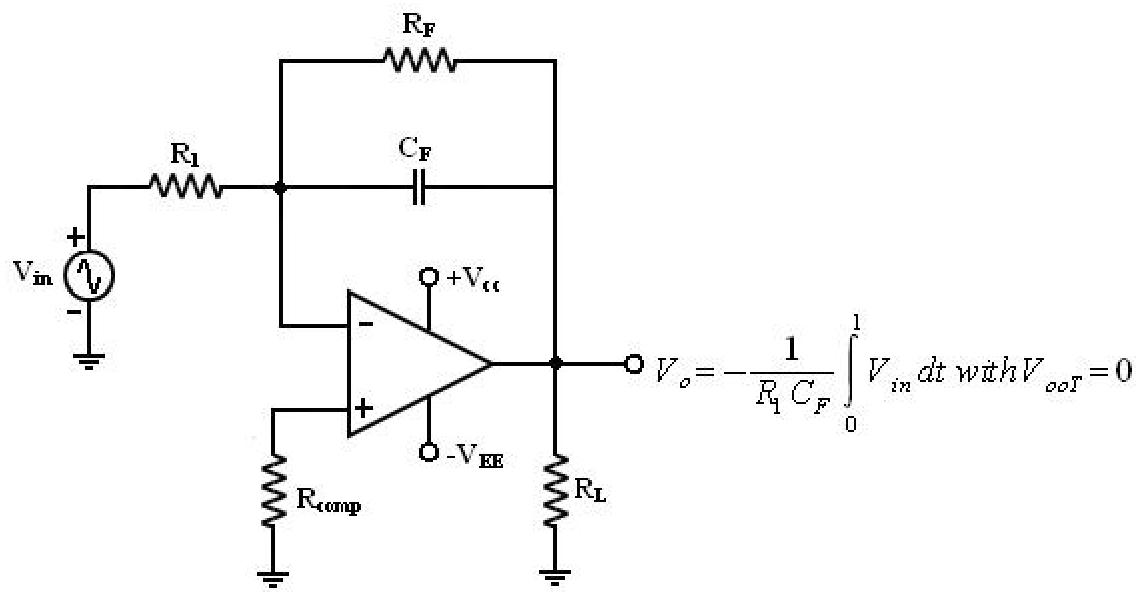


Fig. 2.23 Practical Integrator Circuit

- Both the stability and low frequency roll-off problems can be corrected by the addition of a resistor R_F in the practical integrator.
- Stability refers to a constant gain as frequency of an input signal is varied over a certain range.
- Low frequency -> refers to the rate of decrease in gain roll off at lower frequencies.
- From the fig of practical Integrators, f is some relative operating frequency and for frequencies f to fa to gain R_F / R_1 is constant. After fa the gain decreases at a rate of 20dB/decade or between fa and fb the circuit act as an integrator.

- The gain limiting frequency fa is given by

$$f_a = \frac{1}{2\pi R_1 C_F}$$

- The value of fa and R_1 C_F and R_F C_F values should be selected such that fa<fb.
- The input signal will be integrated properly if the time period T of the signal is larger than or equal to R_F C_F,

$$f_b = \frac{1}{2\pi R_F C_F}$$

Uses:

Most commonly used in

- ✓ analog computers
- ✓ ADC
- ✓ Signal wave shaping circuits.

2.10 Differentiator:

The circuit performs the mathematical operation of differentiation (i.e.) the output waveform is the derivative of the input waveform. The differentiator may be constructed from a basic inverting amplifier if an input resistor $R_1$is replaced by a capacitor C_1. Since the differentiator performs the reverse of the integrator function. Thus the output V_0 is equal to R_F C_1 times the negative rate

of change of the input voltage Vin with time. The –sign indicates a 180° phase shift of the output waveform V0 with respect to the input signal. The below circuit will not do this because it has some practical problems.
The gain of the circuit (RF /XC1) R with R in frequency at a rate of 20dB/decade. This makes the circuit unstable. Also input impedance XC1s with R in frequency which makes the circuit very susceptible to high frequency noise.

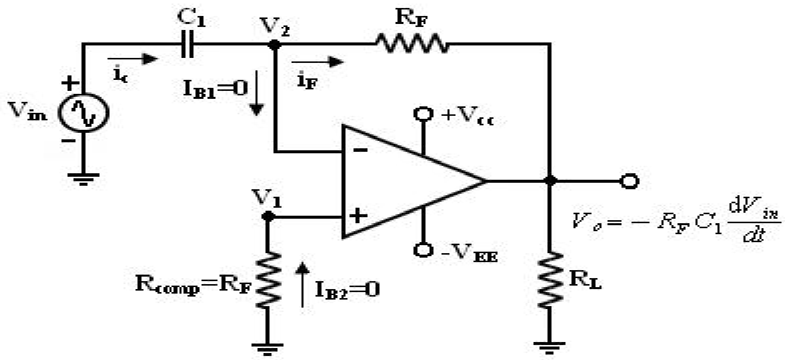


Fig 2.24 Basic Differentiator

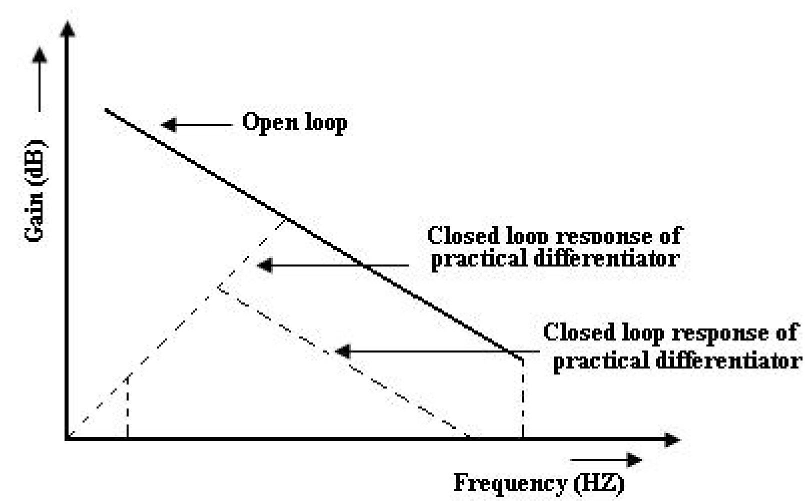


Fig. 2.25 Frequency response of differentiator

From the above fig. fa = frequency at which the gain is 0dB and is given by

$$f_a = \frac{1}{2\pi R_f C_1}$$

Both stability and high frequency noise problems can be corrected by the addition of two components. R_1 and C_F. This circuit is a practical differentiator.
From Frequency fa to feedback the gain Rs at 20dB/decade after feedback the gain S at 20dB/

decade. This 40dB/ decade change in gain is caused by the R_1 C_1 and R_F C_F combinations.
The gain limiting frequency f_b is given by,

$$f_b = \frac{1}{2\pi R_1 C_1}$$

Where R_1 C_1 = R_F C_F
R_1 C_1 and R_F C_F help to reduce the effect of high frequency input, amplifier noise and offsets.
All R1 C1 and RF CF make the circuit more stable by preventing the R in gain with frequency.
The input signal will be differentiated properly, if the time period T of the input signal is larger than or equal to R_F C_1 (i.e) T > R_F C_1 generally, the value of Feedback and in turn R_1 C_1 and R_F C_F values should be selected such that

$$R_F C_1 >> R_1 C_1$$

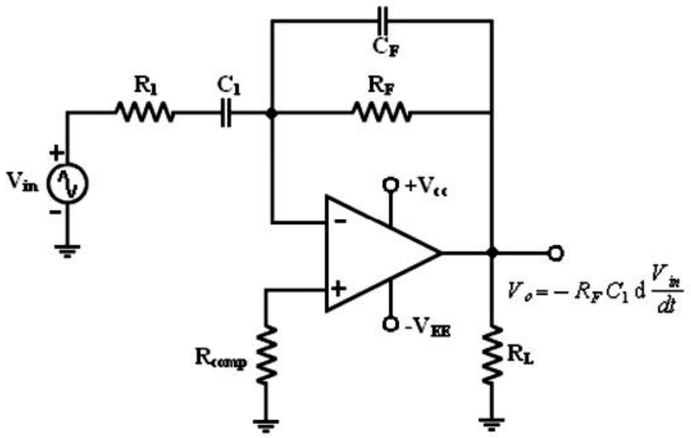


Fig 2.26 Practical Differentiator

A workable differentiator can be designed by implementing the following steps.

1. Select f_a equal to the highest frequency of the input signal to be differentiated then assuming a value of C_1 < 1µf. Calculate the value of R_F.
2. Choose f_b = $20f_a$ and calculate the values of R_1 and C_F so that R_1 C_1 = R_F C_F.

Uses:

It is used in wave shaping circuits to detect high frequency components in an input signal and also as a rate of change and detector in FM modulators.

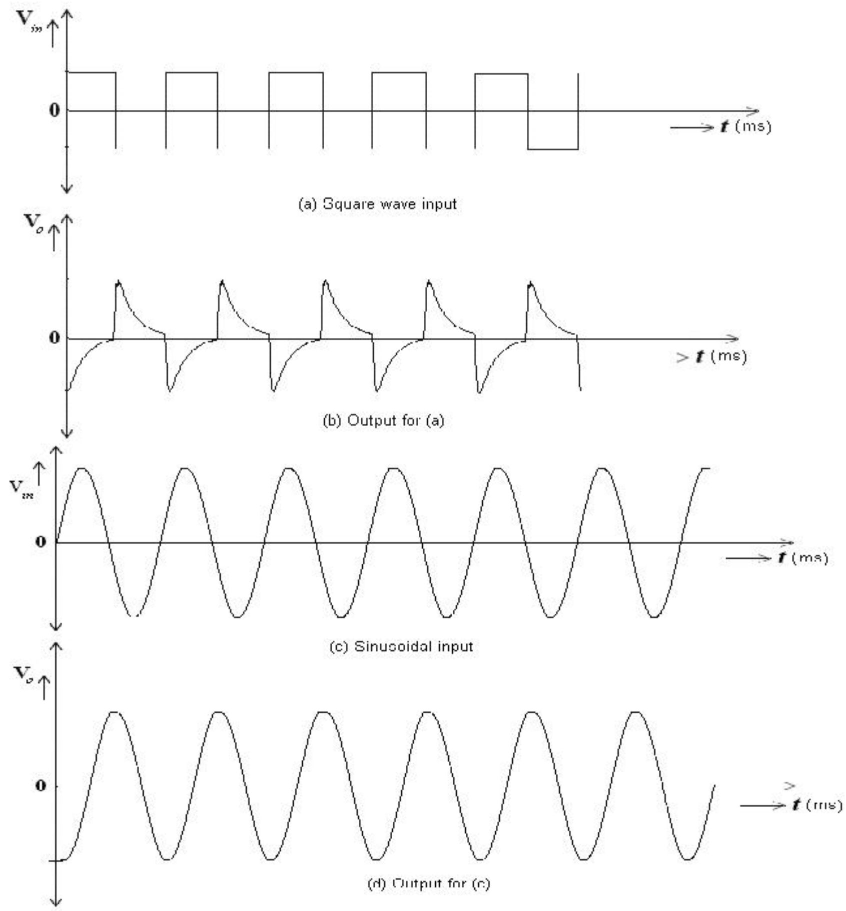


Fig.2.27 Output for practical differentiator.

2.11 Log and Antilog Amplifier:

Log Amplifier:

$$V_o = V_i ln \left(I_f / I_i\right)$$

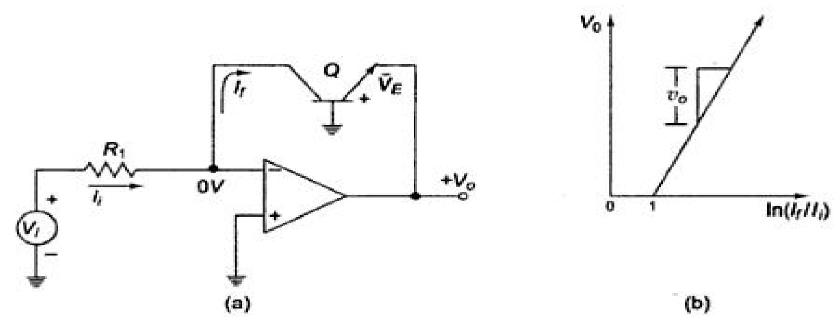


Fig 2.28 Fundamental log-amp Circuit and its characteristics

There are several applications of log and antilog amplifiers.
Antilog computation may require functions such as ln x, log x or sin hx.

Uses:

Direct dB display on a digital Voltmeter and Spectrum analyzer.

Log-amp can also be used to compress the dynamic range of a signal.

A grounded base transistor is placed in the feedback path. Since the collector is placed in the feedback path. Since the collector is held at virtual ground and the base is also grounded, the transistor's voltage-current relationship becomes that of a diode and is given by,

$I_E = I_s[e^{\frac{qV_{BE}}{kT}} - 1]$ and since $I_c = I_E$ for a grounded base transistor $I_C = I_S$ e kT

Is-emitter saturation current $\approx 10^{-13}$A

k=Boltzmann's constant

T=absolute temperature (in ^{o} K)

$$V_o = -\frac{kT}{q} ln\left(\frac{V_i}{R_1 I_S}\right) = -\frac{kT}{q} ln\left(\frac{V_i}{V_R}\right)$$

where $V_{ref} = R_1 I_S$

The output voltage is thus proportional to the logarithm of input voltage.

Although the circuit gives natural log (ln), one can find $\log_{10}$, by proper scaling

Log10X=0.4343 ln X

The circuit has one problem.

The emitter saturation current Is varies from transistor to transistor and with temperature. Thus a stable reference voltage V ref cannot be obtained. This is eliminated by the circuit given below

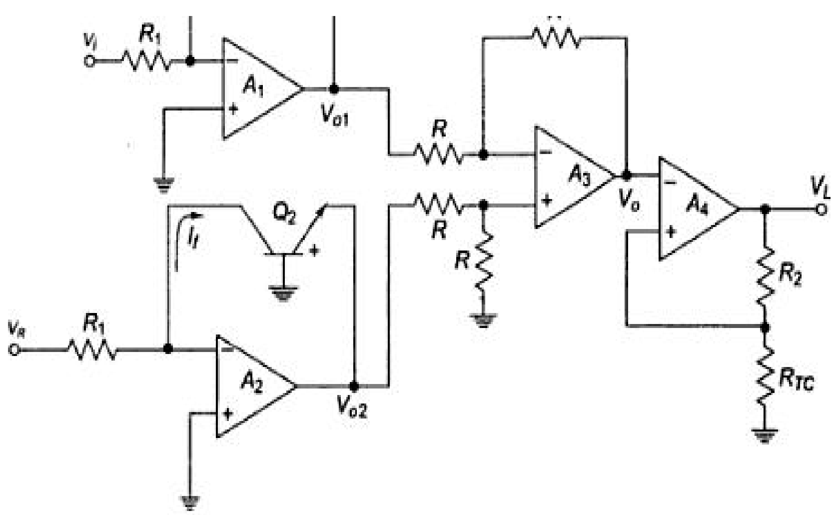


Fig. 2.29 Logarithmic amplifier with compensation of emitter saturation current

The input is applied to one log-amp, while a reference voltage is applied to one log-amp, while a reference voltage is applied to another log-amp. The two transistors are integrated close together in the same silicon wafer. This provides a close match of saturation currents and ensures good thermal tracking.

Assume $I_{S1} = I_{S2} = I_S$

Thus the reference level is now set with a single external voltage source. Its dependence on device and temperature has been removed. The voltage V_o is still dependent upon temperature and is directly proportional to T. This is compensated by the last op-amp stage A4 which provides a non-inverting gain of $(1+R_2/R_{TC})$. Temperature compensated output voltage V_L

$$V_L = \left(1+\frac{R_2}{R_{TC}}\right)\frac{kT}{q} ln\left(\frac{V_i}{V_R}\right)$$

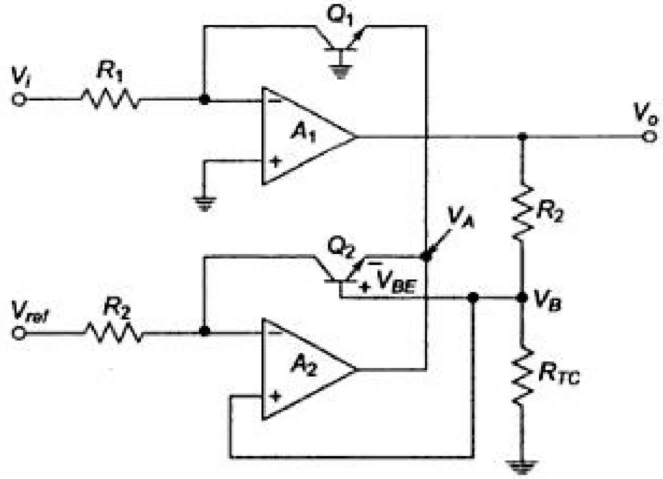


Fig.2.30 Logarithmic amplifier using two op amps

Where R_{TC} is a temperature-sensitive resistance with a positive coefficient of temperature (sensor) so that the slope of the equation becomes constant as the temperature changes.

2.12 Antilog Amplifier

A circuit to convert logarithmically encoded signal to real signals. Transistor in inverting input converts input voltage into logarithmically varying currents

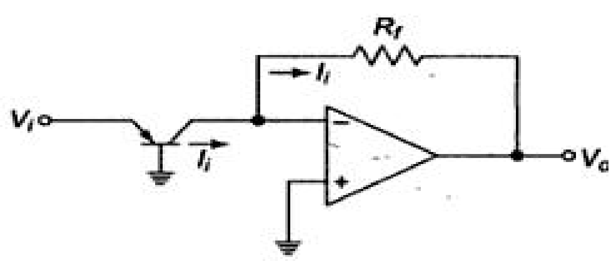


Fig. 2.31 Antilog amplifier

$$I_i = I_c = I_s\left(e^{\frac{\eta V_{BE}}{kT}}\right) \text{ and } V_0 = R_f I_s\left(e^{\frac{\eta V_{BE}}{kT}}\right)$$

The circuit is shown in figure below. The input V_i for the antilog-amp is fed into the temperature compensating voltage divider R_2 and R_{TC} and then to the base of Q2. The output of A2 is fed back to R1 at the inverting input of op amp A_1. The non-inverting inputs are grounded

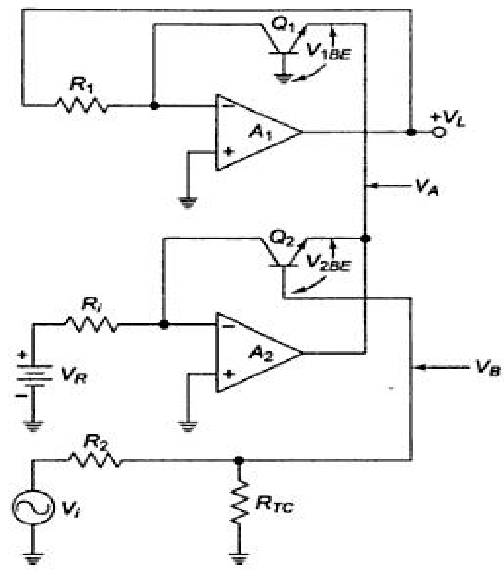


Fig 2.32 Antilog amplifier

$V_{1BE} = \frac{kT}{q}\ln[\frac{V_L}{R_1 I_S}]$ and $V_{2BE} = \frac{kT}{q}\ln[\frac{V_B}{R_1 I_S}]$ and $V_A = -V_{1BE}$ and $V_B = R_{TC}/(R_2+R_{TC})\, V_i$

$V_{Q2E} = V_B + V_{2BE} = R_{TC}/(R_2+R_{TC})\, V_i - \frac{kT}{q}\ln[\frac{V_B}{R_1 I_S}]$

$V_{Q2E} = V_A$

Therefore,
$$-\frac{kT}{q}ln\left(\frac{V_L}{R_1 I_S}\right) = \frac{R_{TC}}{R_2 + R_{TC}}V_i + \frac{kT}{q}ln\left(\frac{V_R}{R_1 I_S}\right)$$

Rearranging, we get

$$\frac{R_{TC}}{R_2 + R_{TC}}V_i = -\frac{kT}{q}ln\left(\frac{V_L}{R_1 I_S}\right) - \frac{kT}{q}ln\left(\frac{V_R}{R_1 I_S}\right)$$
$$= -\frac{kT}{q}ln\left(\frac{V_L}{V_R}\right)$$

We know that $\log_{10} x = 0.4343\, ln\, x$.

Therefore,
$$-0.4343\left(\frac{q}{kT}\right)\left(\frac{R_{TC}}{R_2 + R_{TC}}\right)V_i = 0.4343\, ln\, \left(\frac{V_L}{V_R}\right)$$

$$-0.4343\left(\frac{q}{kT}\right)\left(\frac{R_{TC}}{R_2 + R_{TC}}\right)V_i = \log_{10}\, \left(\frac{V_L}{V_R}\right)$$

$$-KV_i = \log\left(\frac{V_L}{V_R}\right)$$

$$K = 0.4343\left(\frac{q}{kT}\right)\left(\frac{R_{TC}}{R_2 + R_{TC}}\right)$$

$$V_L = V_R 10^{-KV_i}$$

The output V_O of the antilog- amp is fed back to the inverting input of A_1 through the resistor R_1. Hence an increase of input by one volt causes the output to decrease by a decade.

2.13 Comparator

A comparator compares a signal voltage on one input of an op-amp with a known voltage called a reference voltage on the other input. Comparators are used in circuits such as,

- ✓ Digital Interfacing
- ✓ Schmitt Trigger
- ✓ Discriminator
- ✓ Voltage level detector and oscillators

2.13.1 Non-inverting Comparator:

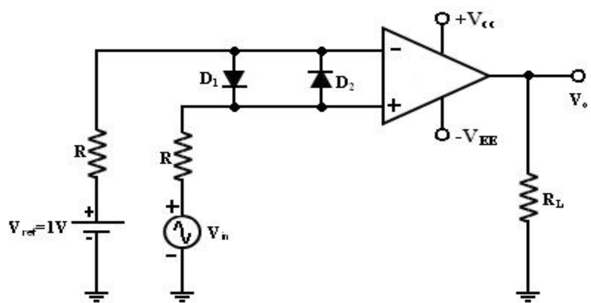


Fig. 2.33 non-inverting comparator circuit

A fixed reference voltage V_{ref} of 1 V is applied to the negative terminal and time varying signal voltage Vin is applied to the positive terminal.

When Vin is less than Vref the output becomes V_0 at –Vsat

[Vin < Vref => V_0 (-Vsat)].

When Vin is greater than Vref, the (+) input becomes positive, the V_0 goes to +Vsat.

[Vin > Vref => V_0 (+Vsat)].

Thus the V_0 changes from one saturation level to another.

The diodes D_1 and D_2 protect the op-amp from damage due to the excessive input voltage Vin. Because of these diodes, the difference input voltage Vid of the op-amp diodes are called clamp diodes.

The resistance R in series with Vin is used to limit the current through D_1 and D_2. To reduce offset problems, a resistance Rcomp = R is connected between the (-ve) input and Vref.

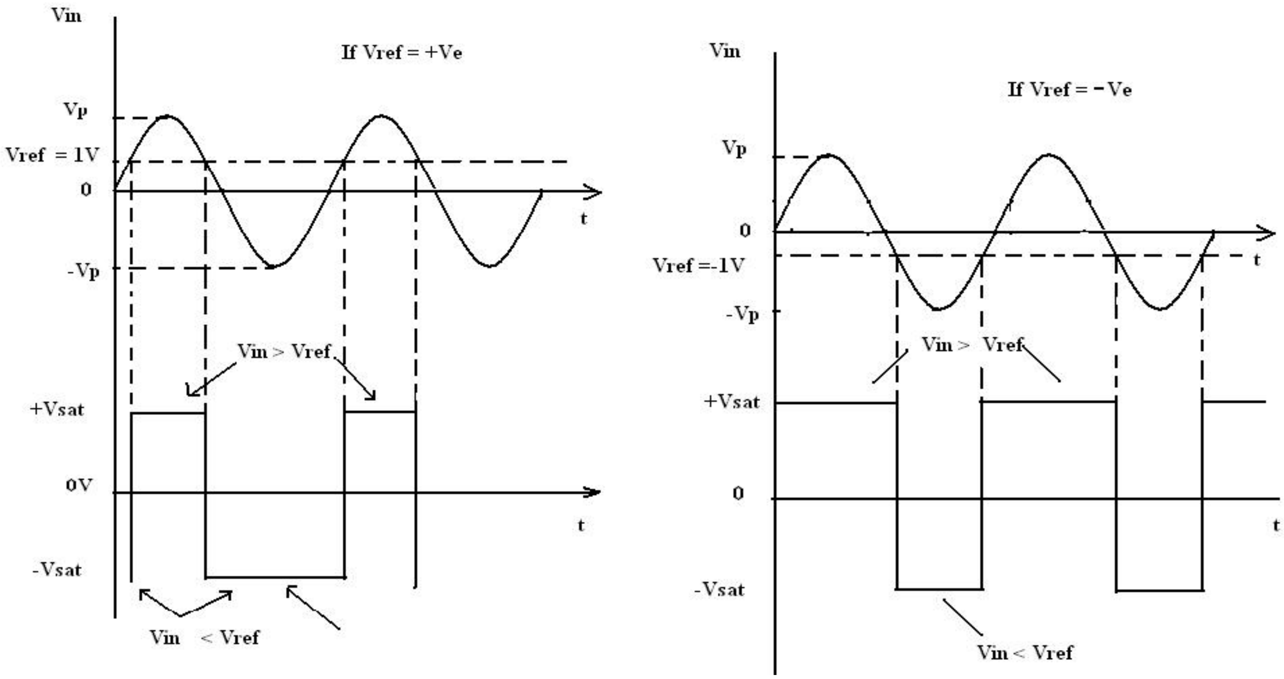


Fig. 2.34 Input and Output Waveforms of non-inverting comparator

2.13.2 Inverting Comparator:

This fig shows an inverting comparator in which the reference voltage Vref is applied to the (+) input terminal and Vin is applied to the (-) input terminal.

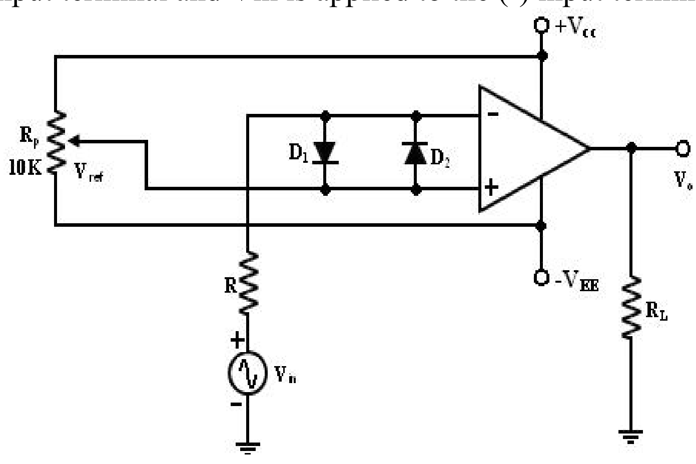


Fig. 2.35 Inverting comparator circuit

In this circuit Vref is obtained by using a 10K potentiometer that forms a voltage divider with DC supply volt +Vcc and -1 and the wiper connected to the input. As the wiper is moved towards +Vcc, Vref becomes more positive. Thus a Vref of a desired amplitude and polarity can be got by simply adjusting the 10k potentiometer.

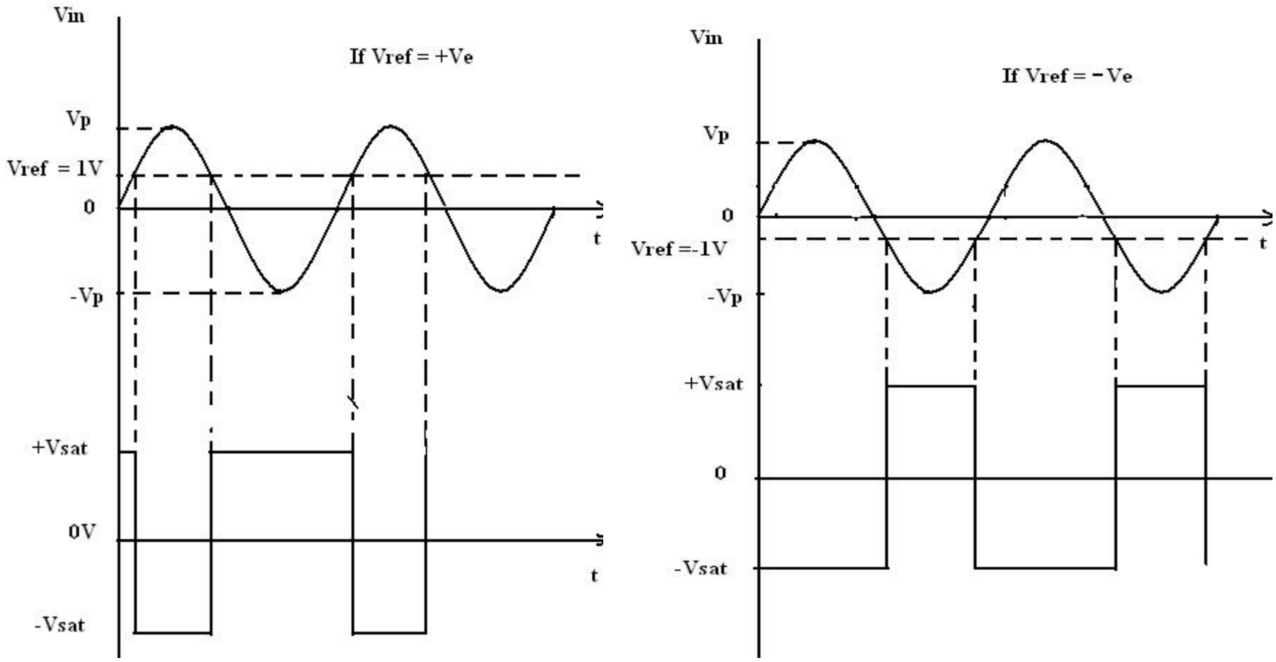


Fig. 2.36 Input and Output Waveforms of non-inverting comparator

Applications:

- ✓ **Zero Crossing Detector: [Sine wave to Square wave converter]**

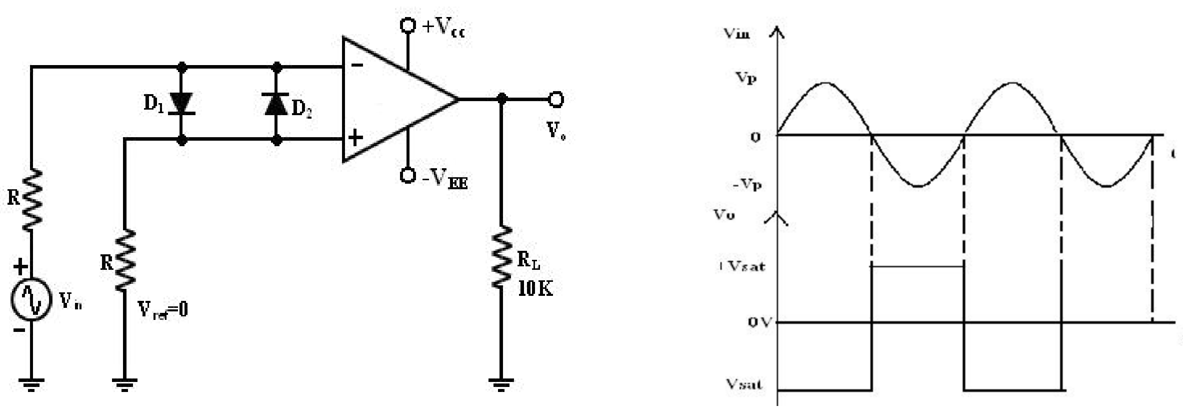


Fig. 2.37 Zero crossing detector circuit and input-output waveforms

One of the applications of comparator is the zero crossing detector or –sine wave to Square wave Converter. The basic comparator can be used as a zero crossing detector by setting Vref is set to Zero.

This Fig shows when in what direction an input signal Vin crosses zero volts. (i.e.) the o/p V_0 is driven into negative saturation when the input the signal Vin passes through zero in positive direction. Similarly, when Vin passes through Zero in negative direction the output V_0 switches and saturates positively.

- ✓ Drawbacks of Zero- crossing detector:

In some applications, the input Vin may be a slowly changing waveform, (i.e) a low frequency

signal. It will take Vin more time to cross 0V, therefore V_0 may not switch quickly from one saturation voltage to the other.
Because of the noise at the op-amp‘s input terminals the output V_0 may fluctuate between 2 saturations voltages +Vsat and –Vsat. Both of these problems can be cured with the use of regenerative or positive feedback that cause the output V_0 to change faster and eliminate any false output transitions due to noise signals at the input Inverting comparator with positive feedback This is known as Schmitt Trigger.

2.14 Schmitt Trigger: [Square Circuit]

This circuit converts an irregular shaped waveform to a square wave or pulse. The circuit is known as Schmitt Trigger or squaring circuit. The input voltage Vin triggers (changes the state of) the o/p V_0 every time it exceeds certain voltage levels called the upper threshold Vut and lower threshold voltage.

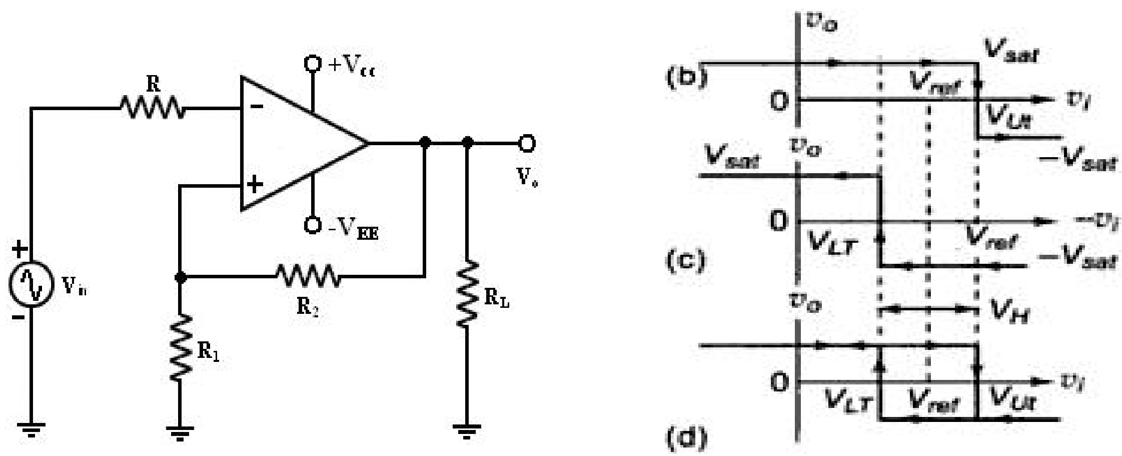


Fig.2.38 Schmitt Trigger circuit and hysteresis phenomenon

These threshold voltages are obtained by using the voltage divider $R_1 – R_2$, where the voltage across R_1 is feedback to the (+) input. The voltage across R_1 is variable reference threshold voltage that depends on the value of the output voltage. When V_0 = +Vsat, the voltage across R_1 is called upper threshold voltage Vut.

The input voltage Vin must be more positive than Vut in order to cause the output V_0 to switch from +Vsat to -Vsat using voltage divider rule,

Voltage at (+) input terminal is $V_{UT}=V_{ref}+ R_2 (V_{sat}-V_{ref})/(R_1+R_2)$ when $V_0= +v_{sat.}$

When v0= -vsat. Hysteris width VH= $V_{UT} - V_{LT} = 2 R_2 (V_{sat})/ (R_1+R_2)$

When V_0 = -Vsat, the voltage across R_1 is called lower threshold voltage V_{lt} .the Vin must be more negative than V_{lt} in order to cause V_0 to switch from –Vsat to +Vsat.

for Vin > V_{lt} , V_0 is at –Vsat.

Voltage at (+) terminal is $V_{LT}=V_{ref} - R_2 (V_{sat}+V_{ref})/(R_1+R_2)$.

- If the threshold voltages Vut and Vlt are made larger than the input noise voltages, the positive feedback will eliminate the false o/p transitions.
- Also the positive feedback, because of its regenerative action, will make V_0 switch faster

between +Vsat and –Vsat.

- Resistance Rcomp=R1 || R2 is used to minimize the offset problems.
- The comparator with positive feedback is said to exhibit hysteresis, a dead band condition. (i.e) when the input of the comparator exceeds Vut its output switches from +Vsat to –Vsat and reverts to its original state, +Vsat when the input goes below V_{LT}. The hysteresis voltage is equal to the difference between Vut and Vlt. Therefore

  $$V_H = Vut - Vlt.$$
- If Vref=0, Vut= $-V_{LT} = 2\, R_2(V_{sat})/(R_1+R_2)$

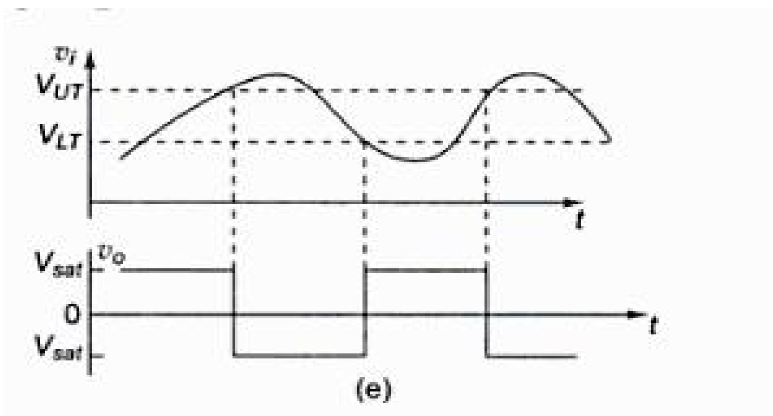


Fig. 2.39 Schmitt Trigger as squarer

2.15 Precision Rectifier:

The ordinary diodes cannot rectify voltages below the cut-in-voltage of the diode. A circuit which can act as an ideal diode or precision signal – processing rectifier circuit for rectifying voltages which are below the level of cut-in voltage of the diode can be designed by placing the diode in the feedback loop of an op-amp.

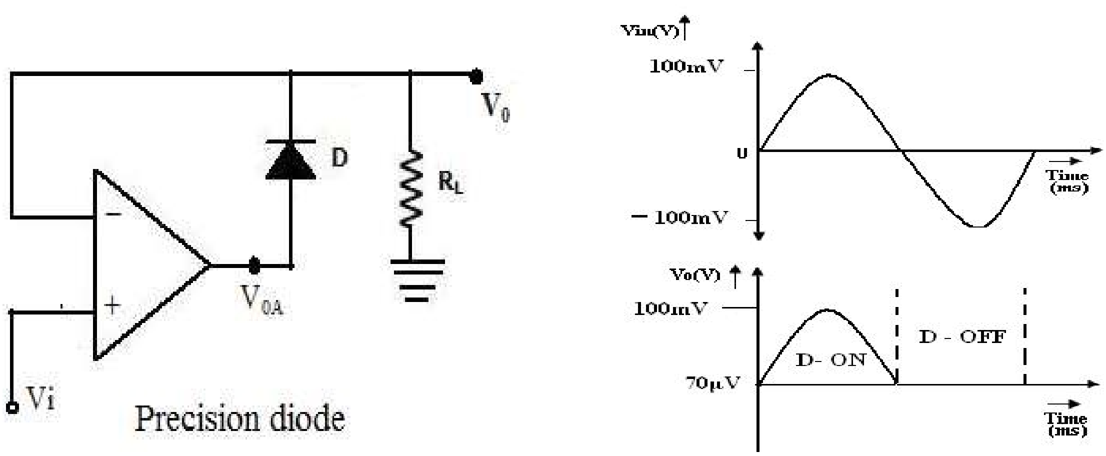


Fig.2.40 Precision diode and its waveform

Precision diodes:

Figure shows the arrangement of a precision diode. It is a single diode arrangement and functions as a non-inverting precision half– wave rectifier circuit. If V_1 in the circuit of figure is positive, the op-amp output V_{OA} also becomes positive. Then the closed loop condition is achieved for the op-amp and the output voltage $V_0 = V_i$. When $V_i < 0$, the voltage V_{0A} becomes negative and the diode is reverse biased. The loop is then broken and the output $V_0 = 0$.

Consider the open loop gain A_{OL} of the op-amp is approximately 10^4 and the cut-in voltage V_γ for silicon diode is $\approx 0.7V$. When the input voltage $V_i > V_\gamma / A_{OL}$, the output of the op-amp V_{OA} exceeds V_γ and the diode D conducts.

Then the circuit acts like a voltage follower for input voltage level $V_i > V_\gamma / A_{OL}$,(i.e. when $V_i > 0.7/10^4 = 70\mu V$), and the output voltage V_0 follows the input voltage during the positive half cycle for input voltages higher than $70\mu V$ as shown in figure.

When V_i is negative or less than V_γ / A_{OL}, the output of op-amp V_{OA} becomes negative, and the diode becomes reverse biased. The loop is then broken, and the op-amp swings down to negative saturation. However, the output terminal is now isolated from both the input signal and the output of the op-amp terminal thus $V_0 = 0$.

No current is then delivered to the load R_L except for the small bias current of the op-amp and the reverse saturation current of the diode.

This circuit is an example of a non-linear circuit, in which linear operation is achieved over the remaining region ($V_i < 0$). Since the output swings to negative saturation level when $V_i < 0$, the circuit is basically of saturating form. Thus the frequency response is also limited.

Applications: The precision diodes are used in

- ✓ half wave rectifier,
- ✓ Full-wave rectifier,
- ✓ peak value detector,
- ✓ Clipper and clamper circuits.

Disadvantage:

It can be observed that the precision diode as shown in figure operated in the first quadrant with $V_i > 0$ and $V_0 > 0$. The operation in third quadrant can be achieved by connecting the diode in reverse direction.

2.15.1 Half – wave Rectifier:

A non-saturating half wave precision rectifier circuit is shown in figure. When $V_i >$ 0V, the voltage at the inverting input becomes positive, forcing the output V_{OA} to go negative. This results in forward biasing the diode D_1 and the op-amp output drops only by $\approx 0.7V$ below the inverting input voltage. Diode D_2 becomes reverse biased. The output voltage V_0 is zero when the input is positive.

When $V_i > 0$, the op-amp output V_{OA} becomes positive, forward biasing the diode D_2 and reverse biasing the diode D_1. The circuit then acts like an inverting amplifier circuit with a non-linear diode in the forward path. The gain of the circuit is unity when $R_f = R_i$.

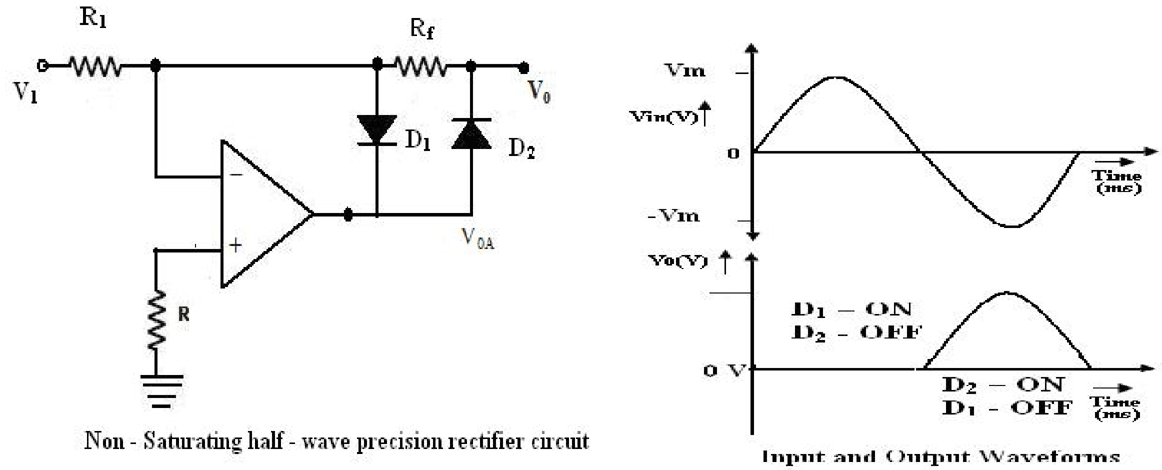


Fig. 2.41 Half wave rectifier and its operation

The circuit operation can mathematically be expressed as

$V_0 = 0$ when $V_i > 0$ and

$V_0 = R_f/R_i V_1$ for $V_i < 0$

The voltage V_{OA} at the op amp output is $V_{OA} = -0.7V$ for $V_i > 0$

$V_{OA} = R_f/R_i V_1 + 0.7V$ for $V_i < 0$

Advantages:

- ✓ it is a precision half wave rectifier and
- ✓ it is a non saturating one.

The inverting characteristics of the output V_0 can be circumvented by the use of an additional inversion for achieving a positive output.

2.15.2 Full wave Rectifier:

The first part of the Full wave circuit is a half wave rectifier circuit. The second part of the circuit is an inverting amplifier.

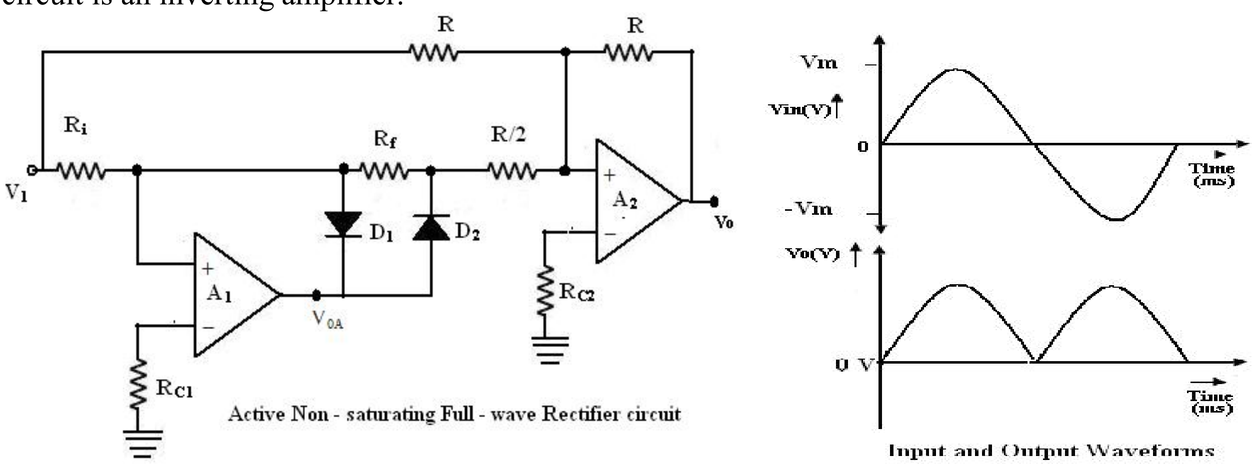


Fig. 2.42 Full wave rectifier and its operation

For positive input voltage $V_i > 0V$ and assuming that $R_F = R_i = R$, the output voltage $V_{OA} = V_i$. The voltage V_0 appears as (-) input to the summing op-amp circuit formed by A_2, The gain for the input V'_0 is R/(R/2), as shown in figure.

The input V_i also appears as an input to the summing amplifier. Then, the net output is $V_0 = -V_i - 2V_0 = -V_i - 2(-V_i) = V_i$. Since $V_i > 0V$, V_0 will be positive, with its input output characteristics in first quadrant. For negative input $V_i < 0V$, the output V'_0 of the first part of rectifier circuit is zero. Thus, one input of the summing circuit has a value of zero. However, V_i is also applied as an input to the summer circuit formed by the op-amp A_2.

The gain for this input id (-R/R) = -1, and hence the output is $V_0 = -V_i$. Since V_i is negative, V_0 will be inverted and will thus be positive. This corresponds to the second quadrant of the circuit.

To summarize the operation of the circuit,

$V_0 = V_i$ when $V_i < 0V$ and

$V_0 = V_i$ for $V_i > 0V$, and hence

$V_0 = |V_i|$

2.16 Peak Detector

Square, Triangular, Saw tooth and pulse waves are typical examples of non-sinusoidal waveforms. A conventional AC voltmeter cannot be used to measure these sinusoidal waveforms because it is designed to measure the RMS value of the pure sine wave. One possible solution to this problem is to measure the peak values of the non-sinusoidal waveforms. Peak detector measures the +ve peak value of the square wave input.

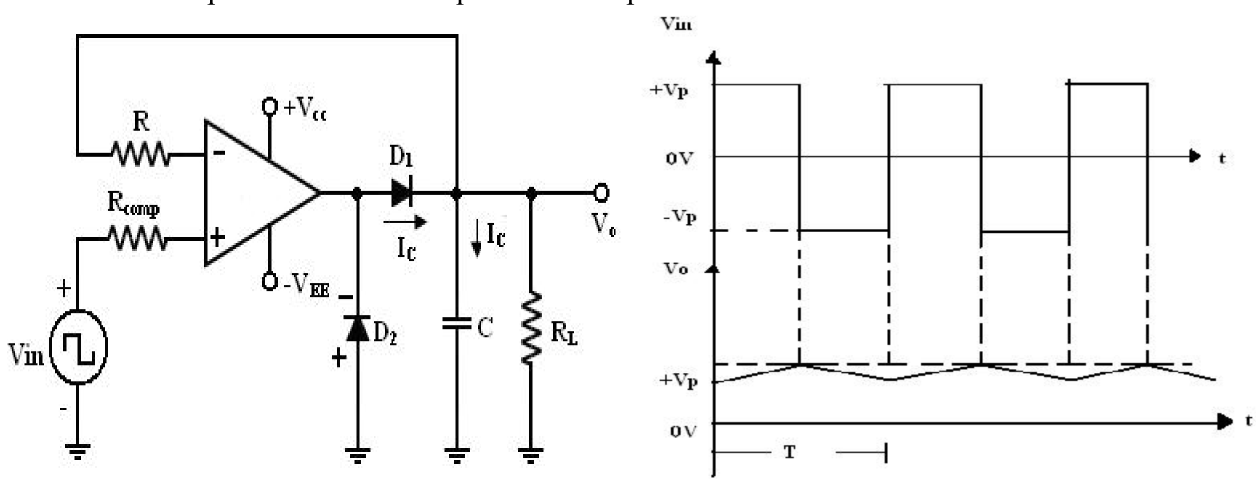


Fig. 2.43 Peak detector circuit and input and output waveforms

i) During the positive half cycle of Vin:

the o/p of the op-amp drives D_1 on. (Forward biased)

Charging capacitor C to the positive peak value V_p of the input volt Vin.

ii) During the negative half cycle of Vin:

D_1 is reverse biased and voltage across C is retained.

The only discharge path for C is through R_L since the input bias I_B is negligible.

For proper operation of the circuit, the charging time constant (CR_d) and discharging time constant (CR_L) must satisfy the following condition.

$CR_d <= T/10$

Where R_d = Resistance of the forward-biased diode.

T = time period of the input waveform.

$CR_L >= 10T$ (2)

Where R_L = load resistor.

If R_L is very small so that eqn. (2) cannot be satisfied.

- Use a (buffer) voltage follower circuit between capacitor C and R_L load resistor.
- R is used to protect the op-amp against the excessive discharge currents.
- Rcomp = minimizes the offset problems caused by input current
- D_2 conducts during the –ve half cycle of Vin and prevents the op-amp from going into negative saturation.

Note: -ve peak of the input signal can be detected simply by reversing diode D_1 and D_2

2.17 Clipper and clipper

Applications:

Wave shaping circuits are commonly used in digital computers and communication such as TV and FM receiver.
Wave shaping technique include clipping and clamping.

In op-amp clipper circuits a rectifier diode may be used to clip off a certain portion of the input signal to obtain a desired o/p waveform.

The diode works as an ideal diode (switch) because when on, the voltage drop across the diode is divided by the open loop gain of the op-amp. When off (reverse biased) the diode is an open circuit. In an op-amp clamper circuits, however a predetermined dc level is deliberately inserted in the o/p volt. For this reason, the clamper is sometimes called a dc inverter.

2.17.1 Positive and Negative

Clipper: Positive Clipper:

A circuit that removes positive parts of the input signal can be formed by using an op-amp with a rectifier diode. The clipping level is determined by the reference voltage Vref, which should less than the i/p range of the op-amp ($V_{ref} < V_{in}$). The Output voltage has the portions of the positive half cycles above V_{ref} clipped off.
The circuit works as follows:

During the positive half cycle of the input, the diode D_1 conducts only until Vin = Vref. This happens because when Vin <Vref, the output volts V_0 of the op-amp becomes negative to device D_1 into conduction when D_1 conducts it closes feedback loop and op-amp operates as a voltage follower. (i.e.) Output V_0 follows input until Vin = Vref.

When Vin > Vref => the V_0 becomes +ve to derive D_1 into off. It opens the feedback loop and op- amp operates open loop. When Vin drops below Vref (Vin<Vref) the o/p of the op-amp V_0 again becomes –ve to device D_1 into conduction. It closes the feedback path. (o/p follows the i/p).
Thus diode D_1 is on for vin<Vref (o/p follows the i/p) and D_1 is off for Vin>Vref.

The op-amp alternates between open loop (off) and closed loop operation as the D_1 is turned off and on respectively. For this reason the op-amp used must be high speed and preferably compensated for unity gain.

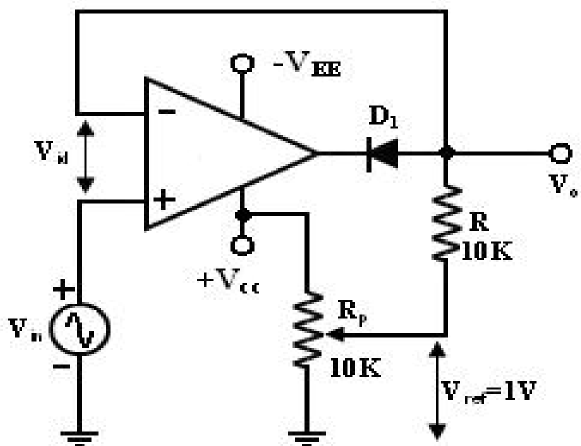


Fig. 2.44 Positive Clipper

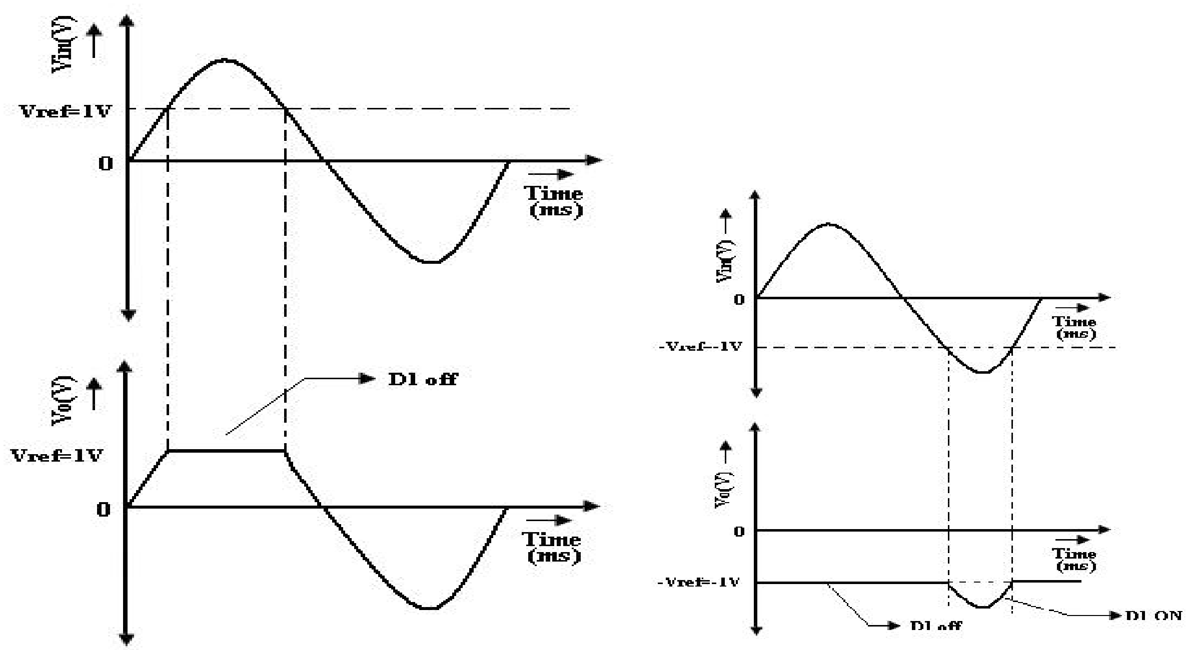


Fig 2.45 Positive clipper input output waveforms

Ex: for high speed op-amp HA 2500, LM310, µA 318. In addition the difference input voltage (Vid=high) is high during the time when the feedback loop is open (D_1 is off) hence an op-amp with a high difference input voltage is necessary to prevent input breakdown. If R_p (pot) is connected to $-V_{EE}$ instead of +Vcc, the ref voltage Vref will be negative (Vref = -ve). This will cause the entire o/p waveform above –Vref to be clipped off.

2.17.2 Negative Clipper:

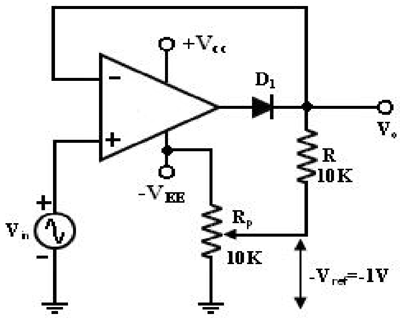


Fig.2.46 Negative clipper

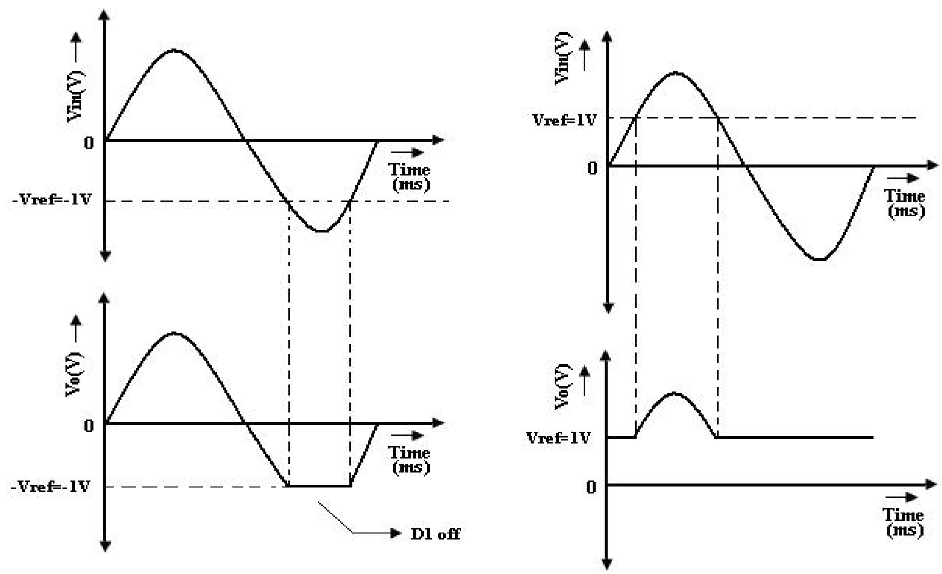


Fig. 2.47 Input output waveforms

The positive clipper is converted into a –ve clipper by simply reversing diode D_1 and changing the polarity of Vref voltage. The negative clipper clips off the –ve parts of the input signal below the reference voltage. Diode D_1 conducts -> when Vin > -Vref and therefore during this period o/p volt V_0 follows the i/p volt Vin. The –Ve portion of the output volt below –Vref is clipped off because (D_1 is off) Vin<-Vref. If –Vref is changed to –Vref by connecting the potentiometer R_p to the +Vcc, the V_0 below +Vref will be clipped off. The diode D_1 must be on for Vin > Vref and off for Vin.

2.17.3 Positive and Negative Clampers:

In clamper circuits a predetermined dc level is added to the output voltage. (or) The output is clamped to a desired dc level.

1. If the clamped dc level is +ve, the clamper is positive clamper
2. If the clamped dc level is –ve, the clamper is negative clamper.

Other equivalent terms used for clamper are dc inserter or restorer. Inverting and Non-Inverting that uses this technique.

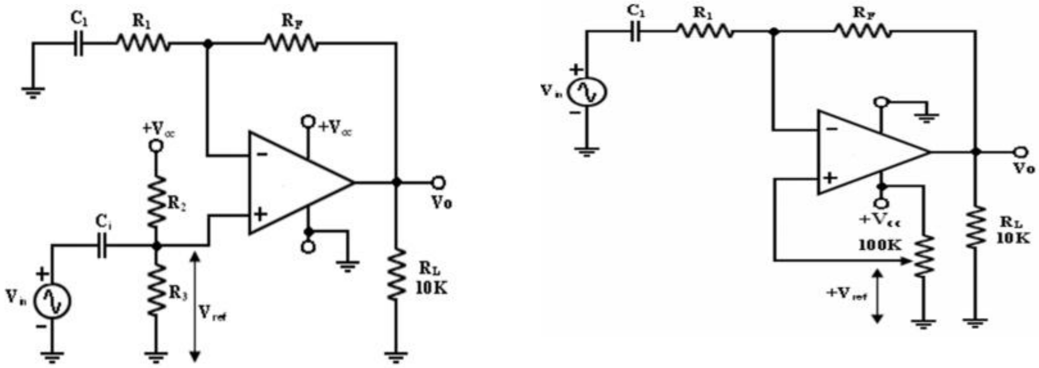


Fig.2.48 Positive –Negative campers

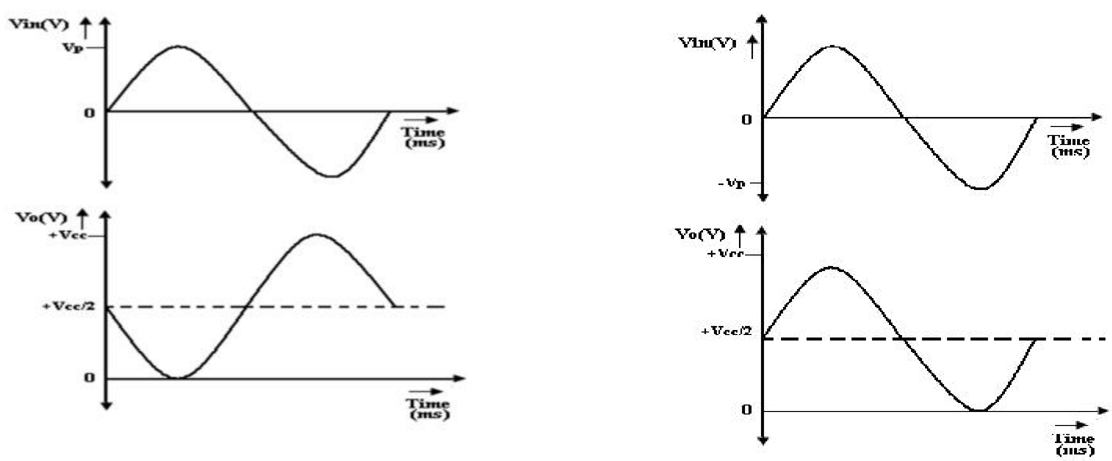


Fig.2.49 Input **and output waveform with +Vref**

Capacitor:

The Value of the capacitors in these circuits depends on different input rates and pulse widths.

1. In both circuits the dc level added to the o/p voltage is approximately equal to Vcc/2.
2. This +ve fixed dc level is needed to obtain a maximum undistorted symmetrical sine wave.

Peak clamper circuit:

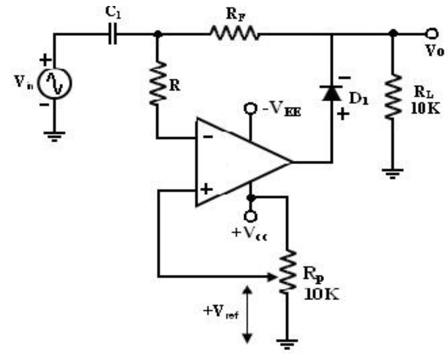


Fig.2.50 Peak clamper circuit

In this circuit, the input waveform peak is clamped at Vref. For this reason, the circuit is called the peak clamper.

First consider the input voltage Vref at the (+) input: since this volt is +ve, V'_0 is also +ve which forward biases D_1. This closed the feedback loop.

Voltage Vin at the (-) input: During its –ve half cycle, diode D_1 conducts, charging c; to the –ve peak value of V_p. During the +ve half cycle, diode D_1 in reverse biased. Since this voltage V_p is in series with the +ve peak volt V_p the o/p volt $V_0 = 2\ V_p$. Thus the nett o/p is Vref plus 2 V_p. So the – ve peak of 2 V_p is at Vref. For precision clamping, $C_iR_d << T/2$

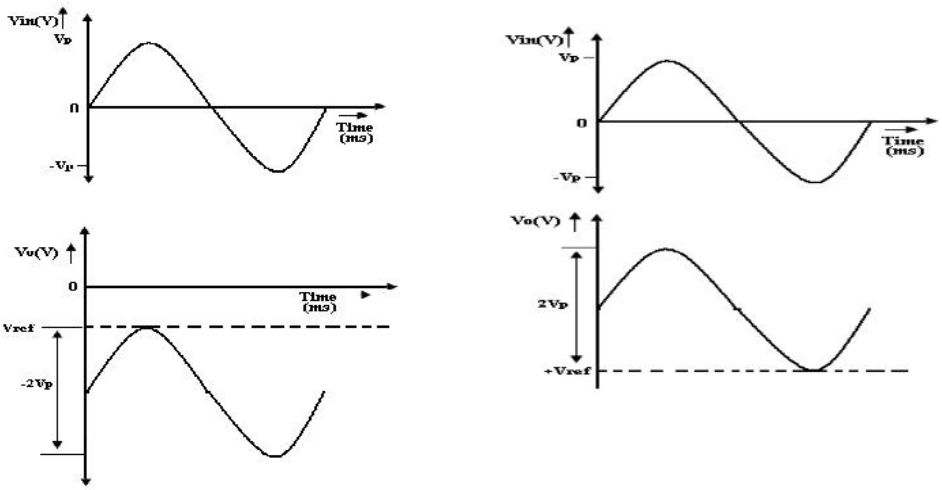


Fig. 2.51 Input and Output Waveform with –Vref

Where R_d = resistance of diode D_1 when it is forward biased.

T = time period of the input waveform.

Resistor R is used to protect the op-amp against excessive discharge currents from capacitor C_i especially when the dc supply voltages are switched off. A +ve peak clamping is accomplished by reversing D_1 and using –ve reference voltage (-Vref).

Note:

Inv and Non-Inv clamper – Fixed dc level

Peak clamper – Variable dc level

2.18 Active filters:

An electric filter is often a frequency selective circuit that passes a specified band of frequencies and blocks or alternates signal and frequencies outside this band.

Filters may be classified as

1. Analog or digital.
2. Active or passive
3. Audio (AF) or Radio Frequency (RF)

1. Analog or digital filters:

Analog filters are designed to process analog signals, while digital filters process analog signals using digital technique.

2. Active or Passive:

Depending on the type of elements used in their construction, filter may be classified as passive or Active elements used in passive filters are Resistors, capacitors, inductors. Elements used in active filters are transistor, or op-amp.

Active filters offer the following advantages over passive filters:

1. Gain and Frequency adjustment flexibility:

    Since the op-amp is capable of providing gain, the i/p signal is not attenuated as it is in a passive filter. [Active filter is easier to tune or adjust].

2. No loading problem:

    Because of the high input resistance and low o/p resistance of the op-amp, the active filter does not cause loading of the source or load.

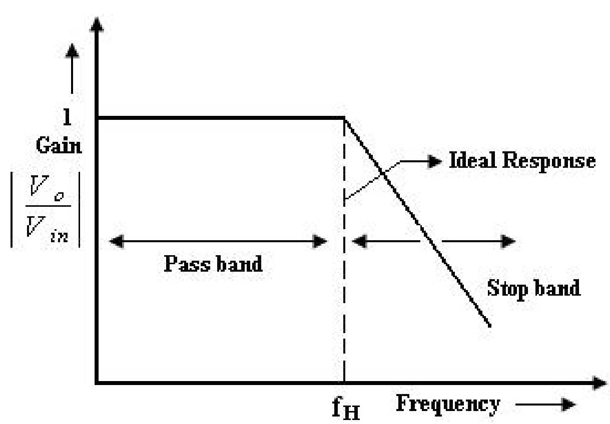
3. Cost:

    Active filters are more economical than passive filter. This is because of the variety of cheaper op-amps and the absence of inductors.

✓ The most commonly used filters are these:

1. Low pass Filters
2. High pass Filters
3. Band pass filters
4. Band –reject filters
5. All pass filters.

Frequency response of the active filters:



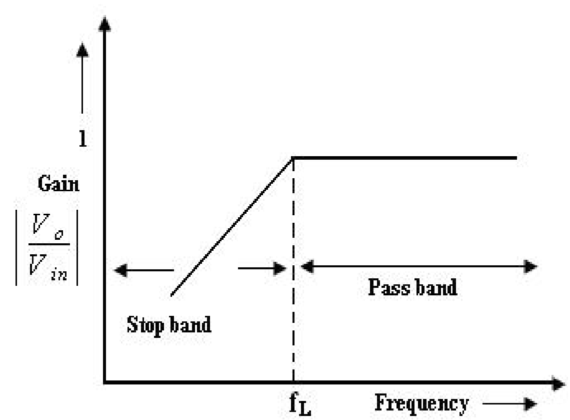


Fig 2.52 Frequency response of Low Pass filter and High pass Filter

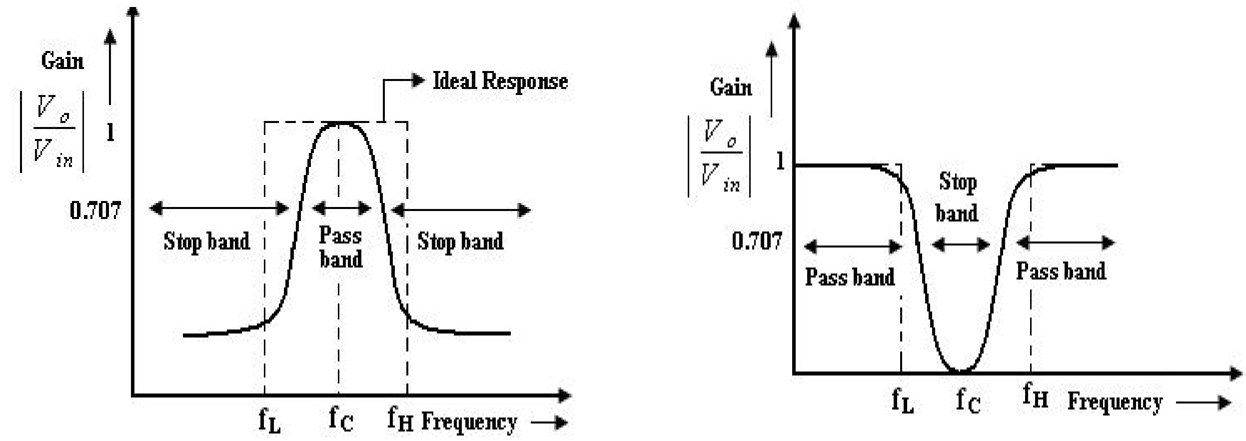


Fig 2.53 Frequency response of Band Pass filter and Band reject Filter

Low pass filters:

1. It has a constant gain from 0 Hz to a high cutoff frequency f_1.
2. At f_H the gain in down by 3db.
3. The frequency between 0 Hz and f_H are known as the pass band frequencies where as the range of frequencies those beyond f_H, that are attenuated includes the stop band frequencies.
4. Butterworth, Chebyshev and Cauer filter are some of the most commonly used practical filters.
5. The key characteristics of the butter worth filter are that it has a flat pass band as well as stop band. For this reason, it is sometimes called flat- flat filters.
6. Chebyshev filter -> has a ripple pass band & flat stop band.
7. Causer Filter -> has a ripple pass band & ripple stop band. It gives best stop band response among the three.

High pass filter:

High pass filter with a stop band $0<f< f_L$ and a pass band $f> f_L$

f_L -> low cut off frequency

f -> operating frequency.

Band pass filter:

It has a pass band between 2 cut off frequencies f_H and f_L where $f_H > f_L$ and two, stop bands: $0<f< f_L$ and $f > f_H$ between the band pass filter (equal to f_H - f_L.

Band –reject filter: (Band stop or Band elimination)

It performs exactly opposite to the band pass.

It has a band stop between 2 cut-off frequency f_L and f_H and 2 pass bands: $0<f< f_L$ and $f> f_H$

f_C -> center frequency.

Note:

The actual response curves of the filters in the stop band either Ror S or both with Rin

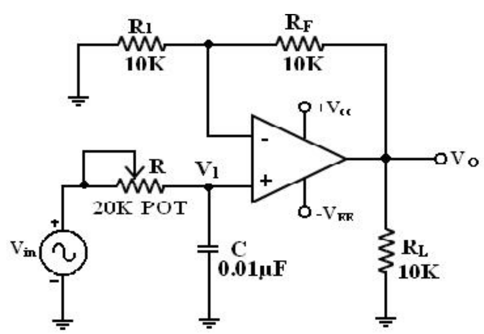
frequencies.
The rate at which the gain of the filter changes in the stop band is determined by the order of the filter.

Ex: 1st order low pass filter the gain rolls off at the rate of 20dB/decade in the stop band. (i.e) for $f > f_H$.

2nd order LPF -> the gain roll off rate is 40dB/decade.

1st order HPF -> the gain rolls off at the rate of 20dB (i.e.) until f:fL
2nd order HPF -> the gain rolls off at the rate of 40dB/decade

First order LPF Butterworth filter:
First order LPF that uses an RC for filtering op-amp is used in the non inverting configuration. Resistor R1 & R_f determine the gain of the filter. According to the voltage –divider rule, the voltage at the non-inverting terminal (across capacitor) C is,



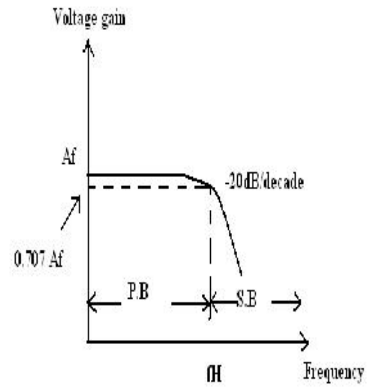


Fig. 2.54 First order LPF Butterworth filter

Gain $A = (1+R_f/R_1)$
Voltage across capacitor $V_1 = V_i / (1+j2\pi fRC)$
Output voltage V0 for non inverting amplifier $= AV_1$
$= (1+R_f/R_1)\ V_i/(1+j2\pi fRC)$
Overall gain $V0/Vi = (1+R_f/R_1)\ V_i/(1+j2\pi fRC$
Transfer function $H(s) = A/(jf/f_h+1)$ if $f_h = 1/2\pi RC$
$H(j\omega) = A/(j\omega RC+1) = A/(j\omega RC+1)$.

The gain magnitude and phase angle of the equation of the LPF can be obtained by converting eqn. (1) b into its equivalent polar form as follows.

1. At very low frequency, $f < f_H$
$|H(j\omega)| = A$

2. At $f = f_H$

   $|H(j\omega)| = A/\sqrt{2} = 0.707A$

3. At $f > f_H$

   $|H(j\omega)| << A \cong 0$

When the frequency increases by tenfold (one decade), the volt gain is divided by 10. The gain falls by 20 dB (=20log10) each time the frequency is reduces by 10. Hence the rate at which the gain rolls off f_H = 20 dB or 6dB/octave (twofold R in frequency). The frequency $f = f_H$ is called the cut off frequency because the gain of the filter at this frequency is down by 3 dB (=20 log 0.707).

Filter design:

A LPF can be designed by implementing the following steps.
1. Choose a value of high cut off frequency f_H.
2. Select a value of C less than or equal to 1μf.
3. Choose the value of R using $f_h = 1/2\pi RC$
4. Finally select values of R1 and RF dependent on the desired pass band gain AF

Using A= (1+Rf/R1)

Second order LP Butterworth filter:

A second order LPF having a gain 40dB/decade in stop band. A First order LPF can be converted into a II order type simply by using an additional RC network.
The gain of the II order filter is set by R1 and R_F, while the high cut off frequency f_H is determined by R2, C2, R3 and C3.

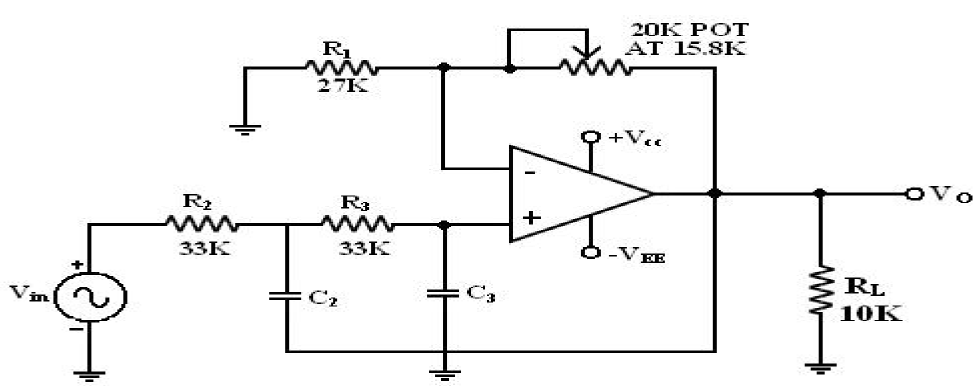


Fig. 2.55 second order LP Butterworth filter

Let $Y_1 = Y_2 = 1/R$, $Y_3 = sC_3$ and $Y_4 = sC_4$. Then the transfer function is

$$H(s) = \frac{\frac{1}{R^2}}{\frac{1}{R^2} + sC_4\left(\frac{2}{R} + sC_3\right)} = \frac{1}{1 + sRC_4\left(2 + sRC_3\right)}$$

Let the time constant $\tau_1 = RC_3$ and $\tau_2 = RC_4$. Substituting $s = j\omega$, we get

$$H(j\omega) = \frac{1}{1 + j\omega\tau_2\left(2 + j\omega\tau_1\right)} = \frac{1}{\left(1 - \omega^2\tau_1\right) + j\left(2\omega\tau_2\right)}$$

Therefore, its magnitude is

$$\left|H(j\omega)\right| = \left[\left(1 - \omega^2\tau_1\tau_2\right)^2 + \left(2\omega\tau_2\right)^2\right]^{-1/2}$$

A maximally flat Butterworth filter will have a minimum rate of change. Therefore,

$$\frac{d|H|}{d\omega}\Big|_{\omega=0} = 0$$

Differentiating $\left|H(j\omega)\right|$, we obtain

$$\frac{d|H|}{d\omega} = -\frac{1}{2}\left[\left(1 - \omega^2\tau_1\tau_2\right)^2 + \left(2\omega\tau_2\right)^2\right]^{-3/2}\left[-4\,\omega\tau_1\tau_2\left(1 - \omega^2\tau_1\tau_2\right) + 8\,\omega\,\tau_2^2\right]$$

Letting the derivative to zero at $\omega = 0$, we get

$$\frac{d|H|}{d\omega}\Big|_{\omega=0} = \left[-4\,\omega\tau_1\tau_2\left(1 - \omega^2\tau_1\tau_2\right) + 8\,\omega\tau_2^2\right]$$

$$= 4\omega\tau_2\left[-\tau_1\left(1 - \omega^2\tau_1\tau_2\right) + 2\tau_2\right]$$

The above equation is satisfied when $2\tau_2 = \tau_1$. That is, $C_3 = 2C_4$. Therefore the magnitude of the transfer function becomes

$$|H| = \frac{1}{\left[1 + 4\left(\omega\tau_2\right)^4\right]^{1/2}}$$

The cut-off frequency occurs when $|H| = \frac{1}{\sqrt{2}}$, or $4\left(\omega_{3dB}\tau_2\right)^4 = 1$. Therefore,

$$\omega_{3dB} = 2\pi f_{3dB} = \frac{1}{\tau_2\sqrt{2}} = \frac{1}{\sqrt{2}RC_4}$$

We know that the cut-off frequency is $\omega_H = \omega_{3dB} = \frac{1}{RC}$.

Comparing the above equations, we get

$$C_4 = 0.707C$$

$$C_3 = 1.414C$$

The magnitude of the voltage transfer function for the second order low-pass Butterworth filter is $\left|H(jf)\right| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_H}\right)^4}}$

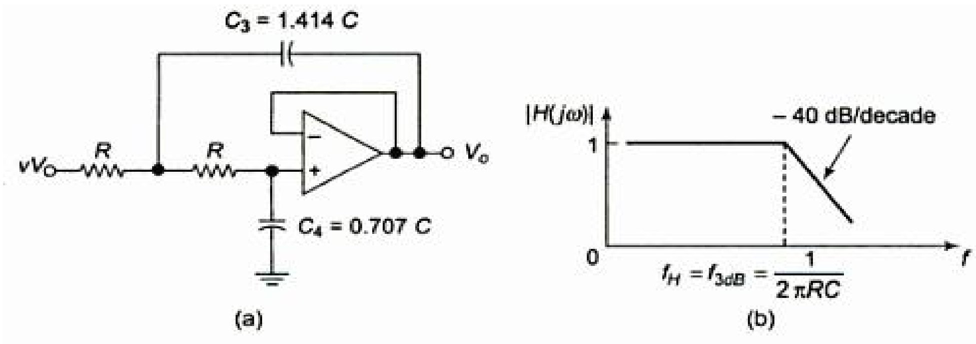


Fig. 2.56 Second order Low pass Butterworth and filter with unity gain and its transfer function

Filter Design:

1. Choose a value for a high cut off freq. (f_H).
2. To simplify the design calculations, set $R_2 = R_3 = R$ and $C_2 = C_3 = C$ then choose a value of $C <= 1\mu f$.
3. Calculate the value of R $R = 1/2\pi f_h C$

4. Finally, because of the equal resistor ($R_2 = R_3$) and capacitor ($C_2 = C_3$) values, the pass band volt gain $A_F = 1 + R_F / R_1$ of the second order had to be = to 1.586. $R_F = 0.586 R_1$. Hence choose a value of R_1 <=100kΩ.
5. Calculate the value of R_F.

First order HP Butterworth filter:

High pass filters are often formed simply by interchanging frequency-determining resistors and capacitors in low-pass filters.
(i.e) I order HPF is formed from a I order LPF by interchanging components R & C. Similarly II order HPF is formed from a II order LPF by interchanging R & C.

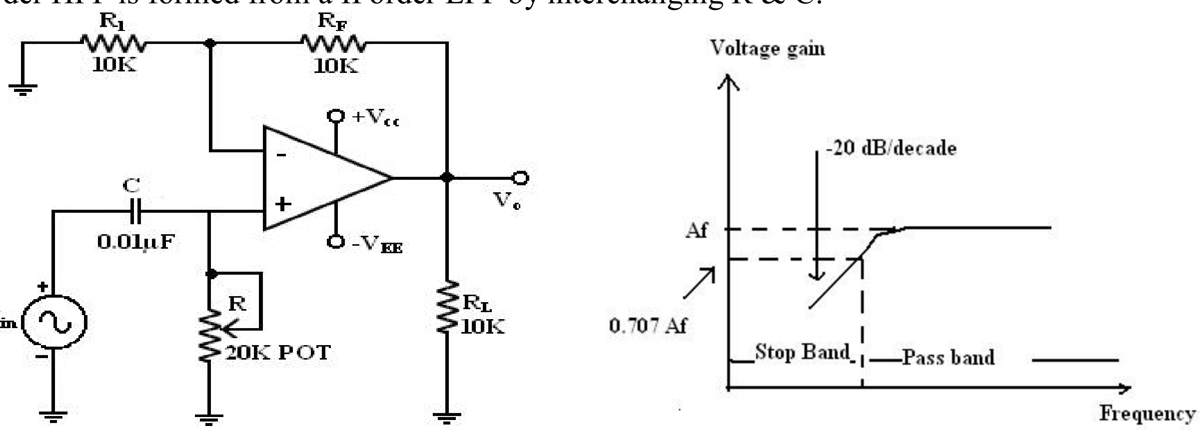


Fig. 2.57 I order HPF and its frequency response

Here I order HPF with a low cut off frequency of f_L. This is the frequency at which the magnitude of the gain is 0.707 times its passband value.
Here all the frequencies higher than f_L are passband frequencies.
The output voltage V0 of the first order active high pass filter is

$$V_o = \left(1 + \frac{R_f}{R_i}\right)\frac{j2\pi fRC}{1 + j2\pi fRC} V_i$$

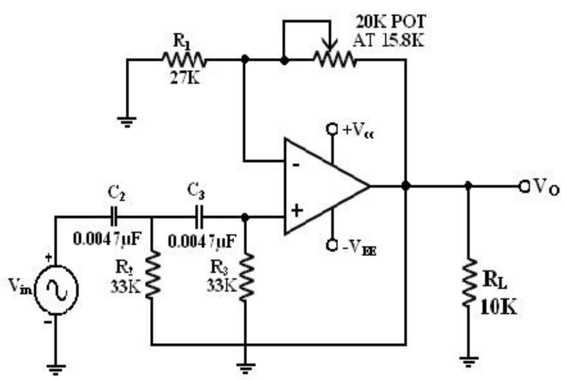
The gain of the filter:

$$\frac{V_o}{V_i} = A\left(\frac{j\left(\frac{f}{f_L}\right)}{1 + j\left(\frac{f}{f_L}\right)}\right)$$

Frequency response of the filter $|H(jf)| = \left|\frac{V_o}{V_i}\right| = \frac{A\left(\frac{f}{f_L}\right)}{\sqrt{1+\left(\frac{f}{f_L}\right)^2}} = \frac{A}{\sqrt{1+\left(\frac{f_L}{f}\right)^2}}$ is

- At high frequencies f>fL gain = A.
- At f= fL gain = 0.707 A.
- At f < fL the gain decreases at a rate of -20 db /decade. The frequency below cutoff frequency is stop band.

✓ **Second – order High Pass Butterworth Filter**:
I order Filter, II order HPF can be formed from a II order LPF by interchanging the frequency



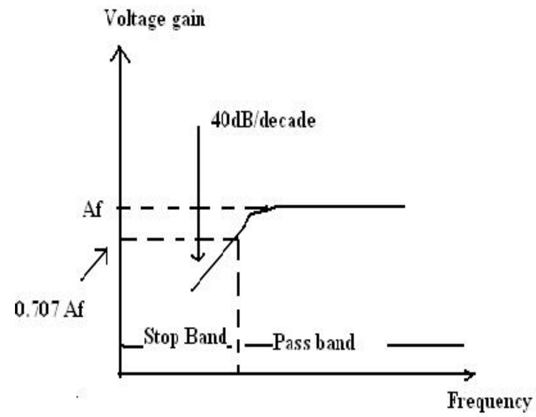


Fig. 2.58 II order HPF and its frequency response

2.20 Band pass filters

- Filters that pass band of frequencies and attenuates others. Its high cutoff frequency and low cutoff frequency are related as $f_H > f_L$ and maximum gain at resonant frequency
- $f_r = \sqrt{f_H f_L}$
- Figure of merit $Q = f_r / (f_H - f_L) = f_r / B$ where B= bandwidth.
- 2 types of filters are Narrow band pass and wide band pass filters

Wide band pass filter:
It is connection of a low pass filter and a high pass filter in cascade.
The f_H of low pass filter and f_L of high pass filter are related as $f_H > f_L$

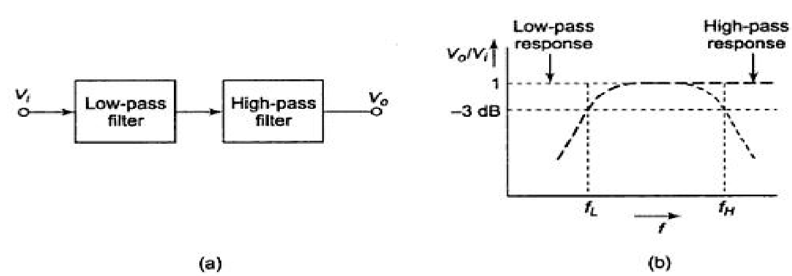


Fig. 2.59 (a) Wide band pass filter and (b) its frequency response

UNIT III

ANALOG MULTIPLIER AND PLL

Analog Multipliers:

A multiple produces an output V0 which is proportional to the product of two inputs Vx and Vy. V0 = KVxVy where K is the scaling factor = (1/10) V^{-1}.

There are various methods available for performing analog multiplication. Four of such techniques, namely,

1. Logarithmic summing technique
2. Pulse height/width modulation Technique
3. Variable trans conductance Technique
4. Multiplication using Gilbert cell and
5. Multiplication using variable trans conductance technique.

An actual multiplier has its output voltage V0 defined by

$$V_0 = \frac{(V_x+\phi_x)(V_y+\phi_y)}{10(1+\varepsilon)}+\phi_0$$

where φx and φy are the offsets associated with signals Vx and Vy, ε is the error signal associated with K and φ0 is the offset voltage of the multiplier output.

Terminologies associated voltage of the multiplier characteristics:

- ✓ **Accuracy:**

This specifies the derivation of the actual output from the ideal output, for any combination of X and Y inputs falling within the permissible operating range of the multiplier.

- ✓ **Linearity:**

This defines the accuracy of the multiplier. The Linearity Error can be defined as the maximum absolute derivation of the error surface. This linearity error imposes a lower limit on the multiplier accuracy.

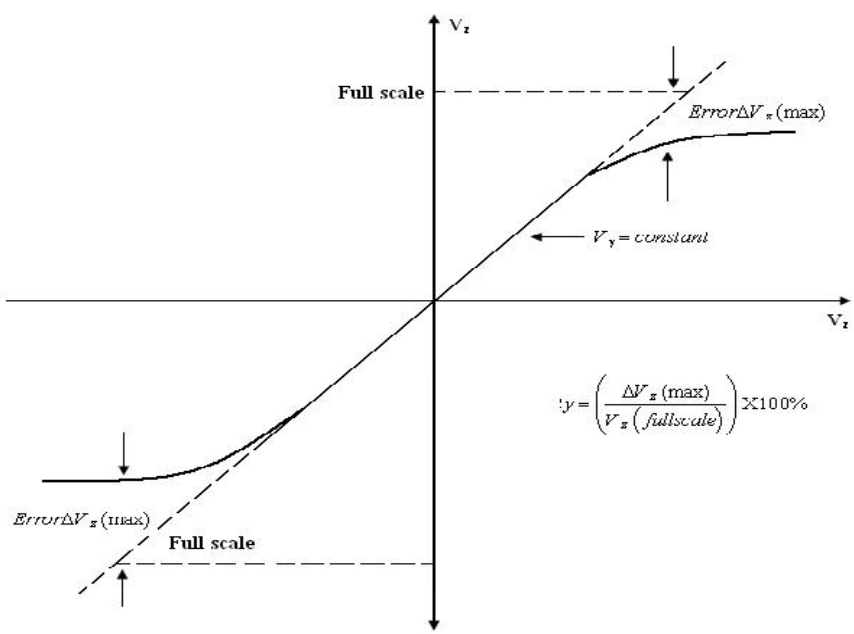


Fig.3.1 Linearity of the multiplier

The figure shows the response of the output as a function of one input voltage Vx when the other Vy is assumed constant. It represents the maximum percentage derivation from the ideal straight line output. An error surface is formed by plotting the output for different combinations of X and Y inputs.

- **Square law accuracy:**

The Square – law curve is obtained with the X and Y inputs connected together and applied with the same input signal. The maximum derivation of the output voltage from an ideal square –law curve expresses the squaring mode accuracy.

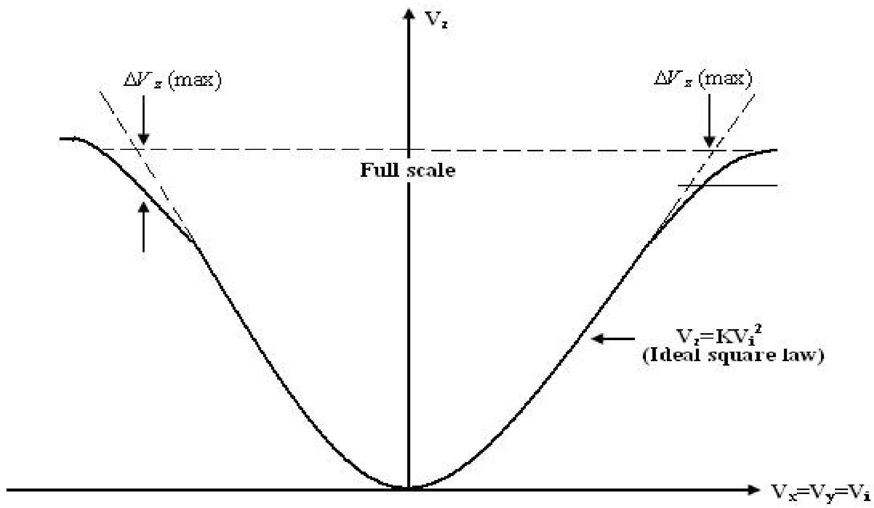


Fig. 3.2 squaring mode accuracy

- **Bandwidth:**

The Bandwidth indicates the operating capability of an analog multiplier at higher frequency values. Small signal 3 dB bandwidth defines the frequency f0 at which the output reduces by 3dB from its low frequency value for a constant input voltage. This is identified individually for the X and Y input channels normally.

The transconductance bandwidth represents the frequency at which the transconductance of the multiplier drops by 3dB of its low frequency value. This characteristic defines the application frequency ranges when used for phase detection or AM detection.

- **Quadrant:**

The quadrant defines the applicability of the circuit for bipolar signals at its inputs. First – quadrant device accepts only positive input signals, the two quadrant device accepts one bipolar signal and one unipolar signal and the four quadrant device accepts two bipolar signals.

- **Logarithmic summing Technique:**

This technique uses the relationship $\ln V_x + \ln V_y = \ln(V_x V_y)$

As shown in figure the input voltages Vx and Vy are converted to their logarithmic equivalent, which are then added together by a summer. An antilogarithmic converter produces the output voltage of the summer. The output is given by,

$$V_z = \ln^{-1} (\ln(V_x V_y)) = V_x V_y .$$

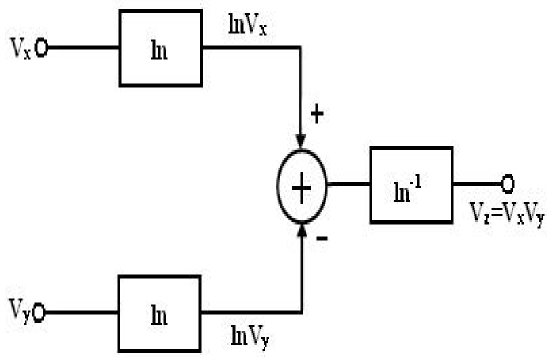


Fig. 3.3 logarithmic summing method

The relationship between I0 and VBE of the transistor is given by $I_C = I_0 e^{(V_{BE}/V_T)}$. It is found that the transistor follows the relationship very accurately in the range of 10nA to 100mA. Logarithmic multiplier has low accuracy and high temperature instability. This method is applicable only to positive values of Vx and Vy.

Limitation: this type of multiplier is restricted to one quadrant operation only.

- ✓ **Pulse Height/ Width Modulation Technique:**

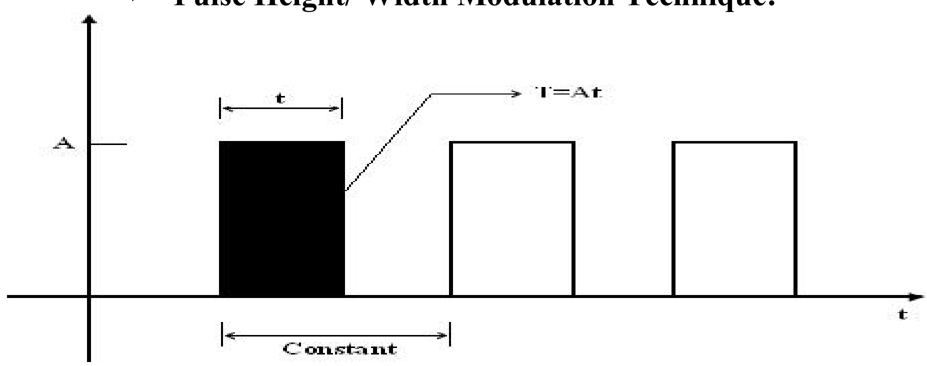


Fig.3.4 Pulse Height/ Width Modulation Technique

In this method, the pulse width of a pulse train is made proportional to one input voltage and the pulse amplitude is made proportional to the second input voltage. Therefore, $V_x = K_x A$, $V_y = K_y t$, and $V_z = K_z T$ where Kx, Ky, Kz are scaling factors. In figure A is the amplitude of the pulse, t is the pulse width and T is the area of the pulse. Therefore,

$$V_z = K_z T = \frac{V_x V_y}{k_x k_y}$$

The modulated pulse train is passed through an integrated circuit. Therefore, the input of the integrator is proportional to the area of pulse, which in turn is proportional to the product of two input voltages.

3.1 Analog multiplier using an Emitter coupled Transistor pair:

The output currents IC1 and IC2 are related to the differential input voltage V1 by

$$I_{C1} = \frac{I_{EE}}{1+e^{V_1/V_T}} \quad \text{and} \quad I_{C2} = \frac{I_{EE}}{1+e^{V_2/V_T}}$$

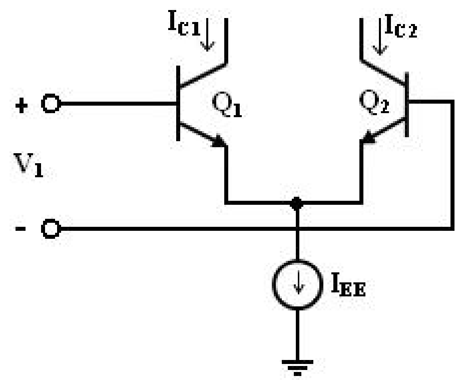


Fig.3.5 multiplier circuit using an emitter coupled pair

where V is thermal voltage and the base currents have been neglected. Combining above eqn., difference between the two output currents as

$$\Delta I_C = I_{C1} - I_C = \frac{I_{EE}}{1+e^{V_1/V_T}} - \frac{I_{EE}}{1+e^{V_2/V_T}} = I_{EE} \tanh (V_1/2V_T)$$

The dc transfer characteristics of the emitter – coupled pair is shown in figure. It shows that the emitter coupled pair can be used as a simple multiplier using this configuration. When the differential input voltage V1 << VT, we can appropriate as given by

$$\Delta I_C = I_{EE} (V_1/2V_T)$$

The current IEE is the bias current for the emitter – coupled pair. If the current IEE is made proportional to a second input signal V2, then

$$I_{EE} = K_0 (V_2 - V_{BE})/2V_T$$

Substituting above eqn. , we get $\Delta I_C = K_0 V_1 (V_2 - V_{BE})/2V_T$

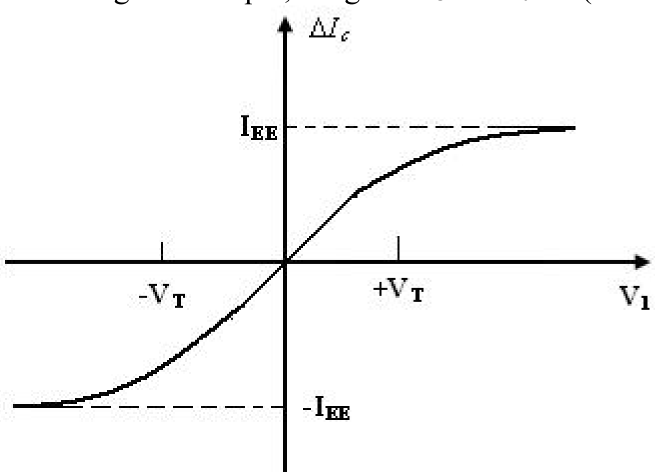


Fig. 3.6 DC Transfer characteristics of emitter coupled pair

This arrangement is shown in figure. It is a simple modulator circuit constructed using a differential amplifier. It can be used as a multiplier, provided V_1 is small and much less than 50mV and V_2 is greater than V_{BE} (on). But, the multiplier circuit shown in figure has several limitations. The first limitation is that V_2 is offset by V_{BE} (on).

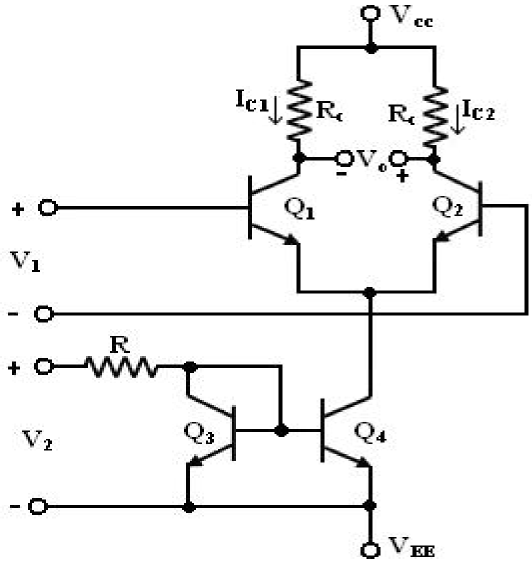


Fig. 3.7 A simple modulator using a differential amplifier

The second is that V2 must always be positive which results in only a two-quadrant multiplier operation. The third limitation is that, the tanh (X) is approximately as X, where X = V1 /2VT. The first two limitations are overcome in the Gilbert cell.

3.2 Gilbert Multiplier cell:

The Gilbert multiplier cell is a modification of the emitter coupled cell and this allows four – quadrant multiplication. Therefore, it forms the basis of most of the integrated circuit balanced Multipliers. Two cross- coupled emitter- coupled pairs in series connection with an emitter coupled pair form the structure of the Gilbert multiplier cell.

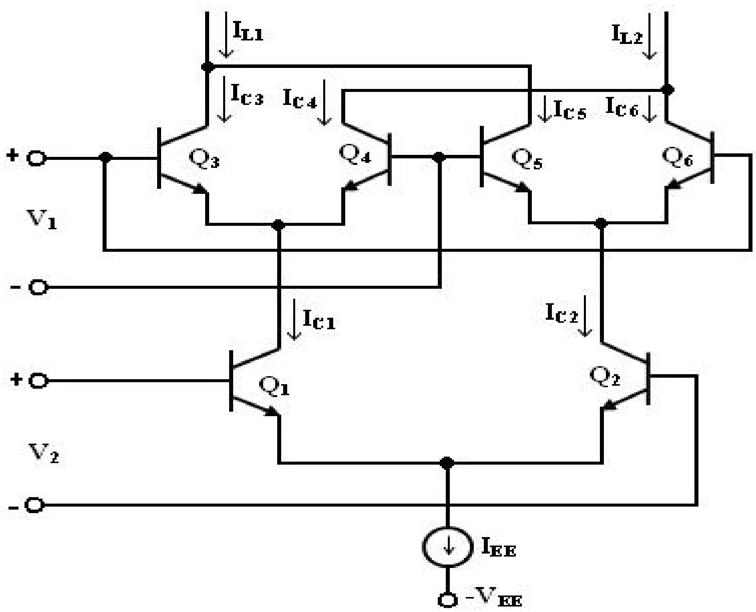


Fig. 3.8 Gilbert multiplier cell

The collector current of Q_3 and Q_4 are given by

$$I_{C3} \frac{I_{C1}}{1+e^{-V_1/V_T}} \text{ and } I_{C4} = \frac{I_{C1}}{1+e^{V_1/V_T}}$$

Similarly, the collector current of Q_5 and Q_6 are given by

The $I_{C5} = \frac{I_{C2}}{1+e^{V_1/V_T}}$ and $I_{C6} = \frac{I_{C2}}{1+e^{-V_1/V_T}}$

collector current I_{C1} and I_{C2} of transistors Q_1 and Q_2 can be expressed as

$$I_{C1} = \frac{I_{EE}}{1+e^{-V_1/V_T}} \text{ and } I_{C2} = \frac{I_{EE}}{1+e^{V_2/V_T}}$$

Substituting the above equation in I_{C3} and I_{C4}, we get

$$I_{C3} = \frac{I_{EE}}{\left[1+e^{\frac{-V_2}{V_T}}\right]\left[1+e^{-\frac{V_1}{V_T}}\right]} \text{ and } I_{C4} = \frac{I_{EE}}{\left[1+e^{\frac{-V_2}{V_T}}\right]\left[1+e^{\frac{V_1}{V_T}}\right]}$$

Similarly substituting I_{c2} in I_{c5} and I_{C6}, we get,

$$I_{C5} = \frac{I_{EE}}{\left[1+e^{\frac{V_2}{V_T}}\right]\left[1+e^{\frac{V_1}{V_T}}\right]} \text{ and } I_{C6} = \frac{I_{EE}}{\left[1+e^{\frac{V_2}{V_T}}\right]\left[1+e^{-\frac{V_1}{V_T}}\right]}$$

The differential output current □I is given by

$$\Delta I = I_{L1} - I_{L2}$$
$$= I_{C3} + I_{C5}) - (I_{C4} + I_{C6})$$
$$= IC3 - IC6) - (IC4 - IC5)$$
$$\Delta I = I_{EE} \tanh(V_1/2V_T) \tanh(V_2/2V_T)$$

3.3 Variable Transconductance Technique:

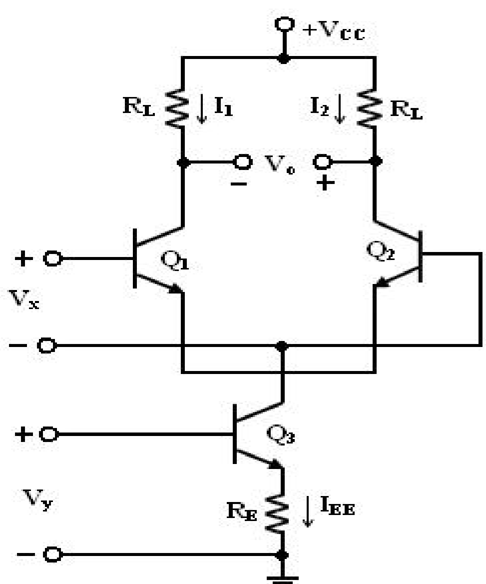


Fig. 3.9 Differential stage of the Tran conductance multiplier

- The variable transconductance technique makes use of the dependence characteristic of the transistor transconductance parameter on the emitter current bias applied. A simple differential circuit arrangement depicting the principle is shown in figure.

- The relationship between V_0 and V_x is given by $V_0 = g_m R_L V_X$ where $g_m = I_{EE} / V_T$ is the transconductance of the stage.
- Application of a second input V_y to the reference current source of the differential amplifier can vary g_m.
- Thus, if $R_E I_{EE} >> V_{BE}$, the bias voltage V_y is related to I_{EE} by the relation $V_y = I_{EE} R_E$.
- Then, the overall voltage transfer expression is given by
  - $V_0 = g_m R_L V_x = (V_y/V_T R_E) V_x R_L$

$$= V_x V_y \; R_L / V_T R_E$$

3.4 Analog Multiplier ICs

Analog multiplier is a circuit whose output voltage at any instant is proportional to the product of instantaneous value of two individual input voltages.

Important applications of these multipliers are multiplication, division, squaring and square – rooting of signals, modulation and demodulation.

These analog multipliers are available as integrated circuits consisting of op-amps and other circuit elements. The Schematic of a typical analog multiplier, namely, AD633 is shown in figure.

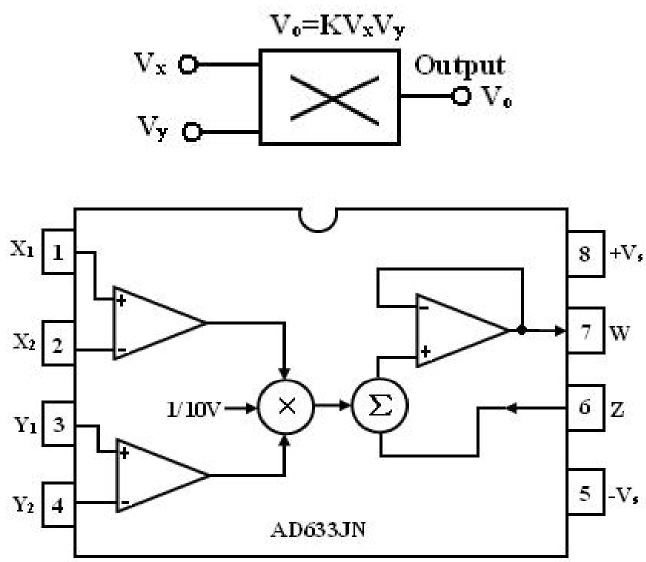


Fig. 3.10 Multiplier IC and its symbol

- The AD633 multiplier is a four – quadrant analog multiplier.
- It possesses high input impedance; this characteristic makes the loading effect on the signal source negligible.
- It can operate with supply voltages ranging from ±18V. The

- IC does not require external components.
- The typical range of the two input signals is ±10V.

Schematic representation of a multiplier:

The schematic representation of an analog multiplier is shown in figure. The output V0 is the product of the two inputs Vx and Vy is divided by a reference voltage Vref. Normally, the reference voltage Vref is internally set to 10V. Therefore, V0 =VxVy/10. In other words, the basic input – output relationship can be defined by KVx Vy when K = 1/10, a constant. Thus for peak input voltages of 10V, the peak magnitude of output voltage is 1/10 *10 *10 =10V. Thus, it can be noted that, as long as Vx < 10V and Vy < 10V, the multiplier output will not saturate.

Multiplier quadrants:

The transfer characteristics of a typical four-quadrant multiplier are shown in figure. Both the inputs can be positive or negative to obtain the corresponding output as shown in the transfer characteristics.

Applications of Multiplier ICs:

The multiplier ICs are used for the following purposes:

1. Voltage Squarer
2. Frequency doublers
3. Voltage divider
4. Square rooter
5. Phase angle detector
6. Rectifier

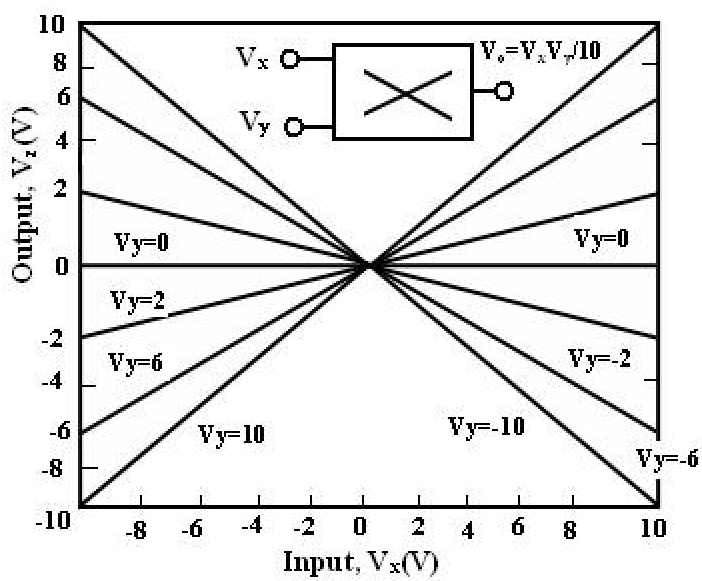


Fig.3.11 Transfer characteristics of a typical four-quadrant multiplier

Voltage Squarer:

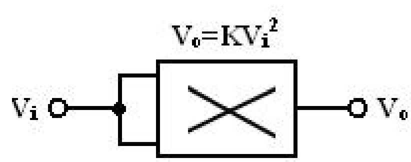


Fig. 3.12 voltage squarer circuit

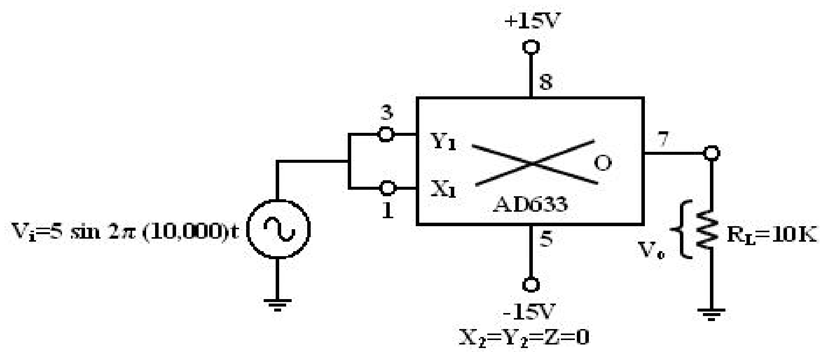
Figure shows the multiplier IC connected as a squaring circuit. The inputs can be positive or negative, represented by any corresponding voltage level between 0 and 10V. The input voltage Vi to be squared is simply connected to both the input terminals, and hence we have, Vx = Vy = Vi and the output is V0 = KVi2. The circuit thus performs the squaring operation. This application can be extended for frequency doubling applications.

Frequency doublers:

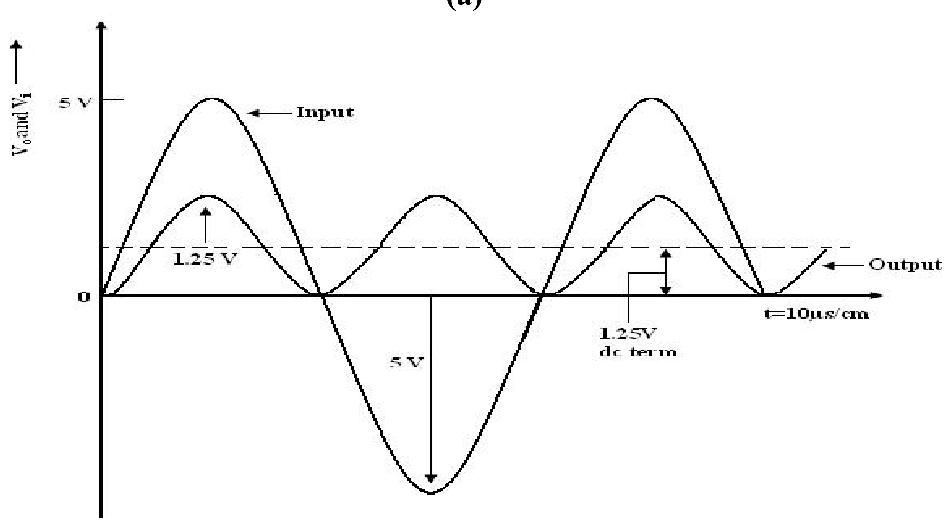
Figure shows the squaring circuit connected for frequency doubling operation. A sine-wave signal Vi has a peak amplitude of Av and frequency of f Hz. Then, the output voltage of the doublers circuit is given by

$$v_0 = \frac{A_v \sin 2\pi ft * A_v \sin 2\pi ft}{10} = \frac{A_v^2}{10} \sin^2 2\pi\, ft = \frac{A_v^2}{20}(1 - \cos 4\,\pi\, ft\,)$$

Assuming a peak amplitude Av of 5V and frequency f of 10KHz, V0 =1.25–1.25 cos2π 20000) t. The first term represents the dc term of 1.25V peak amplitude. The input and output waveforms are shown in figure. The output waveforms ripple with twice the input frequency in the rectified output of the input signal. This forms the principle of application of analog multiplier as rectifier of ac signals.



(a)



(b)

Fig. 3.13 (a) circuit diagram and (b) input –output waveform of frequency doubler

The dc component of output V0 can be removed by connecting a 1µF coupling capacitor between the output terminal and a load resistor, across which the output can be observed.

Voltage Divider:

In voltage divider circuit the division is achieved by connecting the multiplier in the feedback loop of an op-amp.
The voltages Vden and Vnum represent the two input voltages, Vdm forms one input of the multiplier, and output of op-amp VoA forms the second input.
The output VOA forms the second input. The output VOM of the multiplier is connected back of op- amp in the feedback loop. Then the characteristic operation of the multiplier gives
Vom = KVOA Vdm (1)

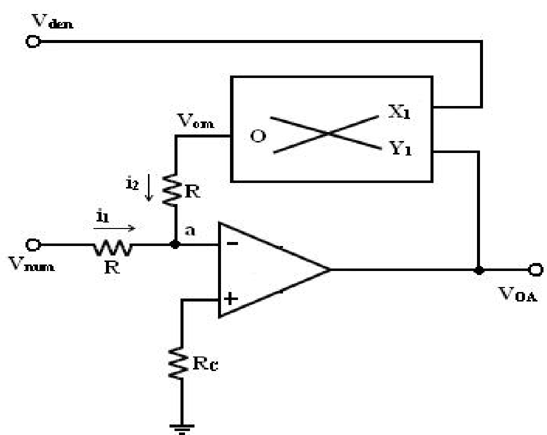


Fig 3.14 divider circuit

As shown in figure, no input signal current can flow into the inverting input terminal of op-amp, which is at virtual ground. Therefore, at the junction a, i1 +i2 =0, the current i1 = Vnum / R, where R is the input resistance and the current i2 = Vom /R. With virtual ground existing at a,

$$i_1+i_2 = \text{Vnum} / R + \text{Vom} /R = 0$$

$$KV_{OA} V_{den} = - V_{num} \quad or$$

$$v_{OA} = - v_{num}/Kv_{den}$$

where *Vnum* and *Vden* are the numerator and denominator voltages respectively. Therefore, the voltage division operation is achieved. *Vnum* can be a positive or negative voltage and *Vden* can have only positive values to ensure negative feedback. When *Vdm* is changed, the gain 10/*Vdm* changes, and this feature is used in automatic gain control (AGC) circuits.

- ✓ **Square Rooter:**

The divider voltage can be used to find the square root of a signal by connecting both inputs of the multiplier to the output of the op-amp. *Substituting* equal in magnitude but opposite in polarity (with respect to ground) to Vi. But we know that Vom is one- term (Scale factor) of V0 * V0 or -Vi = Vom = $V^2/1\ 0$

Solving for V0 and eliminating √-1 yields. V0 = √10|Vi |

Eqn. states that V0 equals the square root of 10 times the absolute magnitude of Vi.

The input voltage Vi must be negative, or else, the op-amp saturates.
The range of Vi is between -1 and -10V. Voltages less than -1V will cause inaccuracies in the result.
The diode prevents negative saturation for positive polarity Vi signals. For positive values of Vi the diode connections are reversed.

- ✓ **Phase Angle detector:**

The multiplier configured for phase angle detection measurement is shown in figure. When two sine-waves of the same frequency are applied to the inputs of the multiplier, the output V0 has a dc component and an AC component.

The trigonometric identity shows that Sin A sin B =1/2 (cos (A-B) – cos (A+B)).
When the two frequencies are equal, but with different phase angles, e.g. A=2πft +θ for signal Vx and B= 2πft for signal Vy, then using the identity

$[\sin(2\pi ft+\theta)][\sin 2\pi ft)]=1/2[\cos\theta-\cos(4\pi ft+\theta)]=1/2(\text{dc- the double frequency term})$

Therefore, when the two input signals Vx and Vy are applied to the multiplier, V0 (dc) is given by

$$v_v(dc)=\frac{v_{xp}v_{yp}}{20}\cos\theta$$

where *Vxp* and *Vyp* are the peak voltage amplitudes of the signals Vx and Vy. Thus, the output V0(dc) depends on the factor cos θ. A dc voltmeter can be calibrated as a phase angle meter when the product of Vxp and Vyp is made equal to 20. Then, a (0-1) V range dc voltmeter can directly read cos θ, with the meter calibrated directly in degrees from a cosine table. The input and output waveforms are shown in figure.
Then the above eqn becomes V0 (dc) = cos θ, if we make the product Vxp Vyp = 20 or in other words, Vxp – Vyp = 4.47V.

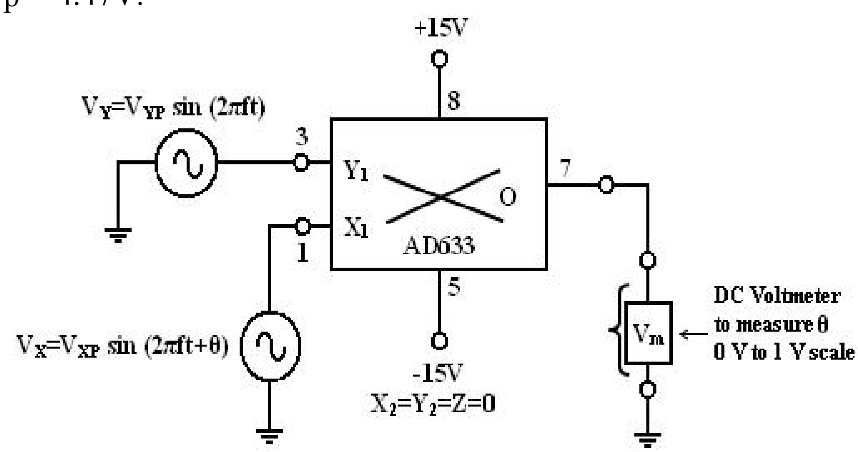


Fig 3.15 Phase angle measurement circuit diagram

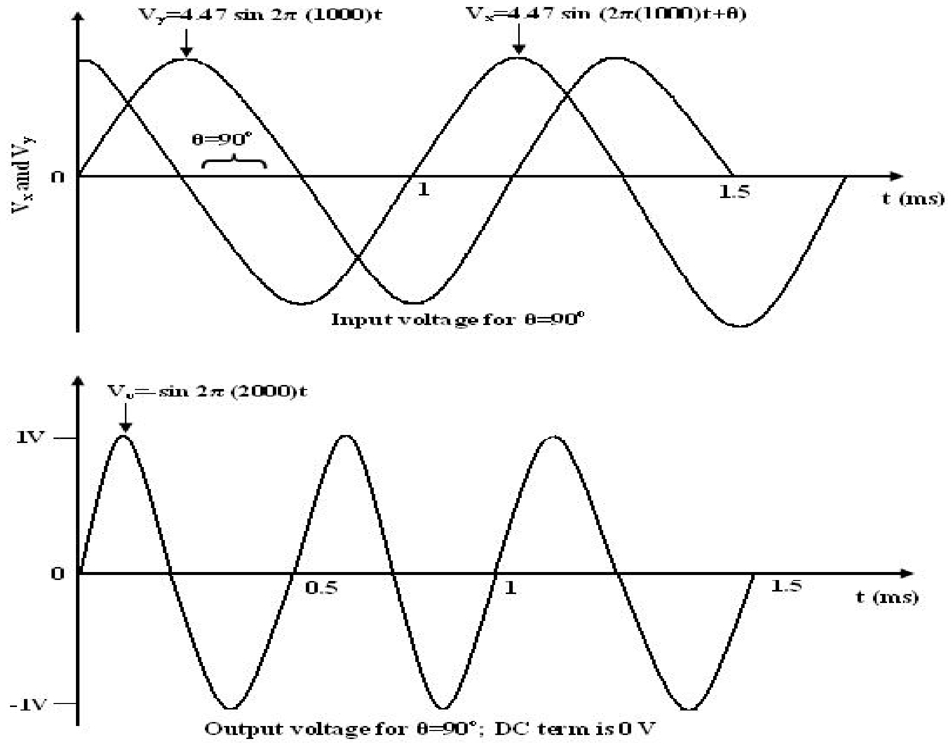


Fig. 3.16 input- output waveforms of phase angle detector

3.5 Operation of Basic Phase Locked Loop

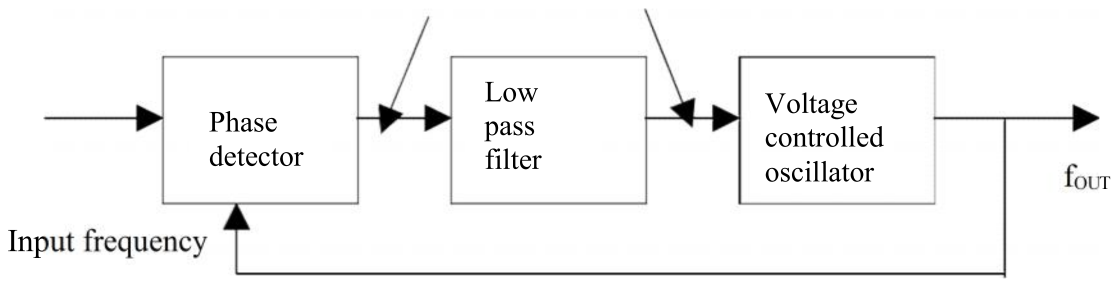


Fig 3.17 Block diagram of PLL

The PLL consists of i) Phase detector ii) LPF iii) VCO. The phase detector or comparator compares the input frequency fIN with feedback frequency fOUT.

- The output of the phase detector is proportional to the phase difference between fIN & fOUT. The output of the phase detector is a dc voltage & therefore is often referred to as the error voltage.
- The output of the phase detector is then applied to the LPF, which removes the high frequency noise and produces a dc level. This dc level in turn, is input to the VCO.
- The output frequency of VCO is directly proportional to the dc level. The VCO frequency is compared with input frequency and adjusted until it is equal to the input frequencies.
- PLL goes through 3 states, i) free running ii) Capture iii) Phase lock.

Before the input is applied, the PLL is in free running state. Once the input frequency is applied the VCO frequency starts to change and PLL is said to be in the capture mode. The VCO frequency continuous to change until it equals the input frequency and the PLL is in phase lock mode. When Phase locked, the loop tracks any change in the input frequency through its repetitive action.

If an input signal v_s of frequency f_s is applied to the PLL, the phase detector compares the phase and frequency of the incoming signal to that of the output v_o of the VCO. If the two signals differ in frequency of the incoming signal to that of the output v_o of the VCO.

The phase detector is basically a multiplier and produces the sum $(f_s + f_o)$ and difference $(f_s - f_o)$ components at its output.

The high frequency component $(f_s + f_o)$ is removed by the low pass filter and the difference frequency component is amplified then applied as control voltage v_c to VCO.

The signal v_c shifts the VCO frequency in a direction to reduce the frequency difference between f_s and f_o. Once this action starts, we say that the signal is in the capture range. The VCO continues to change frequency till its output frequency is exactly the same as the input signal frequency. The circuit is then said to be locked. Once locked, the output frequency f_o of VCO is identical to f_s except for a finite phase difference φ. This phase difference φ generates a corrective control voltage v_c to shift the VCO frequency from f_0 to f_s and thereby maintain the lock. Once locked, PLL tracks the frequency changes of the input signal. Thus, a PLL goes through three stages (i) free running, (ii) capture and (iii) locked or tracking.

Capture range: the range of frequencies over which the PLL can acquire lock with an input signal is called the capture range. This parameter is also expressed as percentage of f_o.

Pull-in time: the total time taken by the PLL to establish lock is called pull-in time. This depends on the initial phase and frequency difference between the two signals as well as on the overall loop gain and loop filter characteristics.

3.5.1 Phase Detector

Phase detector compares the input frequency and VCO frequency and generates DC voltage i.e., proportional to the phase difference between the two frequencies. Depending on whether the analog/digital phase detector is used, the PLL is called either an analog/digital type respectively. Even though most monolithic PLL integrated circuits use analog phase detectors.

Ex for Analog: Double-balanced mixer

Ex for Digital: Ex-OR, Edge trigger, monolithic Phase detector.

- ✓ Ex-OR Phase Detector:

This uses an exclusive OR gate. The output of the Ex-OR gate is high only when fIN or f_{OUT} is high.

The DC output voltage of the Ex-OR phase detector is a function of the phase difference between its two outputs. The maximum dc output voltage occurs when the phase difference is Π radians or

180 degrees. The slope of the curve between 0 or Π radians is the conversion gain kp of the phase detector for eg; if the Ex-OR gate uses a supply voltage Vcc = 5V, the conversion gain Kp is

$$k_{p=\frac{5}{\pi}=1.59\,rad}$$

Advantages of Edge Triggered Phase Detector over Ex-OR are

i) The dc output voltage is linear over 2Π radians or 360 degrees, but in Ex-OR it is Π radians or 180 degrees.

ii) Better Capture, tracking & locking characteristics.

Edge triggered type of phase detector using RS Flip – Flop. It is formed from a pair of cross coupled NOR gates.

RS FF is triggered, i.e., the output of the detector changes its logic state on the positive edge of the inputs fIN & fOUT

- ✓ Monolithic Phase detector:

- It consists of 2 digital phase detector, a charge pump and an amplifier.
- Phase detector 1 is used in applications that require zero frequency and phase difference at lock.
- Phase detector 2, if quadrature lock is desired, when detector 1 is used in the main loop, detector can also be used to indicate whether the main loop is in lock or out of lock.

3.5.2 Low – Pass filter

The function of the LPF is to remove the high frequency components in the output of the phase detector and to remove the high frequency noise. LPF controls the characteristics of the phase locked loop. i.e., capture range, lock ranges, bandwidth

- Lock range(Tracking range):

The lock range is defined as the range of frequencies over which the PLL system follows the changes in the input frequency fIN.

- Capture range:
  Capture range is the frequency range in which the PLL acquires phase lock. Capture range is always smaller than the lock range.
- Filter Bandwidth:
  Filter Bandwidth is reduced, its response time increases. However reduced Bandwidth reduces the capture range of the PLL. Reduced Bandwidth helps to keep the loop in lock through momentary losses of signal and also minimizes noise.

3.5.4 Voltage Controlled Oscillator (VCO):

The third section of PLL is the VCO; it generates an output frequency that is directly proportional to its input voltage. The maximum output frequency of NE/SE 566 is 500 Khz.

3.5.5 Feedback path and optional divider:

Most PLLs also include a divider between the oscillator and the feedback input to the phase detector to produce a frequency synthesizer. A programmable divider is particularly useful in radio Transmitter applications, since a large number of transmit frequencies can be produced from a single stable, accurate, but expensive, quartz crystal–controlled reference oscillator.

Some PLLs also include a divider between the reference clock and the reference input to the phase detector. If this divider divides by *M*, it allows the VCO to multiply the reference frequency by *N* / *M*. It might seem simpler to just feed the PLL a lower frequency, but in some cases the reference frequency may be constrained by other issues, and then the reference divider is useful.

Frequency multiplication in a sense can also be attained by locking the PLL to the 'N'th harmonic of the signal.

The equations governing a phase-locked loop with an analog multiplier as the phase detector may be derived as follows. Let the input to the phase detector be $x_c(t)$ and the output of the voltage- controlled oscillator (VCO) is $x_r(t)$ with frequency $\omega_r(t)$, then the output of the phase detector $x_m(t)$ is given by

$$x_m(t) = x_c(t) \cdot x_r(t)$$

the VCO frequency may be written as a function of the VCO input $y(t)$ as

$$\omega_r(t) = \omega_f + g_v y(t)$$

where g_v is the *sensitivity* of the VCO and is expressed in Hz / V.

Hence the VCO output takes the form

$$x_r(t) = A_r \cos\left(\int_0^t \omega_r(\tau)\, d\tau\right) = A_r \cos(\omega_f t + \varphi(t))$$

where

$$\varphi(t) = \int_0^t g_v y(\tau)\, d\tau$$

The loop filter receives this signal as input and produces an output

$$x_f(t) = F_{\text{filter}}(x_m(t))$$

where Filter is the operator representing the loop filter transformation.

When the loop is closed, the output from the loop filter becomes the input to the VCO thus

$$y(t) = x_f(t) = F_{\text{filter}}(x_m(t))$$

We can deduce how the PLL reacts to a sinusoidal input signal:

$$x_C(t) = A_C\sin(\omega_C t).$$

The output of the phase detector then is:

$$x_m(t) = A_c \sin(\omega_c t) A_r \cos(\omega_f t + \varphi(t)).$$

This can be rewritten into sum and difference components using trigonometric identities:

$$x_m(t) = \frac{A_c A_f}{2} \sin(\omega_c t - \omega_f t - \varphi(t)) + \frac{A_c A_f}{2} \sin(\omega_c t + \omega_f t + \varphi(t))$$

As an approximation to the behaviour of the loop filter we may consider only the difference frequency being passed with no phase change, which enables us to derive a small-signal model Of the phase-locked loop. If we can make $\omega_f \approx \omega_c$ then the $\sin(.)$ can be approximated by its argument resulting in: $y(t) = x_f(t) \simeq -A_c A_f \varphi(t)/2$. The phase-locked loop is said to be locked if this is the case.

3.6 Control System Analysis/ Closed Loop Analysis Of PLL

Phase locked loops can also be analyzed as control systems by applying the Laplace transform. The loop response can be written as:

$$\frac{\theta_o}{\theta_i} = \frac{K_p K_v F(s)}{s + K_p K_v F(s)}$$

Where

- θ_o is the output phase in radians
- θ_i is the input phase in radians
- K_p is the phase detector gain in volts per radian
- K_v is the VCO gain in radians per volt-second
- $F(s)$ is the loop filter transfer function (dimensionless)

The loop characteristics can be controlled by inserting different types of loop filters. The simplest filter is a one-pole RC circuit. The loop transfer function in this case is:

$$F(s) = \frac{1}{1 + sRC}$$

The loop response becomes:

$$\frac{\theta_o}{\theta_i} = \frac{\frac{K_p K_v}{RC}}{s^2 + \frac{s}{RC} + \frac{K_p K_v}{RC}}$$

This is the form of a classic harmonic oscillator. The denominator can be related to that of a second order system:

$$s^2 + 2s\zeta\omega_n + \omega_n^2$$

Where

- ζ is the damping factor
- ω_n is the natural frequency of the loop.

For the one-pole RC filter,

$$\omega_n - \sqrt{\frac{K_p K_v}{RC}}$$

$$\zeta = \frac{1}{2\sqrt{K_p K_v RC}}$$

The loop natural frequency is a measure of the response time of the loop, and the damping factor is a measure of the overshoot and ringing. Ideally, the natural frequency should be high and the damping factor should be near 0.707 (critical damping). With a single pole filter, it is not possible to control the loop frequency and damping factor independently. For the case of critical damping,

$$RC = \frac{1}{2K_p K_v}$$
$$\omega_c = K_p K_v \sqrt{2}$$

A slightly more effective filter, the lag-lead filter includes one pole and one zero. This can be realized with two resistors and one capacitor. The transfer function for this filter is

$$F(s) = \frac{1 + sCR_2}{1 + sC(R_1 + R_2)}$$

This filter has two time constants

$\tau 1 = C(R1 + R2)$ $\tau 2$
$= CR2$

Substituting above yields the following natural frequency and damping factor

$$\omega_n - \sqrt{\frac{K_p K_v}{\tau_1}}$$

$$\zeta = \frac{1}{2\omega_n \tau_1} + \frac{\omega_n \tau_2}{2}$$

The loop filter components can be calculated independently for a given natural frequency and damping factor

$$\tau_1 = \frac{K_p K_v}{\omega_n^2}$$

$$\tau_2 = \frac{2\zeta}{\omega_n} - \frac{1}{K_p K_v}$$

Real world loop filter design can be much more complex eg using higher order filters to reduce various types or source of phase noise.

Applications of PLL:

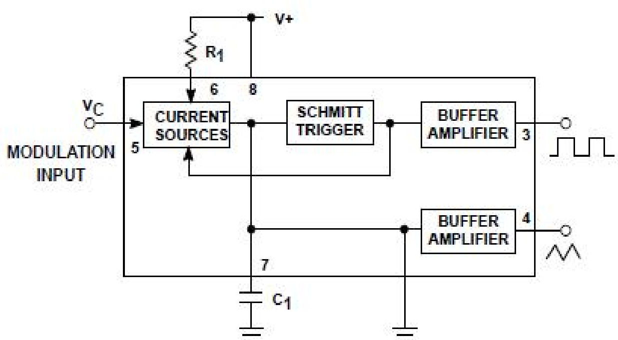
The PLL principle has been used in applications such as

- FM stereo decoders

- motor speed control
- tracking filters
- FM modulation and demodulation
- FSK modulation
- Frequency multiplier
- Frequency synthesis etc.,

Example PLL ICs:560 series (560, 561, 562, 564, 565 & 567)

3.7 Voltage Controlled Oscillator:



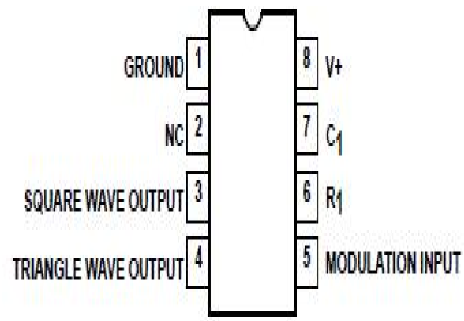


Fig. 3.18 Pin diagram and block diagram of VCO

Referring to the circuit in the above figure, the capacitor c1 is linearly charged or discharged by a constant current source/sink. The amount of current can be controlled by changing the voltage vc applied at the modulating input (pin 5) or by changing the timing resistor R1 external to the IC chip. The voltage at pin 6 is held at the same voltage as pin 5.

Thus, if the modulating voltage at pin 5 is increased, the voltage at pin 6 also increases, resulting in less voltage across R1 and thereby decreasing the charging current.

The voltage across the capacitor C1 is applied to the inverting input terminal of Schmitt trigger via buffer amplifier. The output voltage swing of the Schmitt trigger is designed to Vcc and 1.5 Vcc. If Ra = Rb in the positive feedback loop, the voltage at the non-inverting input terminal of Schmitt trigger swings from 0.5 Vcc to 0.25 Vcc.

When the voltage on the capacitor c1 exceeds 0.5 Vcc during charging, the output of the Schmitt trigger goes LOW (0.5 Vcc). The capacitor now discharges and when it is at 0.25 Vcc, the output of Schmitt trigger goes HIGH (Vcc). Since the source and sink currents are equal, capacitor charges and discharges for the same amount of time. This gives a triangular voltage waveform across c1 which is also available at pin 4.

The square wave output of the Schmitt trigger is inverted by buffer amplifier at pin 3. The output waveforms are shown near the pins 4 and 3.

The output frequency of the VCO can be given as follows:

$$f_o = \frac{2\,[(V+) - (V_C)]}{R_1\,C_1\,V+}$$

where V+ is Vcc.

The output frequency of the VCO can be changed either by (i) R1, (ii) c1 or (iii) the voltage vc at the modulating input terminal pin 5. The voltage vc can be varied by connecting a

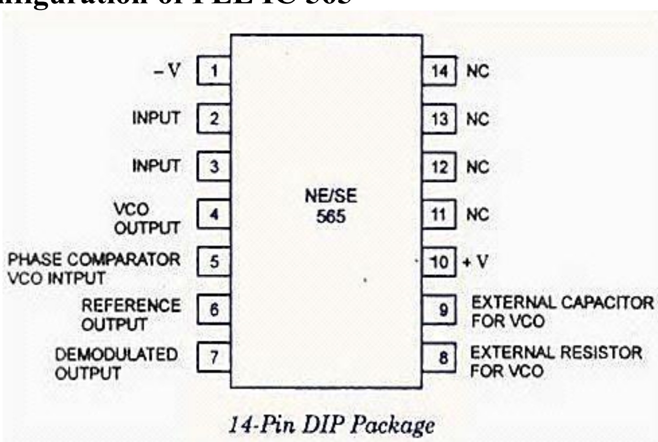
R1R2 circuit as shown in the figure below. The components R1and c1 are first selected so that VCO output frequency lies in the centre of the operating frequency range.

Now the modulating input voltage is usually varied from 0.75 Vcc to Vcc which can produce a frequency variation of about 10 to 1.

The signetics NE/SE 560 series is monolithic phase locked loops. The SE/NE 560, 561, 562, 564, 565 & 567 differ mainly in operating frequency range, poser supply requirements & frequency & bandwidth adjustment ranges.

3.8 Monolithic Phase Locked Loops (PLL IC 565):

Pin Configuration of PLL IC 565



Basic Block Diagram Representation of IC 565

The important electrical characteristics of the 565 PLL are,

- Operating frequency range: 0.001Hz to 500 Khz.
- Operating voltage range: ±6 to ±12v
- Input level required for tracking: 10mv rms min to 3 Vpp max
- Input impedance: 10 K ohms typically.
- Output sink current: 1mA
- Output source current: 10 mA

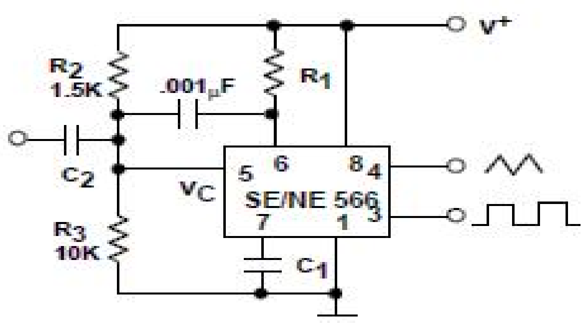
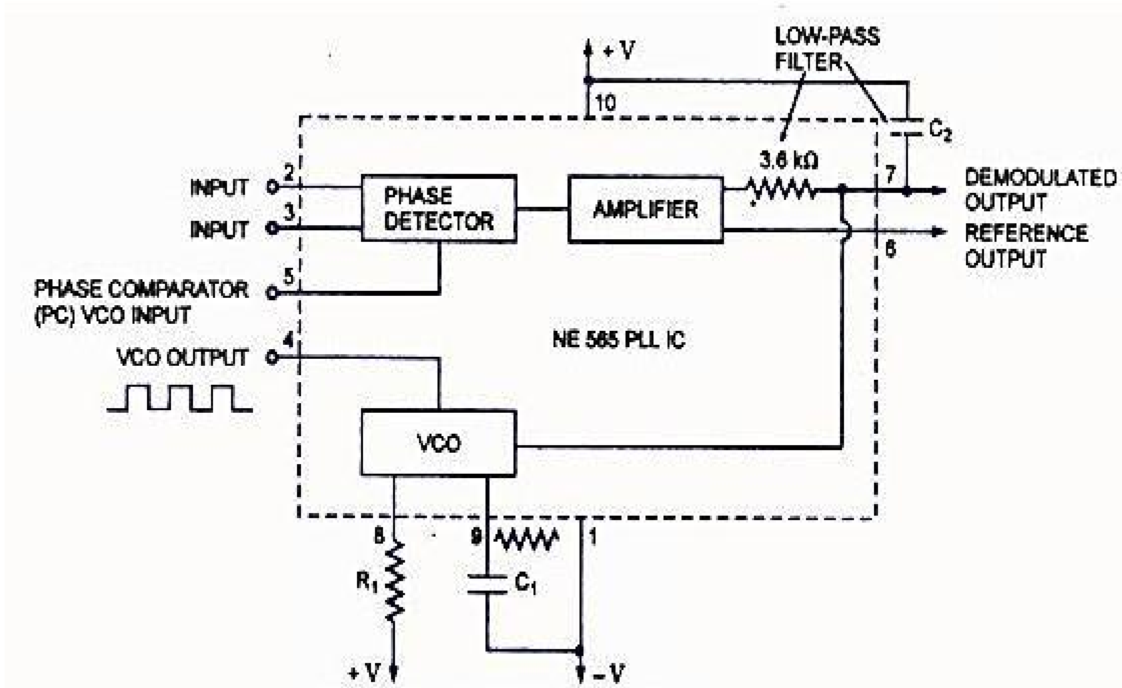


Fig. 3.19 External connections of VCO



NE/SE 565 PLL Block Diagram

The center frequency of the PLL is determined by the free running frequency of the VCO, which is given by

$f_{OUT} = 1.2/4R_1C_1$

where R1&C1 are an external resistor & a capacitor connected to pins 8 & 9.

- The VCO free-running frequency f_{OUT} is adjusted externally with R1 & C1 to be at the center of the input frequency range.
- C1 can be any value; R1 must have a value between 2 k ohms and 20 K ohms.
- Capacitor C2 connected between 7 & +V.
- The filter capacitor C2 should be large enough to eliminate variations in the demodulated output voltage in order to stabilize the VCO frequency.

The lock range fL & capture range fc of PLL is given by,

$$\Delta f_L = \pm 7.8 \; f_{out}/V \quad \text{Hz}$$

Where f_{OUT} = free running frequency of VCO (Hz)

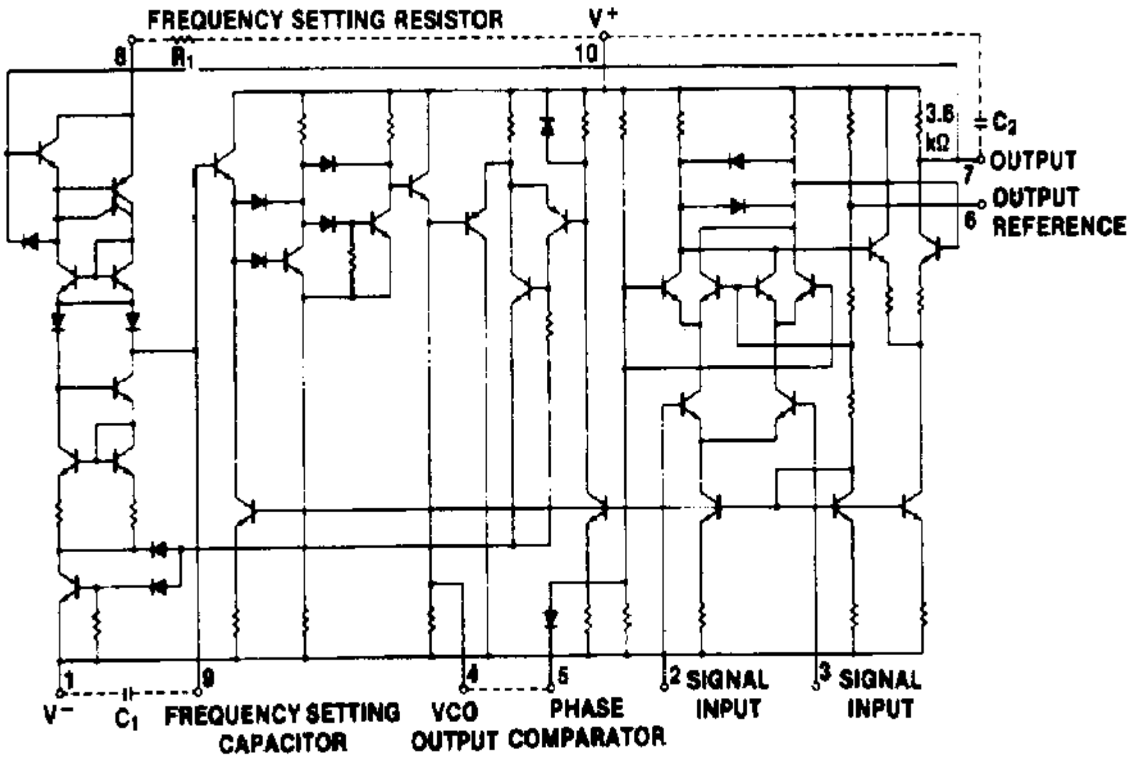
V = (+Vcc)-(-Vcc) volts

$\Delta f_C = \pm[\,\Delta f_L/(2\Pi)(3.6)(10^3)C2\,]^{1/2}$

5.9 Monolithic PLL IC 565 applications:

The output from a PLL system can be obtained either as the voltage signal vc(t) corresponding to the error voltage in the feedback loop, or as a frequency signal at VCO output terminal. The voltage output is used in frequency discriminator applications whereas the frequency output is used in signal conditioning, frequency synthesis or clock recovery applications.

The circuit diagram of LM565 PLL



When PLL is locked to an input frequency, the error voltage vc(t) is proportional to (fs-fo). If the input frequency is varied as in the case of FM signal vc will also vary in order to maintain the lock. Thus the voltage output serves as a frequency discriminator which converts the input frequency changes to voltage changes.

In the case of frequency output, if the input signal is comprised of many frequency components corrupted with noise and other disturbances, the PLL can be made to lock, selectively on one particular frequency component at the input. The output of VCO would then regenerate that particular frequency (because of LPF which gives output for beat frequency) and attenuate heavily other frequencies. VCO output thus can be used for regenerating or reconditioning a desired frequency signal (which is weak and buried in noise) out of many undesirable frequency signals.
Some of the typical applications of PLL are discussed below.

- ✓ **Frequency Multiplier:**

Frequency divider is inserted between the VCO & phase comparator. Since the output of the divider is locked to the fIN, VCO is actually running at a multiple of the input frequency.
The desired amount of multiplication can be obtained by selecting a proper divide-by-N network, where N is an integer.

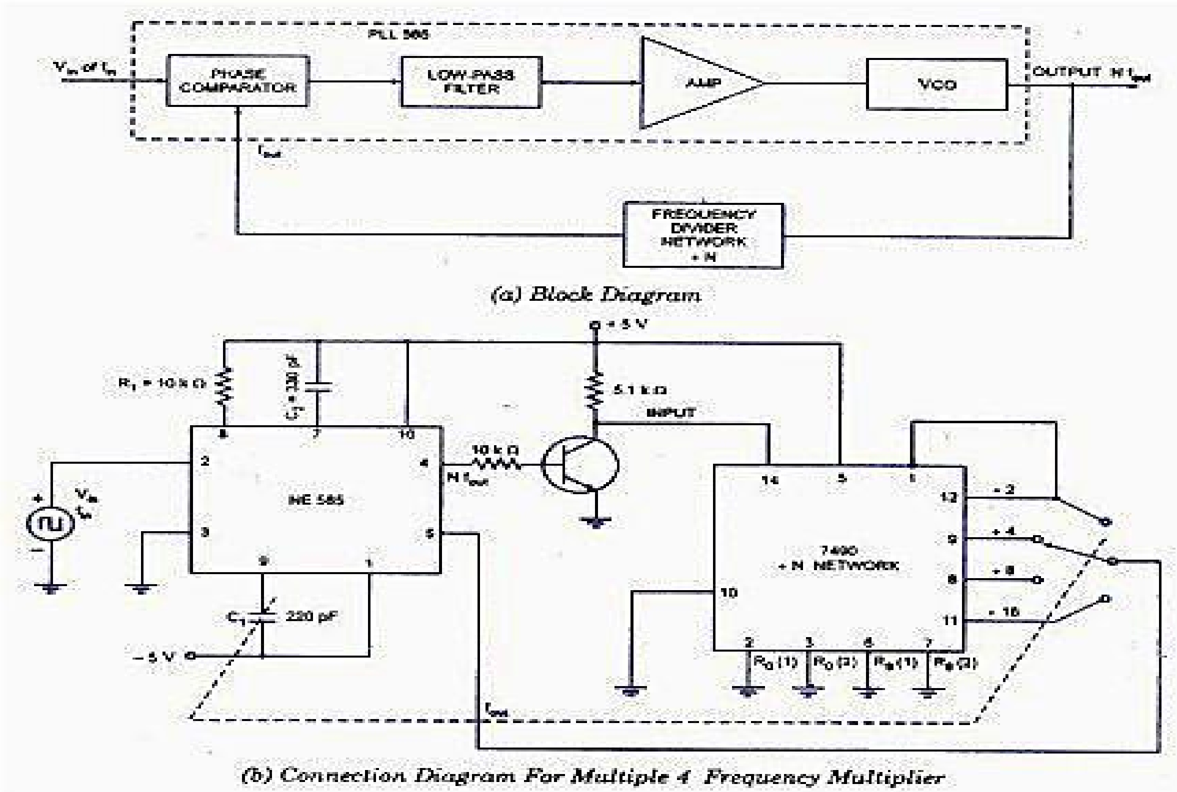
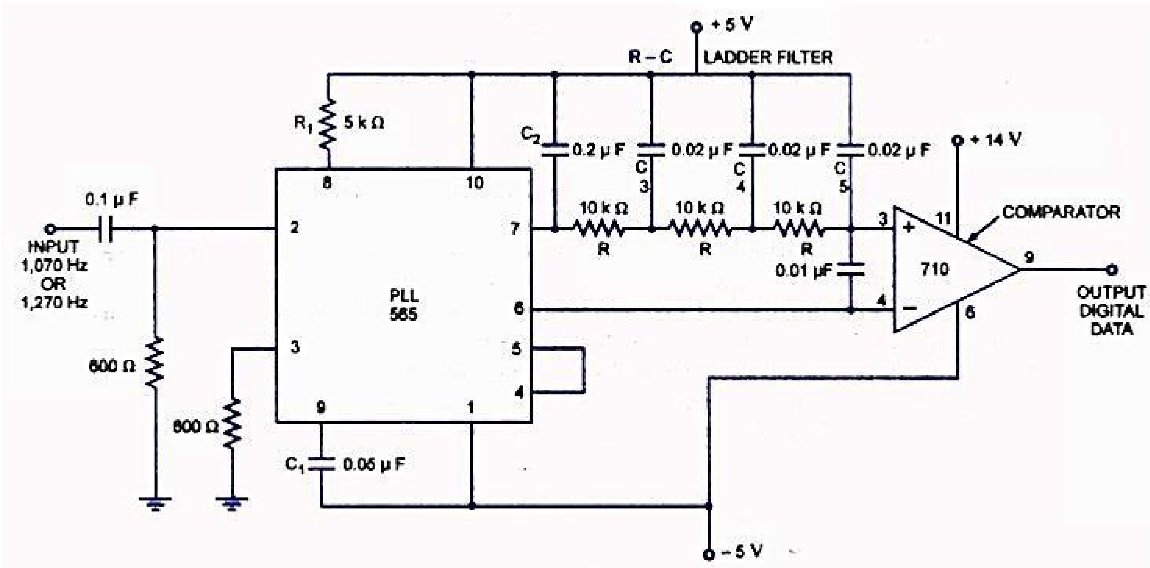


Fig. 3.20 Frequency multiplier using PLL

✓ **Frequency Shift Keying (FSK) demodulator:**

In computer peripheral & radio (wireless) communication the binary data or code is transmitted by means of a carrier frequency that is shifted between two preset frequencies. Since a carrier frequency is shifted between two preset frequencies, the data transmission is said to use a FSK. The frequency corresponding to logic 1 & logic 0 states are commonly called the mark & space frequency.

For example, When transmitting teletype writer information using a modulator-demodulator (modem) a 1070-1270 (mark-space) pair represents the originate signal, while a 2025-2225 Hz (mark-space) pair represents the answer signal.



565 As An FSK Demodulator

✓ **FSK Generator:**

- The FSK generator is formed by using a 555 as an astable multivibrator, whose frequency is controlled by the state of transistor Q1.
- In other words, the output frequency of the FSK generator depends on the logic state of the digital data input.
- 150 Hz is one the standards frequencies at which the data are commonly transmitted.
- When the input is logic 1, the transistor Q1 is off. Under the condition, 555 timer works in its normal mode as an astable multivibrator i.e., capacitor C charges through RA & RB to 2/3 Vcc & discharges through RB to 1/3 Vcc. Thus capacitor C charges & discharges between 2/3 Vcc & 1/3 Vcc as long as the input is logic 1.
- The frequency of the output waveform is given by,

$$f_o = \frac{1.45}{(R_A + 2R_B)C} = 1070 \text{ Hz (mark frequency)}$$

- When the input is logic 0, (Q1 is ON saturated) which in turn connects the resistance Rc across RA. This action reduces the charging time of capacitor C1 increases the output frequency, which is given by,

$$f_o = \frac{1.45}{(R_A \parallel R_C + 2 R_B)C} = 1270 \text{ Hz (space frequency)}$$

- By proper selection of resistance Rc, this frequency is adjusted to equal the space frequency of 1270 Hz. The difference between the FSK signals of 1070 Hz & 1270 Hz is 200 Hz, this difference is called “frequency shift”.
- The output 150 Hz can be made by connecting a voltage comparator between the output of the ladder filter and pin 6 of PLL.
- The VCO frequency is adjusted with R1 so that at f_{IN} = 1070 Hz.

✓ **FSK Demodulator:**

- The output of 555 FSK generators is applied to the 565 FSK demodulator.
- Capacitive coupling is used at the input to remove dc line.
- At the input of 565, the loop locks to the input frequency & tracks it between the 2 frequencies.
- R1 & C1 determine the free running frequency of the VCO, 3 stages RC ladder filter is used to remove the carrier component from the output.

Applications:

In digital data communication and computer peripheral, binary data is transmitted by means of a carrier frequency which is shifted between two preset frequencies. This type of data transmission is called frequency shift keying (FSK) technique. The binary data can be retrieved using FSK demodulator. The figure below shows FSK demodulator using PLL for tele-typewriter signals of 1070 Hz and 1270 Hz. As the signal appears at the input, the loop locks to the input frequency and tracks it between the two frequencies with a corresponding dc shift at the output. A three stage filter removes the carrier component and the output signal is made logic compatible by a voltage comparator.

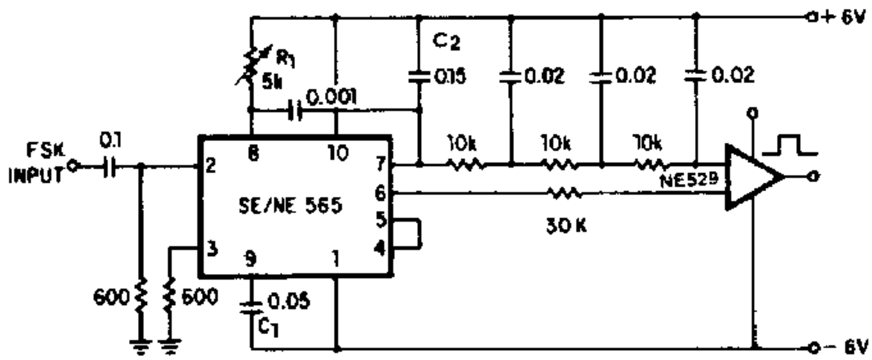


Fig. 3.21 FSK demodulator circuit

✓ **AM Demodulation:**

A PLL may be used to demodulate AM signals as shown in the figure below. The PLL is locked to the carrier frequency of the incoming AM signal. The output of VCO which has the same frequency as the carrier, but unmodulated is fed to the multiplier. Since VCO output is always 90^0 before being fed to the multiplier. This makes both the signals applied to the multiplier and the difference signals, the demodulated output is obtained after filtering high frequency components by the LPF. Since the PLL responds only to the carrier frequencies which are very close to the VCO output, a PLL AM detector exhibits high degree of selectivity and noise immunity which is not possible with conventional peak detector type AM modulators.

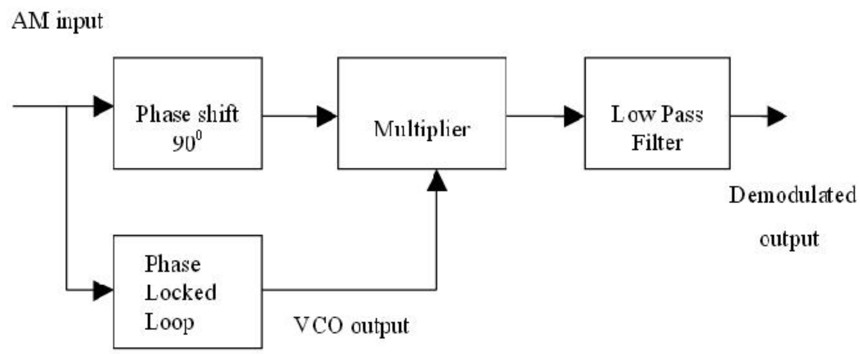


Fig.3.22 AM demodulator

✓ **FM Demodulation:**

If PLL is locked to a FM signal, the VCO tracks the instantaneous frequency of the input signal. The filtered error voltage which controls the VCO and maintains lock with the input signal is the demodulated FM output.

The VCO transfer characteristics determine the linearity of the demodulated output. Since, VCO used in IC PLL is highly linear, it is possible to realize highly linear FM demodulators.

✓ **Frequency multiplication/division:**

The block diagram shown below shows a frequency multiplier/divider using PLL. A divide by N network is inserter between the VCO output and the phase comparator input. In the locked state, the VCO output frequency fo is given by fo = Nfs. The multiplication factor can be obtained by selecting a proper scaling factor N of the counter.

Frequency multiplication can also be obtained by using PLL in its harmonic locking mode. If the input signal is rich in harmonics e.g. square wave, pulse train etc., then the VCO can be directly locked to the n-th harmonic of the input signal without connecting any frequency divider in between. However, as the amplitude of the higher order harmonics becomes less, effective locking may not take place for high values of n. Typically n is kept less than 10.

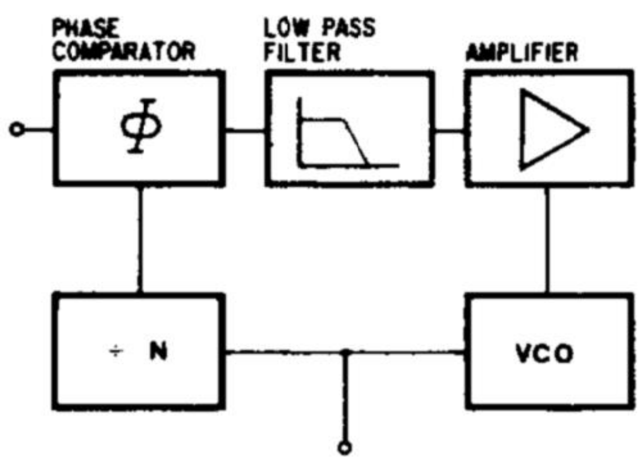


Fig. 3.23 Frequency Divider

The circuit of the figure above can also be used for frequency division. Since the VCO output (a square wave) is rich in harmonics, it is possible to lock the m-th harmonic of the VCO output with the input signal fs. The output f_o of VCO is now given by $f_o = f_s/m$

- ✓ PLL Frequency Synthesis:

In digital wireless communication systems (GSM, CDMA etc), PLL's are used to provide the Local Oscillator (LO) for up-conversion during transmission, and down-conversion during reception. In most cellular handsets this function has been largely integrated into a single integrated circuit to reduce the cost and size of the handset.

However due to the high performance required of base station terminals, the transmission and reception circuits are built with discrete components to achieve the levels of performance required. GSM LO modules are typically built with a Frequency Synthesizer integrated circuit, and discrete resonator VCO's.

Principle of PLL synthesizers

A phase locked loop does for frequency what the Automatic Gain Control does for voltage. It compares the frequencies of two signals and produces an error signal which is proportional to the difference between the input frequencies.

The error signal is then low pass filtered and used to drive a voltage-controlled oscillator (VCO) which creates an output frequency. The output frequency is fed through a frequency divider back to the input of the system, producing a negative feedback loop.

If the output frequency drifts, the error signal will increase, driving the frequency in the opposite direction so as to reduce the error. Thus the output is *locked* to the frequency at the other input. This input is called the reference and is derived from a crystal oscillator, which is very stable in frequency.

The block diagram below shows the basic elements and arrangement of a PLL based frequency synthesizer.

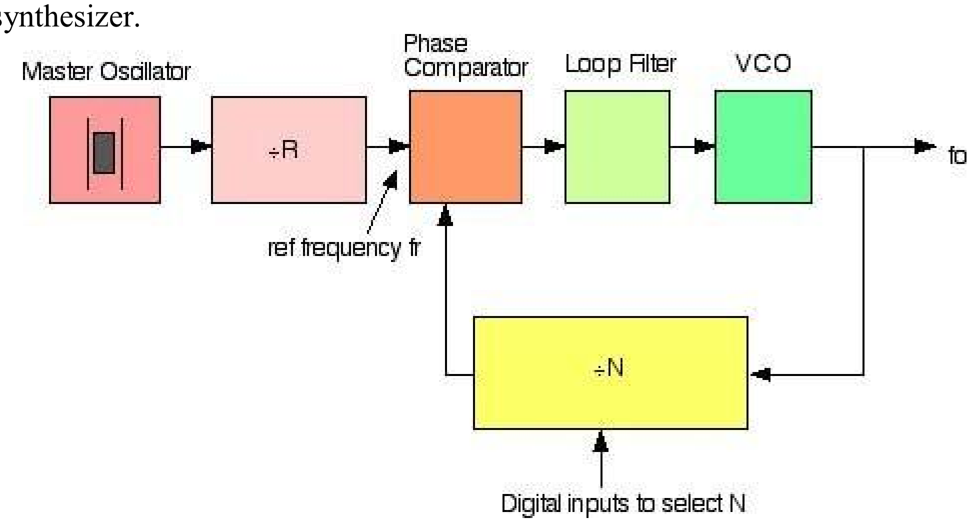


Fig. 2.24 PLL based frequency synthesizer

The key to the ability of a frequency synthesizer to generate multiple frequencies is the divider placed between the output and the feedback input. This is usually in the form of a digital counter, with the output signal acting as a clock signal.

The counter is preset to some initial count value, and counts down at each cycle of the clock signal. When it reaches zero, the counter output changes state and the count value is reloaded.

This circuit is straightforward to implement using flip-flops, and because it is digital in nature, is very easy to interface to other digital components or a microprocessor. This allows the frequency output by the synthesizer to be easily controlled by a digital system.

Example:

Suppose the reference signal is 100 kHz, and the divider can be preset to any value between 1 and 100. The error signal produced by the comparator will only be zero when the output of the divider is also 100 kHz. For this to be the case, the VCO must run at a frequency which is 100 kHz x the divider count value.

Thus it will produce an output of 100 kHz for a count of 1, 200 kHz for a count of 2, 1 MHz for a count of 10 and so on. Note that only whole multiples of the reference frequency can be obtained with the simplest integer N dividers. Fractional N dividers are readily available

Practical considerations:

In practice this type of frequency synthesizer cannot operate over a very wide range of frequencies, because the comparator will have a limited bandwidth and may suffer from aliasing problems. This would lead to false locking situations, or an inability to lock at all. In addition, it is hard to make a high frequency VCO that operates over a very wide range.

This is due to several factors, but the primary restriction is the limited capacitance range of varactor diodes. However, in most systems where a synthesizer is used, we are not after a huge range, but rather a finite number over some defined range, such as a number of radio channels in a specific band.

Many radio applications require frequencies that are higher than can be directly input to the digital counter. To overcome this, the entire counter could be constructed using high-speed logic such as ECL, or more commonly, using a fast initial division stage called a *prescaler* which reduces the frequency to a manageable level.

Since the prescaler is part of the overall division ratio, a fixed prescaler can cause problems designing a system with narrow channel spacing's - typically encountered in radio applications. This can be overcome using a dual-modulus prescaler.

Further practical aspects concern the amount of time the system can switch from channel to channel, time to lock when first switched on, and how much noise there is in the output. All of these are a function of the *loop filter* of the system, which is a low-pass filter placed between the output of the frequency comparator and the input of the VCO.

Usually the output of a frequency comparator is in the form of short error pulses, but the input of the VCO must be a smooth noise- free DC voltage. (Any noise on this signal naturally causes frequency modulation of the VCO.).

Heavy filtering will make the VCO slow to respond to changes, causing drift and slow response time, but light filtering will produce noise and other problems with harmonics. Thus the design of the filter is critical to the performance of the system and in fact the main area that a designer will concentrate on when building a synthesizer system.

UNIT IV
ANALOG TO DIGITAL & DIGITAL TO ANALOG CONVERTERS

4. 1 1Analog To Digital Conversion

The natural state of audio and video signals is analog. When digital technology was not yet around, they are recorded or played back in analog devices like vinyl discs and cassette tapes. The storage capacity of these devices is limited and doing multiple runs of re-recording and editing produced poor signal quality. Developments in digital technology like the CD, DVD, Blu-ray, flash devices and other memory devices addressed these problems.

For these devices to be used, the analog signals are first converted to digital signals using analog to digital conversion (ADC). For the recorded audio and video signals to be heard and viewed again, the reverse process of digital to analog conversion (DAC) is used.

ADC and DAC are also used in interfacing digital circuits to analog systems. Typical applications are control and monitoring of temperature, water level, pressure and other real-world data.

An ADC inputs an analog signal such as voltage or current and outputs a digital signal in the form of a binary number. A DAC, on the other hand, inputs the binary number and outputs the corresponding analog voltage or current signal.

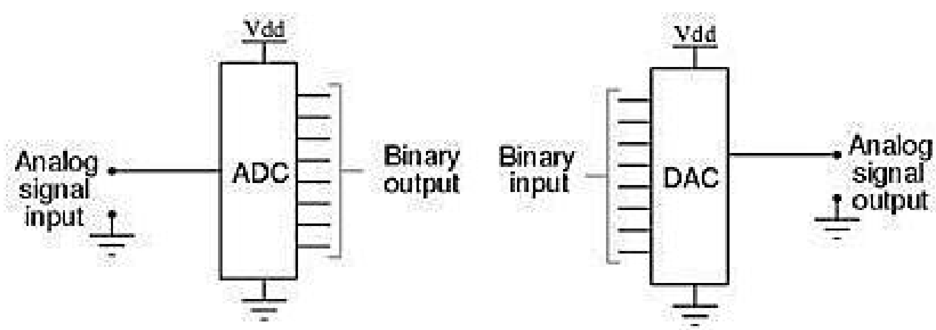


Fig 4.1 ADC and DAC circuits

- ✓ **Sampling rate**

The analog signal is continuous in time and it is necessary to convert this to a flow of digital values. It is therefore required to define the rate at which new digital values are sampled from the analog signal. The rate of new values is called the *sampling rate* or *sampling frequency* of the converter.

A continuously varying band limited signal can be sampled (that is, the signal values at intervals of time T, the sampling time, are measured and stored) and then the original signal can be *exactly* reproduced from the discrete-time values by an interpolation formula. The accuracy is limited by quantization error. However, this faithful reproduction is only possible if the sampling rate is higher than twice the highest frequency of the signal. This is essentially what is embodied in the Shannon-Nyquist sampling theorem.

Since a practical ADC cannot make an instantaneous conversion, the input value must necessarily be held constant during the time that the converter performs a conversion (called the *conversion* *time*). An input circuit called a sample and hold performs this task—in most cases by using a capacitor to store the analog voltage at the input, and using an electronic switch or gate to disconnect the capacitor from the input. Many ADC integrated circuits include the sample and hold subsystem internally.

- ✓ **Accuracy**

An ADC has several sources of errors. Quantization error and (assuming the ADC is intended to be linear) non-linearity is intrinsic to any analog-to-digital conversion. There is also a so-called *aperture error* which is due to a clock jitter and is revealed when digitizing a time-variant signal (not a constant value).

These errors are measured in a unit called the *LSB*, which is an abbreviation for least significant bit. In the above example of an eight-bit ADC, an error of one LSB is 1/256 of the full signal range, or about 0.4%.

- ✓ **Quantization error**

Quantization error is due to the finite resolution of the ADC, and is an unavoidable imperfection in all types of ADC. The magnitude of the quantization error at the sampling instant is between zero and half of one LSB.

In the general case, the original signal is much larger than one LSB. When this happens, the quantization error is not correlated with the signal, and has a uniform distribution.

Its RMS value is the standard deviation of this distribution, given by $\frac{1}{\sqrt{12}}\mathrm{LSB} \approx 0.289\,\mathrm{LSB}$. In the eight-bit ADC example, this represents 0.113% of the full signal range.

At lower levels the quantizing error becomes dependent of the input signal, resulting in distortion. This distortion is created after the anti-aliasing filter, and if these distortions are above 1/2 the sample rate they will alias back into the audio band. In order to make the Quantizing error independent of the input signal, noise with amplitude of 1 quantization step is added to the signal. This slightly reduces signal to noise ratio, but completely eliminates the distortion. It is known as dither.

- ✓ **Non-linearity**

All ADCs suffer from non-linearity errors caused by their physical imperfections, resulting in their output to deviate from a linear function (or some other function, in the case of a deliberately non-linear ADC) of their input. These errors can sometimes be mitigated by calibration, or prevented by testing.

Important parameters for linearity are integral non-linearity (INL) and differential non- linearity (DNL). These non-linear ties reduce the dynamic range of the signals that can be digitized by the ADC, also reducing the effective resolution of the ADC.

D To A Converter- Specifications

D/A converters are available with wide range of specifications specified by manufacturer. Some of the important specifications are Resolution, Accuracy, linearity, monotonicity, conversion time, settling time and stability.

✓ **Resolution:**

Resolution is defined as the number of different analog output voltage levels that can be provided by a DAC. Or alternatively resolution is defined as the ratio of a change in output voltage resulting for a change of 1 LSB at the digital input. Simply, resolution is the value of LSB.

Resolution (Volts) = VoFS / (2^n - 1) = 1 LSB increment Where 'n' is the number of input bits
'VoFS' is the full scale output voltage.

Example:
Resolution for an 8 – bit DAC for example is said to have
: 8 – bit resolution
: A resolution of 0.392 of full-Scale (1/255)
: A resolution of 1 part in 255.

Thus resolution can be defined in many different ways.

he following table shows the resolution for 6 to 16 bit DACs

Table 4.1 Resolution for DAC

S.No.	**Bits**	**Intervals**	**LSB size (% of full-scale)**	**LSB size (For a 10 V full-scale)**
1.	6	63	1.588	158.8 mV
2.	8	255	0.392	39.2 mV
3.	10	1023	0.0978	9.78 mV
4.	12	4095	0.0244	2.44 mV
5.	14	16383	0.0061	0.61 mV
6.	16	65535	0.0015	0.15 mV

✓ **Accuracy:**

Absolute accuracy is the maximum deviation between the actual converter output and the ideal converter output. The ideal converter is the one which does not suffer from any problem. Whereas, the actual converter output deviates due to the drift in component values, mismatches, aging, noise and other sources of errors.

The relative accuracy is the maximum deviation after the gain and offset errors have been removed. Accuracy is also given in terms of LSB increments or percentage of full-scale voltage. Normally, the data sheet of a D/A converter specifies the relative accuracy rather than absolute accuracy.

✓ **Linearity:**

Linearity error is the maximum deviation in step size from the ideal step size. Some D/A converters are having a linearity error as low as 0.001% of full scale. The linearity of a D/A converter is defined as the precision or exactness with which the digital input is converted into analog output. An ideal D/A converter produces equal increments or step sizes at output for every change in equal increments of binary input.

- ✓ **Monotonicity:**

A Digital to Analog converter is said to be monotonic if the analog output increases for an increase in the digital input. A monotonic characteristic is essential in control applications. Otherwise it would lead to oscillations. If a DAC has to be monotonic, the error should be less than ± (1/2) LSB at each output level. Hence all the D/A converters are designed such that the linearity error satisfies the above condition.
When a D/A Converter doesn't satisfy the condition described above, then, the output voltage may decrease for an increase in the binary input.

- ✓ **Conversion Time:**

It is the time taken for the D/A converter to produce the analog output for the given binary input signal. It depends on the response time of switches and the output of the Amplifier. D/A converters speed can be defined by this parameter. It is also called as setting time.

- ✓ **Settling time:**

It is one of the important dynamic parameter. It represents the time it takes for the output to settle within a specified band ± (1/2) LSB of its final value following a code change at the input (Usually a full-scale change). It depends on the switching time of the logic circuitry due to internal parasitic capacitances and inductances. A typical settling time ranges from 100 ns to 10 μs depending on the word length and type of circuit used.

- ✓ **Stability:**

The ability of a DAC to produce a stable output all the time is called as Stability. The performance of a converter changes with drift in temperature, aging and power supply variations. So all the parameters such as offset, gain, linearity error & monotonicity may change from the values specified in the datasheet. Temperature sensitivity defines the stability of a D/A converter.

4.2 Digital To Analog Conversion

A DAC converts an abstract finite-precision number (usually a fixed-point binary number) into a concrete physical quantity (e.g., a voltage or a pressure). In particular, DACs are often used to convert finite-precision time series data to a continually-varying physical signal.

A typical DAC converts the abstract numbers into a concrete sequence of impulses that are then processed by a reconstruction filter using some form of interpolation to fill in data between the impulses.

Other DAC methods (e.g., methods based on Delta-sigma modulation) produce a pulse-density modulated signal that can then be filtered in a similar way to produce a smoothly-varying signal.By the Nyquist–Shannon sampling theorem, sampled data can be reconstructed perfectly provided that its bandwidth meets certain requirements (e.g., a baseband signal with bandwidth less than the Nyquist frequency). However, even with an ideal reconstruction filter, digital sampling introduces quantization that makes perfect reconstruction practically impossible. Increasing the digital resolution (i.e., increasing the number of bits used in each sample) or introducing sampling dither can reduce this error.

DACs are at the beginning of the analog signal chain, which makes them very important to system performance. The most important characteristics of these devices are:

4.3 Specifications:

- ✓ **Resolution**: This is the number of possible output levels the DAC is designed to reproduce. This is usually stated as the number of bits it uses, which is the base two logarithm of the number of levels. For instance a 1 bit DAC is designed to reproduce 2 (2^1) levels while an 8 bit DAC is designed for 256 (2^8) levels. Resolution is related to the
- ✓ **Effective number of bits** (ENOB) which is a measurement of the actual resolution attained by the DAC.
- ✓ **Maximum sampling frequency**: This is a measurement of the maximum speed at which the DACs circuitry can operate and still produce the correct output. As stated in the Nyquist–Shannon sampling theorem, a signal must be sampled at over twice the frequency of the desired signal. For instance, to reproduce signals in all the audible spectrum, which includes frequencies of up to 20 kHz, it is necessary to use DACs that operate at over 40 kHz. The CD standard samples audio at kHz, thus DACs of this frequency are often used. A common frequency in cheap computer sound cards is 48 kHz—many work at only this frequency, offering the use of other sample rates only through (often poor) internal resampling.
- ✓
- ✓ **Monotonicity**: This refers to the ability of a DAC's analog output to move only in the direction that the digital input moves (i.e., if the input increases, the output doesn't dip before asserting the correct output.) This characteristic is very important for DACs used as a low frequency signal source or as a digitally programmable trim element.
- ✓ **THD+N**: This is a measurement of the distortion and noise introduced to the signal by the DAC. It is expressed as a percentage of the total power of unwanted harmonic distortion and noise that accompany the desired signal. This is a very important DAC characteristic for dynamic and small signal DAC applications.
- ✓ **Dynamic range**: This is a measurement of the difference between the largest and smallest signals the DAC can reproduce expressed in decibels. This is usually related to DAC resolution and noise floor.

Other measurements, such as phase distortion and sampling period instability, can also be very important for some applications.

4.4 Binary-Weighted Resistor DAC

The binary-weighted-resistor DAC employs the characteristics of the inverting summer Op Amp circuit. In this type of DAC, the output voltage is the inverted sum of all the input voltages. If the input resistor values are set to multiples of two: 1R, 2R and 4R, the output voltage would be equal to the sum of V1, V2/2 and V3/4. V1 corresponds to the most significant bit (MSB) while V3 corresponds to the least significant bit (LSB).

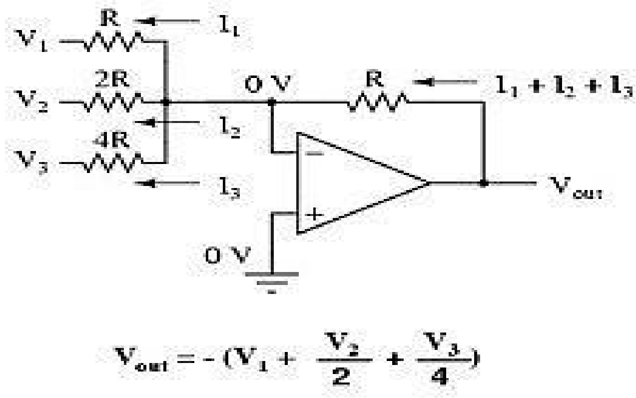


Fig. 4.2 Binary weighted DAC

The circuit for a 4-bit DAC using binary weighted resistor network is shown below:

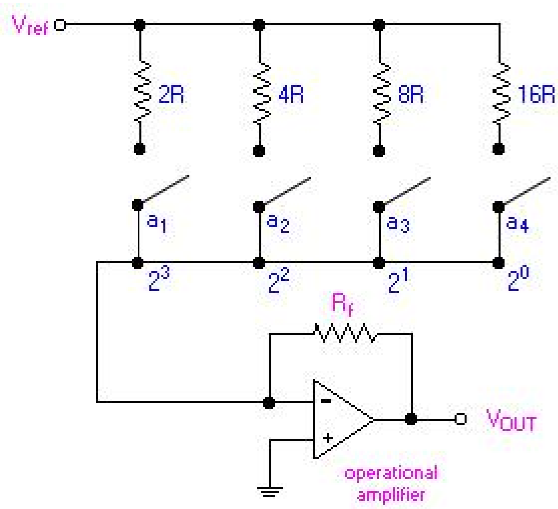


Fig. 4.3 weighted resistor DAC using Op-amp

The binary inputs, ai (where i = 1, 2, 3 and 4) have values of either 0 or 1. The value, 0, represents an open switch while 1 represents a closed switch.

The operational amplifier is used as a summing amplifier, which gives a weighted sum of the binary input based on the voltage, Vref.

For a 4-bit DAC, the relationship between Vout and the binary input is as follows:

$$
\begin{aligned}
V_{OUT} &= -\,iR_f \\
&= -\left[V_{ref}\left(\frac{a_1}{2R} + \frac{a_2}{4R} + \frac{a_3}{8R} + \frac{a_4}{16R}\right)\right]R_f \\
&= -\frac{V_{ref}R_f}{R}\left(\frac{a_1}{2} + \frac{a_2}{4} + \frac{a_3}{8} + \frac{a_4}{16}\right) \\
&= -\frac{V_{ref}R_f}{R}\left(\frac{a_1}{2^1} + \frac{a_2}{2^2} + \frac{a_3}{2^3} + \frac{a_4}{2^4}\right)
\end{aligned}
$$

The negative sign associated with the analog output is due to the connection to a summing amplifier, which is a polarity-inverting amplifier. When a signal is applied to the latter type of amplifier, the polarity of the signal is reversed (i.e. a + input becomes -, or vice versa).

For a n-bit DAC, the relationship between Vout and the binary input is as follows:

$$V_{OUT} = -\frac{V_{ref} R_f}{R} \sum_{i=1}^{n} \frac{a_i}{2^i}$$

The LSB, which is also the incremental step, has a value of - 0.625 V while the MSB or the full scale has a value of - 9.375 V.

Practical Limitations:

- The most significant problem is the large difference in resistor values required between the LSB and MSB, especially in the case of high resolution DACs (i.e. those that has large number of bits). For example, in the case of a 12-bit DAC, if the MSB is 1 kΩ, then the LSB is a staggering 2 MΩ.
- The maintenance of accurate resistances over a large range of values is problematic. With the current IC fabrication technology, it is difficult to manufacture resistors over a wide resistance range that maintains an accurate ratio especially with variations in temperature.

4.5 R-2R Ladder DAC

An enhancement of the binary-weighted resistor DAC is the R-2R ladder network. This type of DAC utilizes Thevenin's theorem in arriving at the desired output voltages.

A disadvantage of the former DAC design was its requirement of several different precise input resistor values: one unique value per binary input bit.

The R-2R network consists of resistors with only two values - R and 2xR. If each input is supplied either 0 volts or reference voltage, the output voltage will be an analog equivalent of the binary value of the three bits. VS_2 corresponds to the most significant bit (MSB) while VS_0 corresponds to the least significant bit (LSB).

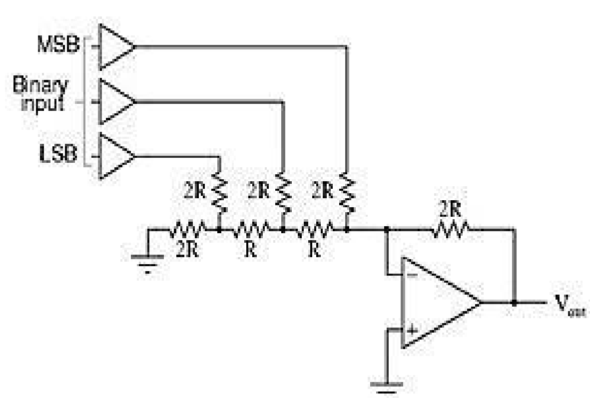


Fig.4.4 Ladder type DAC circuit

Vout = - (VMSB + Vn + VLSB) = - (VRef + VRef/2 + VRef/ 4)

Table 4.2 operation of a R-2R ladder DAC

Binary	Output voltage
000	0.00 V
001	-1.25 V
010	-2.50 V
011	-3.75 V
100	-5.00 V
101	-6.25 V
110	-7.50 V
111	-8.75 V

<u>4.6 Inverted Or Current Mode DAC</u>

Current mode DACs operates based on the ladder currents. The ladder is formed by resistance R in the series path and resistance 2R in the shunt path. Thus the current is divided into $i1$, $i2$, $i3$ …………in. in each arm. The currents are either diverted to the ground bus (io) or to the Virtual-ground bus ($\bar{io}$).

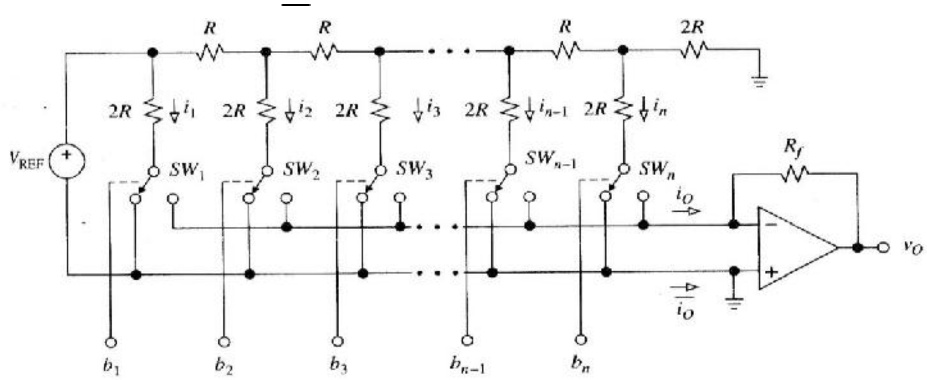


Fig.4.5 Current mode DAC

The currents are given as

$i1 = V_{REF}/2R = (V_{REF}/R)\ 2^{-1}$, $i2 = (V_{REF})/2)/2R = (V_{REF}/R)\ 2^{-2}$………$in = (V_{REF}/R)\ 2^{-n}$.

And the relationship between the currents are given as

$$i2 = i1/2$$
$$i3 = i1/4$$

$$i4 = i1/8$$
$$in = i1/2^{n-1}$$

Using the bits to identify the status of the switches, and letting V0 = -Rf *io* gives

$$V0 = -(Rf/R)\ VREF\ (b1 2^{-1} + b2 2^{-2} + \ldots\ldots\ldots\ldots + bn 2^{-n})$$

The two currents *io* and *io* are complementary to each other and the potential of *io* bus must be sufficiently close to that of ~~the~~ *io* bus. Otherwise, linearity errors will occur. The final op-amp is used as current to voltage converter.

Advantages

1. The major advantage of current mode D/A converter is that the voltage change across each switch is minimal. So the charge injection is virtually eliminated and the switch driver design is made simpler.
2. In Current mode or inverted ladder type DACs, the stray capacitance do not affect the Speed of response of the circuit due to constant ladder node voltages. So improved speed performance.

✓ **Voltage Mode DAC**

This is the alternative mode of DAC and is called so because the 2R resistance in the shunt path is switched between two voltages named as VL and VH. The output of this DAC is obtained from the leftmost ladder node. As the input is sequenced through all the possible binary state starting from All 0s (0…..0) to all 1s (1…..1). The voltage of this node changes in steps of 2^{-n} (VH - VL) from the minimum voltage of Vo = VL to the maximum of Vo = VH - 2^{-n} (VH - VL).

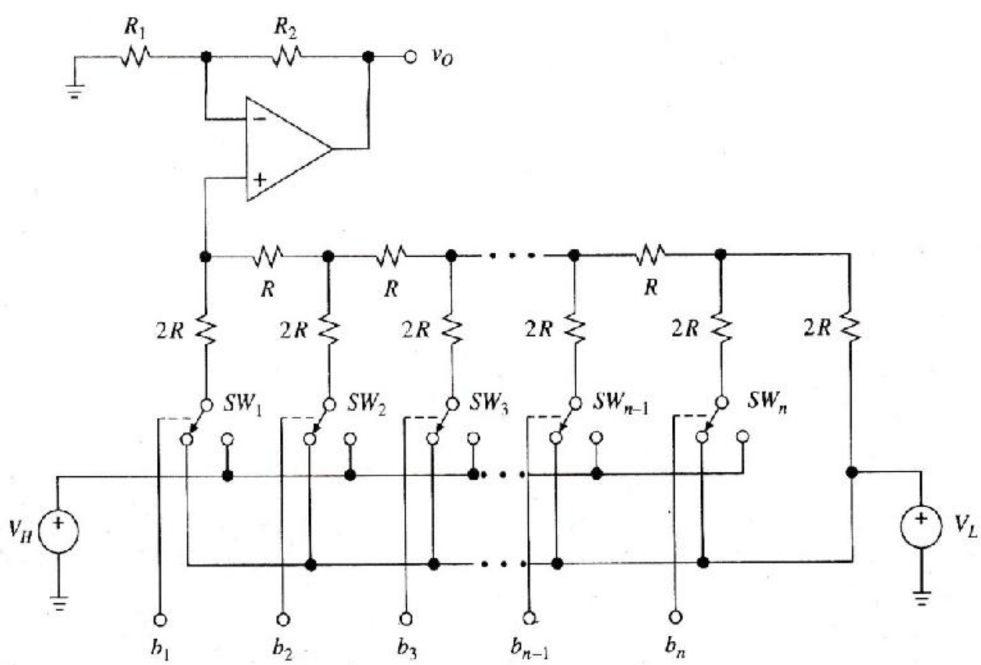


Fig. 4.6 Voltage mode DAC

The diagram also shows a non-inverting amplifier from which the final output is taken. Due to this buffering with a non- inverting amplifier, a scaling factor defined by K = 1 + (R2/R1) results.

Advantages

1. The major advantage of this technique is that it allows us to interpolate between any two voltages, neither of which need not be a zero.
2. More accurate selection and design of resistors R and 2R are possible and simple construction.
3. The binary word length can be easily increased by adding the required number or R-2R sections.

4.7 Switches For DAC

The Switches which connects the digital binary input to the nodes of a D/A converter is an electronic switch. Although switches can be made of using diodes, bipolar junction Transistors, Field Effect transistors or MOSFETs, there are four main configurations used as switches for DACs. They are

i) Switches using overdriven Emitter Followers.
ii) Switches using MOS Transistor- Totem pole MOSFET Switch and CMOS Inverter Switch.
iii) CMOS switch for Multiplying type DACs.
iv) CMOS Transmission gate switches.

These configurations are used to ensure the high speed switching operations for different types of DACs.

✓ **Switches using overdriven Emitter Followers:**

The bipolar transistors have a negligible resistance when they are operated in saturation. The bipolar transistor operating in saturation region indicates a minimum resistance and thus represents ON condition. When they are operating in cut-off region indicates a maximum resistance and thus represents OFF condition.
The circuit shown here is the arrangement of two transistors connected as emitter followers. A silicon transistor operating in saturation will have an offset voltage of 0.2V dropped across them. To have a zero offset voltage condition, the transistors must be overdriven because the saturation factor becomes negative. The two transistors Q1 (NPN) and Q2 (PNP) acts as a double pole switch. The bases of the transistors are driven by +5.75V and -5.75V.
Case 1:
When VB1 = VB2 = +5.75V, Q1 is in saturation and Q2 is OFF. And VE ≈ 5V
with VBE1 = VBE2 = 0.75V
Case 2:
When VB1 = VB2 = -5.75V, Q2 is in saturation and Q1 is OFF. And VE ≈ - 5V
with VBE1 = VBE2 = 0.75V
Thus the terminal B of the resistor Re is connected to either -5V or +5V depending on the input bit.

✓ **Switches using MOS transistor:**

i) Totem pole MOSFET Switch:

As shown in the figure, the totem pole MOSFET Switch is connected in series with resistors of R-2R network. The MOSFET driver is connected to the inverting terminal of the summing op-amp. The complementary outputs Q and $\bar{Q}$ drive the gates of the MOSFET M1 and M2 respectively. The SR flip flop holds one bit of digital information of the binary word under conversion. Assuming the negative logic (-5V for logic 1 and +5V for logic 0) the operation is given as two cases.
Case 1:
When the bit line is 1 with S=1 and R=0 makes Q=1 and $\bar{Q}$ =0. This makes the transistor M1 ON, thereby connecting the resistor R to reference voltage -VR. The transistor M2 remains in OFF condition.
Case 2:
When the bit line is 0 with S=0 and R=1 makes Q=0 and $\bar{Q}$ =1. This makes the transistor M2 ON, thereby connecting the resistor R to Ground. The transistor M1 remains in OFF condition.
ii) CMOS Inverter Switch:
The figure of CMOS inverter is shown here. It consists of a CMOS inverter connected with an op-amp acting as a buffer. The buffer drives the resistor R with very low output impedance. Assuming positive logic (+5V for logic 1 and 0V for logic 0), the operation can be explained in two cases.
Case1:
When the complement of the bit line $\bar{Q}$ is low, M1 becomes ON connecting VR to the non- inverting input of the op-amp. This drives the resistor R HIGH.
Case2:
When the complement of the bit line $\bar{Q}$ is high, M2 becomes ON connecting Ground to the non- inverting input of the op-amp. This pulls the resistor R LOW (to ground).

- ✓ **CMOS switch for Multiplying type DACs:**

The circuit diagram of CMOS Switch is shown here. The heart of the switching element is formed by transistors M1 and M2. The remaining transistors accept TTL or CMOS compatible logic inputs and provides the anti-phase gate drives for the transistors M1 and M2. The operation for the two cases is as follows.
Case 1:
When the logic input is 1, M1 is ON and M2 is OFF. Thus current IK is diverted to Io bus.
Case 2:
When the logic input is 0, M2 is ON and M1 is OFF. Thus current IK is diverted to Io bus.

- ✓ **CMOS Transmission gate switches:**

The disadvantage of using individual NMOS and PMOS transistors are threshold voltage drop (NMOS transistor passing only minimum voltage of VR- VTH and PMOS transistor passing minimum voltage of VTH). This is eliminated by using transmission gates which uses a parallel connection of both NMOS and PMOS. The arrangement shown here can pass voltages from VR to 0V acting as a ideal switch. The following cases explain the operation.
Case 1:
When the bit-line bk is HIGH, both transistors Mn and Mp are ON, offering low resistance over the entire range of bit voltages.

Case 2:
When the bit-line bk is LOW, both the transistors are OFF, and the signal transmission is inhibited (Withdrawn).
Thus the NMOS offers low resistance in the lower portion of the signal and PMOS offers low resistance in the upper portion of the signal. As a combination, they offer a low parallel resistance throughout the operating range of voltage. Wide varieties of these kinds of switches were available. Example: CD4066 and CD4051.

4.8 High Speed Sample and Hold Circuits

Introduction:

Sample-and-hold (S/H) is an important analog building block with many applications, including analog-to-digital converters (ADCs) and switched-capacitor filters. The function of the S/H circuit is to sample an analog input signal and hold this value over a certain length of time for subsequent processing.

Taking advantages of the excellent properties of MOS capacitors and switches, traditional switched capacitor techniques can be used to realize different S/H circuits [1]. The simplest S/H circuit in MOS technology is shown in Figure 1, where *Vin* is the input signal, M1 is an MOS transistor operating as the sampling switch, Ch is the hold capacitor, ck is the clock signal, and Vout is the resulting sample-and-hold output signal.

As depicted by Figure 4. , in the simplest sense, a S/H circuit can be achieved using only one MOS transistor and one capacitor. The operation of this circuit is very straightforward.

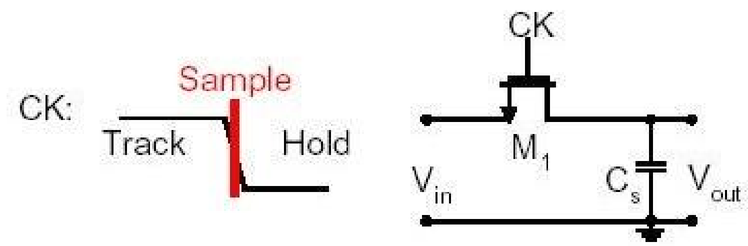


Figure 4.7 Simplest sample-and-hold circuits in MOS technology.

Figure 4, in the simplest sense, a S/H circuit can be achieved using only one MOS transistor and one capacitor. The operation of this circuit is very straightforward. Whenever *ck* is high, the MOS switch is on, which in turn allows *Vout* to track *Vin*. On the other hand, when *ck* is low, the MOS switch is off. During this time, *Ch* will keep *Vout* equal to the value of *Vin* at the instance when *ck* goes low.
Unfortunately, in reality, the performance of this S/H circuit is not as ideal as described above. The two major types of errors occur. They are charge injection and clock feed through, that are associated with this S/H implementation. Three new S/H techniques, all of which try to minimize the errors caused by charge injection and/or clock feed through.

Alternative CMOS Sample-and-Hold Circuits

Three alternative CMOS S/H circuits that are developed with the intention to minimize charge injection and/or clock feed through are

✓ **Series Sampling:**

The S/H circuit of Figure 4. is classified as parallel sampling because the hold capacitor is in parallel with the signal. In parallel sampling, the input and the output are dc-coupled. On the other hand, the S/H circuit shown in Figure 2 is referred to as series sampling because the hold capacitor is in series with the signal.

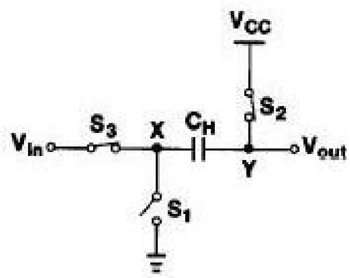


Figure 4.8 Series sampling.

When the circuit is in sample mode, both switches *S2* and *S3* are on, while *S1* is off. Then,*S2* is turned off first, which means *Vout* is equal to *VCC* (or *VDD* for most circuits) and the voltage drop across *Ch* will be *VCC – Vin*. Subsequently, *S3* is turned off and *S1* is turned on simultaneously. By grounding node *X*, *Vout* is now equal to *VCC –Vin*, and the drop from *VCC* to *VCC – Vin* is equal to the instantaneous value of the input.

As a result, this is actually an inverted S/H circuit, which requires inversion of the signal at a later stage. Since the hold capacitor is in series with the signal, series sampling can isolate the common- mode levels of the input and the output.

This is one advantage of series sampling over parallel sampling. In addition, unlike parallel sampling, which suffers from signal-dependent charge injection, series sampling does not exhibit such behavior because *S2* is turned off before *S3*. Thus, the fact that the gate-to-source voltage, *VGS*, of *S2* is constant means that charge injection coming from *S2* is also constant (as opposed to being signal-dependent), which means this error can be easily eliminated through differential operation.

Limitations:

On the other hand, series sampling suffers from the nonlinearity of the parasitic capacitance at node *Y*. This parasitic capacitance introduces distortion to the sample-and hold value, thus mandating that *Ch* be much larger than the parasitic capacitance. On top of this disadvantage, the settling time of the S/H circuit during hold mode is longer for series sampling than for parallel sampling. The reason for this is because the value of *Vout* in series sampling is being reset to *VCC* (or *VDD*) for every sample, but this is not the case for parallel sampling.

✓ **Switched Op-Amp Based Sample-and-Hold Circuit:**

This S/H technique takes advantage of the fact that when a MOS transistor is in the saturation region, the channel is pinched off and disconnected from the drain. Therefore, if the hold capacitor is connected to the drain of the MOS transistor, charge injection will only go to the source junction, leaving the drain unaffected. Based on this concept, a switched op- amp (SOP) based S/H circuit, as shown in Figure 4.9

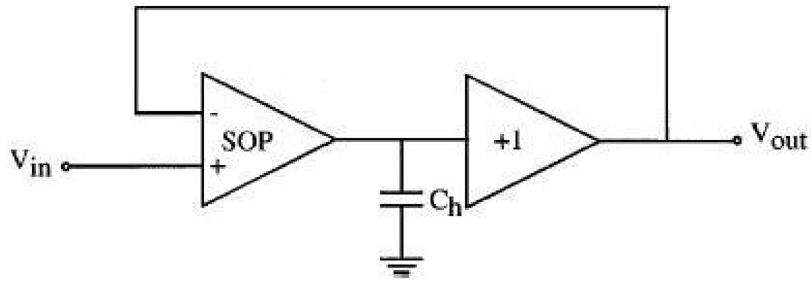


Fig. 4.9 Switched op-amp based sample and hold circuit.

During sample mode, the SOP behaves just like a regular op-amp, in which the value of the output follows the value of the input. During hold mode, the MOS transistors at the output node of the SOP are turned off while they are still operating in saturation, thus preventing any channel charge from flowing into the output of the SOP. In addition, the SOP is shut off and its output is held at high impedance, allowing the charge on *Ch* to be preserved throughout the hold mode. On the other hand, the output buffer of this S/H circuit is always operational during sample and hold mode and is always providing the voltage on *Ch* to the output of the S/H circuit.

S/H circuits that operate in closed loop configuration can achieve high resolution, but their requirements for high gain circuit block, such as an op-amp, limits the speed of the circuits. As a result, better and faster S/H circuits must be developed.

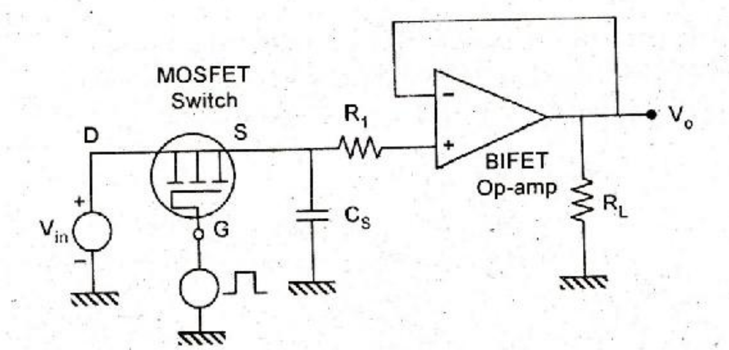


Fig.4.10 High speed Sample and Hold circuit with MOSFET

The above figure shows a sample and holds circuit with MOSFET as Switch acting as a sampling device and also consists of a holding capacitor Cs to store the sample values until the next sample comes in. This is a high speed circuit as it is apparent that CMOS switch has a very negligible propagation delay.

Three S/H circuits to reduce error:

- series sampling,
- SOP based S/H circuit,
- bottom plate S/H circuit with bootstrapped switch

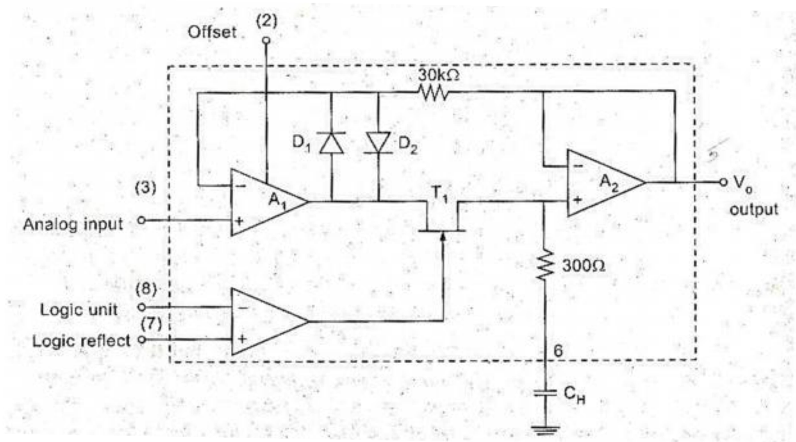


Fig.4.11 LF 398 IC- Functional Diagram

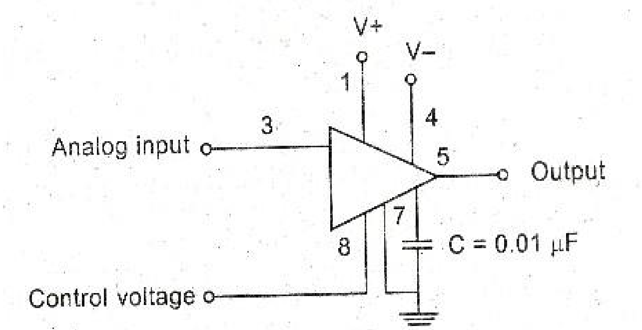


Fig.4.12 Connections of S&H IC

4.9 A to D Converter- Specifications

Like DAC, ADCs are also having many important specifications. Some of them are Resolution, Quantization error, Conversion time, Analog error, Linearity error, DNL error, INL error & Input voltage range.

- ✓ **Resolution:**

The resolution refers to the finest minimum change in the signal which is accepted for conversion, and it is decided with respect to number of bits. It is given as $1/2^n$, where 'n' is the number of bits in the digital output word. As it is clear, that the resolution can be improved by increasing the number of bits or the number of bits representing the given analog input voltage.

Resolution can also be defined as the ratio of change in the value of input voltage Vi, needed to change the digital output by 1 LSB. It is given as

$$\text{Resolution} = V_{iFS} / (2^n - 1)$$

Where 'V_{iFS}' is the full-scale input voltage.
'n' is the number of output bits.

- ✓ **Quantization error:**

If the binary output bit combination is such that for all the values of input voltage Vi between any two voltage levels, there is a unavoidable uncertainty about the exact value of Vi when the output is a particular binary combination. This uncertainty is termed as quantization error. Its value is ± (1/2) LSB. And it is given as,

$$QE = ViFS / 2(2^n - 1)$$

Where 'ViFS' is the full-scale input voltage
'n' is the number of output bits.

Maximum the number of bits selected, finer the resolution and smaller the quantization error.

- ✓ **Conversion Time:**

It is defined as the total time required for an A/D converter to convert an analog signal to digital output. It depends on the conversion technique and propagation delay of the circuit components.

- ✓ **Analog error:**

An error occurring due to the variations in DC switching point of the comparator, resistors, reference voltage source, ripples and noises introduced by the circuit components is termed as Analog error.

- ✓ **Linearity Error:**

It is defined as the measure of variation in voltage step size. It indicates the difference between the transitions for a minimum step of input voltage change. This is normally specified as fraction of LSB.

- ✓ **Differential Non-Linearity(DNL) Error:**

The analog input levels that trigger any two successive output codes should differ by 1 LSB. Any deviation from this 1 LSB value is called as DNL error.

- ✓ **Integral Non-Linearity (INL) Error:**

The deviation of characteristics of an ADC due to missing codes causes INL error. The maximum deviation of the code from its ideal value after nulling the offset and gain errors is called as Integral Non-Linearity Error.

- ✓ **Input Voltage Range:**

It is the range of voltage that an A/D converter can accept as its input without causing any overflow in its digital output.

- ✓ **Analog Switches**

There were two types of analog switches. Series and Shunt switch. The Switch operation is shown for both the cases VGS=0 VGS= VGs (off)

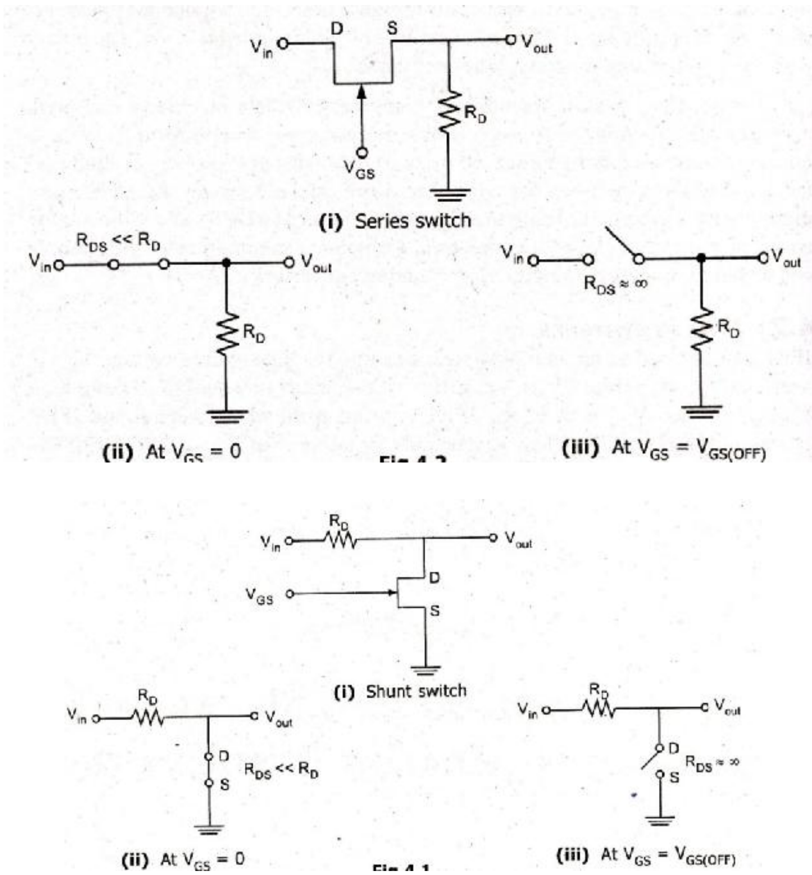


Fig 4.13 Series and shunt Analog switches

4.10 Direct-conversion ADC/Flash type ADC:

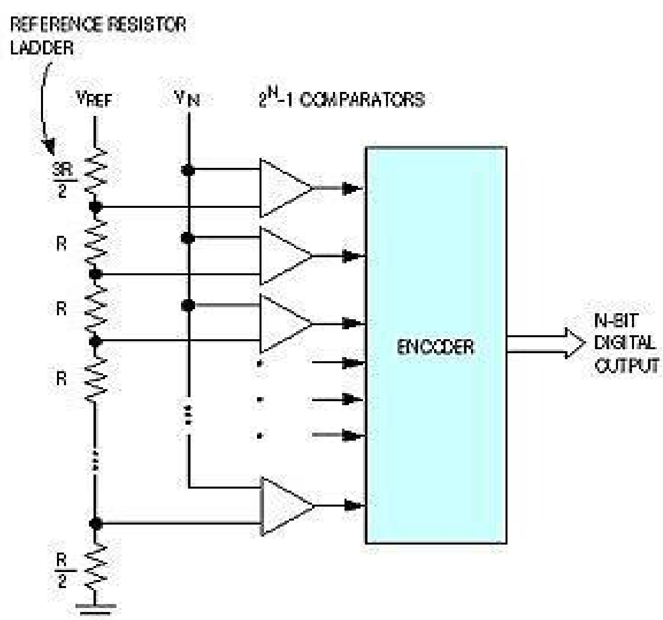


Fig.4.14 Flash ADC

This process is extremely fast with a sampling rate of up to 1 GHz. The resolution is however, limited because of the large number of comparators and reference voltages required. The input signal is fed simultaneously to all comparators. A priority encoder then generates a digital output that corresponds with the highest activated comparator.

4.12 Successive-approximation ADCs

Successive-approximation ADC is a conversion technique based on a successive-approximation register (SAR). This is also called bit-weighing conversion that employs a comparator to weigh the applied input voltage against the output of an N-bit digital-to-analog converter (DAC).

The final result is obtained as a sum of N weighting steps, in which each step is a single-bit conversion using the DAC output as a reference. SAR converters sample at rates up to 1Mbps, requires a low supply current, and the cheapest in terms of production cost.

A successive-approximation ADC uses a comparator to reject ranges of voltages, eventually settling on a final voltage range. Successive approximation works by constantly comparing the input voltage to the output of an internal digital to analog converter (DAC, fed by the current value of the approximation) until the best approximation is achieved.

At each step in this process, a binary value of the approximation is stored in a successive approximation register (SAR). The SAR uses a reference voltage (which is the largest signal the ADC is to convert) for comparisons.

For example if the input voltage is 60 V and the reference voltage is 100 V, in the 1st clock cycle, 60 V is compared to 50 V (the reference, divided by two. This is the voltage at the output of the internal DAC when the input is a '1' followed by zeros), and the voltage from the comparator is positive (or '1') (because 60 V is greater than 50 V). At this point the first binary digit (MSB) is set to a '1'. In the 2nd clock cycle the input voltage is compared to 75 V (being halfway between 100 and 50 V: This is the output of the internal DAC when its input is '11' followed by zeros) because 60 V is less than 75 V, the comparator output is now negative (or '0'). The second binary digit is therefore set to a '0'. In the 3rd clock cycle, the input voltage is compared with 62.5 V (halfway between 50 V and 75 V: This is the output of the internal DAC when its input is '101' followed by zeros). The output of the comparator is negative or '0' (because 60 V is less than 62.5 V) so the third binary digit is set to a 0. The fourth clock cycle similarly results in the fourth digit being a '1' (60 V is greater than 56.25 V, the DAC output for '1001' followed by zeros). The result of this would be in the binary form 1001. This is also called *bit-weighting conversion*, and is similar to a binary.

The analogue value is rounded to the nearest binary value below, meaning this converter type is mid-rise (see above). Because the approximations are successive (not simultaneous), the conversion takes one clock-cycle for each bit of resolution desired.

The clock frequency must be equal to the sampling frequency multiplied by the number of bits of resolution desired. For example, to sample audio at 44.1 kHz with 32 bit resolution, a clock frequency of over 1.4 MHz would be required.

ADCs of this type have good resolutions and quite wide ranges. They are more complex than some other designs.

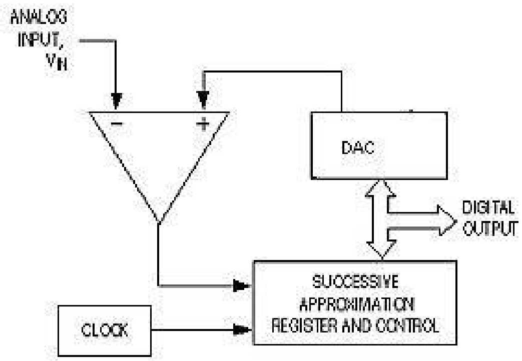


Fig.4.15 Successive approximation ADC

4.13 Dual slope ADC (Integrating ADCs)

In an integrating ADC, a current, proportional to the input voltage, charges a capacitor for a fixed time interval T charge. At the end of this interval, the device resets its counter and applies an opposite-polarity negative reference voltage to the integrator input. Because of this, the capacitor is discharged by a constant current until the integrator output voltage zero again.

The T discharge interval is proportional to the input voltage level and the resultant final count provides the digital output, corresponding to the input signal. This type of ADCs is extremely slow devices with low input bandwidths. Their advantage, however, is their ability to reject high-frequency noise and AC line noise such as 50Hz or 60Hz. This makes them useful in noisy industrial environments and typical application is in multi-meters.

An integrating ADC (also dual-slope or multi-slope ADC) applies the unknown input voltage to the input of an integrator and allows the voltage to ramp for a fixed time period (the run-up period). Then a known reference voltage of opposite polarity is applied to the integrator and is allowed to ramp until the integrator output returns to zero (the run-down period). The input voltage is computed as a function of the reference voltage, the constant run-up time period, and the measured run-down time period.

The run-down time measurement is usually made in units of the converter's clock, so longer integration times allow for higher resolutions. Likewise, the speed of the converter can be improved by sacrificing resolution.

Use: Converters of this type (or variations on the concept) are used in most digital voltmeters for their linearity and flexibility.

4.14 A/D Using Voltage To Time Conversion:

The Block diagram shows the basic voltage to time conversion type of A to D converter. Here the cycles of variable frequency source are counted for a fixed period. It is possible to make an A/D converter by counting the cycles of a fixed-frequency source for a variable period. For this, the analog voltage required to be converted to a proportional time period.

As shown in the diagram a negative reference voltage -VR is applied to an integrator, whose output is connected to the inverting input of the comparator. The output of the comparator is at 1 as long as the output of the integrator Vo is less than Va.

At t = T, Vc goes low and switch S remains open. When VEN goes high, the switch S is closed, thereby discharging the capacitor. Also the NAND gate is disabled. The waveforms are shown here.

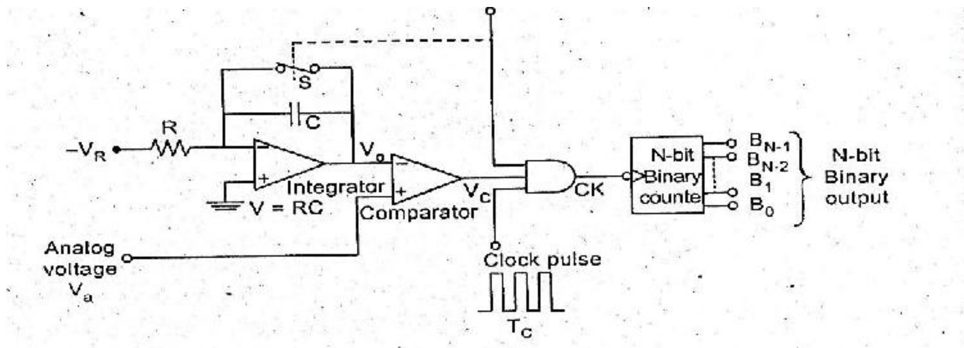


Fig. 4.16 A/D Using Voltage To Time Conversion:

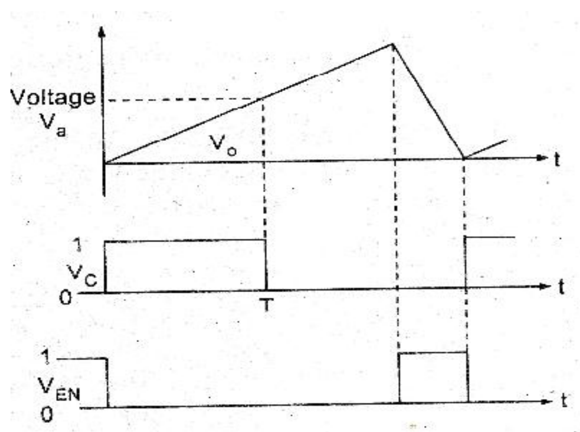


Fig.4.17 Conversion process

4.15 Sigma-delta ADCs/ Over sampling Converters:

It consists of 2 main parts - modulator and digital filter. The modulator includes an integrator and a comparator with a feedback loop that contains a 1-bit DAC. The modulator oversamples the input signal, converting it to a serial bit stream with a frequency much higher than the required sampling rate. This is then transformed by the output filter to a sequence of parallel digital words at the sampling rate. The characteristics of sigma-delta converters are high resolution, high accuracy,

Low noise and low cost.

Typical applications are for speech and audio.

A **Sigma-Delta ADC** (also known as a Delta-Sigma ADC) oversamples the desired signal by a large factor and filters the desired signal band. Generally a smaller number of bits than required are converted using a Flash ADC after the Filter. The resulting signal, along with the error generated by the discrete levels of the Flash, is fed back and subtracted from the input to the filter. This negative feedback has the effect of noise shaping the error due to the Flash so that it does not appear in the desired signal frequencies.

A digital filter (decimation filter) follows the ADC which reduces the sampling rate, filters off unwanted noise signal and increases the resolution of the output. (sigma-delta modulation, also called delta-sigma modulation)

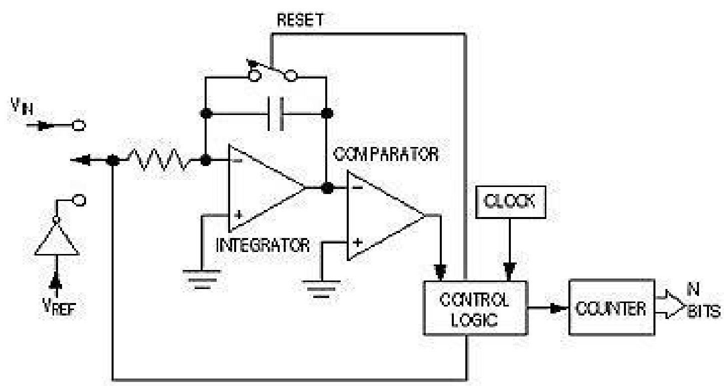


Fig 4.18 Sigma-delta ADCs/ Over sampling Converters:

UNIT V

WAVEFORM GENERATORS AND SPECIAL FUNCTION ICs

Basics Of Oscillators: Criteria for oscillation:

The canonical form of a feedback system is shown in Figure 5.1, and Equation 1 describes the performance of any feedback system (an amplifier with passive feedback Components constitute a feedback system).

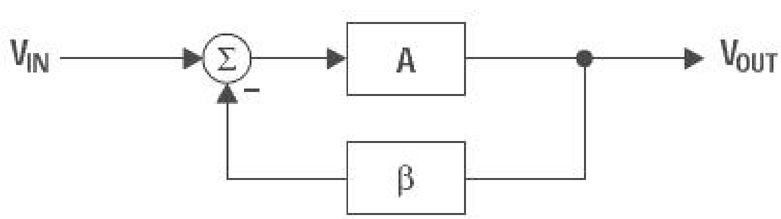


Fig. 5.1 Canonical form of feedback circuit

$$\frac{V_{OUT}}{V_{IN}} = \frac{A}{1 + A\beta} \qquad \textbf{(1)}$$

Oscillation results from an unstable state; i.e., the feedback system can't find a stable state because its transfer function can't be satisfied. Equation 1 becomes unstable when $(1+A\beta) = 0$ because A/0 is an undefined state. Thus, the key to designing an oscillator is to insure that $A\beta = -1$ (called the Barkhausen criterion), or using complex math the equivalent expression is $A\beta = 1 \angle -180°$. The 180° phase shift criterion applies to negative feedback systems, and 0° phase shift applies to positive feedback systems.

The output voltage of a feedback system heads for infinite voltage when $A\beta = -1$. When the output voltage approaches either power rail, the active devices in the amplifiers change gain, causing the value of A to change so the value of $A\beta \neq 1$; thus, the charge to infinite voltage slows down and eventually halts. At this point one of three things can occur.

First, nonlinearity in saturation or cutoff can cause the system to become stable and lock up.

Second, the initial charge can cause the system to saturate (or cut off) and stay that way for a long time before it becomes linear and heads for the opposite power rail.

Third, the system stays linear and reverses direction, heading for the opposite power rail. Alternative two produces highly distorted oscillations (usually quasi square waves), and the resulting oscillators are called relaxation oscillators. Alternative three produces sine wave oscillators.

- ✓ **Phase Shift in Oscillators:**

The 180° phase shift in the equation $A\beta = 1 \angle -180°$ is introduced by active and passive components. The phase shift contributed by active components is minimized because it varies with temperature, has a wide initial tolerance, and is device dependent.

Amplifiers are selected such that they contribute little or no phase shift at the oscillation frequency. A single pole RL or RC circuit contributes up to 90° phase shift per pole, and because 180° is required for oscillation, at least two poles must be used in oscillator design.

An LC circuit has two poles; thus, it contributes up to 180° phase shift per pole pair, but LC and LR oscillators are not considered here because low frequency inductors are expensive, heavy, bulky, and non-ideal. LC oscillators are designed in high frequency applications beyond the frequency range of voltage feedback op amps, where the inductor size, weight, and cost are less significant.

Multiple RC sections are used in low-frequency oscillator design in lieu of inductors. Phase shift determines the oscillation frequency because the circuit oscillates at the frequency that accumulates –180° phase shift. The rate of change of phase with frequency, dS/dt, determines frequency stability.

When buffered RC sections (an op amp buffer provides high input and low output impedance) are cascaded, the phase shift multiplies by the number of sections, n (see Figure 2).

Although two cascaded RC sections provide 180° phase shift, dS/dt at the oscillator frequency is low, thus oscillators made with two cascaded RC sections have poor frequency stability. Three equal cascaded RC filter sections have a higher dS/dt, and the resulting oscillator has improved frequency stability.

Adding a fourth RC section produces an oscillator with an excellent dS/dt, thus this is the most stable oscillator configuration. Four sections are the maximum number used

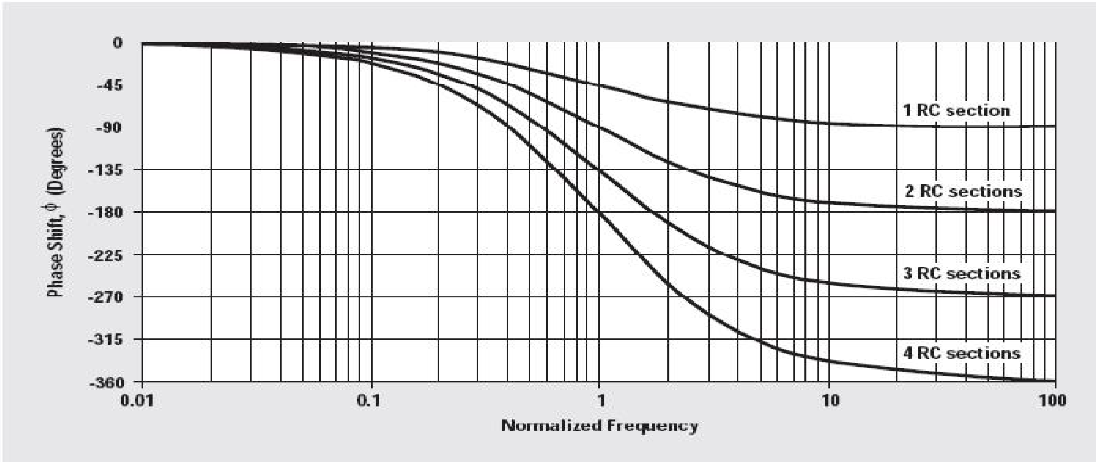


Figure 5.2 Phase plot of RC sections

because op amps come in quad packages, and the four-section oscillator yields four sine waves that are 45° phase shifted relative to each other, so this oscillator can be used to obtain sine/cosine or quadrature sine waves.

Applications

Crystal or ceramic resonators make the most stable oscillators because resonators have an extremely high dS/dt resulting from their non-linear properties.

Resonators are used for high- frequency oscillators, but low-frequency oscillators do not use resonators because of size, weight,
and cost restrictions.

Op amps are not used with crystal or ceramic resonator oscillators because op amps have low bandwidth. It is more cost-effective to build a high- frequency crystal oscillator and count down the output to obtain a low frequency than it is to use a low-frequency resonator.

Gain in Oscillators:

The oscillator gain must equal one ($A\beta = 1\angle -180°$) at the oscillation frequency. The circuit becomes stable when the gain exceeds one and oscillations cease. When the gain exceeds one with a phase shift of –180°, the active device non-linearity reduces the gain to one.

The non-linearity happens when the amplifier swings close to either power rail because cutoff or saturation reduces the active device (transistor) gain. The paradox is that worst-case design practice requires nominal gains exceeding one for manufacturability, but excess gain causes more distortion of the output sine wave.

When the gain is too low, oscillations cease under worst-case conditions, and when the gain is too high, the output wave form looks more like a square wave than a sine wave.
Distortion is a direct result of excess gain overdriving the amplifier; thus, gain must be carefully controlled in low distortion oscillators.

Phase-shift oscillators have distortion, but they achieve low-distortion output voltages because cascaded RC sections act as distortion filters. Also, buffered phase-shift oscillators have low distortion because the gain is controlled and distributed among the buffers.

5.1 Sine Wave Generators (Oscillators)

Sine wave oscillator circuits use phase shifting techniques that usually employ

- Two RC tuning networks, and
- Complex amplitude limiting circuitry

5.1.1 RC Phase Shift Oscillator

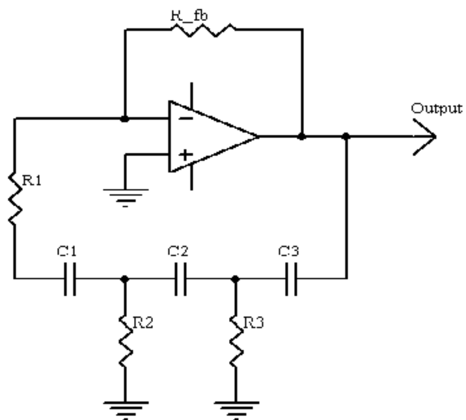


Fig.5.3 RC Phase shift oscillator

RC phase shift oscillator using op-amp in inverting amplifier introduces the phase shift of 180° between input and output. The feedback network consists of 3 RC sections each producing 60° phase shift. Such a RC phase shift oscillator using op-amp is shown in the figure.
The output of amplifier is given to feedback network. The output of feedback network drives the amplifier. The total phase shift around a loop is 1800 of amplifier and 180° due to 3 RC sections, thus 360°. This satisfies the required condition for positive feedback and circuit works as an oscillator.

$$f_{\text{oscillation}} = \frac{1}{2\pi\sqrt{R_2R_3(C_1C_2+C_1C_3+C_2C_3)+R_1R_3(C_1C_2+C_1C_3)+R_1R_2C_1C_2}}$$

Oscillation criterion:

$$R_{\text{feedback}} = 2(R_1+R_2+R_3)+\frac{2R_1R_3}{R_2}+\frac{C_2R_2+C_2R_3+C_3R_3}{C_1}$$

$$+\frac{2C_1R_1+C_1R_2+C_3R_3}{C_2}+\frac{2C_1R_1+2C_2R_1+C_1R_2+C_2R_2+C_2R_3}{C_3}$$

$$+\frac{C_1R_1^2+C_3R_1R_3}{C_2R_2}+\frac{C_2R_1R_3+C_1R_1^2}{C_3R_2}+\frac{C_1R_1^2+C_1R_1R_2+C_2R_1R_2}{C_3R_3}$$

$$A\beta = A\left(\frac{1}{RCs+1}\right)^3 \qquad \textbf{(3)}$$

The loop phase shift is –180° when the phase shift of each section is –60°, and this occurs when $\omega = 2\pi f = 1.732/RC$ because the tangent 60° = 1.73. The magnitude of β at this point is $(1/2)^3$, so the gain, A, must be equal to 8 for the system gain to be equal to 1.

5.1.2 Wien Bridge Oscillator:

Figure 5. 3 give the Wien-bridge circuit configuration. The loop is broken at the positive input, and the return signal is calculated in Equation 2 below.

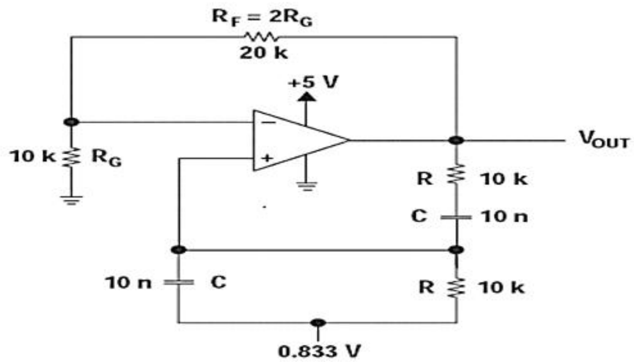


Fig.5.4 Wien Bridge Oscillator

$$\frac{V_{RETURN}}{V_{OUT}} = \frac{\dfrac{R}{RCs+1}}{\dfrac{R}{RCs+1} + R + \dfrac{1}{Cs}} = \frac{1}{3 + RCs + \dfrac{1}{RCs}} = \frac{1}{3 + j\left(RC\omega - \dfrac{1}{RC\omega}\right)}. \quad (2)$$

where $s = j\omega$ and $j = \sqrt{1}$.

When $\omega = 2\pi f = 1/RC$, the feedback is in phase (this is positive feedback), and the gain is 1/3, so oscillation requires an amplifier with a gain of 3. When RF = 2RG, the amplifier gain is 3 and oscillation occurs at $f = 1/2\pi RC$. The circuit oscillated at 1.65 kHz rather than 1.59 kHz with the component values shown in Figure 3, but the distortion is noticeable.

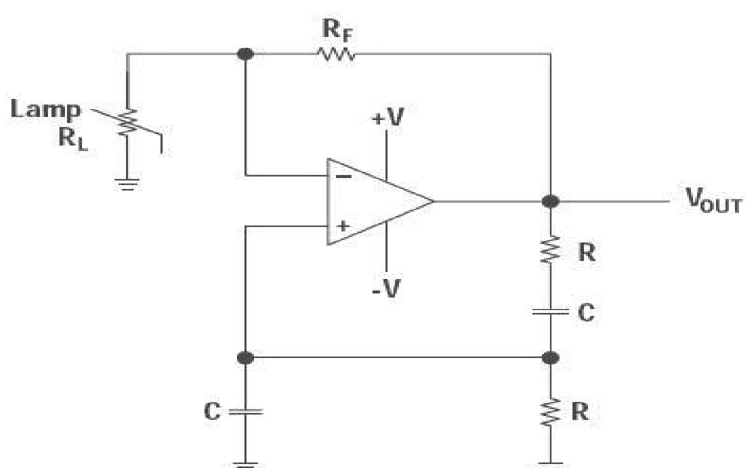


Fig.5.5 Wien Bridge Circuit Schematic with non-linear feedback

Figure 4 shows a Wien-bridge circuit with non-linear feedback. The lamp resistance, RL, is nominally selected as half the feedback resistance, RF, at the lamp current established by RF and RL. The non-linear relationship between the lamp current and resistance keeps output voltage changes small.

If a voltage source is applied directly to the input of an **ideal** amplifier with feedback, the input current will be:

$$i_{in} = \frac{v_{in} - v_{out}}{Z_f}$$

Where *vin* is the input voltage, *vout* is the output voltage, and *Zf* is the feedback impedance. If the voltage gain of the amplifier is defined as:

$$A_v = \frac{v_{out}}{v_{in}}$$

And the input admittance is defined as:

$$Y_i = \frac{i_{in}}{v_{in}}$$

Input admittance can be rewritten as:

$$Y_i = \frac{1 - A_v}{Z_f}$$

For the Wien Bridge, *Zf* is given by:

$$Z_f = R + \frac{1}{j\omega C}$$

$$Y_i = \frac{(1 - A_v)(\omega^2 C^2 R + j\omega C)}{1 + (\omega C R)^2}$$

If Av is greater than 1, the input admittance is a negative resistance in parallel with an inductance. The inductance is:

$$L_{in} = \frac{\omega^2 C^2 R^2 + 1}{\omega^2 C (A_v - 1)}$$

If a capacitor with the same value of C is placed in parallel with the input, the circuit has a natural resonance at:

$$\omega = \frac{1}{\sqrt{L_{in} C}}$$

Substituting and solving for inductance yields:

$$L_{in} = \frac{R^2 C}{A_v - 2}$$

If Av is chosen to be 3: $Lin = R^2 C$

Substituting this value yields:

$$\omega = \frac{1}{RC} \quad \text{Or} \quad f = \frac{1}{2\pi RC}$$

Similarly, the input resistance at the frequency above is: $R_{in} = \frac{-2R}{A_v - 1}$

For $Av = 3$: $Rin = -R$

If a resistor is placed in parallel with the amplifier input, it will cancel some of the negative resistance. If the net resistance is negative, amplitude will grow until clipping occurs.

Similarly, if the net resistance is positive, oscillation amplitude will decay. If a resistance

is added in parallel with exactly the value of *R*, the net resistance will be infinite and the circuit can sustain stable oscillation at any amplitude allowed by the amplifier.

Increasing the gain makes the net resistance more negative, which increases amplitude. If gain is reduced to exactly 3 when suitable amplitude is reached, stable, low distortion oscillations will result.

Amplitude stabilization circuits typically increase gain until suitable output amplitude is reached. As long as *R*, *C*, and the amplifier are linear, distortion will be minimal.

5.2 Multivibrators

5.2.1 Astable Multivibrator

The two states of circuit are only stable for a limited time and the circuit switches between them with the output alternating between positive and negative saturation values.

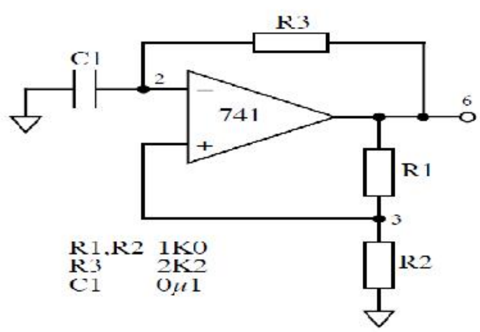


Fig. 5.6 Astable multivibrator circuit

Analysis of this circuit starts with the assumption that at time t=0 the output has just switched to state 1, and the transition would have occurred.

An op-amp Astable multivibrator is also called as free running oscillator. The basic principle of generation of square wave is to force an op-amp to operate in the saturation region (±Vsat).

A fraction β =R2/(R1+R2) of the output is feedback to the positive input terminal of op-amp. The charge in the capacitor increases & decreases upto a threshold value called ±βVsat. The charge in the capacitor triggers the op-amp to stay either at +Vsat or –Vsat.

Asymmetrical square wave can also be generated with the help of Zener diodes. Astable multi vibrator do not require a external trigger pulse for its operation & output toggles from one state to another and does not contain a stable state.

Astable multi vibrator is mainly used in timing applications & waveforms generators.

Design

1. The expression of fo is obtained from the charging period t1 & t2 of capacitor as

$$T=2RC\ln (R_1+2R_2)/R_1$$

2. To simplify the above expression, the value of R1 & R2 should be taken as R2 = 1.16R Such that fo simplifies to fo =1/2RC.
3. Assume the value of R1 and find R2.

4. Assume the value of C & Determine R from fo =1/2 RC
5. Calculate the threshold point ~~from βV~~SATl = R1lVTl/ R1 -R2 l/βVSATl w h e r e β is the feedback ratio.

5.2.2 Monostable Multivibrator using Op-amp: circuit diagram:

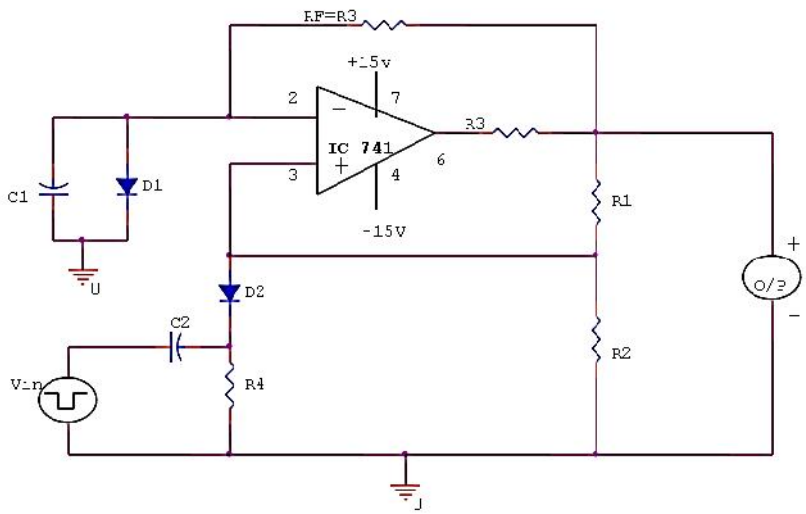


Fig.5.7 Mono stable Multi vibrator using Op-amp

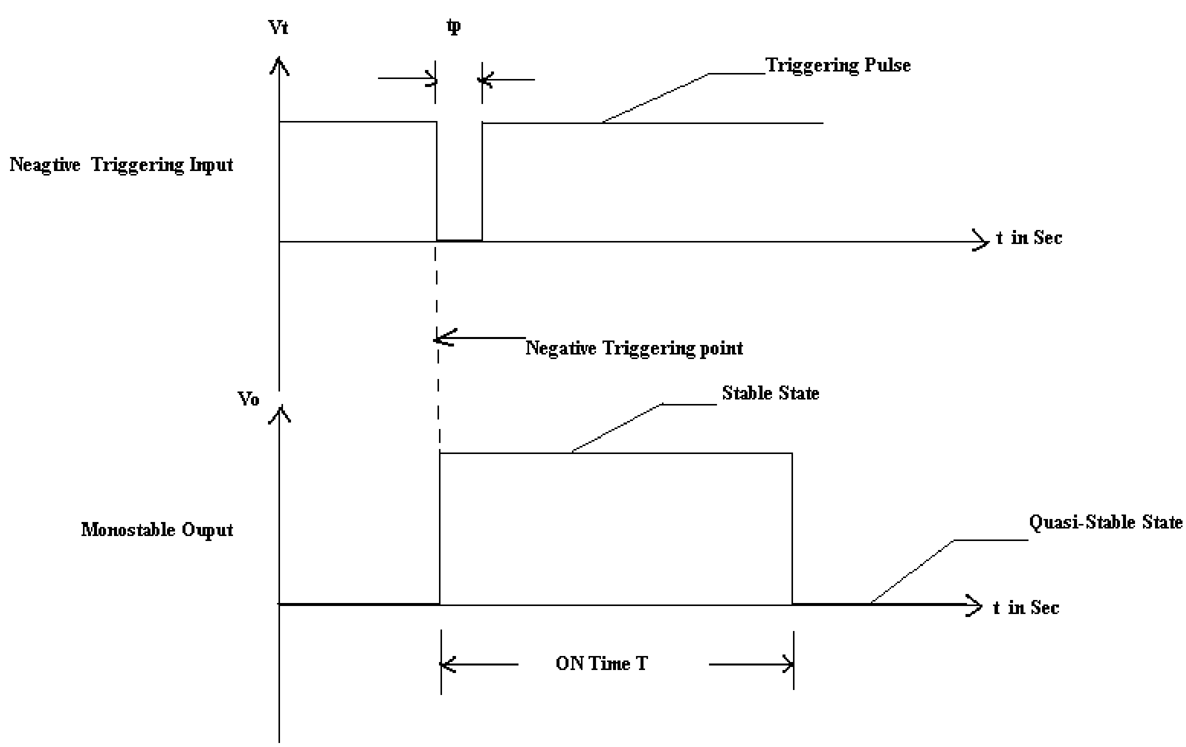


Fig.5.8 Input Output Waveform:

A multivibrator which has only one stable and the other is quasi stable state is called as Monostable multivibrator or one-short multivibrator. This circuit is useful for generating signal output pulse of adjustable time duration in response to a triggering signal. The width of the output pulse depends only on the external components connected to the op-amp. Usually a negative trigger pulse is given to make the output switch to other state. But, it then return to its stable state after a time interval determining by circuit components. The pulse width T can be given as T = 0.69RC. For Monostable operation the triggering pulse width Tp should be less then T, the pulse width of Monostable multivibrator. This circuit is also called as time delay circuit or gating circuit.

Design:

1. Calculating β from expression

$$\beta = \frac{R1}{R1 + R2}$$

2. The value of R & C from the pulse width time expression.

$$T = RC \ln \frac{(1 + V_D / V_{sat})}{1 - \beta}$$

$$T = RC \ln \frac{(1 + V_D / V_{sat})}{0.5}$$

$$T \equiv 0.69RC.$$

3. Triggering pulse width T_p must be much smaller then T. $T_p < T$.

5.2.3 Triangular Wave Generator Circuit:

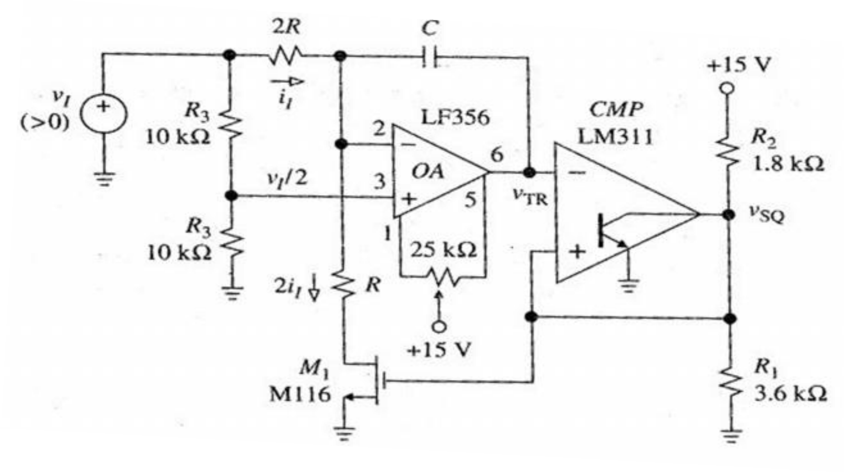


Fig. 5.9 Circuit diagram of Triangular waveform generator

This signal generator gives two waveforms: a triangle-wave and a square- wave. The central component of this circuit is the integrator capacitor CI. Basically we are interested in performing two functions on CI: *charge it, discharge it - repeat indefinitely.* The output waveforms are shown here and it is apparent that a square wave generator followed by an integrator acts as a triangular wave generator.

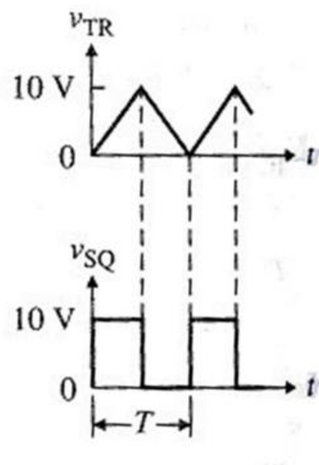


Fig.5.10 Output waveforms from generator

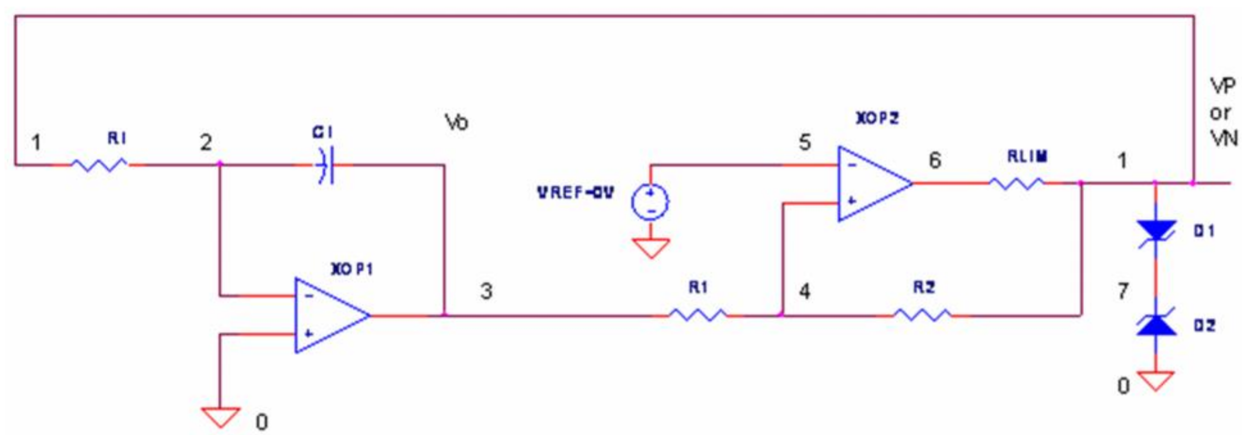


Fig. 5.11 Basic triangular waveform generator

The triangle peaks and period may not accurately meet +/-10V swing at 100 us. The main reason is that current source and thresholds are derived from Zener diodes - not exactly the most accurate reference.

5.2.4 Linear Ramp Generator

A triangle wave implies that the circuit generates a linear voltage ramp. One way to achieve this goal is by charging discharging CI with a constant current. The Op Amp Integrator provides for this.

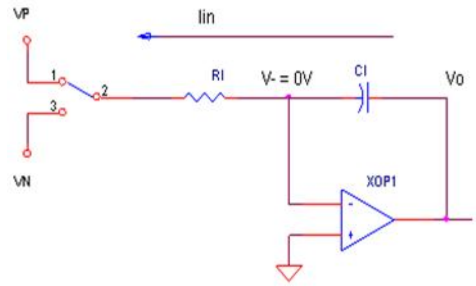


Fig. 5.12 Linear Ramp Generator

Ramp Up

Connect R_I to V_N and With V- held at the virtual ground (0V), a constant current flows from V- to V_N.

$$I_{in} = V_N / R_I.$$

CI integrates Iin creating a positive linear ramp at Vo. The ramp is linear because Vo changes proportionally to the time elapsed ΔT.

$$\Delta V_o = - V_N / (C_I \cdot R_I) \cdot \Delta T$$

Ramp Down Connect RI to V_P and constant current flows from V_P to V-,

$$Iin = - V_P / R_I.$$

Now Vo ramps down linearly $\Delta Vo = - V_P / (C_I \cdot R_I) \cdot \Delta T$

$$\text{Ramp Up:} \Delta V_o/\Delta T = -V_N/(C_I R_I)$$

$$\text{Ramp Down:} \Delta Vo / \Delta T = - V_P / (C_I \cdot R_I)$$

These equations show the parameters available to control the ramp up / down speeds. Asymmetrical voltage swings are got by including a reference voltage VREF to the comparator's negative input.

$$V_{th}+ = V_{REF} \cdot (R_1+R_2)/R_2 - V_N \cdot R_1/R_2$$

$$V_{th}- = V_{REF} \cdot (R_1+R_2)/R_2 - V_P \cdot R_1 / R_2$$

Upper and Lower Bounds

When do we switch from charging to discharging CI? Basically, there is a need to pick two levels - *an upper and a lower threshold* - to define the bounds of the triangle wave. The circuit ramps up or down, reversing at the upper and lower thresholds.

- With one leg of RI at VN, the output **ramps up** until the **Upper Threshold (Vth+)** is reached. Then RI is switched from VN to VP.

- With one leg of RI at VP, the output **ramps down** until the **Lower Threshold (Vth-)** is reached. Then RI is switched from VP to VN.

Comparator:

An Op Amp Comparator with two thresholds. Produce circuit changes in output state from VN to VP (or vice-versa) depending on the upper Vth+ and lower Vth- thresholds.

$$Vth+ = -V_N \cdot R_1/R_2$$

$$Vth- = -V_P \cdot R_1 / R_2$$

Comparator Working:

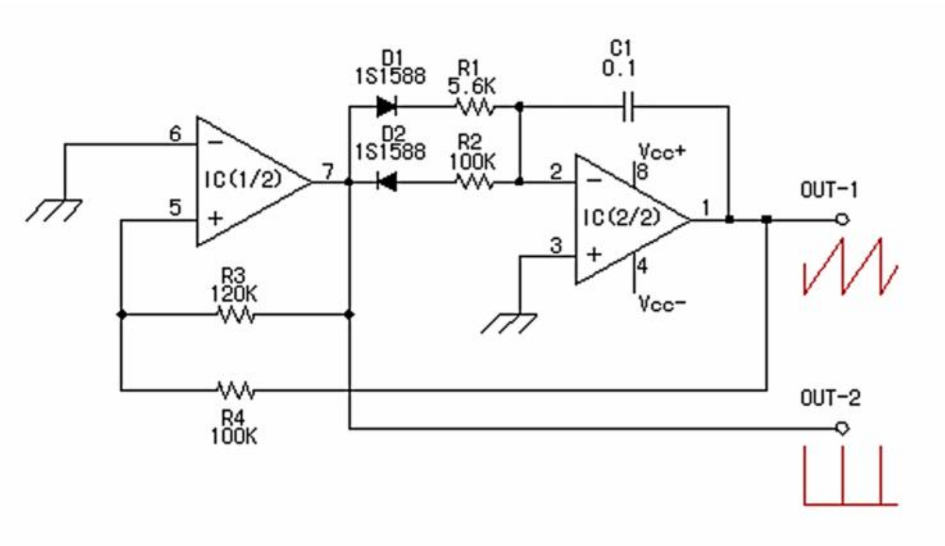
- When Vin > Vth+, the output switches to VP, the POSITIVE output state.
- When Vin < Vth-, the output switches to VN, the NEGATIVE output state.

Zener diodes D1 and D2 set the positive and negative output levels:

$$V_P = V_{fD1} + V_{ZD2} \quad V_N = V_{fD2} + V_{ZD1}.$$

These output levels do double duty - they set the comparator thresholds, and set the voltage levels for the next stage - the integrator.

5.3 Saw-Tooth Wave Generator



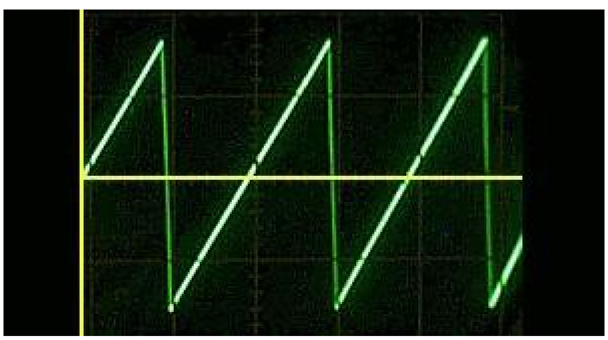


Fig. 5.13 Saw-Tooth Wave Generator Circuit Diagram and output waveform

The saw tooth wave oscillator which used the operational amplifier. The composition of this circuit is the same as the triangular wave oscillator basically and is using two operational amplifiers.

At the circuit diagram above, IC(1/2) is the Schmitt circuit and IC(2/2) is the Integration circuit. The difference with the triangular wave oscillator is to be changing the time of the charging and the discharging of the capacitor. When the output of IC (1/2) is positive voltage, it charges rapidly by the small resistance (R1) value.

(When the integration output voltage falls) When the output of IC(1/2) is negative voltage, it is made to charge gradually at the big resistance(R2) value. The output waveform of the integration circuit becomes a form like the tooth of the saw. Such voltage is used for the control of the electron beam (the scanning line) of the television,

When picturing a picture at the cathode-ray tube, an electron beam is moved comparative slow. (When the electron beam moves from the left to the right on the screen). When turning back, it is rapidly moved.(When moving from the right to the left).

Like the triangular wave oscillator, the line voltage needs both of the positive power supply and the negative power supply. Also, to work in the oscillation, the condition of R3>R4 is necessary. However, when making the value of R4 small compared with R3, the output voltage becomes small. The near value is good for R3 and R4

The oscillation frequency can be calculated by the following formula.

$$f = \frac{1}{2C\,(R_1+R_2)}\left(\frac{R_3}{R_4}\right)$$

With the circuit diagram,the oscillation frequency is as follows.

$$f = (1/2C\,(R1+R2))*(R3/R4)$$
$$= (1/(2x0.1x10^{-6}x(5.6x10^{3}+100x10^{3}))x(120x10^{3}/100x10^{3})$$
$$= (1/(21.12x10^{-3}))x1.2$$
$$= 56.8\ Hz$$

5.4 Function Generator IC 8038:

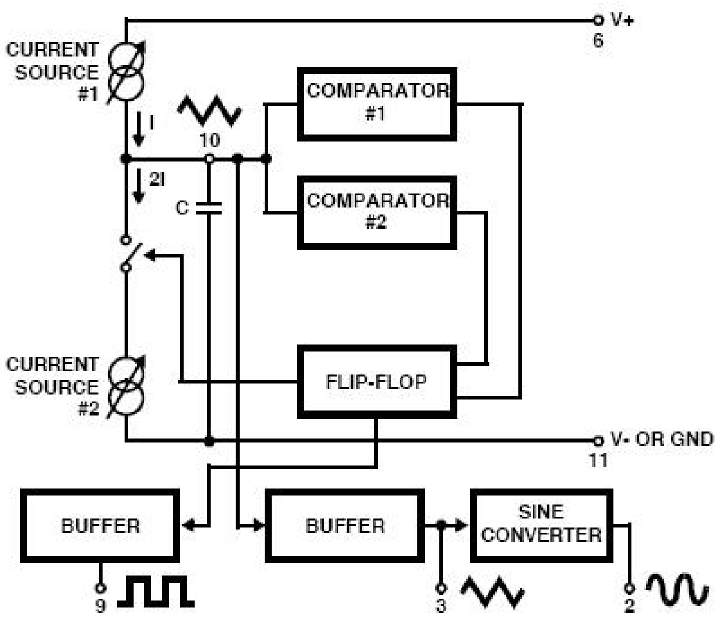


Fig.5.14 Functional block diagram of Function generator

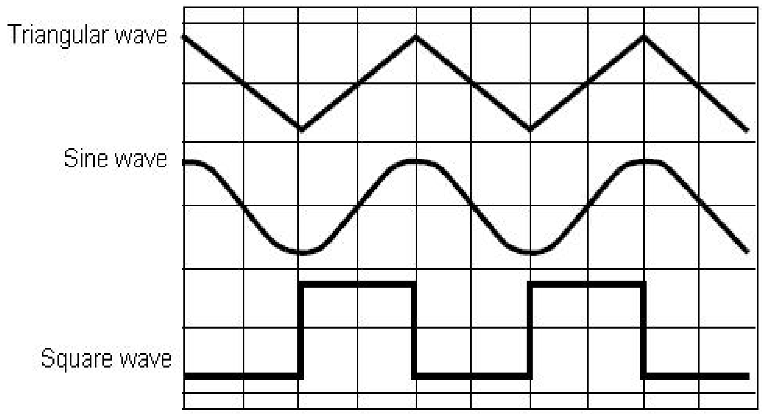


Fig. 5.15 Output Waveforms from Function Generator IC 8038

It consists of two current sources, two comparators, two buffers, one FF and a sine wave converter.

Pin description:

- Pin 1 & Pin 12: Sine wave adjusts:

  The distortion in the sine wave output can be reduced by adjusting the 100KΩ pots connected between pin12 & pin11 and between pin 1 & 6.
- Pin 2 Sine Wave Output:

  Sine wave output is available at this pin. The amplitude of this sine wave is 0.22 Vcc. Where $\pm$ 5V $\leq$ Vcc $\leq$ $\pm$ 15 V.

- Pin 3 Triangular Wave output:

  Triangular wave is available at this pin. The amplitude of the triangular wave is 0.33Vcc. Where $\pm$ 5V $\leq$ Vcc $\leq$ $\pm$ 15 V.

- Pin 4 & Pin 5 Duty cycle / Frequency adjust:

  The symmetry of all the output wave forms & 50% duty cycle for the square wave output is adjusted by the external resistors connected from Vcc to pin 4. These external resistors & capacitors at pin 10 will decide the frequency of the output wave forms.
- Pin 6 + Vcc:

  Positive supply voltage the value of which is between 10 & 30V is applied to this pin.
- Pin 7 : FM Bias:

  This pin along with pin no8 is used to TEST the IC 8038.
- Pin9 : Square Wave Output:

  A square wave output is available at this pin. It is an open collector output so that this pin can be connected through the load to different power supply voltages. This arrangement is very useful in making the square wave output.
- Pin 10 : Timing Capacitors:

  The external capacitor C connected to this pin will decide the output frequency along with the resistors connected to pin 4 & 5.
- Pin 11 : -VEE or Ground:

  If a single polarity supply is to be used then this pin is connected to supply ground & if ($\pm$) supply voltages are to be used then (-) supply is connected to this pin.
- Pin 13 & Pin 14: NC (No Connection)

Important features of IC 8038:

1. All the outputs are simultaneously available.
2. Frequency range : 0.001Hz to 500kHz
3. Low distortion in the output wave forms.
4. Low frequency drifts due to change in temperature.
5. Easy to use.

Parameters:

(i) Frequency of the output wave form:

The output frequency dependent on the values of resistors R1 & R2 along with the external capacitor C connected at pin 10.

If RA= RB = R & if RC is adjusted for 50% duty cycle then f_0= 0.3/RC; RA = R1, RB = R3, RC= R2.

(ii) Duty cycle / Frequency Adjust : (Pin 4 & 5):

Duty cycle as well as the frequency of the output wave form can be adjusted by external resistors at pin 4 & 5.

The values of resistors RA & RB connected between Vcc pin 4 & 5 respectively along with the capacitor connected at pin 10 decide the frequency of the wave form. The values of RA & RB should be in the range of 1kΩ to 1MΩ.

(iii)FM Bias:

- The FM Bias input (pin7) corresponds to the junction of resistors R1 & R2.
- The voltage Vin is the voltage between Vcc & pin8 and it decides the output frequency.
- The output frequency is proportional to Vin as given by the following expression. For RA = RB (50% duty cycle).
- f_0= 5 $V_{in}/CRAV_{cc}$; where C is the timing capacitor.
- With pin 7 & 8 connected to each other the output frequency is given by f_0= 0.3/RC where R = RA = RB for 50% duty cycle.
- This is because M Sweep input (pin 8):
- Vin=R1Vcc/R1+R2
- This input should be connected to pin 7, if we want a constant output frequency.
- But if the output frequency is supposed to vary, then a variable dc voltage should be applied to this pin.
- The voltage between Vcc & pin 8 is called Vin and it decides the output frequency as, $f_0=1.5\ V_{in}/CR_AV_{CC}$

A potentiometer can be connected to this pin to obtain the required variable voltage required to change the output frequency.

5.5 The 555 Timer IC

The 555 is a monolithic timing circuit that can produce accurate & highly stable time delays or oscillation. The timer basically operates in one of two modes: either

(i) Monostable (one - shot) multivibrator or
(ii) Astable (free running) multivibrator

The important features of the 555 timer are these:

(i) It operates on +5v to +18 v supply voltages
(ii) It has an adjustable duty cycle
(iii) Timing is from microseconds to hours
(iv) It has a current o/p

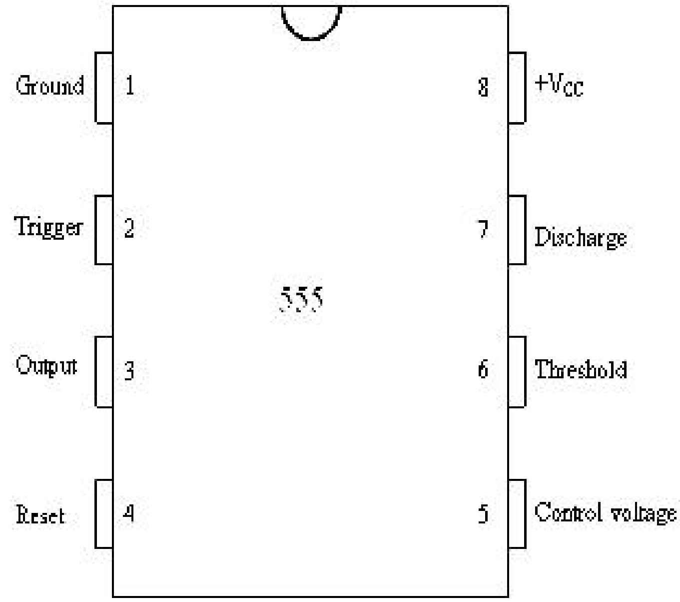


Fig. 5.16 Pin configuration of 555 timer

Pin description:

Pin 1: Ground**:**

All voltages are measured with respect to this terminal.

Pin 2: Trigger:

The o/p of the timer depends on the amplitude of the external trigger pulse applied to this pin.

Pin 3: Output:

There are 2 ways a load can be connected to the o/p terminal either between pin3 & ground or between pin 3 & supply voltage

(Between Pin 3 & Ground ON load) (Between Pin 3 & + Vcc OFF load)

(i) When the input is low:

The load current flows through the load connected between Pin 3 & +Vcc in to the output terminal & is called the sink current.

(ii) When the output is high:

The current through the load connected between Pin 3 & +Vcc (i.e. ON load) is zero. However the output terminal supplies current to the normally OFF load. This current is called the source current.

Pin 4: Reset:

The 555 timer can be reset (disabled) by applying a negative pulse to this pin. When the reset function is not in use, the reset terminal should be connected to +Vcc to avoid any false triggering.

Pin 5: Control voltage:

An external voltage applied to this terminal changes the threshold as well as trigger voltage. In other words by connecting a potentiometer between this pin & GND, the pulse width of the output waveform can be varied. When not used, the control pin should be bypassed to ground with 0.01 capacitor to prevent any noise problems.

Pin 6: Threshold:

This is the non inverting input terminal of upper comparator which monitors the voltage across the external capacitor.

Pin 7: Discharge:

This pin is connected internally to the collector of transistor Q1.

When the output is high Q1 is OFF.

When the output is low Q is (saturated) ON.

Pin 8: +Vcc:

The supply voltage of +5V to +18V is applied to this pin with respect to ground.

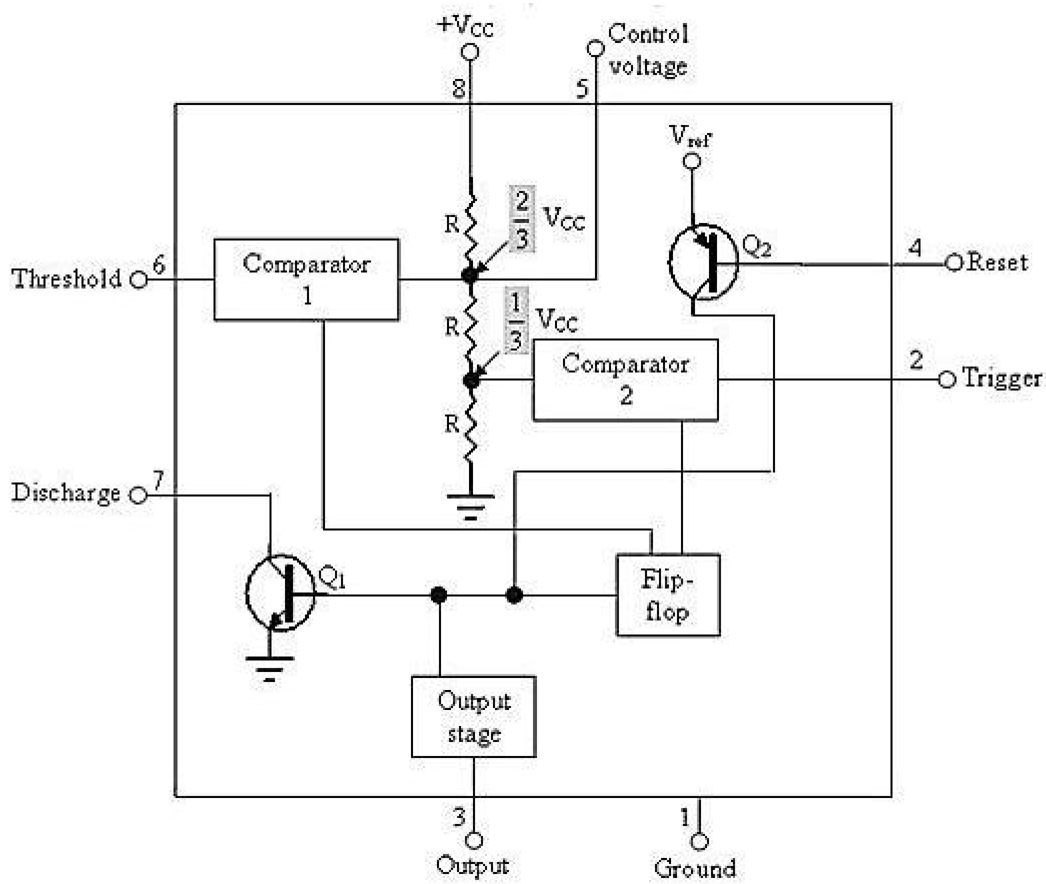


Fig.5.17 Block **Diagram of 555 Timer IC**

From the above figure, three 5k internal resistors act as voltage divider providing bias voltage of 2/3 Vcc to the upper comparator & 1/3 Vcc to the lower comparator. It is possible to vary time electronically by applying a modulation voltage to the control voltage input terminal (5).

(i) In the Stable state:

The output of the control FF is high. This means that the output is low because of power amplifier which is basically an inverter. Q = 1; Output = 0

(ii) At the Negative going trigger pulse:

The trigger passes through (Vcc/3) the output of the lower comparator goes high & sets the FF. Q = 1; Q = 0

(iii) At the Positive going trigger pulse: It passes through 2/3Vcc, the output of the upper comparator goes high and resets the FF. Q = 0; Q = 1

The reset input (pin 4) provides a mechanism to reset the FF in a manner which overrides the effect of any instruction coming to FF from lower comparator.

- **Monostable Operation:**

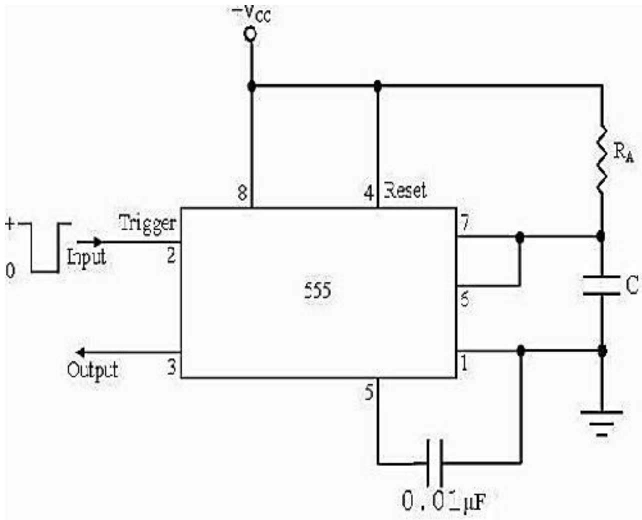


Fig. 5.18 555 connected as a Monostable Multivibrator

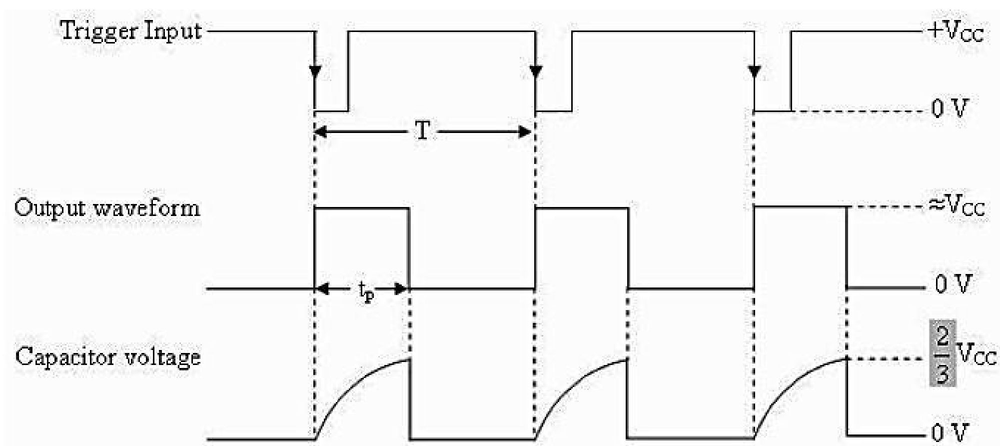


Fig. 5.19 Waveforms of monostable multivibrators

Initially when the output is low, i.e. the circuit is in a stable state, transistor Q1 is ON & capacitor C is shorted to ground. The output remains low. During negative going trigger pulse, transistor Q1 is OFF, which releases the short circuit across the external capacitor C & drives the output high. Now the capacitor C starts charging toward Vcc through RA. When the voltage across the capacitor equals 2/3 Vcc, upper comparator switches from low to high. i.e. Q = 0, the transistor Q1 = OFF ; the output is high.

Since C is unclamped, voltage across it rises exponentially through R towards Vcc with a time constant RC (fig b) as shown in below. After the time period, the upper comparator resets the FF,

i.e. Q = 1, Q1 = ON; the output is low.[i.e discharging the capacitor C to ground potential (fig c)]. The voltage across the capacitor as in fig (b) is given by

$$V_c = V_{cc}\,(1-e^{-t/RC}) \ldots\ldots (1)$$

Therefore At $t = T$, $V_c = 2/3\ V_{cc}$

$$2/3\ V_{cc} = V_{cc}(1-e^{-T/RC})$$

or

$$T = RC \ln (1/3)$$

Or

$$T = 1.1RC \text{ seconds} \qquad \ldots\ldots\ldots\ldots\ldots (2)$$

If the reset is applied Q2 = OFF, Q1 = ON, timing capacitor C immediately discharged. The output now will be as in figure (d & e). If the reset is released output will still remain low until a negative going trigger pulse is again applied at pin 2.

Applications of Monostable Mode of Operation:

(a) Frequency Divider:

The 555 timer as a monostable mode. It can be used as a frequency divider by adjusting the length of the timing cycle tp with respect to the time period T of the trigger input. To use the monostable multivibrator as a divide by 2 circuit, the timing interval tp must be a larger than the time period of the trigger input. [Divide by 2, □tp > T of the trigger]

By the same concept, to use the monostable multivibrator as a divide by 3 circuit, tp must be slightly larger than twice the period of the input trigger signal & so on, [divide by 3 tp > 2T oftrigger]

(b) Pulse width modulation:

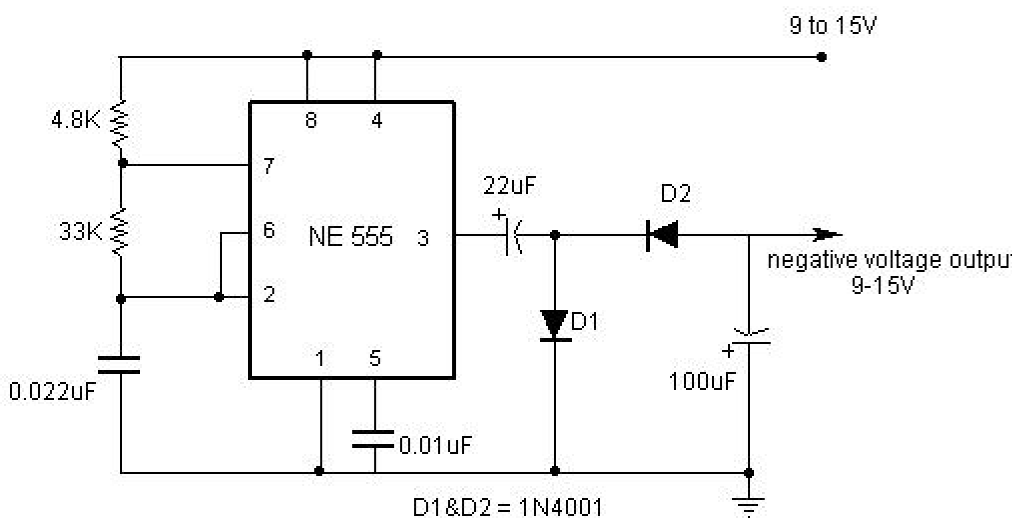


Fig.5.20 Pulse Width Modulation

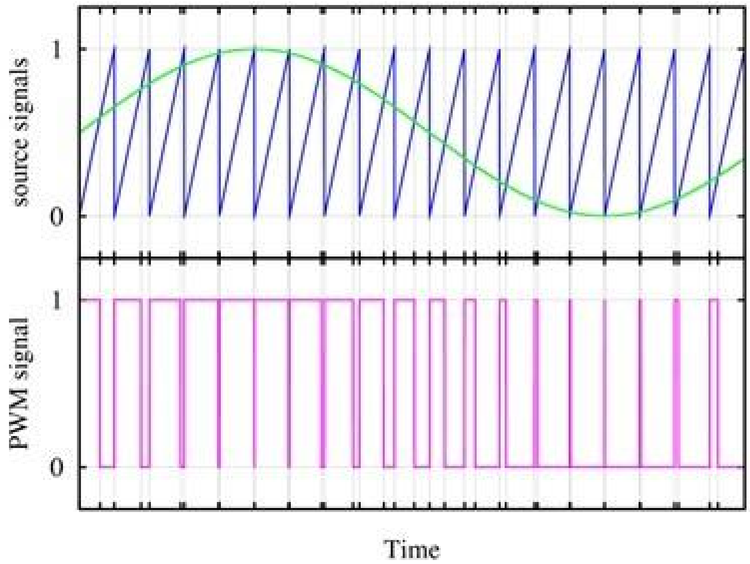


Fig. 5.21 Output Waveform

Pulse width of a carrier wave changes in accordance with the value of a incoming (modulating signal) is known as PWM. It is basically monostable multivibrator. A modulating signal is fed in to the control voltage (pin 5). Internally, the control voltage is adjusted to 2/3 Vcc externally applied modulating signal changes the control voltage level of upper comparator. As a result, the required to change the capacitor up to threshold voltage level changes, giving PWM output.

(c) Pulse Stretcher:

This application makes use of the fact that the output pulse width (timing interval) of the monostable multivibrator is of longer duration than the negative pulse width of the input trigger. As such, the output pulse width of the monostable multivibrator can be viewed as a stretched version of the narrow input pulse, hence the name "Pulse stretcher".

Often, narrow –pulse width signals are not suitable for driving an LED display, mainly because of their very narrow pulse widths. In other words, the LED may be flashing but not be visible to the eye because its on time is infinitesimally small compared to its off time. The 55 pulse stretcher can be used to remedy this problem. The LED will be ON during the timing interval tp = 1.1RAC which can be varied by changing the value of RA & C.

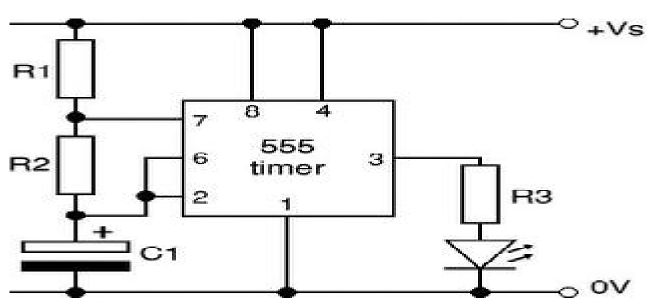


Fig.5.23 Pulse Stretcher

The 555 timer as an Astable Multivibrator:

An Astable multivibrator, often called a free running multivibrator, is a rectangular wave generating circuit. Unlike the monostable multivibrator, this circuit does not require an external trigger to change the state of the output, hence the name free running. However, the time during which the output is either high or low is determined by 2 resistors and capacitors, which are externally connected to the 55 timer.

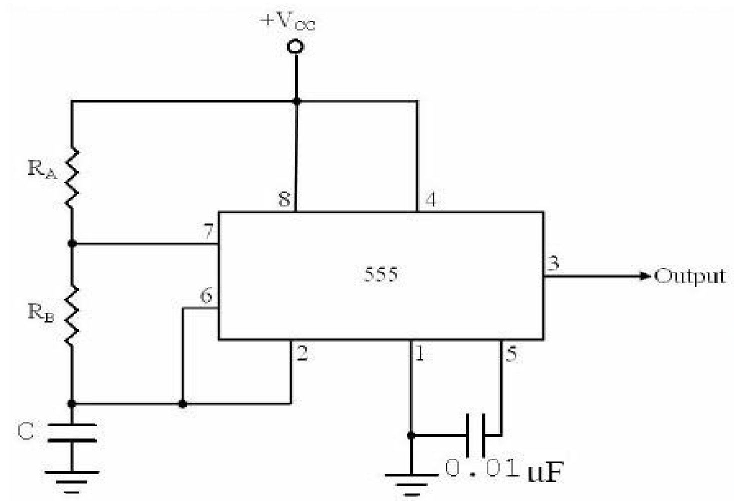


Fig.5.24 Astable Multivibrator

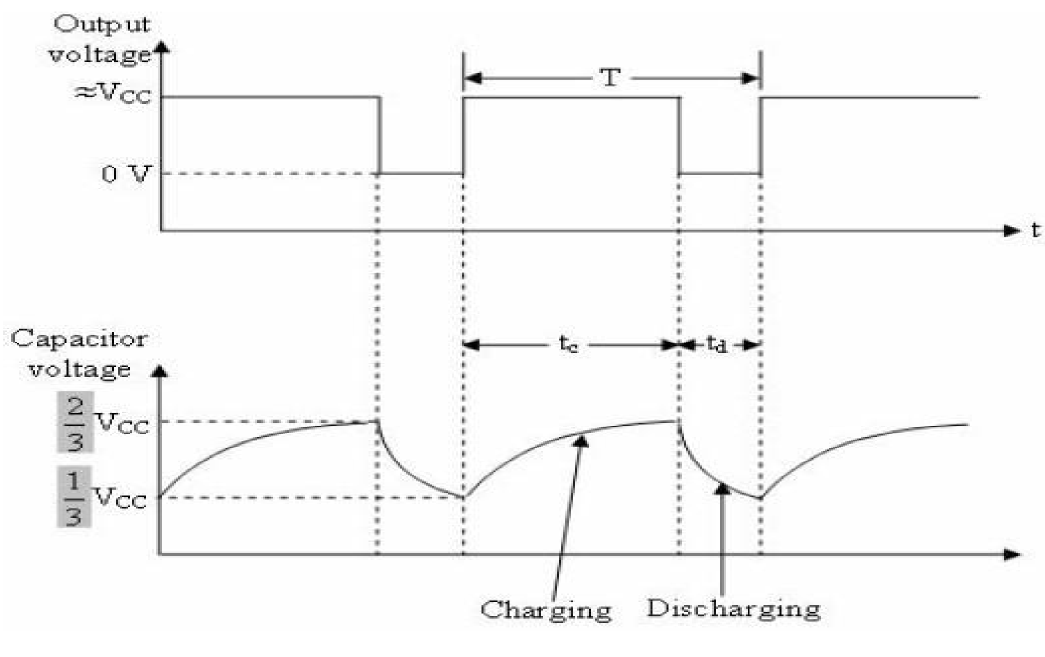


Fig. 5.25 Waveforms of Astable multivibrator

The above figures show the 555 timer connected as an astable multivibrator and its model graph

Initially, when the output is high :

Capacitor C starts charging toward Vcc through RA & RB. However, as soon as voltage across the capacitor equals 2/3 Vcc. Upper comparator triggers the FF & output switches low.

When the output becomes Low:

Capacitor C starts discharging through RB and transistor Q1, when the voltage across C equals 1/3 Vcc, lower comparator output triggers the FF & the output goes high. Then cycle repeats. The capacitor is periodically charged & discharged between 2/3 Vcc & 1/3 Vcc respectively. The time during which the capacitor charges from 1/3 Vcc to 2/3 Vcc equal to the time the output is high & is given by

$$t_c = (R_A+R_B)C \ln 2 \ldots\ldots\ldots\ldots\ldots(1) \text{ Where } [\ln 2 = 0.69]$$
$$= 0.69\,(R_A+R_B)\,C$$

Where RA & RB are in ohms. And C is in farads.

Similarly, the time during which the capacitors discharges from 2/3 Vcc to 1/3 Vcc is equal to the time, the output is low and is given by,

$$t_c = R_B\, C \ln 2$$
$$t_d = 0.69\, R_B\, C \ldots\ldots\ldots\ldots\ldots\ldots\ldots..(2)$$ where RB is in ohms and C is in farads.

Thus the total period of the output waveform is

$$T = t_c + t_d = 0.69\,(R_A+2R_B)\,C \ldots\ldots\ldots\ldots\ldots.(3)$$

This, in turn, gives the frequency of oscillation as, $f_0 = 1/T = 1.45/(R_A+2R_B)C \ldots\ldots\ldots(4)$

Equation 4 indicates that the frequency f_0 is independent of the supply voltage Vcc. Often the term duty cycle is used in conjunction with the astable multivibrator. The duty cycle is the ratio of the time tc during which the output is high to the total time period T. It is generally expressed as a percentage.

$$\%\text{ duty cycle} = (t_c / T) * 100$$
$$\%\, DC = [(R_A+R_B)/ /(R_A+2R_B)] * 100$$

Astable Multivibrator Applications:

(a) Square wave oscillator:

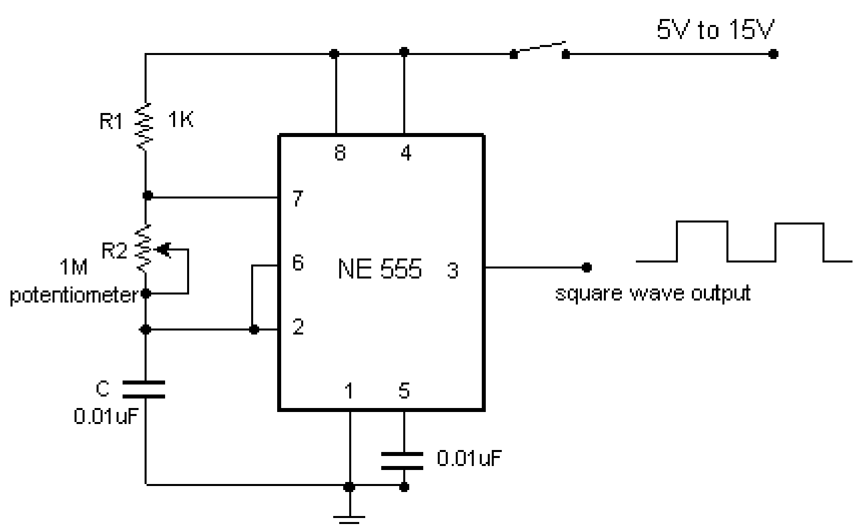
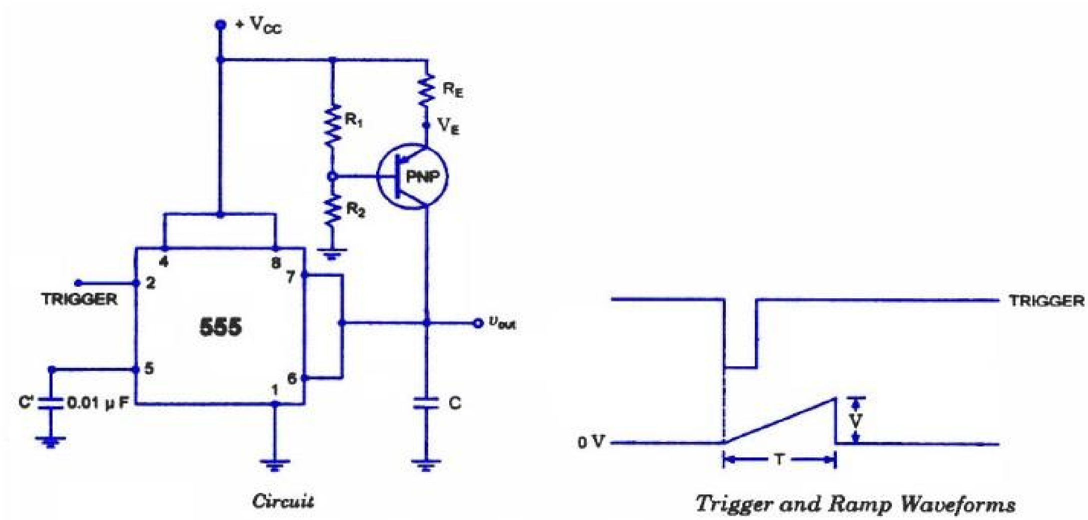


Fig.5.26 Square Wave Oscillator

Without reducing RA = 0 ohm, the astable multivibrator can be used to produce square wave output. Simply by connecting diode D across Resistor RB. The capacitor C charges through RA & diode D to approximately
2/3Vcc & discharges through RB & Q1 until the capacitor voltage equals approximately 1/3Vcc, then the cycle repeats.
To obtain a square wave output, RA must be a combination of a fixed resistor & potentiometer so that the potentiometer can be adjusted for the exact square wave.

(b) Free – running Ramp generator:



Ramp Generator Using The Timer 555

- The astable multivibrator can be used as a free – running ramp generator when resistor RA
  & RB is replaced by a current mirror.
- The current mirror starts charging capacitor C toward Vcc at a constant rate.
- When voltage across **C equals to 2/3 Vcc**, upper comparator turns transistor Q1 ON & C rapidly discharges through transistor Q1.
- When voltage across **C equals to 1/3 Vcc**, lower comparator switches transistor OFF & then capacitor C starts charging up again.
- Thus the charge – discharge cycle keeps repeating.
- The discharging time of the capacitor is relatively negligible compared to its charging time.
- The time period of the ramp waveform is equal to the charging time & is approximately is given by,

  T = VccC/3IC

  (1) IC = (Vcc - VBE)/R = constant current

Therefore the free – running frequency of ramp generator is

f0 = 3IC/ Vcc C

5.6 Linear Regulators

- All electronic circuits need a dc power supply for their operation. To obtain this dc voltage from 230 V ac mains supply, we need to use rectifier.
- Therefore the filters are used to obtain a “steady” dc voltage from the pulsating one.
- The filtered dc voltage is then applied to a regulator which will try to keep the dc output voltage constant in the event of voltage fluctuations or load variation.
  The combination of rectifier & filter can produce a dc voltage. But the problem with this type of dc power supply is that its output voltage will not remain constant in the event of fluctuations in an AC input or changes in the load current(IL).
- The output of unregulated power supply is connected at the input of voltage regulator circuit.
- The voltage regulator is a specially designed circuit to keep the output voltage constant. It does not remain exactly constant. It changes slightly due to changes in certain parameters.

Factors affecting the output voltage:

i) IL (Load Current)
ii) VIN (Input Voltage)
iii) T (Temperature)

IC Voltage Regulators:

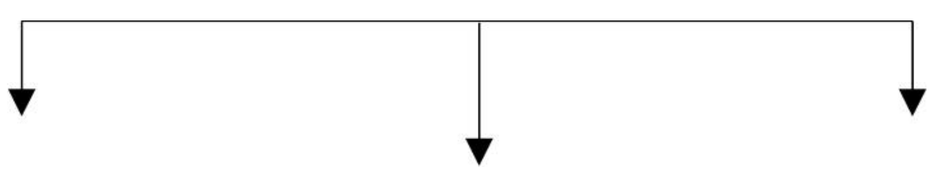
They are basically series regulators.

✓ Important features of IC Regulators:

1. Programmable output
2. Facility to boost the voltage/current
3. Internally provided short circuit current limiting
4. Thermal shutdown
5. Floating operation to facilitate higher voltage output

✓ Classifications of IC voltage regulators:

IC Voltage Regulator



Fixed Volt Reg. Positive/negative | Adjustable O/P Volt Reg | Switching Reg

- Fixed & Adjustable output Voltage Regulators are known as Linear Regulator.
- A series pass transistor is used and it operates always in its active region.

Switching Regulator:

1. Series Pass Transistor acts as a switch.
2. The amount of power dissipation in it decreases considerably.
3. Power saving result is higher efficiency compared to that of linear.

Adjustable Voltage Regulator:

Advantages of Adjustable Voltage Regulator over fixed voltage regulator are,

1. Adjustable output voltage from 1.2v to 57 v
2. Output current 0.10 to 1.5 A
3. Better load & line regulation
4. Improved overload protection
5. Improved reliability under the 100% thermal overloading

Adjustable Positive Voltage Regulator (LM317):

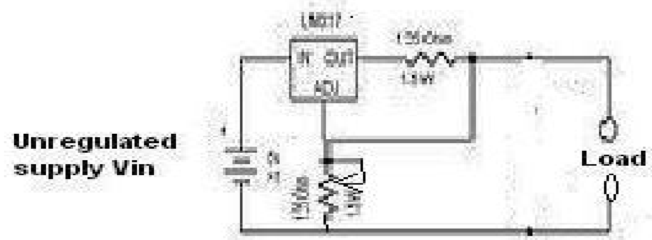


Fig. 5.27 Circuit diagram of LM317 regulator

- LM317 series adjustable 3 terminal positive voltage regulator, the three terminals are Vin, Vout & adjustment (ADJ).
- LM317 requires only 2 external resistors to set the output voltage.
- LM317 produces a voltage of 1.25v between its output & adjustment terminals. This voltage is called as Vref.
- Vref (Reference Voltage) is a constant, hence current I1 flows through R1 will also be constant. Because resistor R1 sets current I1. It is called "current set" or "program resistor".
- Resistor R2 is called as "Output set" resistors, hence current through this resistor is the sum of I1 & Iadj
- LM317 is designed in such as that Iadj is very small & constant with changes in line voltage & load current.
- The output voltage Vo is, $V_o=R_1I_1+(I_1+I_{adj})R_2$ ------------- (1)

Where $I_1= V_{ref}/R_1$

$$V_o = (V_{ref}/R_1)\, R_1 + V_{ref}/R_1 + I_{adj}\, R_2$$
$$= V_{ref} + (V_{ref}/R_1)\, R_2 + I_{adj}\, R_2$$
$$V_o = V_{ref}\,[1 + R_2/R_1] + I_{adj}\, R_2 \quad \text{------------- (2)}$$

R_1 = Current (I_1) set resistor

R_2 = output (V_o) set resistor.

Vref = 1.25v which is a constant voltage between output and ADJ terminals.

- Current Iadj is very small. Therefore the second term in (2) can be neglected.
- Thus the final expression for the output voltage is given by

$$V_o= 1.25v[1 + R2/R1] \quad \text{------------- (3)}$$

Eqn (3) indicates that we can vary the output voltage by varying the resistance R2. The value of R1 is normally kept constant at 240 ohms for all practical applications.

Practical Regulator using LM317:

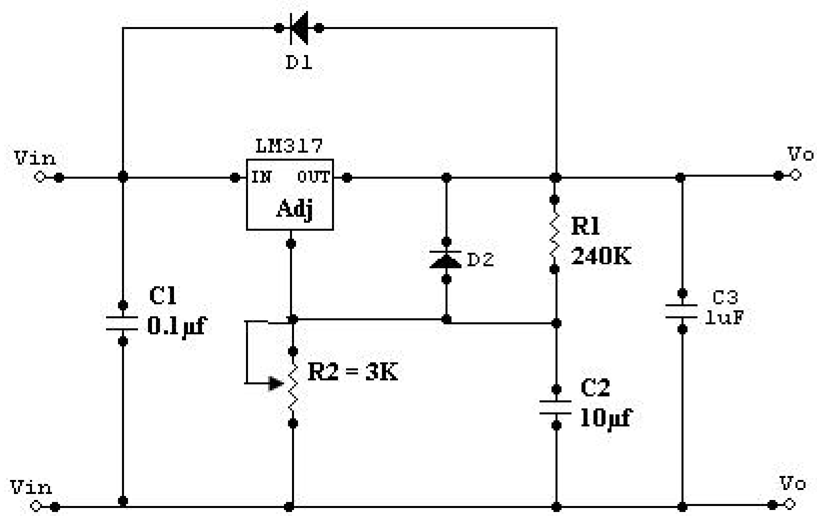


Fig. 5.28 Practical regulator

- If LM317 is far away from the input power supply, then 0.1µf disc type or 1µf tantalum capacitor should be used at the input of LM317.
- The output capacitor Co is optional. Co should be in the range of 1 to 1000µf.
- The adjustment terminal is bypassed with a capacitor C2 this will improve the ripple rejection ratio as high as 80 dB is obtainable at any output level.
- When the filter capacitor is used, it is necessary to use the protective diodes.
- These diodes do not allow the capacitor C2 to discharge through the low current point of the regulator.
- These diodes are required only for high output voltages (above 25v) & for higher values of output capacitance 25µf and above.

5.7 IC 723 – General Purpose Regulator

✓ Disadvantages of fixed voltage regulator:

1. Do not have the shot circuit
2. Output voltage is not adjustable

These limitations can be overcomes in IC723.

✓ Features of IC723:

1. Unregulated dc supply voltage at the input between 9.5V & 40V
2. Adjustable regulated output voltage between 2 to 3V.
3. Maximum load current of 150 mA (ILmax = 150mA).
4. With the additional transistor used, ILmax upto 10A is obtainable.
5. Positive or Negative supply operation
6. Internal Power dissipation of 800mW.
7. Built in short circuit protection.
8. Very low temperature drift.
9. High ripple rejection.

✓ The simplified functional block diagram can be divided in to 4 blocks.

1. Reference Generating block:

   The temperature compensated Zener diode, constant current source & voltage reference amplifier together from the reference generating block. The Zener diode is used to generate a fixed reference voltage internally. Constant current source will make the Zener diode to operate at affixed point & it is applied to the Non – inverting terminal of error amplifier. The Unregulated input voltage ±Vcc is applied to the voltage reference amplifier as well as error amplifier.

2. Error Amplifier:

   Error amplifier is a high gain differential amplifier with 2 input (inverting & Non-inverting). The Non-inverting terminal is connected to the internally generated reference voltage. The Inverting terminal is connected to the full regulated output voltage.

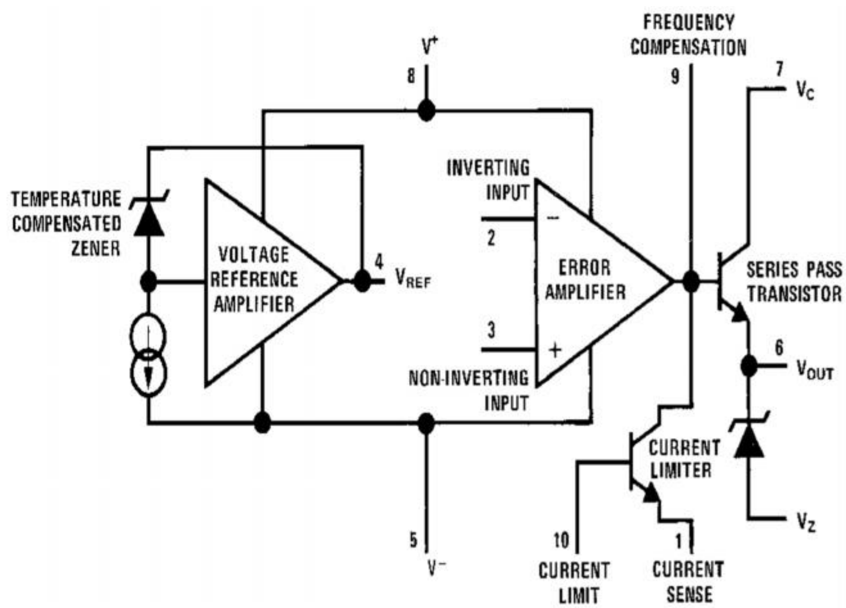


Fig. 5.29 Functional block diagram of IC723

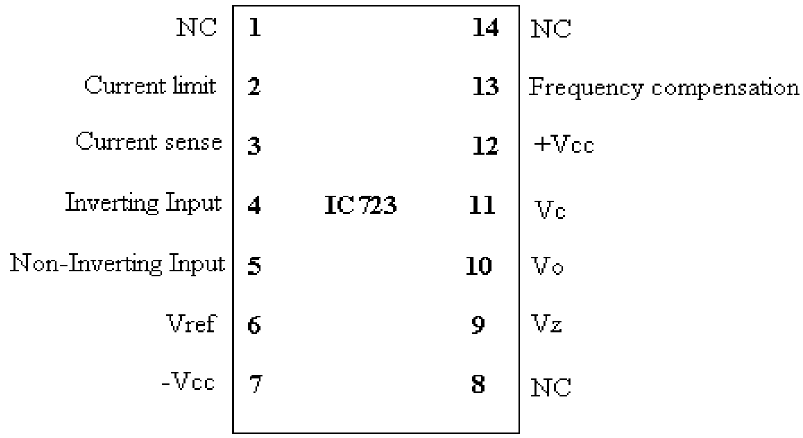


Fig.5.30 Pin diagram of IC723

3. Series Pass Transistor:
   Q1 is the internal series pass transistor which is driven by the error amplifier. This transistor actually acts as a variable resistor & regulates the output voltage. The collector of transistor Q1 is connected to the Un-regulated power supply. The maximum collector voltage of Q1 is limited to 36Volts. The maximum current which can be supplied by Q1 is 150mA.
4. Circuitry to limit the current:

The internal transistor Q2 is used for current sensing & limiting. Q2 is normally OFF transistor. It turns ON when the IL exceeds a predetermined limit.

Low voltage, Low current is capable of supplying load voltage which is equal to or between 2 to 7Volts.

Vload = 2 to 7V and Iload= 50mA

5.7.1 IC723 as a LOW voltage LOW current:

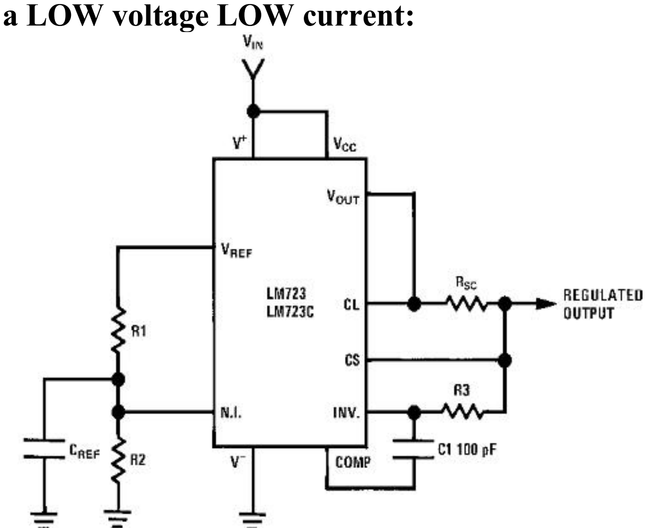


Fig.5.31 Typical circuit connection diagram

- R_1 & R_2 from a potential divider between Vref & Gnd.
- The Voltage across R2 is connected to the Non – inverting terminal of the regulator IC

$V_{non-inv} = R_2/(R_1+R_2)\ V_{ref}$

- Gain of the internal error amplifier is large

$$V_{non-inv} = V_{in}$$

- Therefore the Vo is connected to the Inverting terminal through R3 & RSC must also be equal to Vnon-inv

$$V_o = V_{non-inv} = R_2/(R_1+R_2)\ V_{ref}$$

R1 & R2 can be in the range of 1 KΩ to 10KΩ & value of R3 is given by

$$R_3 = R_1 || R_2 = R_1R_2/(R_1+R_2)$$

Rsc (current sensing resistor) is connected between Cs & CL. The voltage drop across R_{sc} is proportional to the IL.

- This resistor supplies the output voltage in the range of 2 to 7 volts, but the load current can be higher than 150mA.
- The current sourcing capacity is increased by including a transistor Q in the circuit.
- The output voltage , $V_o = R_2/(R_1+R_2)\ V_{ref}$

5.7.2 IC723 as a HIGH voltage LOW Current:

- This circuit is capable of supplying a regulated output voltage between the ranges of 7 to 37 volts with a maximum load current of 150 mA.
- The Non – inverting terminal is now connected to Vref through resistance R3.
- The value of R1 & R2 is adjusted in order to get a voltage of Vref at the inverting terminal at the desired output.

$$V_{in} = V_{ref} = R_2/(R_1+R_2)\ V_0$$
$$V_o = [1+R_1/R_2]\ V_{in}$$

- Rsc is connected between CL & Cs terminals as before & it provides the short Circuit current limiting $Rsc = 0.6/I_{limit}$
- The value of resistors R3 is given by ,

$$R3 = R_1 || R_2 = R_1R_2/(R_1+R_2)$$

5.7.3 IC723 as a HIGH voltage HIGH Current:

- An external transistor Q is added in the circuit for high voltage low current regulator to improve its current sourcing capacity.

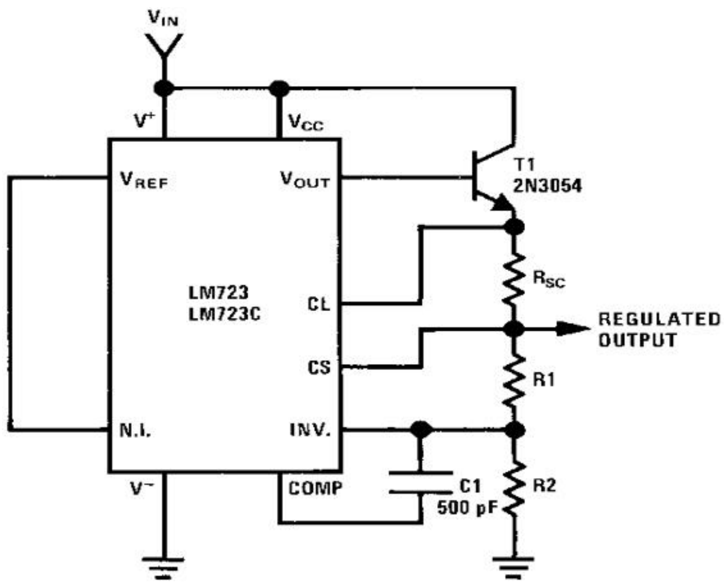


Fig.5.32 Typical circuit connection diagram

- For this circuit the output voltage varies between 7 & 37V.
- Transistor Q increase the current sourcing capacity thus IL (MAX) is greater than 150mA.
- The output voltage Vo is given by ,

$V0= V_o = [1+R_1/R_2]\ V_{in}$

$Rsc = 0.6/I_{limit}$

5.8 Switching Regulators

Introduction

The switching regulator offers the advantages

- higher power conversion efficiency
- Increased design flexibility (multiple output voltages of different polarities can be generated from a single input voltage).
- a lot less heat and
- Smaller size.

The primary filter capacitor is placed on the input to the regulator to help filter out the 60 cycle ripple. If the output voltage is 12 volts and the input voltage is 24 volts then we must drop 12 volts across the regulator. At output currents of 10 amps this translates into 120 watts (12 volts times 10 amps) of heat energy that the regulator must dissipate into heat.

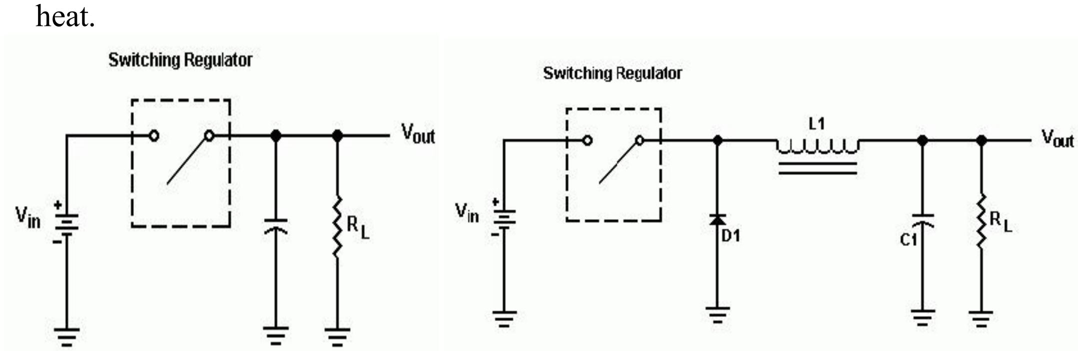


Fig.5.33 switching regulator principle

The switching regulator is much more efficient than the linear regulator achieving efficiencies as high as 80% to 95% in some circuits. The obvious result is smaller heat sinks, less heat and smaller overall size of the power supply.

The switching regulator is really nothing more than just a simple switch. This switch goes on and off at a fixed rate usually between 50 Khz to 100Khz as set by the circuit.

Operation:

Diode D1 has to be a Schottky or other very fast switching diode. Inductor L1must be a type of core that does not saturate under high currents. Capacitor C1 is normally a low ESR (Equivalent Series Resistance) type.

To understand the action of D1 and L1, let's look at what happens when S1 is closed as indicated below:

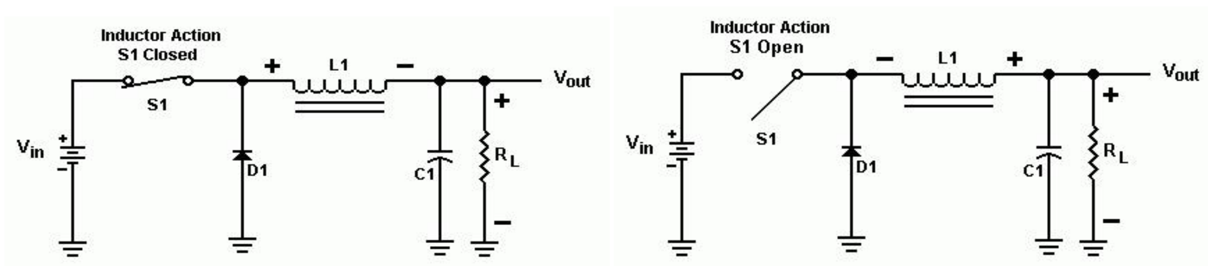


Fig.5.34 switching operation of regulator

L1, which tends to oppose the rising current, begins to generate an electromagnetic field in its core. Diode D1 is reversed biased and is essentially an open circuit at this point.

When S1 opens, the electromagnetic field that was built up in L1 is now discharging and generating a current in the reverse polarity. As a result, D1 is now conducting and will continue until the field in L1 is diminished. This action is similar to the charging and discharging of capacitor C1. The use of this inductor/diode combination gives us even more efficiency and augments the filtering of C1.

Because the switching system operates in the 50 to 100 kHz region and has an almost square waveform, it is rich in harmonics way up into the HF and even the VHF/UHF region

Four most commonly used switching converter types:

Buck: used the reduce a DC voltage to a lower DC voltage.

Boost: provides an output voltage that is higher than the input.

Buck-Boost (invert): an output voltage is generated opposite in polarity to the input.

Fly back: an output voltage that is less than or greater than the input can be generated, as well as multiple outputs.

Converters:

Push-Pull: A two-transistor converter that is especially efficient at low input voltages.

Half-Bridge: A two-transistor converter used in many off-line applications.

Full-Bridge: A four-transistor converter (usually used in off-line designs) that can generate the highest output power of all the types listed.

Switching Regulator:

An example of general purpose regulator is Motorola's MC1723. It can be used in many different ways, for example, as a fixed positive or negative output voltage regulator, variable regulator or switching regulator because of its flexibility.

To minimize the power dissipation during switching, the external transistor used must be a switching power transistor.

To improve the efficiency of a regulator, the series pass transistor is used as a switch rather than as a variable resistor as in the linear mode.

- A regulator constructed to operate in this manner is called a series switching regulator. In such regulators the series pass transistor is switched between cut off & saturation at a high frequency which produces a pulse width modulated (PWM) square wave output.
- This output is filtered through a low pass LC filter to produce an average dc output voltage.
- Thus the output voltage is proportional to the pulse width and frequency.
- The efficiency of a series switching regulator is independent of the input & output differential & can approach 95%

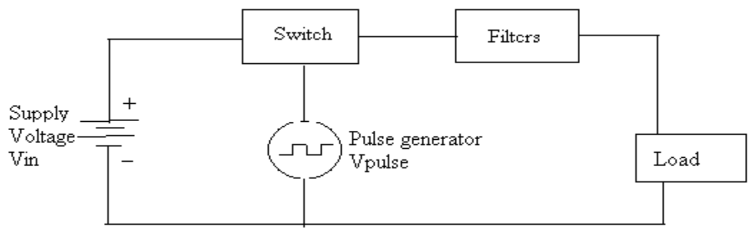


Fig.5.35 Basic Switching regulator

A basic switching regulator consists of 4 major components,

1. Voltage source Vin
2. Switch S1
3. Pulse generator Vpulse
4. Filter F1

1. Voltage Source Vin:

It may be any dc supply – a battery or an unregulated or a regulated voltage. The voltage source must satisfy the following requirements.

- It must supply the required output power & the losses associated with the switching regulator.
- It must be large enough to supply sufficient dynamic range for line & load regulations.
- It must be sufficiently high to meet the minimum requirement of the regulator system to be designed.
- It may be required to store energy for a specified amount of time during power failures.

2. Switch S1:

It is typically a transistor or thyristor connected as a power switch & is operated in the saturated mode. The pulse generator output alternately turns the switch ON & OFF

3. Pulse generator Vpulse:

It provides an asymmetrical square wave varying in either frequency or pulse width called frequency modulation or pulse width modulation respectively. The most effective frequency range for the pulse generator for optimum efficiency 20 KHz. This frequency is inaudible to the human ear & also well within the switching speeds of most inexpensive transistors & diodes.

- The duty cycle of the pulse wave form determines the relationship between the input & output voltages. The duty cycle is the ratio of the on time ton, to the period T of the pulse waveform.

  Duty cycle = $t_{on}/(t_{on}+t_{off}) = t_{on}/T = t_{on}.f$

  Where ton = On-time of the pulse waveform toff=off-time of the pulse wave form

  T = time period = ton + toff

  = 1/frequency or T = 1/f

- Typical operating frequencies of switching regulator range from 10 to 50 kHz.
- Lower operating frequency improve efficiency & reduce electrical noise, but require large filter components (inductors & capacitors).

4. Filter F1:

It converts the pulse waveform from the output of the switch into a dc voltage. Since this switching mechanism allows a conversion similar to transformers, the switching regulator is often referred to as a dc transformer.

The output voltage Vo of the switching regulator is a function of duty cycle & the input voltage Vin.

Vo is expressed as follows,

$$V_o = t_{on} V_{in}/T$$

- This equation indicates that, if time period T is constant, Vo is directly proportional to the ON-time, ton for a given value of Vin. This method of changing the output voltage by varying ton is referred to as a pulse width modulation.
- Similarly, if ton is held constant, the output voltage Vo is inversely proportional to the period T or directly proportional to the frequency of the pulse waveform. This method of varying the output voltage is referred to as frequency modulation (FM).
- Switching regulator can operate in any of 3 modes
  - i) Step – Down
  - ii) Step – Up
  - iii) Polarity inverting

5.8 Monolithic Switching Regulator [µa78s40]:

The µA78S40 consists of a temperature compensated voltage reference, duty cycle controllable oscillator with an active current limit circuit, a high gain comparator, a high- current, high voltage output switch, a power switching diode & an uncommitted op-amp.

Important features of the µA78S40 switching regulators are:

- Step up, down & Inverting operation

- Operation from 2.5 to 40V input
- 80dB line & load regulations
- Output adjustable from 1.3 to 40V
- Peak current to 1.5A without external resistors
- Variable frequency, variable duty cycle device

The internal switching frequency is set by the timing capacitor CT, connected between pin12 & ground pin 11. The initial duty cycle is 6:1. The switching frequency & duty cycle can be modified by the current limit circuitry, IPK sense, pin14, 7 the comparator, pin9 & 10.

Comparator:

The comparator modifies the OFF time of the output switch transistor Q1 & Q2. In the step – up & step down modes, the non-inverting input(pin9) of the comparator is connected to the voltage reference of 1.3V (pin8) & the inverting input (pin10) is connected to the output terminal via the voltage divider network.

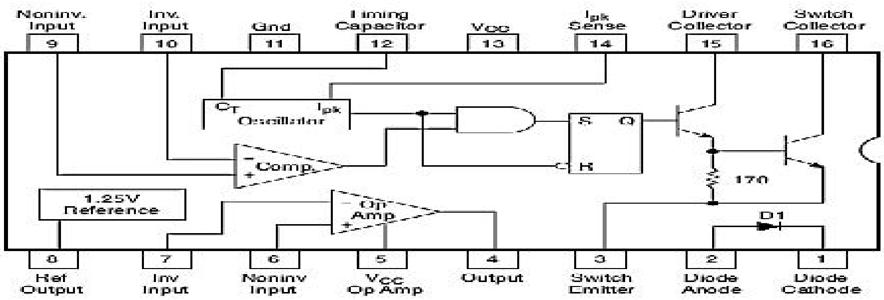


Fig.5.36 Functional block diagram of µA78S40

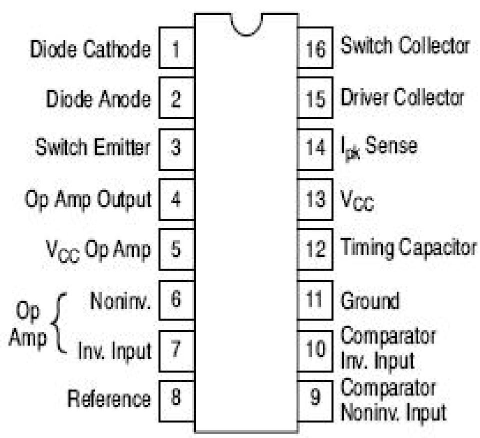


Fig.5.37 Pin diagram of Monolithic Switching Regulator [µa78s40]:

- In the Inverting mode the non – inverting input is connected to both the voltage reference & the output terminal through 2 resistors & the inverting terminal is connected to ground.
- When the output voltage is correct, the comparator output is in high state & has no effect on the circuit operation.

- However, if the output is too high & the voltage at the inverting terminal is higher than that at the non-inverting terminal, then the comparator output goes low.
- In the LOW state the comparator inhibits the turn on of the output switching transistors. This means that, as long as the comparator output is low, the system is in off time.
- As the output current rises or the output voltage falls, the off time of the system decreases.
- Consequently, as the output current nears its maximum IoMAX, the off time approaches its minimum value.

In all 3 modes (Step down, step up, Inverting), the current limit circuit is completed by connecting a sense resistor Rsc, between IPK sense & Vcc.

- The current limit circuit is activated when a 330mV potential appears across Rsc.
- Rsc is selected such that 330mV appears across it when the desired peak current IPK, flows through it.
- When the peak current is reached, the current limit circuit is turned on.

The forward voltage drop, VD, across the internal power diode is used to determine the value of inductor L off time & efficiency of the switching regulator.

Another important quantity used in the design of a switching regulator is the saturation voltage Vs:

In the step down mode an "output saturation volt" is 1.1V typical, 1.3VMAX.

In the step up mode an "Output saturation volt" is 0.45V typical, 0.7 maximum.

The desired peak current value is reached; the current limiting circuit turns ON & immediately terminates the ON time & starts OFF time.

- As we increase IL (load current), Vout will decrease, to compensate for this, the ON time of the output is increased automatically.

If the I_L decreased then Vout increased, to compensate for this, the OFF time of the output is increased automatically.

(i) Step – Down Switching Regulator:

- CT is the timing capacitor which decides the switching frequency.
- Rsc is the current sensing resistance. Its value is given by
- The Non-inverting terminal of the internal op-amp (pin9) is connected to the 1.3V reference (pin8).
- Resistances R1 & R2 from a potential divider, across the output voltage Vo. Their value should be such that the potential at the inverting input of the op-amp should be equal to 1.3V ref when Vo is at its desired level.

The output capacitance Co is used for reducing the ripple contents in the output voltage. It acts as a filter along with the inductor L.

- The inductor L is a part of filter connected on the output side, to reduce the ripple percentage.
- The 0.1µF capacitor connected between pin8 & ground bypasses any noise voltage coupled to the reference (pin8).

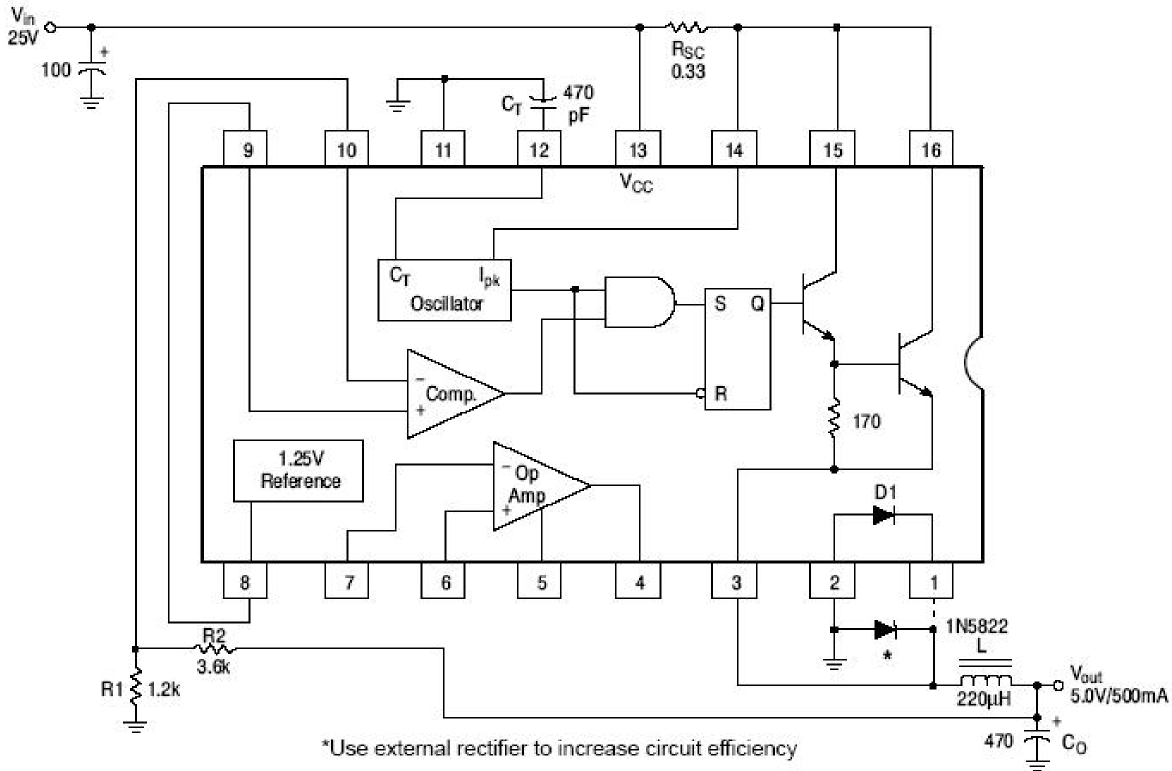


Fig. 5.38 Step down convertor

(ii) Step – Up Switching Regulator:

- Inductor is connected between the collectors of Q1 & Q2.
- When Q1 is ON, the output is shorted & the collector current of Q1 flows through L.
- The diode D1 is reverse biased & Co supplies the load current.
- The inductor stores the energy. When the Q1 is turned OFF, there is a self induced emf that appears across the inductor with polarities.
- The output voltage is given by,

$$Vo = Vin + VL$$

- Hence it will be always higher than Vin & step up operation is achieved.

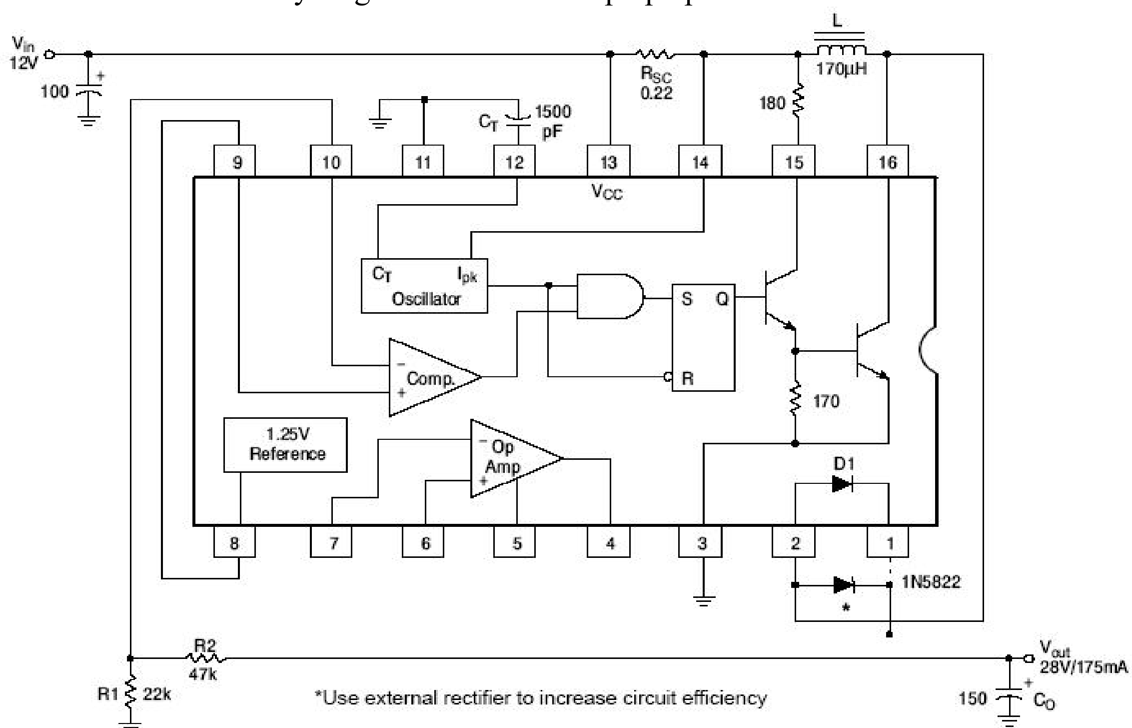
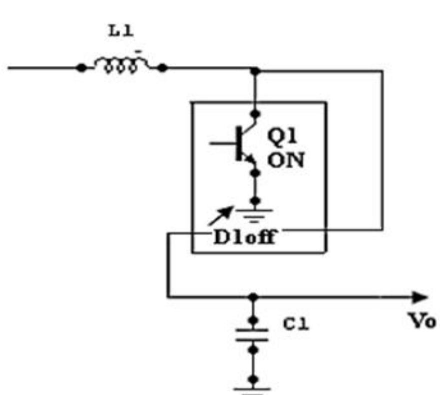
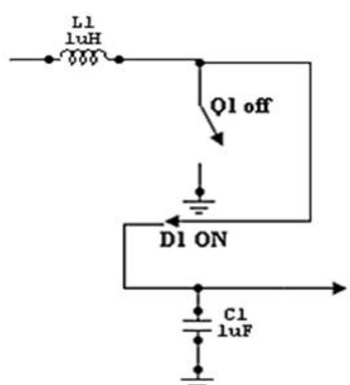


Fig.5.39 Step up convertor





With Q1 ON with Q1 oFF

(iii) Inverting Switching Regulator:

Inverting switching regulator converts a positive input voltage into a negative output voltage which is higher in magnitude.

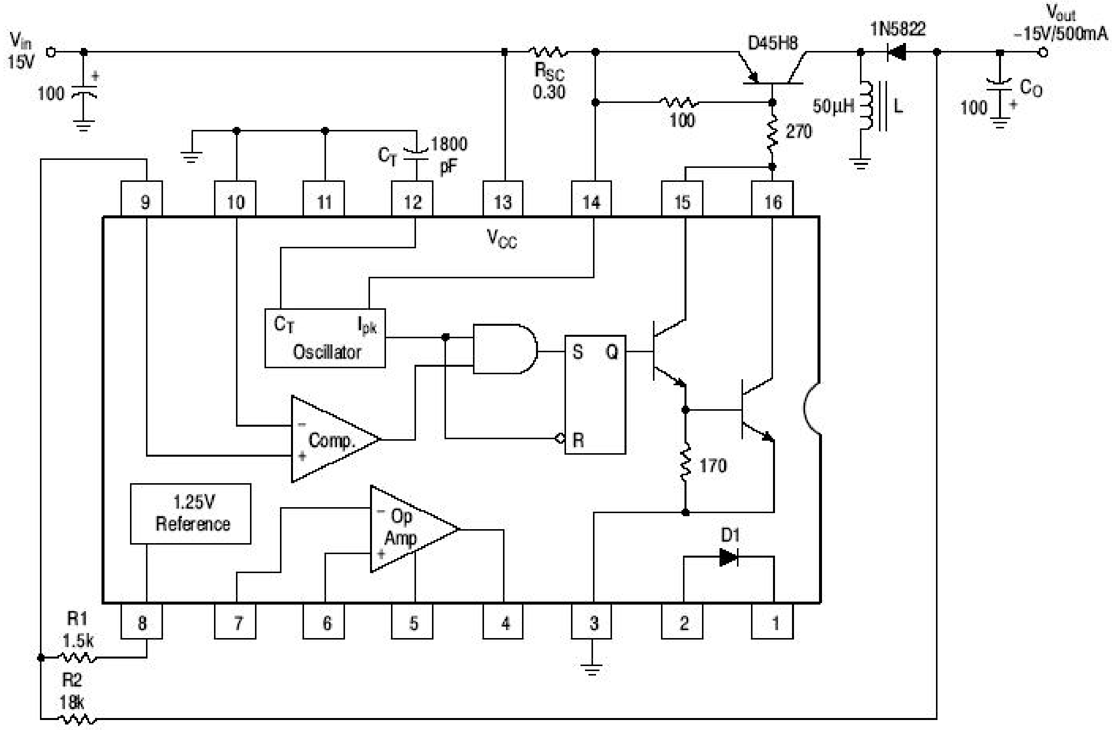


Fig.5.40 Circuit diagram of inverting switching regulator

5.9 The Switched Capacitor Filter

Basic Representation:

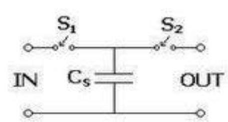


Fig 5.41 Switched-capacitor resistor

The simplest switched capacitor (SC) circuit is the switched capacitor resistor, made of one capacitor C and two switches S1 and S2 which connect the capacitor with a given frequency alternately to the input and output of the SC. Each switching cycle transfers a charge q from the input to the output at the switching frequency f. Recall that the charge q on a capacitor C with a voltage V between the plates is given by:

$$q = CV$$

where V is the voltage across the capacitor. Therefore, when S1 is closed while S2 is open, the charge transferred from the source to CS is:

$$q_{IN} = C_S V_{IN}$$

And when S2 is closed while S1 is open, the charge transferred from CS to the load is:

$$q_{OUT} = C_S V_{OUT}$$

Thus, the charge transferred in each cycle is:

$$q = q_{OUT} - q_{IN} = C_S(V_{OUT} - V_{IN})$$

Since a charge q is transferred at a rate f, the rate of transfer of charge per unit time is:

$$I = qf$$

Note that we use I, the symbol for electric current, for this quantity. This is to demonstrate that a continuous transfer of charge from one node to another is equivalent to a current. Substituting for q in the above, we have:

$$I = C_S(V_{OUT} - V_{IN})f$$

Let us define V, the voltage across the SC from input to output, thus:

$$V = V_{OUT} - V_{IN}$$

We now have a relationship between I and V, which we can rearrange to give an equivalent resistance R:

$$R = \frac{V}{I} = \frac{1}{C_S f}$$

Thus, the SC behaves like a resistor whose value depends on CS and f.

The SC resistor is used as a replacement for simple resistors in integrated circuits because it is easier to fabricate reliably with a wide range of values. It also has the benefit that its value can be adjusted by changing the switching frequency. See also: operational amplifier applications.

This same circuit can be used in discrete time systems (such as analog to digital converters) as a track and hold circuit. During the appropriate clock phase, the capacitor samples the analog voltage through switch one and in the second phase presents this held sampled value to an electronic circuit for processing.

Switched Capacitor Circuits:

The switched capacitor filter allows for very sophisticated, accurate, and tuneable analog circuits to be manufactured without using resistors.

Advantages: resistors are hard to build on integrated circuits (they take up a lot of room), and the circuits can be made to depend on ratios of capacitor values (which can be set accurately), and not absolute values (which vary between manufacturing runs).

The Switched Capacitor Resistor:

Consider the circuit shown with a capacitor connected to two switches and two different voltages.

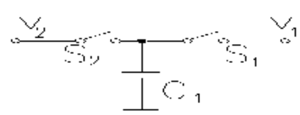


Fig.5.42 Example circuit

If S2 closes with S1 open, then S1 closes with switch S2 open, a charge (q is transferred from v2 to v1 with

$$\Delta q = C_1(v_2 \quad v_1)$$

If this switching process is repeated N times in a time (t, the amount of charge transferred per unit time is given by

$$\frac{\Delta q}{\Delta t} = C_1(v_2 - v_1)\frac{N}{\Delta t}$$

the number of cycles per unit time is the switching frequency (or clock frequency, fCLK)

$$i = C_1(v_2 - v_1)f_{CLK}$$

Rearranging we get

$$\frac{(v_2 \quad v_1)}{i} - \frac{1}{C_1 f_{CLK}} - R$$

Which states that the switched capacitor is equivalent to a resistor? The value of this resistor decreases with increasing switching frequency or increasing capacitance, as either will increase the amount of charge transferred from v2 to v1 in a given time.

The Switched Capacitor Integrator:

Now consider the integrator circuit. You have shown (in a previous lab) that the input-output relationship for this circuit is given by (neglecting initial conditions):

$$v_o(t) = -\frac{1}{RC_2}\int v_i(t)dt = -\omega'\int v_i(t)dt$$

We can also write this with the "s" notation (assuming a sinusoidal input, Ae^{st}, s=jω)

$$V_o(s) = -\frac{\omega'}{s}$$

If you replaced the input resistor with a switched capacitor resistor, you would get

$$\omega' = \frac{1}{RC_2} = f_{CLK}\frac{C_1}{C_2}$$

Thus, you can change the equivalent ω' of the circuit by changing the clock frequency. The value of ω' can be set very precisely because it depends only on the ratio of C1 and C2, and not their absolute value.

- ✓ **Switched Capacitor Filter ICs**:

Some of the Switched capacitor filter ICs is MF 5, MF10 and MF100

- ✓ **MF10:**

The MF10 contains two of the second-order universal filter sections found in the MF5. Therefore with MF10, two second order filters or one fourth-order filter can be built. As the MF5 and MF10 have similar filter sections, the design procedure for them is same.

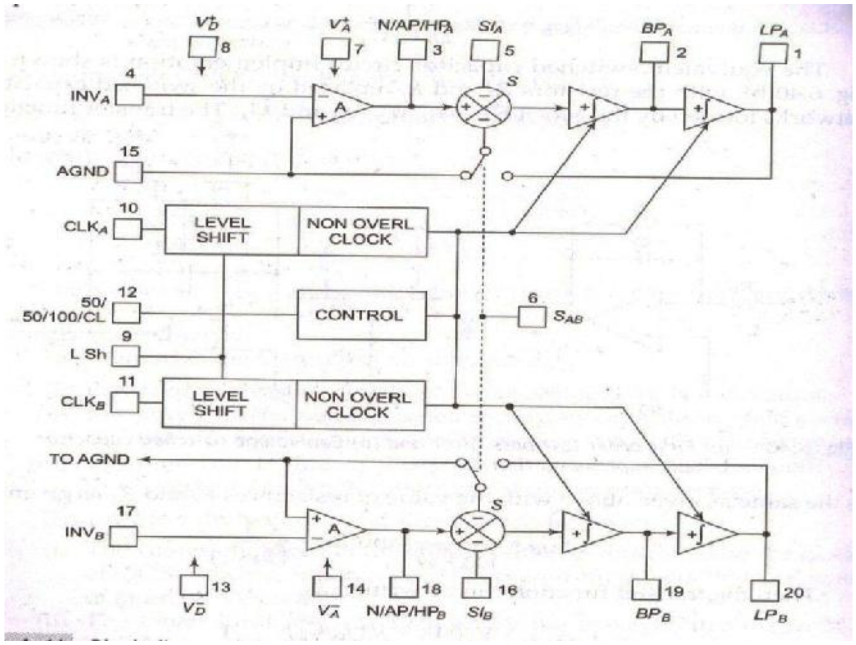


Fig.5.43 circuit connections of MF10

5.10 Frequency to Voltage (F-V) and voltage to frequency convertors (V-F)

- F-V convertors applications: Tachometer in motor speed control
  Rotational speed measurement.
- Two types of it: Pulse integrating
  Phase locked loop

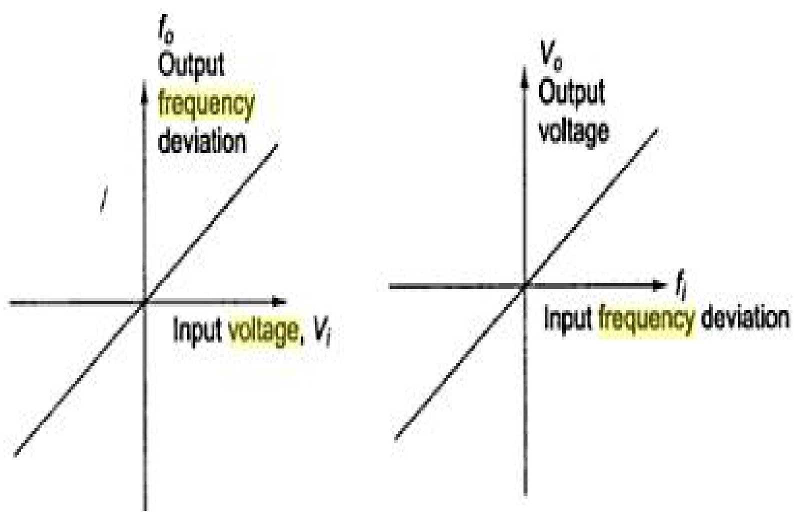


Fig. 5.44 Ideal characteristics of V-F convertor and F-V convertor

- F-V convertor produces an output voltage whose amplitude is a function of input signal frequency.

- $V0=k_f f_i$ kf is sensitivity of F-V convertor
- It is basically a FM discriminator.

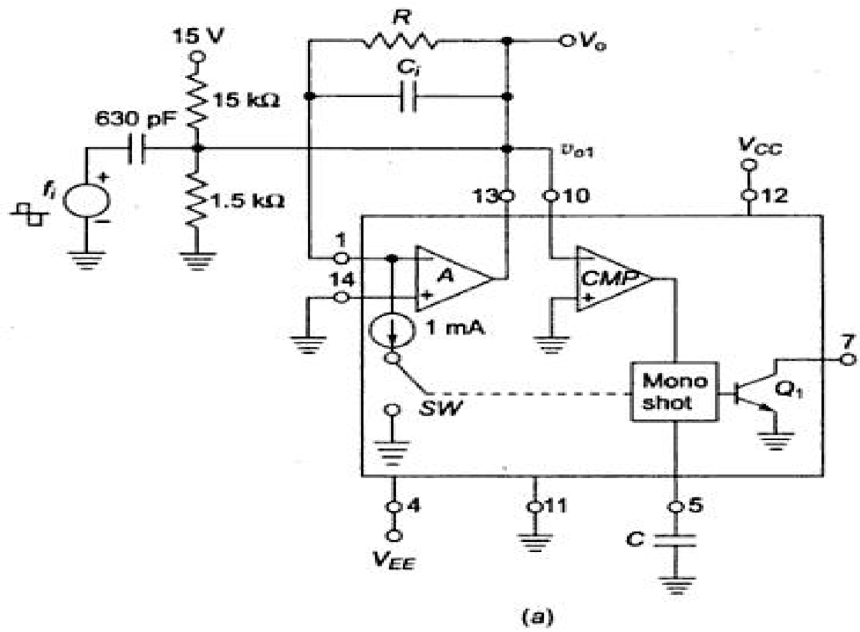


Fig.5.45 Frequency To Voltage Convertor using VFC32 (V-F)

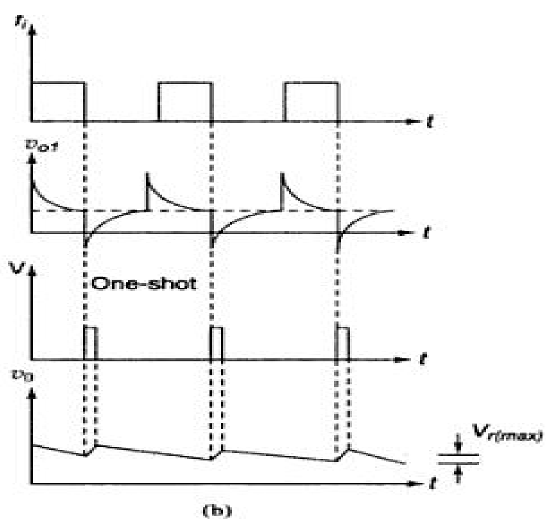


Fig.5.46 F-V Convertor using VF32 and input and output characteristics

Input frequency is applied to comparator A.
Resistor R acts as feedback element.
Capacitor Ci enables charge-balancing,
High pass network conditions input signal
For negative spike of V_{01}, comparator COMP triggers one shot multivibrator with threshold 7.5V The output of multivibrator closes the switch SW, for a time T_H, this causes voltage Vo to build up and inject thru R and this continues until current out of summing input of opamp is equal to that injected by Vo through R continuously.
$V_o=10^{-3} *T_H *R*f_i$ as $T_H=7.5\ C/1X10^{-3}$
Ripple Voltage, $Vr(max) = 7.5\ C/C_i$

5.10.2 Voltage to frequency convertor

Principle: Charge balancing technique-the process of charging and discharging results in frequency proportional to input signal $F_0= k\ V_i$

Operation: Op-amp A converts input V_i to current $I_i = V_i/R$ into summing junction.
When switch SW is open the current flows into capacitor Ci and charges it, and node voltage Vo1 produce ramp down.
When $V_{01}=0$ CMP triggers and sends a triggering signal to one shot multivibrator that closes the switch SW and turns transistor Q ON for time T_H.
The threshold of mono shot = 7.5 V and $T_H= 7.5\ C/10^{-3}$
During T_H , V_{01} ramps upward by amount $\Delta V01=(1mA-I_i)\ T_H/Ci$
Time duration TL for vo1 to return to 0 is $TL = C\Delta V01/I_i$

$$T_L+T_H = 1mA\ T_H/I_i = T$$

$$F_0=V_i/7.5\ RC$$

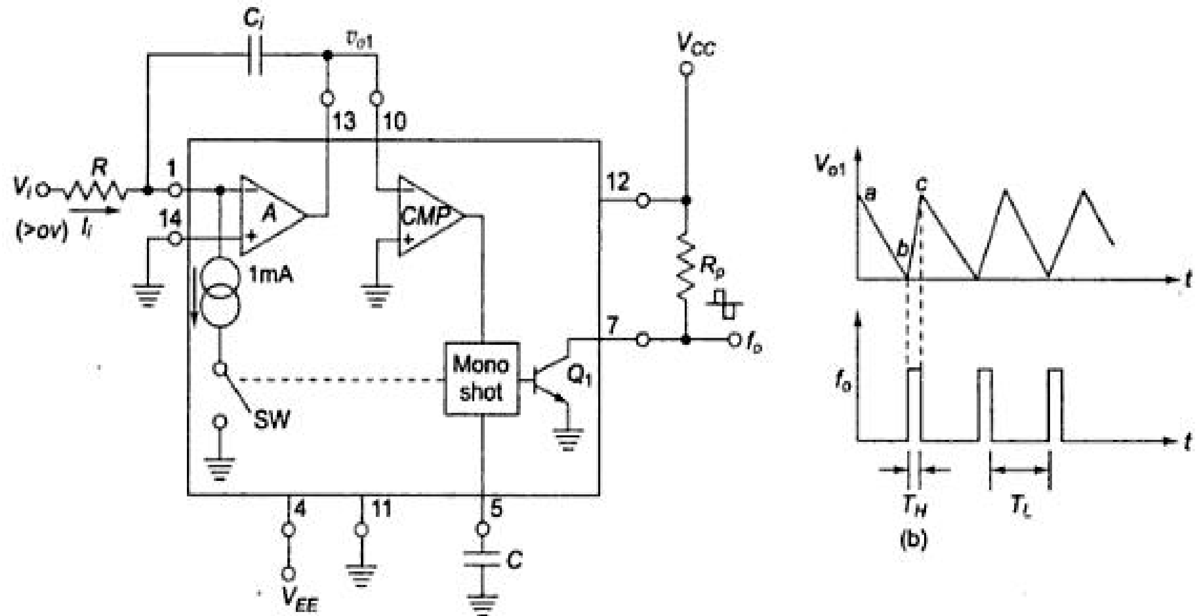


Fig.5.47 Voltage-Frequency convertor using VF32 and its input output characteristics

5.11 Power Audio Amplifier IC LM 380:

Introduction:

Small signal amplifiers are essentially voltage amplifier that supplies their loads with larger amplifier signal voltage.

On the other hand, large signal or power amplifier supply a large signal current to current operated loads such as speakers & motors.

In audio applications, however, the amplifier called upon to deliver much higher current than that supplied by general purpose op-amps. This means that loads such as speakers & motors requiring substantial currents cannot be driven directly by the output of general purpose op-amps.

To handle it following is done

- To use discrete or monolithic power transistors called power boosters at the output of the op-amp
- To use specialized ICs designed as power amplifiers like LM 380.

✓ Features of LM380:

1. Internally fixed gain of 50 (34dB)
2. Output is automatically self centering to one half of the supply voltage.
3. Output is short circuit proof with internal thermal limiting.
4. Input stage allows the input to be ground referenced or ac
5. Wide supply voltage range (5 to 22V).
6. High peak current capability.
7. High impedance.

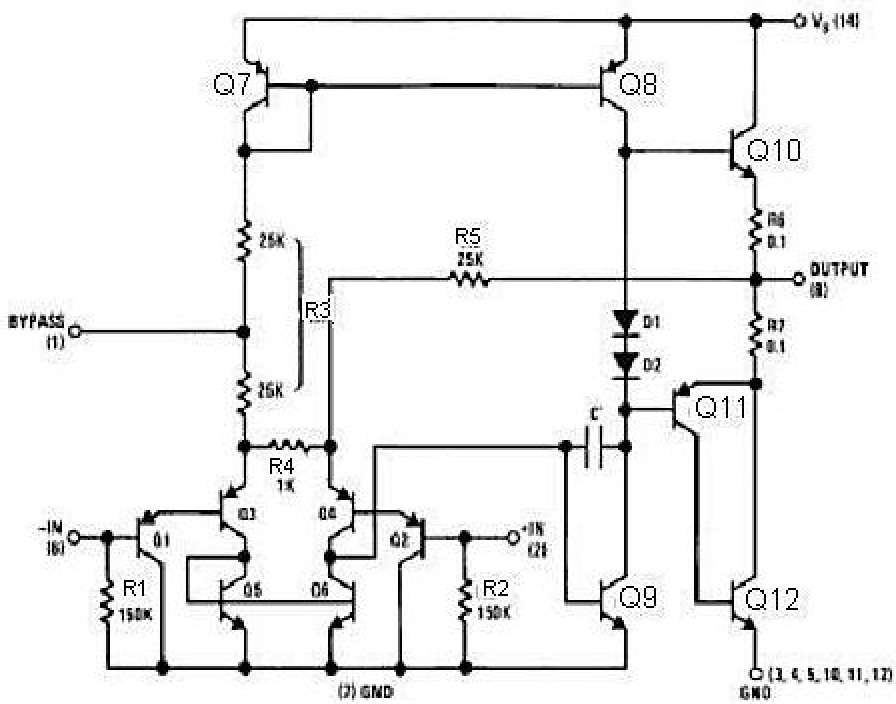


Fig.5.48 Functional block diagram of Audio Power Amplifier

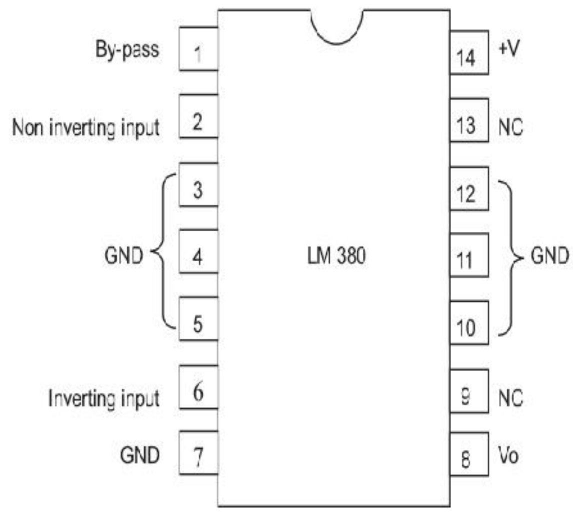


Fig.5.49 Pin diagram Of Power amplifier LM380

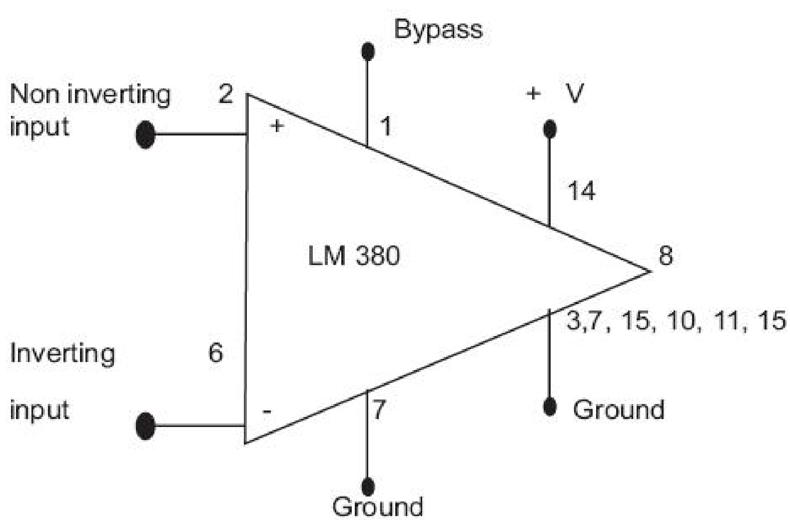


Fig.5.50 Block diagram of LM380

LM380 circuit description:

It is connected of 4 stages,

(i) PNP emitter follower
(ii) Different amplifier
(iii) Common emitter
(iv) Emitter follower

(i) PNP Emitter follower:

- The input stage is emitter follower composed of PNP transistors Q1 & Q2 which drives the PNP Q3-Q4 differential pair.
- The choice of PNP input transistors Q1 & Q2 allows the input to be referenced to ground i.e., the input can be direct coupled to either the inverting & non-inverting terminals of the

amplifier.

(ii) Differential Amplifier:

- The current in the PNP differential pair Q3-Q4 is established by Q7, R3 & +V.
- The current mirror formed by transistor Q7, Q8 & associated resistors then establishes the collector current of Q9.
- Transistor Q5 & Q6 constitute of collector loads for the PNP differential pair.
- The output of the differential amplifier is taken at the junction of Q4 & Q6 transistors & is applied as an input to the common emitter voltage gain.

(iii) Common Emitter amplifier stage:

- Common Emitter amplifier stage is formed by transistor Q9 with D1, D2 & Q8 as a current source load.
- The capacitor C between the base & collector of Q9 provides internal compensation & helps to establish the upper cutoff frequency of 100 KHz.
- Since Q7 & Q8 form a current mirror, the current through D1 & D2 is approximately the same as the current through R3.
- D1 & D2 are temperature compensating diodes for transistors Q10 & Q11 in that D1 & D2 have the same characteristics as the base-emitter junctions of Q11. Therefore the current through Q10 & (Q11-Q12) is approximately equal to the current through diodes D1 & D2.

(iv) (Output stage) - Emitter follower:

- Emitter follower formed by NPN transistor Q10 & Q11. The combination of PNP transistor Q11 & NPN transistor Q12 has the power capability of NPN transistors but the characteristics of a PNP transistor.
- The negative dc feedback applied through R5 balances the differential amplifier so that the dc output voltage is stabilized at +V/2;
- To decouple the input stage from the supply voltage +V, by pass capacitor in order of micro farad should be connected between the bypass terminal (pin 1) & ground (pin 7).
- The overall internal gain of the amplifier is fixed at 50. However gain can be increased by using positive feedback.

✓ **Applications:**

(i) Audio Power Amplifier:

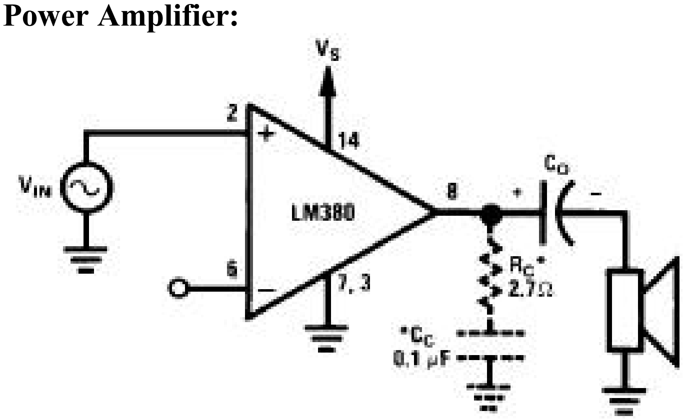


Fig. 5.51 Connections of audio power amplifier

- Amplifier requires very few external components because of the internal biasing, compensation & fixed gain.
- When the power amplifier is used in the non inverting configuration, the inverting terminal may be either shorted to ground, connected to ground through resistors & capacitors.
- Similarly when the power amplifier is used in the inverting mode, the non inverting terminal may be either shorted to ground or returned to ground through resistor or capacitor.
- Usually a capacitor is connected between the inverting terminal & ground if the input has a high internal impedance.
- As a precautionary measure, an RC combination should be used at the output terminal (pin 8) to eliminate 5-to-10 MHz oscillation.
- C1 is coupling capacitor which couples the output of the amplifier to the 8 ohms loud speaker which acts as a load. The amplifier will amplify the Vin applied at the non-inverting terminal.

(ii) LM 380 as a High gain amplifier:

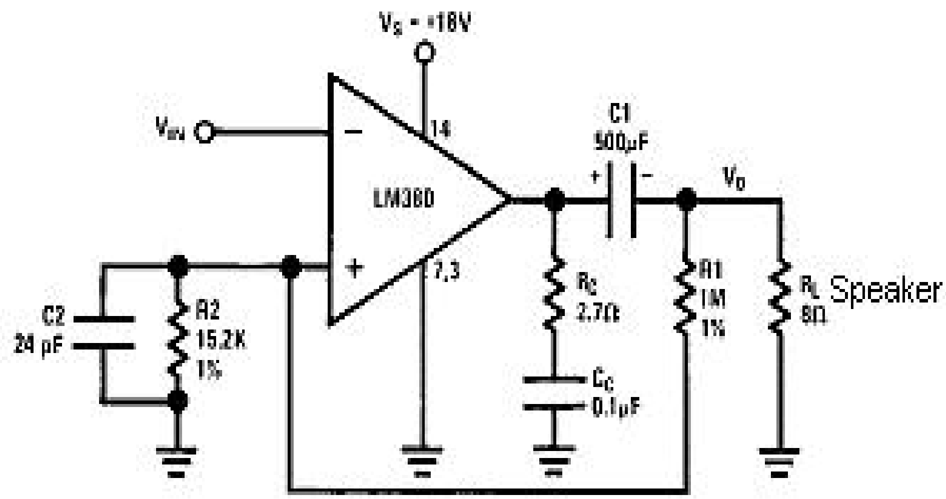


Fig.5.52 Circuit connections

- The gain of LM380 is internally fixed at 50. But it can be increased by using the external components.
- The increase in gain is possible due to the use of positive feedback, this setup to obtain a gain 200.

(iii) LM 380 as a variable Gain:

- Instead of getting a fixed gain of 50, it is possible to obtain a variable gain up to 50 by connecting a potentiometer between the input terminals.

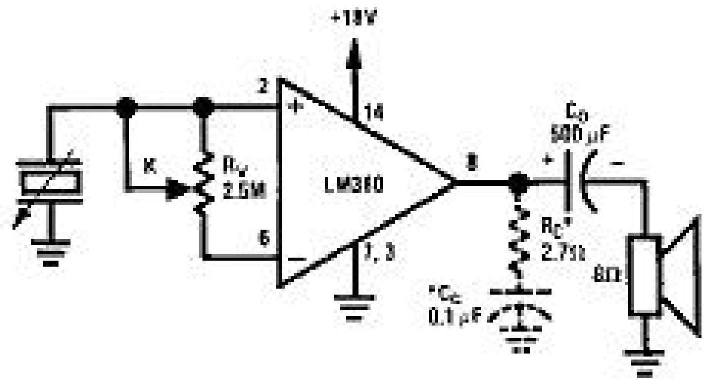


Fig. 5.53 Circuit connections

(iv) LM 380 as a Bridge Audio Power Amplifier:

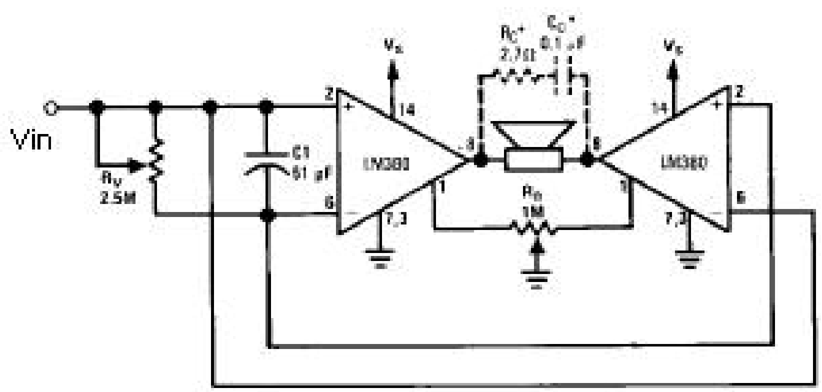


Fig. 5.54 Circuit connections

- If a certain application requires more power than what is provided by a single LM380 amplifier, then 2 LM380 chips can be used in the bridge configuration.
- With this arrangement we get an output voltage swing which is twice that of a single LM380 amplifier.
- As the voltage is doubled, power output will increase by four times that of a single LM380 amplifier. The pot R4 is used to balance the output offset voltages of the two chips.

(v) Intercom system using LM 380:

- When the switch is in Talk mode position, the master speaker acts as a microphone.
- When the switch is in Listen position, the remote speaker acts as a microphone.
- In either phone the overall gain of the circuit is the same depends on the turns of transformer T.

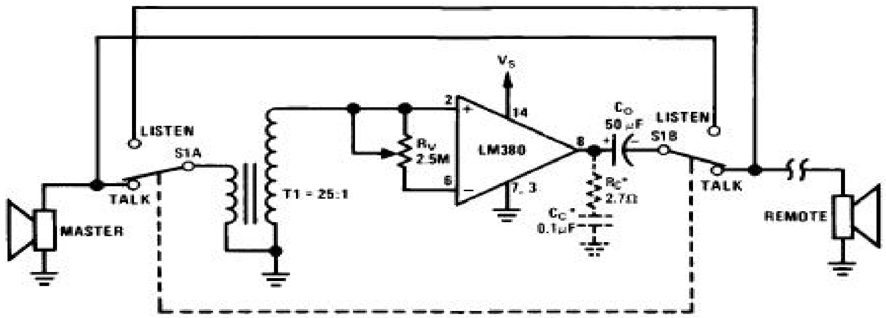


Fig. 5.55 Talk mode

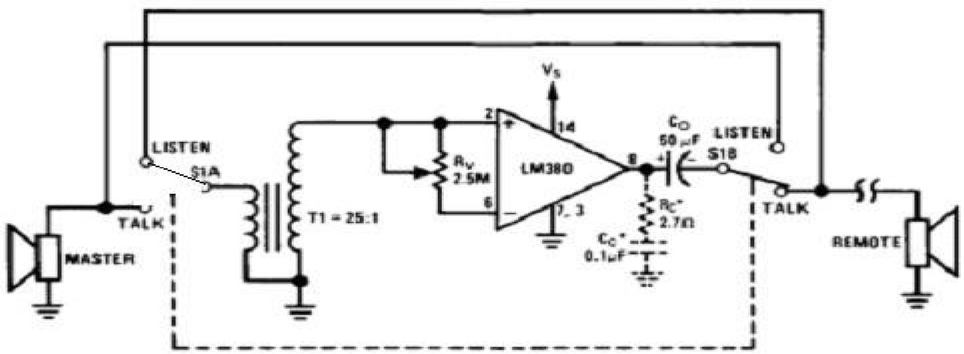


Fig.5.56 Listen mode

5.12 Monolithic video amplifier

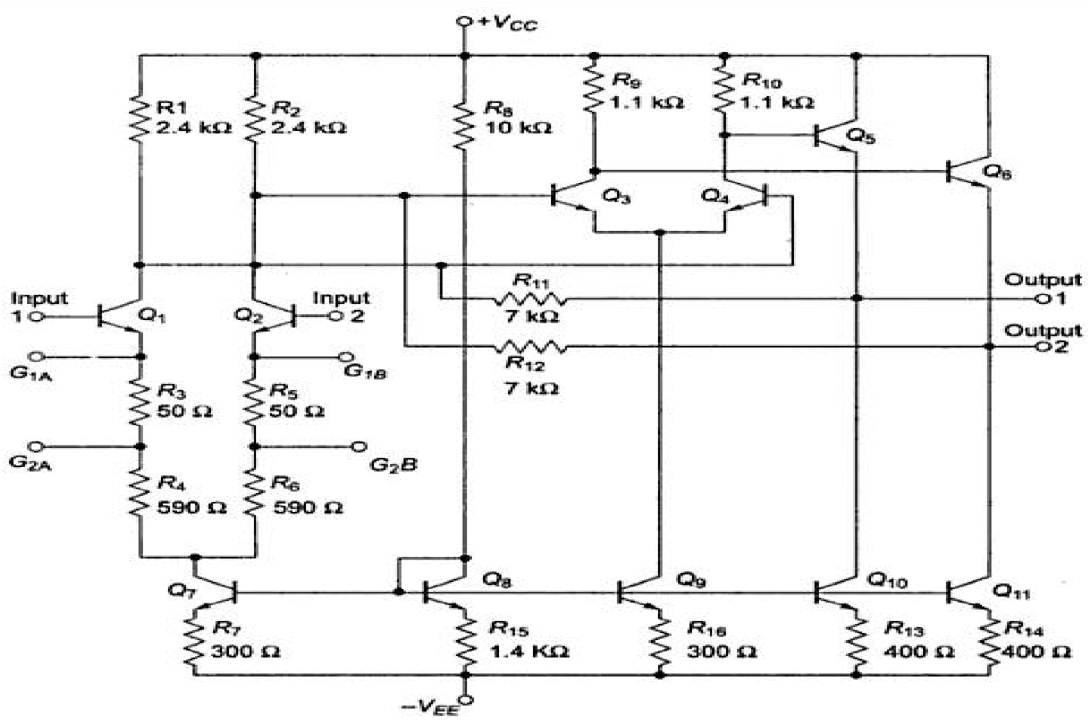


Fig.5.57 Monolithic video amplifier

5.13 Opto couplers/Opto Isolators and fibre optic IC

- Opto couplers or Opt isolators is a combination of light source & light detector in the same package.
- They are used to couple signal from one point to other optically, by providing a complete electric isolation between them. This kind of isolation is provided between a low power control circuit & high power output circuit, to protect the control circuit.
- Characteristics of opto coupler:
  (i) Current Transfer Ratio:
  It is defined as the ratio of output collector current (Ic) to the input forward current (If)
  CTR = Ic/If * 100%. Its value depends on the devices used as source & detector.
  (ii) Isolation voltage between input & output:
  It is the maximum voltage which can exist differentially between the input & output without affecting the electrical isolation voltage is specified in K Vrms with a relative humidity of 40 to 60%.
  (iii) Response Time:
  Response time indicates how fast an opto coupler can change its output state. Response time largely depends on the detector transistor, input current & load resistance.
  (iv) Common mode Rejection:
  Even though the opto couplers are electrically isolated for dc & low frequency signals, an impulsive input signal (the signal which changes suddenly) can give rise to a displacement current Ic= Cf*dv/dt. This current can flow between input & output due to the capacitance Cf existing between input & output. This allows the noise to appear in the output. Depending on the type of light source & detector used we can get a variety of opto couplers.

They are as follows,
(i) LED – Photodiode opto coupler:

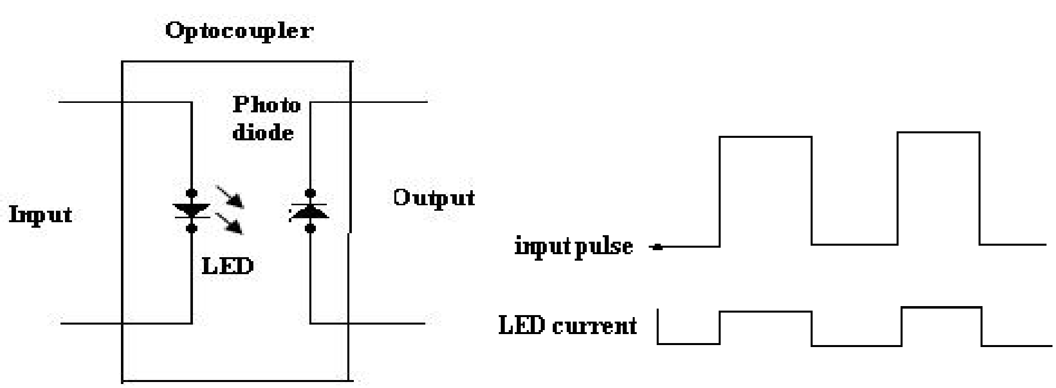


Fig.5.58 Schematic symbol and waveforms

- LED photodiode shown in figure, here the infrared LED acts as a light source & photodiode is used as a detector.
- The advantage of using the photodiode is its high linearity. When the pulse at the input goes high, the LED turns ON. It emits light. This light is focused on the photodiode.
- In response to this light the photocurrent will start flowing though the photodiode. As soon as the input pulse reduces to zero, the LED turns OFF & the photocurrent through the photodiode reduces to zero. Thus the pulse at the input is coupled to the output side.

(ii) LED – Phototransistor Opto coupler:

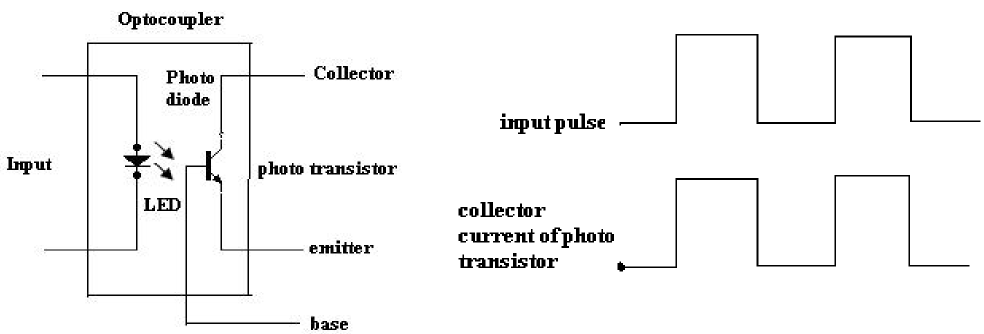


Fig.5.59 Schematic symbol and waveforms

- The LED phototransistor opto coupler shown in figure. An infrared LED acts as a light source and the phototransistor acts as a photo detector.
- This is the most popularly used opto coupler, because it does not need any additional amplification.
- When the pulse at the input goes high, the LED turns ON. The light emitted by the LED is focused on the CB junction of the phototransistor.
- In response to this light photocurrent starts flowing which acts as a base current for the phototransistor.
- The collector current of phototransistor starts flowing. As soon as the input pulse reduces to zero, the LED turns OFF & the collector current of phototransistor reduces to zero. Thus the pulse at the input is optically coupled to the output side.
- The input & output waveforms are 180º out of phase as the output is taken at the collector of the phototransistor

Advantages of Opto coupler:

- ✓ Control circuits are well protected due to electrical isolation.
- ✓ Wideband signal transmission is possible.
- ✓ Due to unidirectional signal transfer, noise from the output side does not get coupled to the input side.

- ✓ Interfacing with logic circuits is easily possible.
- ✓ It is small size & light weight device.

Disadvantages:

- ✓ Slow speed.
- ✓ Possibility of signal coupling for high power signals.

Applications:

Opto couplers are used basically to isolate low power circuits from high power circuits.

- At the same time the control signals are coupled from the control circuits to the high power circuits.
- Some of such applications are,
  - (i) AC to DC converters used for DC motor speed control
  - (ii) High power choppers
  - (iii) High power inverters
- One of the most important applications of an opto coupler is to couple the base driving signals to a power transistor connected in a DC-DC chopper.

Opto coupler IC:

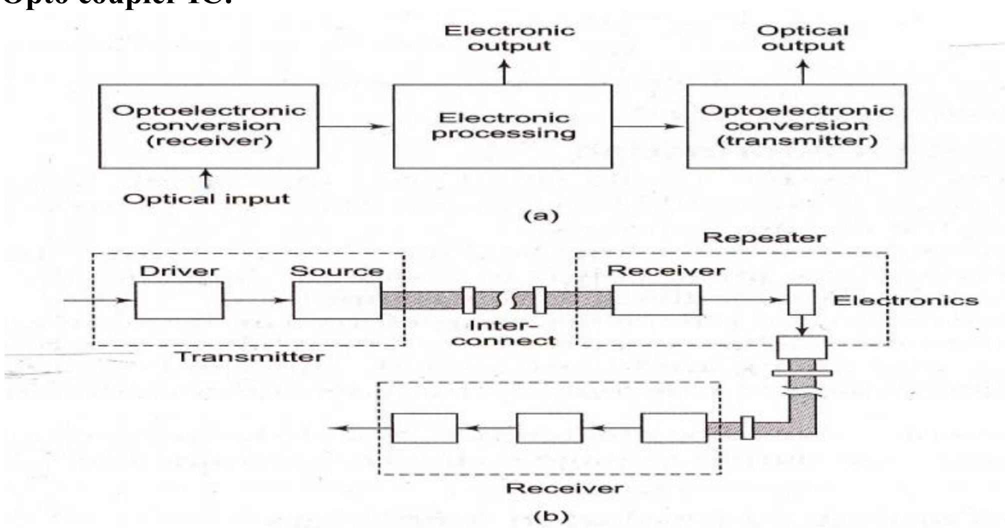


Fig. 5.60 The block diagram of opto-electronic-integrated circuit (OEIC)

The optocouplers are available in the IC form MCT2E is the standard optocoupler IC which is used popularly in many electronic application.

- This input is applied between pin 1& pin 2. An infrared light emitting diode is connected between these pins.
- The infrared radiation from the LED gets focused on the internal phototransistor.
- The base of the phototransistor is generally left open. But sometimes a high value pull down resistance is connected from the Base to ground to improve the sensitivity.
- The block diagram shows the opto-electronic-integrated circuit (OEIC) and the major components of a fiber-optic communication facility.
- The block diagram shows the opto-electronic-integrated circuit (OEIC) and the major components of a fiber-optic communication facility.

Question Bank

UNIT-I BASICS OF OPERATIONAL AMPLIFIERS

2 marks questions

1. **What do you mean by a band-gap referenced biasing circuit?**
The biasing sources referenced to VBE has a negative temperature coefficient and VT has a positive temperature co-efficient. Band gap reference circuit is one in which the output current is referenced to a composite voltage that is a weighted sum of VBE and VT so that by proper weighting, zero temperature coefficient can be achieved.

2. Define thermal drift.
The bias current, offset current & offset voltage change with temperature. A circuit carefully nulled at 25°C may not remain so when the temperature raises to 35°C.This is called thermal drift. Often, offset current drift is expressed in nA/ °C and offset voltage drift in mV/ ^{0}C.

3. Define supply voltage rejection ratio (SVRR)
The change in OPAMP's input offset voltage due to variations in supply voltage is called the supply voltage rejection ratio. It is also called Power Supply Rejection Ratio (PSRR) or Power Supply Sensitivity (PSS)
.

4. **Define an operational amplifier.**
An operational amplifier is a direct-coupled, high gain amplifier consisting of one or more differential amplifier. By properly selecting the external components, it can be used to perform a variety of mathematical operations.

5. **Mention the characteristics of an ideal op-amp.**
- Open loop voltage gain is infinity.
- Input impedance is infinity.
- Output impedance is zero.
- Bandwidth is infinity.
- Zero offset.

6. **Define input offset voltage.**
A small voltage applied to the input terminals to make the output voltage as zero when the two input terminals are grounded is called input offset voltage.

7. **Define CMRR of an op-amp.**
The relative sensitivity of an op-amp to a difference signal as compared to a common –mode signal is called the common –mode rejection ratio. It is expressed in decibels. CMRR= Ad/Ac

8. What is frequency response of Op-amp?
The plot showing the variations in magnitude and phase angle of the gain due to change in frequency is called frequency response of Op-amp. The plot is used to find the bandwidth and cut-off

frequencies of Op-amp.

9. Define Unity Gain Bandwidth of Op-amp.
For a certain frequency of the input signal, the gain of the Op-amp reduces to 0 dB. This means 20 log |AOL (f) | is 0dB i.e. |AOL (f) | = 1. Such a frequency is called gain cross over frequency or unity gain bandwidth (UGB).

10. **Define slew rate.**
The slew rate is defined as the maximum rate of change of output voltage caused by a step input voltage. An ideal slew rate is infinite which means that opamp's output voltage should change instantaneously in response to input step voltage.

16 marks questions
1. Wilson current source circuit and widlar current source circuit.
2. Explain Block diagram of op-amp (8) Internal Stages of Op-amp.
3. Explain the operation of differential amplifier, DC analysis of differential amplifier, Differential amplifier with constant current source, Differential amplifier active load.
4. Elaborate on AC & DC characteristics of Op-amp.
5. Analyse the open loop and closed loop characteristics of Op amp

UNIT II - APPLICATION OF OP – AMPS

1. **Mention some of the linear applications of op – amps.**
Adder, subtractor, voltage –to- current converter, current –to- voltage converters, instrumentation amplifier, analog computation ,power amplifier, etc are some of the linear op-amp circuits.

2. **Mention some of the non – linear applications of op-amps.**
Rectifier, peak detector, clipper, clamper, sample and hold circuit, log amplifier, anti –log amplifier, multiplier are some of the non – linear op-amp circuits.

3. Define virtual ground property of Op-amp.
Concept of virtual ground says that the two input terminals of the Op-amp are always at the same potential. Thus if one terminal is grounded the other can be assumed to be at ground potential, which is called virtual ground.

4. What is Voltage follower?
- A circuit in which the output voltage follows the input voltage is called voltage follower Circuit.
- In Op-amp if the inverting input and the output terminals are shorted and if any signal is Applied at the non-inverting terminal, it appears at the output without any change.
- It is also called as source follower, unity gain amplifier, buffer amplifier or isolation amplifier.

1. **Calculate the output voltage V_0 of the circuit shown in fig. 1**

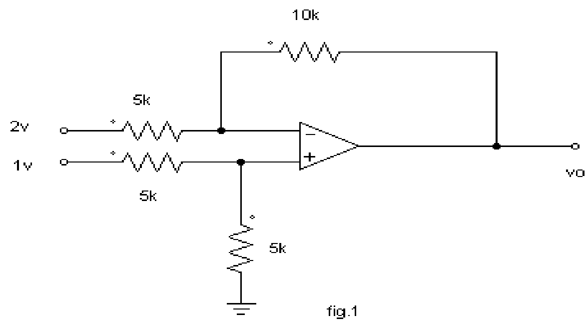


fig.1

6. Draw the circuit diagram of voltage follower using IC 741.

7. For the op-amp shown, determine the voltage gain.

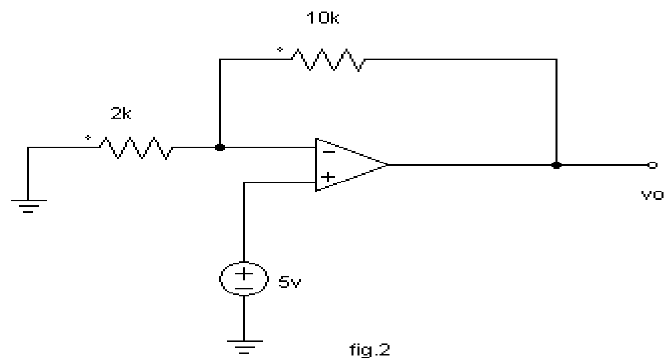


fig.2

8. List the features of instrumentation amplifier.

- High gain accuracy
- High CMRR
- High gain stability with low temperature co-efficient
- Low dc offset
- Low output impedance

9. What are the applications of V-I converter?

- Low voltage dc and ac voltmeter
- LED
- Zener diode tester

10. What do you mean by a precision diode?

The major limitation of ordinary diode is that it cannot rectify voltages below the cut – in voltage of the diode. A circuit designed by placing a diode in the feedback loop of an op – amp is called the precision diode and it is capable of rectifying input signals of the order of milli volt.

11.What are the limitations of the basic differentiator circuit?

At high frequency, a differentiator may become unstable and break into oscillations. The input impedance decreases with increase in frequency , thereby making the circuit sensitive to high frequency noise.

12. Write down the condition for good differentiation.
For good differentiation, the time period of the input signal must be greater than or equal to $R_f C_1$
$T > R_f C_1$ Where, R_f is the feedback resistance C_f is the input capacitance

13. What are the applications of comparator?
- Zero crossing detector
- Window detector
- Time marker generator
- Phase detector

14. What is a Schmitt trigger?
Schmitt trigger is a regenerative comparator. It converts sinusoidal input into a square wave output. The output of Schmitt trigger swings between upper and lower threshold voltages, which are the reference voltages of the input waveform.

32.Differentiate Schmitt trigger and comparator.

S.No	Comparator	Schmitt trigger
1.	Feedback is not used that is Op-amp is used in open loop mode	Feedback is used that is Op-amp is used in closed loop mode
2.	False triggering due to noise voltages is possible.	False triggering due to noise voltages is not possible.
3.	A single reference voltage exists which acts as triggering voltage.i.e. V_{ref} or $-V_{ref}$	Two different threshold voltages exists as V_{UT} and V_{LT}
4.	The hysteresis exists with a width $H=V_{UT} - V_{LT}$	The hysteresis does not exists

16. Define logarithmic and antilogarithmic amplifier.
- The Op-amp circuit in which the output is proportional to the logarithmic of the input is called logarithmic amplifier. It employs a diode or a transistor in the negative feedback path.
- The Op-amp circuit in which the output is proportional to the antilogarithmic of the input is called logarithmic amplifier. It employs a diode or a transistor in the input stage.

17. List the applications of Log amplifiers.
- Analog computation may require functions such as ln x, log x, sin h x etc.

These functions can be performed by log amplifiers
- Log amplifier can perform direct dB display on digital voltmeter and spectrum analyzer
- Log amplifier can be used to compress the dynamic range of a signal

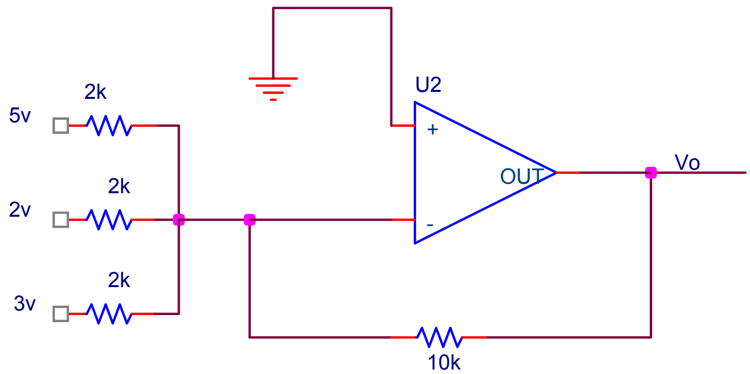
18. What is a filter?
Filter is a frequency selective circuit that passes signal of specified band of frequencies and attenuates the signals of frequencies outside the band.

19. What are the advantages of active filters?
- Active filters used op- amp as the active element and resistors and capacitors as passive elements.
- By enclosing a capacitor in the feedback loop , inductor less active filters can be obtained
- Op-amp used in non – inverting configuration offers high input impedance and low output impedance, thus improving the load drive capacity.

20.What are the requirements for producing sustained oscillations in feedback circuits?
For sustained oscillations,
- The total phase shift around the loop must be zero at the desired frequency of oscillation, fo. ie, AB =0 (or) 360°
- At fo, the magnitude of the loop gain | A β | should be equal to unity

16 marks questions

1. What is meant by voltage follower?
2. Determine the output voltage v_o for the following circuit.



3. Explain the operation of the Schmitt trigger.
4. Describe Logarithmic and anti-logarithmic amplifier
5. Explain Precision rectifier and its applications -Clipper and clamper circuits
6. Analyse of low pass Butterworth filters.
7. Elaborate on I to V and V to I converters and applications
8. Explain application of op-amp as integrator and differentiator
9. Discuss the need for an instrumentation amplifier? Explain three op-amp instrumentation amplifier with diagram (16)

UNIT –III ANALOG MULTIPLIER AND PLL

2 marks questions
1. What is a four quadrant multiplier?
In a multiplier circuit, if both the inputs are allowed to swing in both positive and negative directions the multiplier is called as a four quadrant multiplier.

2. Mention some areas where PLL is widely used.
- Radar synchronization
- Satellite communication systems
- Air borne navigational systems
- FM communication systems

- Computers.

3. List the basic building blocks of PLL
- Phase detector/comparator
- Low pass filter
- Error amplifier
- Voltage controlled oscillator

4. What are the three stages through which PLL operates?
- Free running
- Capture
- Locked/ tracking

5. Define lock-in range of a PLL.
The range of frequencies over which the PLL can maintain lock with the incoming signal is called the lock-in range or tracking range. It is expressed as a percentage of the VCO free running frequency.

6. Define capture range of PLL.
The range of frequencies over which the PLL can acquire lock with an input signal is called the capture range. It is expressed as a percentage of the VCO free running frequency.

7. Define Pull-in time.
The total time taken by the PLL to establish lock is called pull-in time. It depends on the initial phase and frequency difference between the two signals as well as on the overall loop gain and loop filter characteristics.

8. Mention some typical applications of PLL:
- Frequency multiplication/division
- Frequency translation
- AM detection
- FM demodulation
- FSK demodulation.

9. What is a voltage controlled oscillator?
Voltage controlled oscillator is a free running multivibrator operating at a set frequency called the free running frequency. This frequency can be shifted to either side by applying a dc control voltage and the frequency deviation is proportional to the dc control voltage.

10. Define VCO.
A voltage controlled oscillator is an oscillator circuit in which the frequency of oscillations can be controlled by an externally applied voltage.

11. On what parameters does the free running frequency of VCO depend on?
- External timing resistor, RT
- External timing capacitor, CT

 The dc control voltage Vc.

12. Give the expression for the VCO free running frequency.
fo = 0.25 / RT CT

13. Define Voltage to Frequency conversion factor.
Voltage to Frequency conversion factor is defined as,
Ky = f**o** / V**c= 8fo /Vcc**
where, Vc is the modulation voltage required to produce the frequency shift fo

14. Define FSK modulation.
In digital data communication, binary data is transmitted by means of a carrier frequency. FSK employs two different carrier frequencies one for logic 1 and other for logic 0 states ofbinary data signal. This process is called FSK modulation.

14. Define free running mode.
In a PLL if the error control voltage is zero then the PLL is said to be operated in free running mode and its output frequency is called its center frequency f0.

16 marks
1. Elaborate on Gilbert multiplier cell/ Four quadrant multiplier.
2. Explain Analog multiplier using emitter coupled transistor.
3.Analyze the analog multiplier IC with a neat circuit diagram. Discuss its applications.
4.Discuss the Operation of PLL & applications of PLL.
5.With a neat functional diagram, explain the operation of VCO. Also derive an expression for fo.

UNIT IV - ANALOG TO DIGITAL AND DIGITAL TO ANALOG CONVERTERS

1. Explain in brief the principle of operation of successive Approximation ADC.
The circuit of successive approximation ADC consists of a successive approximation register (SAR), to find the required value of each bit by trial & error. With the arrival of START command, SAR sets the MSB bit to 1. The O/P is converted into an analog signal & it is compared with I/P signal. This O/P is low or High. This process continues until all bits are checked.

2. Where are the successive approximation type ADC's used?
The Successive approximation ADCs are used in applications such as data loggers & instrumentation where conversion speed is important.

3. What is the main drawback of a dual-slop ADC?
The dual slope ADC has long conversion time. This is the main drawback of dual slope ADC.

4. State the advantages of dual slope ADC:
It provides excellent noise rejection of ac signals whose periods are integral multiples of the

integration time T.

5. Define conversion time.
It is defined as the total time required to convert an analog signal into its digital output. It depends on the conversion technique used & the propagation delay of circuit components. The conversion time of a successive approximation type ADC is given by
T(n+1) where T---clock period
Tc---conversion time n----no. of bits

6.. **Define resolution of a data converter.**
The resolution of a converter is the smallest change in voltage which may be produced at the output or input of the converter.
Resolution (in volts)= VFS/2n-1=1 LSB increment. The resolution of an ADC is defined as the smallest change in analog input for a one bit change at the output.

7. **Define accuracy of converter.**
Absolute accuracy:
It is the maximum deviation between the actual converter output & the ideal converter output.
Relative accuracy:
It is the maximum deviation after gain & offset errors have been removed. The accuracy of a converter is also specified in form of LSB increments or % of full scale voltage.

8. **What is settling time?**
It represents the time it takes for the output to settle within a specified band }.LSB of its final value following a code change at the input (usually a full scale change). It depends upon the switching time of the logic circuitry due to internal parasitic capacitance & inductances. Settling time ranges from 100ns. 10μ s depending on word length & type circuit used.

9. **Explain in brief stability of a converter:**
The performance of converter changes with temperature age & power supply variation . So all the relevant parameters such as offset, gain, linearity error & monotonicity must be specified over the full temperature & power supply ranges to have better stability performances.

10. **What is meant by linearity?**
The linearity of an ADC/DAC is an important measure of its accuracy & tells us how close the converter output is to its ideal transfer characteristics. The linearity error is usually expressed as a fraction of LSB increment or percentage of full-scale voltage. A good converter exhibits a linearity error of less than }.LSB.

11. **What is monotonic DAC?**
A monotonic DAC is one whose analog output increases for an increase in digital input.

12. Find the resolution of an 8-bit DAC.
Resolution =1/(2^8 -1) =1/ 255

13. Which is the fastest A/D converter and why?

`Flash type A/D converter is the fastest ADC, because the fast conversion speed is accomplished by providing 2n-1 comparators and simultaneously comparing the input signal with unique reference levels spaced 1 LSB apart.

14**. What are limitations of Flash type ADC?**

Flash type ADC employs 2n-1 comparators for conversion which makes it costlier which tradeoffs in the speed of conversion.

15. Name the essentials parts of a DAC.

Analog input signal, D/A converter circuit, Switches for DAC.

16. What are advantages and disadvantages of R-2R ladder DAC?

Adv:-

Easier to build accurately as only two precision metal film r istors are required Number of bits can be expanded by adding more sections of same R/2R values.

In inverted R/2R ladder DAC, node voltages remain constant with changing input binary wo This avoids any slowdown effects by stray capacitances.

Disadv:-

With increasing output bits the circuit becomes larger

The switches used are noted for the sources of errors.

17. **What is a sample and hold circuit? Where it is used?**

A sample and hold circuit is one which samples an input signal and holds on to its last sampled value until the input is sampled again. This circuit is mainly used in digital interfacing, analog to digital systems, and pulse code modulation systems.

18.**Define sample period and hold period.**

The time during which the voltage across the capacitor in sample and hold circuit is equal to the input voltage is called sample period. The time period during which the voltage across the capacitor is held constant is called hold period.

19.What are the specifications of data convertors?

conversion time, settling time, accuracy, linearity, monotonic etc.

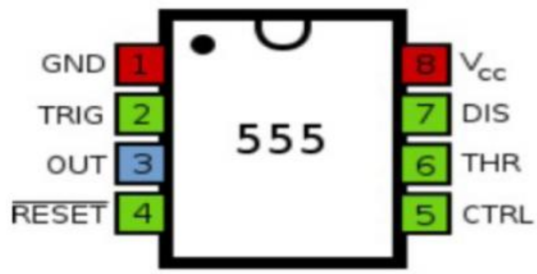
16 marks questions

1. Explain the operation of sample and hold circuit .
2. Explain the operation of FLASH type ADC.
3. Weighted resistor type D/A converter, R-2R ladder type D/A converter – Voltage mode and current mode
4. Dual Slope type A/D converter , Single slope type A/D converter
5. Sample and Hold circuits for A/D converter
6. Explain A/D convertors using any two methods - successive Approximation ADC.

UNIT V – SPECIAL FUNCTION ICs
2 mark questions

1. Define 555 IC?
The 555 timer is an integrated circuit specifically designed to perform signal generation and timing functions.

2. Draw the Pin diagram of IC 555.



Pin	Name	Purpose
1	GND	Ground, low level (0 V)
2	TRIG	OUT rises, and interval starts, when this input falls below 1/3 V_{CC}.
3	OUT	This output is driven to $+V_{CC}$ or GND.
4	$\overline{\text{RESET}}$	A timing interval may be interrupted by driving this input to GND.
5	CTRL	"Control" access to the internal voltage divider (by default, 2/3 V_{CC}).
6	THR	The interval ends when the voltage at THR is greater than at CTRL.
7	DIS	Open collector output; may discharge a capacitor between intervals.
8	$V+$, V_{CC}	Positive supply voltage is usually between 3 and 15 V.

3. List the basic blocks of IC 555 timer?
- **A relaxation oscillator**
- RS flip flop
- Two comparator
- Discharge transistor.

4. Give the classification of voltage regulators:
- Series / Linear regulator
- Switching regulators

5.Mention some applications of 555 timer:
- Oscillator
- pulse generator
- ramp and square wave generator

- mono-shot multivibrator
- burglar alarm
- Traffic light control.

6.List the applications of 555 timer in monostable mode of operation:
- Missing pulse detector
- Linear ramp generator
- Frequency divider
- Pulse width modulation.

7.List the applications of 555 timer in Astable mode of operation:
- FSK generator
- Pulse-position modulator

8. What is a multivibrator?
Multivibrators are a group of regenerative circuits that are used extensively in timing applications. It is a wave shaping circuit which gives symmetric or asymmetric square output. It has two states either stable or quasi- stable depending on the type of multivibrator.

9 .What do you mean by monostable multivibrator?
Monostable multivibrator is one which generates a single pulse of specified duration in response to each external trigger signal. It has only one stable state. Application of a trigger causes a change to the quasi-stable state. An external trigger signal generated due to charging and discharging of the capacitor produces the transition to the original stable state.

10.What is an astable multivibrator?
Astable multivibrator is a free running oscillator having two quasi-stable states. Thus, there is oscillations between these two states and no external signal are required to produce the change in state.

11. Give the classification of voltage regulators: What is a voltage regulator?
A voltage regulator is an electronic circuit that provides a stable dc voltage independent of the load current, temperature, and ac line voltage variations.

12. What is a linear voltage regulator?
Series or linear regulator uses a power transistor connected in series between the unregulated dc input and the load and it conducts in the linear region .The output voltage is controlled by the continuous voltage drop taking place across the series pass transistor.

13. What is a switching regulator?
Switching regulators are those which operate the power transistor as a high frequency on/off switch, so that the power transistor does not conduct current continuously. This gives improved efficiency over series regulators.

14. Give some examples of monolithic IC voltage regulators:
- 78XX series fixed output, positive voltage regulators

- 79XX series fixed output, negative voltage regulators
- 723 general purpose regulator.

15. What is the purpose of having input and output capacitors in three terminal IC regulators?
A capacitor connected between the input terminal and ground cancels the inductive effects due to long distribution leads. The output capacitor improves the transient response.

16. Define line regulation.
Line regulation is defined as the percentage change in the output voltage for a change in the input voltage. It is expressed in milli volts or as a percentage of the output voltage.

17. **Define load regulation.**
Load regulation is defined as the change in output voltage for a change in load current. It is expressed in millivolts or as a percentage of the output voltage.

18. **What is meant by current limiting?**
Current limiting refers to the ability of a regulator to prevent the load current from increasing above a preset value.

19. **Give the drawbacks of linear regulators:**

The input step down transformer is bulky and expensive because of low line frequency.
- Because of low line frequency, large values of filter capacitors are required to decrease the ripple.
- Efficiency is reduced due to the continuous power dissipation by the transistor as it operates in the linear region.

20. What is the advantage of switching regulators?
- Greater efficiency is achieved as the power transistor is made to operate as low impedance switch. Power transmitted across the transistor is in discrete pulses rather than as a steady current flow.
- By using suitable switching loss reduction technique, the switching frequency can be increased so as to reduce the size and weight of the inductors and capacitors.

21. Define drop-out voltage of a fixed voltage regulator.
It is the minimum voltage that must exist between input and output terminals. It is defined as the ratio of the r.m.s input ripple voltage to the r.m.s output ripple voltage. It is expressed in decibels (dB).

22. What are the advantages of switched capacitor filters?
- Very high value of resistors can be easily simulated using small value capacitors, of the order of 10pF.
- The switched capacitor filters require no external reactive components like indı capacitors.
- Completely active filters can be easily obtained on a monolithic IC chip.
- Good temperature stability, high accuracy and low cost.

23. Define video amplifier.

A video amplifier has to amplify signals over a wideband of frequencies, say upto 20MHz. It is a RC coupled amplifier with bandwidth from d.c. to high frequency upto few megahertz.

24. What is opto coupler?

An opto-isolator, also called an optocoupler, photocoupler, or optical isolator, is "an electronic device designed to transfer electrical signals by utilizing light waves to provide coupling with electrical isolation between its input and output". The main purpose of an opto-isolator is "to prevent high voltages or rapidly changing voltages on one side of the circuit from damaging components or distorting transmissions on the other side

25. What is LM380?

It is a power amplifier produced by national semiconductor. It is capable of delivering 2.5 W min, to 8 ohm load.

26. Mention the advantages of opto-couplers:

- Better isolation between the two stages.
- Impedance problem between the stages is eliminated.
- Wide frequency response.
- Easily interfaced with digital circuit.
- Compact and light weight.
- Problems such as noise, transients, contact bounce, are eliminated.

27. What is an isolation amplifier?

An isolation amplifier is an amplifier that offers electrical isolation between its input and output terminals

28. What are the types of optocouplers?

- LED and a photo diode,
- LED and photo transistor,
- LED and Darlington

29. Give two examples of IC optocouplers?

- Examples for opto-coupler IC
- MCT 2F
- MCT 2E

.

30. List the applications of Isolation amplifier.

- Floating pulse amplifier output voltage and current interface
- Instrumentation in high-noise environments.
- Analogue front-end processing.
- Medical instrumentation.

16 marks

1. Wien Bridge and RC phase oscillator using IC741
2. Triangular and saw tooth wave generator

3. Working of ICL 8038 function generator
4. Discuss in detail the operation of Astable multivibrator, Monostable multivibrator.
5. Explain functional block diagram of IC 555 timer.(8)
6. In detail discuss the 723 IC general purpose voltage regulators.
7. Explain the operation of switching regulators. Give its advantages.
8. Explain the functional diagram of LM 380 power amplifier.
9. Explain the operation of ICL 8038 function generator. Give its advantages.
10. Explain the operation of(1) opto electronic ICs, (2) audio amplifier (3) video amplifier (4) Opto-couplers and fiber optic ICs

UNIVERSITY QUESTIONS

Department of Electronics and Communication Engineering

EC2254-Linear Integrated Circuits

April/May 2014

Question Paper

PART A (10*2=20)

1. List any two advantages of ICS over discrete components.

2. Define slew state.

3. What is a voltage follower?

4. Draw the circuit diagram of peak detector.

5. State the operation of a basic PLL.

6. What is the need for frequency synthesizer?

7. What is a sample/hold circuit?

8. Give the two advantages of SA type ADC.

9. List the two type of multi vibrators.

10. What is an opto-coupler?

PART-B (5*16=80)

11. a) Explain the construction of monolithic bipolar transistor, monolithic diode and Integrated resistors. (16)

Or

b) Explain the Integral circuit diagram of IC741.Discuss its AC and DC performance Characteristics. (16)

12 .a) With neat diagram explain logarithmic amplifier and anti logarithmic amplifier.

Or

b) With neat diagram explain the application of op-amp as precision rectifier, clipper and Clamper. (16)

13. a) Explain the working of Analog multiplier using emitter coupled transistor pair. Discuss the applications of analog multiplier IC. (16)

b) Explain the application of PLL as AM detection, FM detection and FSK demodulation. (16)

14. a) Explain weighted resistor type and R-2R Ladder type DAC. (16)

Or

b) Explain Flash type, Single type and Dual slope type ADC. (16)

15. a) With neat diagram explain IC723 General Purpose regulator. (16)

Or

b) Explain in detail voltage to frequency and frequency to voltage converters. (16)

Reg. No. | | | | | | | | | | |

Question Paper Code: 11286

B.E./B.Tech.Degree Examinations, April/May 2011
Regulations 2008

Fourth Semester

Electronics and Communication Engineering

EC 2254 Linear Integrated Circuits
(Common to PTEC 2254 Linear Integrated Circuits for B.E.(Part -Time) Third Semester ECE - Regulations 2009)

Time: Three Hours — Maximum: 100 marks

Answer ALL Questions

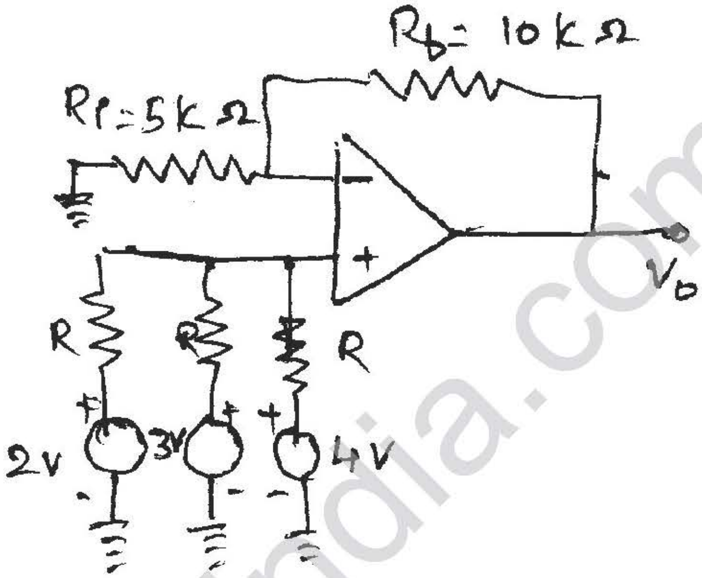
Part A - (10 x 2 = 20 marks)

1. Name the different methods used in fabrication of integrated resistors.
2. Define CMRR of an OP-AMP.
3. Mention two important features of an instrumentation amplifier.
4. How does precision rectifier differ from the conventional rectifier?
5. What are the advantages of emitter coupled transistor pair?
6. With reference to a VCO, define voltage to frequency conversion factor K_v.
7. Define accuracy of a D/A converter
8. Which is the fastest ADC? State the reason.
9. Define the duty cycle in astable multivibrator using IC 555.
10. What is an optocoupler? Mention its applications.

Part B - (5 x 16 = 80 marks)

11. (a) (i) With a neat circuit diagram and with necessary equations, explain the concept of Widlar current source used in op-amp circuit. (10)

(ii) For the non-inverting op-amp shown in figure below, find the output voltage V_0. (6)



OR

11. (b) (i) Define and explain slew rate. What is 'full-power bandwidth'? Also explain the methods adopted to improve slew rate. (10)

(ii) Define output off-set voltage. Explain methods to nullify off-set voltage? (6)

12. (a) With neat circuit diagrams and mathematical expressions, explain the operation of the following op-amp applications:

(i) Scale changer (4)

(ii) Voltage follower (4)

(iii) Non-Inverting adder (4)

(iv) Integrator. (4)

OR

12. (b) With the help of circuits and necessary equations, explain how log and antilog computations are performed using IC 741. (16)

13. (a) Sketch and explain the following applications of multipliers:

(1) Squaring (4)

(2) finding square root (4)

(3) frequency doubler (4)

(4) phase angle detector. (4)

OR

13. (b) (i) Draw the block diagram of VCO and explain its operation. Also derive the frequency of oscillator. (10)

(ii) Draw the circuit of a PLL used as AM detector and explain its operation. (6)

14. (a) (i) Explain the operation of a weighted resistor type D/A converter. (8)

(ii) What are the limitations in weighted resistor type D/A converters and explain how this problem can be solved in R-2R ladder type D/A converters. (8)

OR

14. (b) (i) With the neat block diagram, explain, in detail, the successive approximation type A/D converter. (8)

(ii) Explain the over sampling A/D converter with functional block diagram. (8)

15. (a) (i) Draw the circuit using op-amp to generate triangular wave. Explain its operation. (8)

(ii) With a neat diagram, explain the working of step down switching regulator. (8)

OR

15. (b) (i) With suitable diagram, explain the working of a switched capacitor filter. Also explain how resistor can be realized using switched capacitor filter. (8)

(ii) With necessary diagrams, explain the operation of frequency to voltage converters. (8)

Reg. No. | | | | | | | | | | |

Question Paper Code: 55335

B.E./B.Tech. DEGREE EXAMINATIONS, NOV./DEC. 2011
Regulations 2008

Fourth Semester

Electronics and Communication Engineering

EC 2254 Linear Integrated Circuits
(Common to PTEC 2254 Linear Integrated Circuits for B.E.(Part -Time)
Third Semester ECE - Regulations 2009)

Time: Three Hours Maximum: 100 marks

Answer ALL Questions

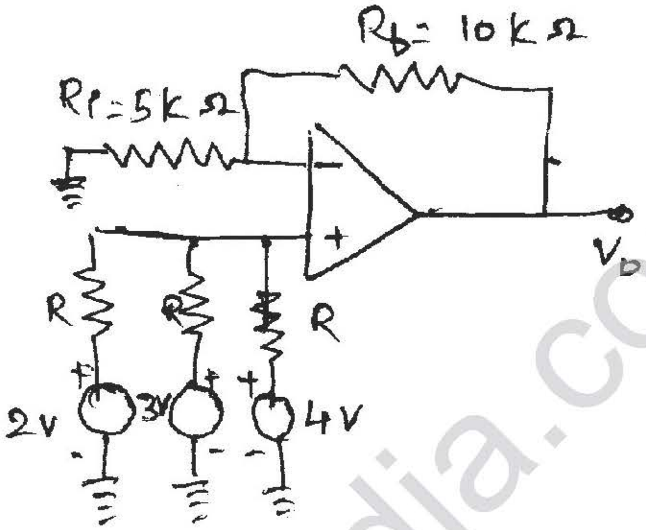
Part A - (10 x 2 = 20 marks)

1. Define current mirror with magnification.
2. Define slew rate.
3. Why *a*re integrators preferred over differentiators?
4. What *i*s comparator?
5. What *a*re the advantages o*f* variable transconductance technique?
6. Define: Capture range of a PLL.
7. What is meant by resolution *o*f a DAC?
8. Which *i*s *t*he fastest ADC? State *t*he reason.
9. Define *t*he duty cycle in astable multivibrator using IC 555.
10. What ar*e* the advantages o*f* Switched capacitor filter over active filters?

Part B - (5 x 16 = 80 marks)

11. (a) (i) With a neat circuit diagram and with necessary equations, explain the concept of Widlar current source used in op-amp circuit. (10)

(ii) For the non-inverting op-amp shown in figure below, find the output voltage V_0. (6)



OR

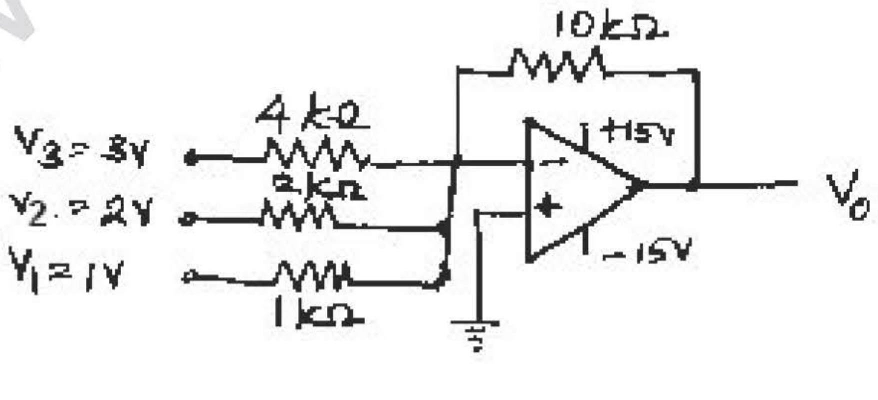
11. (b) (i) With a neat block diagram, explain the general stages of an OP-AMP IC. (6)

(ii) Explain, with a circuit diagram, the working of BJT - emitter coupled differential amplifier. Also explain the concept of active load and sketch the relevant circuit diagram. (10)

12. (a) (i) Explain the construction and working of OP-AMP based instrumentation amplifier. (8)

(ii) Draw an adder-subtractor type of circuit with op-amp to obtain the relation $V_0 = (V_1 + V_2) - (V_3 + V_4)$ (4)

(iii) Calculate the output of the following circuit. (4)



OR

12. (b) (i) Explain the working of OP-AMP based Schmitt trigger circuit. (8)

(ii) Design an OP-AMP based second order active low pass filter with cut off frequency 2 kHz. (8)

13. (a) (i) Sketch and explain the multiplier cell using emitter-coupled transistor pair. Prove that the output voltage is proportional to the product of the two input voltages. (12)

(ii) State the limitations of emitter-coupled pair. (4)

OR

13. (b) (i) With usual notations, show that the 'lock-in-range' of PLL is $\Delta f_L = \pm 7.8 f_0/V$. (10)

(ii) Explain how the IC 565 PLL can be used as a FSK demodulator. (6)

14. (a) Explain the following types of digital to analog converters, with suitable circuit diagrams:

(i) Binary weighted resistor DAC (6)

(ii) R-2R Ladder DAC (5)

(iii) Inverted R-2R Ladder DAC (5)

OR

14. (b) (i) Draw the circuit of a flash type ADC and explain. (8)

(ii) What is the purpose of *'high speed sample and hold circuit'*? Sketch such a circuit and explain. Also name the parameters associated with it. (8)

15. (a) (i) With neat functional block diagram, explain the working of IC 555 in astable mode. (8)

(ii) Describe in detail, the working principle of IC 8038 function generator. (8)

OR

15. (b) (i) With a neat functional diagram, explain the operation of LM 380 power amplifier. (8)

(ii) Explain the operation of switched capacitor filter. What are the advantages and disadvantages of this type of filter? (8)